

AHSME

§1 AHSME 1965

1. (*AHSME 1965*) The number of real values of x satisfying the equation $2^{2x^2-7x+5} = 1$ is:

(A) 0 (B) 1 (C) 2 (D) 3 (E) more than 4

Answer: C

Solution 1

Solution by e_power_pi_times_i

Take the logarithm with a base of 2 to both sides, resulting in the equation $2x^2 - 7x + 5 = 0$. Factoring results in $(2x - 5)(x - 1) = 0$, so there are (C) 2 real solutions.

Solution 2

Notice that $a^0 = 1, a > 0$. So $2^0 = 1$. So $2x^2 - 7x + 5 = 0$. Evaluating the **discriminant**, we see that it is equal to $7^2 - 4 * 2 * 5 = 9$. So this means that the equation has two real solutions. Therefore, select C.

2. (*AHSME 1965*) A regular hexagon is inscribed in a circle. The ratio of the length of a side of the hexagon to the length of the shorter of the arcs intercepted by the side, is:

(A) 1 : 1 (B) 1 : 6 (C) 1 : π (D) 3 : π (E) 6 : π

Answer: D

Solution

Suppose that each side of the hexagon is 3. Then the distance from each vertex of the hexagon to the center is also 3, so that the circle has radius 3. Since the circle has circumference $2\pi(3) = 6\pi$, the arc intercepted by any side (which measures 60°) has length $\frac{1}{6} * 6\pi = \pi$, so that our answer is (D) and we are done.

3. (*AHSME 1965*) The expression $(81)^{-2^{-2}}$ has the same value as:

(A) $\frac{1}{81}$ (B) $\frac{1}{3}$ (C) 3 (D) 81 (E) 81^4

Answer: B

Solution

Let us recall PEMDAS. We calculate the exponent first. $(-2)^{-2} = \frac{1}{(-2)^2} = \frac{1}{4}$

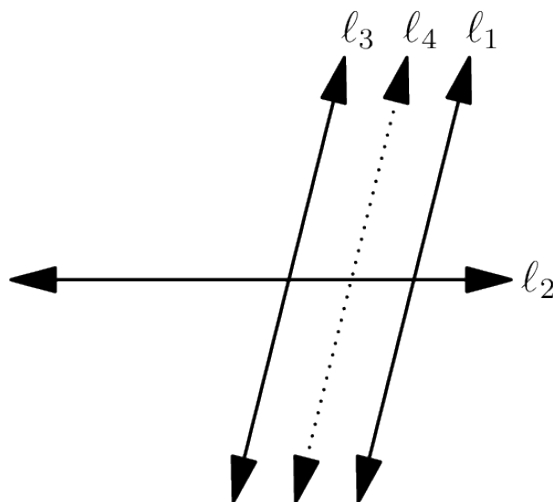
When we substitute, we get $81^{\frac{1}{4}} = \sqrt[4]{81} =$ (C) 3.

4. (AHSME 1965) Line ℓ_2 intersects line ℓ_1 and line ℓ_3 is parallel to ℓ_1 . The three lines are distinct and lie in a plane. The number of points equidistant from all three lines is:

(A) 0 (B) 1 (C) 2 (D) 4 (E) 8

Answer: C

Solution



centering

The lines are coplanar, $\ell_1 \parallel \ell_3$, and ℓ_1 intersects ℓ_2 . Therefore, ℓ_3 also intersects ℓ_2 . The **locus** of all points equidistant from parallel lines ℓ_1 and ℓ_3 is a third parallel line in between them. Let this line be ℓ_4 , and let the distance from ℓ_4 to either ℓ_1 or ℓ_3 be d . The points equidistant from lines ℓ_1 , ℓ_2 , and ℓ_3 must all lie on ℓ_4 and be a distance d from line ℓ_2 . There are only 2 points, on either side of ℓ_2 , which satisfy these conditions. Thus, our answer is (C) 2.

5. (AHSME 1965) When the repeating decimal $0.363636\ldots$ is written in simplest fractional form, the sum of the numerator and denominator is:

(A) 15 (B) 45 (C) 114 (D) 135 (E) 150

Answer: A

Solution

We let $x = 0.\overline{36}$. Thus, $100x = 36.\overline{36}$. We find that $100x - x = 99x = 36.\overline{36} - 0.\overline{36} = 36$, or $x = \frac{36}{99} = \frac{4}{11}$. Since $4 + 11 = 15$, the answer is (A) 15.

6. (AHSME 1965) If $10^{\log_{10} 9} = 8x + 5$ then x equals:

(A) 0 (B) $\frac{1}{2}$ (C) $\frac{5}{8}$ (D) $\frac{9}{8}$ (E) $\frac{2\log_{10} 3 - 5}{8}$

Answer: B

Solution

Notice that $10^{\log_{10} 9} = 9$. Therefore, the condition we are looking for is $9 = 8x + 5$, or $x = \frac{1}{2}$. So the answer is (B)

7. (AHSME 1965) The sum of the reciprocals of the roots of the equation $ax^2 + bx + c = 0$ is:

- (A) $\frac{1}{a} + \frac{1}{b}$ (B) $-\frac{c}{b}$ (C) $\frac{b}{c}$ (D) $-\frac{a}{b}$ (E) $-\frac{b}{c}$

Answer: E

Solution 1

Using **Vieta's formulas**, we can write the sum of the roots of any quadratic equation in the form $ax^2 + bx + c = 0$ as $-\frac{b}{a}$, and the product as $\frac{c}{a}$.

If r and s are the roots, then the sum of the reciprocals of the roots is $\frac{1}{r} + \frac{1}{s} = \frac{r+s}{rs}$.

Applying the formulas, we get $\frac{-\frac{b}{a}}{\frac{c}{a}}$, or $-\frac{b}{c} \Rightarrow \boxed{E}$.

Solution 2

Let $f(x) = ax^2 + bx + c$. $f(x)$ has roots r and s , and so $x^2 f(\frac{1}{x})$ has roots $\frac{1}{r}$ and $\frac{1}{s}$.

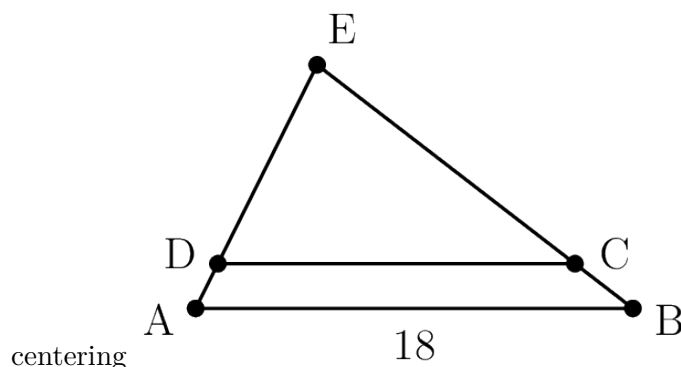
Because $x^2 f(\frac{1}{x}) = cx^2 + bx + a$, by **Vieta's formulas**, $\frac{1}{r} + \frac{1}{s} = \boxed{\frac{-b}{c}}$, which is answer choice $\boxed{E}$.

8. (AHSME 1965) One side of a given triangle is 18 inches. Inside the triangle a line segment is drawn parallel to this side forming a trapezoid whose area is one-third of that of the triangle. The length of this segment, in inches, is:

- (A) $6\sqrt{6}$ (B) $9\sqrt{2}$ (C) 12 (D) $6\sqrt{3}$ (E) 9

Answer: A

Solution



Let the given triangle be $\triangle ABE$ with $AB = 18$. Also, let $\overline{CD} \parallel \overline{AB}$ with C on $\overline{BE}$ and D on $\overline{AE}$. We know from the problem that $[\overline{ABCD}] = \frac{1}{3}[\triangle ABE]$, so $[\triangle DCE] = \frac{2}{3}[\triangle ABE]$. Because $\overline{DC} \parallel \overline{AB}$, $\triangle DCE \sim \triangle ABE$ by **AA similarity**. Because the ratio of these two triangles' areas is $\frac{2}{3}$, the ratio of their sides must be $\sqrt{\frac{2}{3}} = \frac{\sqrt{6}}{3}$. Thus, $CD = \frac{\sqrt{6}}{3}AB = \frac{\sqrt{6}}{3} * 18 = \boxed{6\sqrt{6}}$, which is answer choice $\boxed{A}$.

9. (AHSME 1965) The vertex of the parabola $y = x^2 - 8x + c$ will be a point on the x -axis if the value of c is:

- (A) -16 (B) -4 (C) 4 (D) 8 (E) 16

Answer: E

Solution

Notice that if the vertex of a parabola is on the x -axis, then the x -coordinate of the vertex must be a solution to the quadratic. Since the quadratic is strictly increasing on either side of the vertex, the solution must have double multiplicity,

or the quadratic is a perfect square trinomial. This means that for the vertex of $y = x^2 - 8x + c$ to be on the x-axis, the trinomial must be a perfect square, and have discriminant of zero. So,

$$0 = b^2 - 4ac$$

$$0 = (-8)^2 - 4c$$

$$c = 64$$

$$c = 16$$

Therefore $c = 16$, and our answer is **(E)**.

10. (AHSME 1965) The statement $x^2 - x - 6 < 0$ is equivalent to the statement:

(A) $-2 < x < 3$ (B) $x > -2$ (C) $x < 3$

(D) $x > 3$ and $x < -2$ (E) $x > 3$ or $x < -2$

Answer: A

Solution

To solve this problem, we may begin by factoring $x^2 - x - 6$ as $(x - 3)(x + 2)$. This is an upward opening parabola, therefore the solutions to $(x - 3)(x + 2) < 0$ are between the roots of the equation. That means our solutions are all x such that $-2 < x < 3$, or simply **(A)**.

11. (AHSME 1965) Consider the statements:

I : $(\sqrt{-4})(\sqrt{-16}) = \sqrt{(-4)(-16)}$,

II : $\sqrt{(-4)(-16)} = \sqrt{64}$,

III : $\sqrt{64} = 8$.

Of these the following are incorrect.

(A) none (B) I only (C) II only (D) III only (E) I and III only

Answer: B

Solution

Statement I is incorrect, because $(\sqrt{-4})(\sqrt{-16}) = (i\sqrt{4})(i\sqrt{16}) = i^2(\sqrt{4})(\sqrt{16}) = -\sqrt{64}$.

Statement II is correct, because $(-4)(-16) = 64$, so $\sqrt{(-4)(-16)} = \sqrt{64}$.

Statement III is correct, because $8^2 = 64$ and $8 \geq 0$ (so it is the **principal square root** of 64).

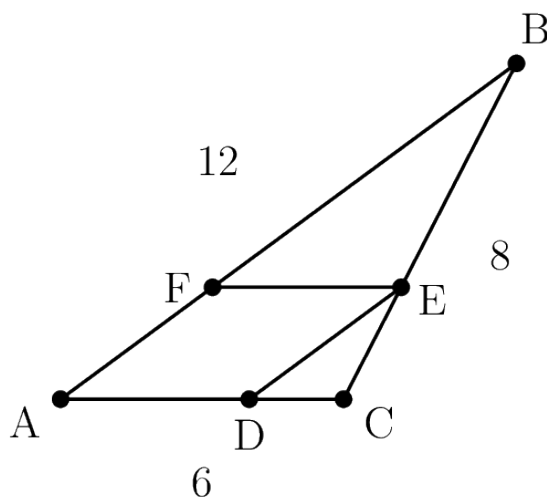
Thus, only statement I is incorrect, so we choose answer **(B)**.

12. (AHSME 1965) A rhombus is inscribed in $\triangle ABC$ in such a way that one of its vertices is A and two of its sides lie along AB and AC . If $\overline{AC} = 6$ inches, $\overline{AB} = 12$ inches, and $\overline{BC} = 8$ inches, the side of the rhombus, in inches, is:

(A) 2 (B) 3 (C) $3\frac{1}{2}$ (D) 4 (E) 5

Answer: D

Solution



centering

As in the diagram, suppose the rhombus $ADEF$ is inscribed in $\triangle ABC$ with D on $\overline{AC}$, E on $\overline{BC}$, and F on $\overline{AB}$. Let the side length of the rhombus be x . Because a rhombus is a parallelogram, its opposite sides are parallel. Thus, by **AA similarity**, $\triangle BFE \sim \triangle EDC$, and so $\frac{FE}{FB} = \frac{DC}{DE}$. Because $\overline{FE}$ and $\overline{DE}$ are sides of the rhombus, they both have length x . Furthermore, $FB = 12 - x$, and $DC = 6 - x$, because, combined with sides of the rhombus, they form sides of the triangle. Thus, by substituting into the proportion derived above, we see that:

$$\begin{aligned}\frac{FE}{FB} &= \frac{DC}{DE} \\ \frac{x}{12-x} &= \frac{6-x}{x} \\ x^2 &= 72 - 12x - 6x + x^2 \\ 18x &= 72 \\ x &= 4\end{aligned}$$

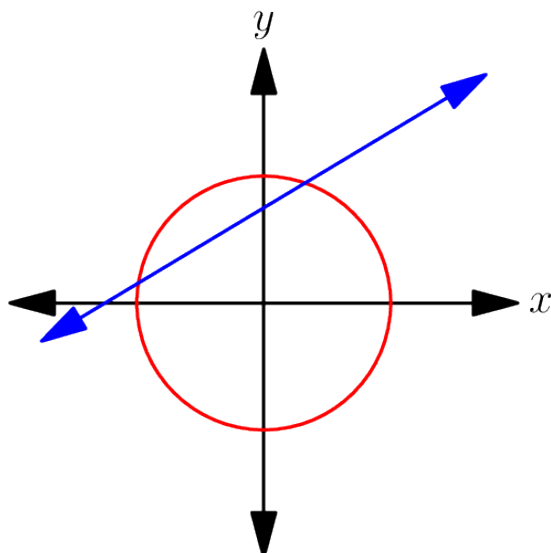
Thus, the side of the rhombus has length (D) 4.

13. (AHSME 1965) Let n be the number of number-pairs (x, y) which satisfy $5y - 3x = 15$ and $x^2 + y^2 \leq 16$. Then n is:

(A) 0 (B) 1 (C) 2 (D) more than two, but finite (E) greater than any finite number

Answer: E

Solution



centering

For an ordered pair (x, y) to satisfy the restrictions, the point (x, y) must lie on the graph of the line $5y - 3x = 15$ (shown in blue in the diagram) and on the closed disk given by $x^2 + y^2 \leq 16$ (whose boundary is red in the diagram). Because the y -intercept of the line, $(0, 3)$, is within the disk (and not on the boundary of the disk), the line cannot be tangent to the disk, and so the disk must contain infinitely many of the points on the line. Thus, our answer is **(E)** greater than any finite number

14. (AHSME 1965) The sum of the numerical coefficients in the complete expansion of $(x^2 - 2xy + y^2)^7$ is:
 (A) 0 (B) 7 (C) 14 (D) 128 (E) 128^2

Answer: A

Solution

Notice that the given equation, $(x^2 - 2xy + y^2)^7$ can be factored into $(x - y)^{14} = \sum_{k=0}^{14} \binom{14}{k} (-1)^k \cdot x^{14-k} \cdot y^k$.

Notice that if we plug in $x = y = 1$, then we are simply left with the sum of the coefficients from each term. Therefore, the sum of the coefficients is $(1 - 1)^7 = 0$, **(A)**.

15. (AHSME 1965) The symbol 25_b represents a two-digit number in the base b . If the number 52_b is double the number 25_b , then b is:
 (A) 7 (B) 8 (C) 9 (D) 11 (E) 12

Answer: B

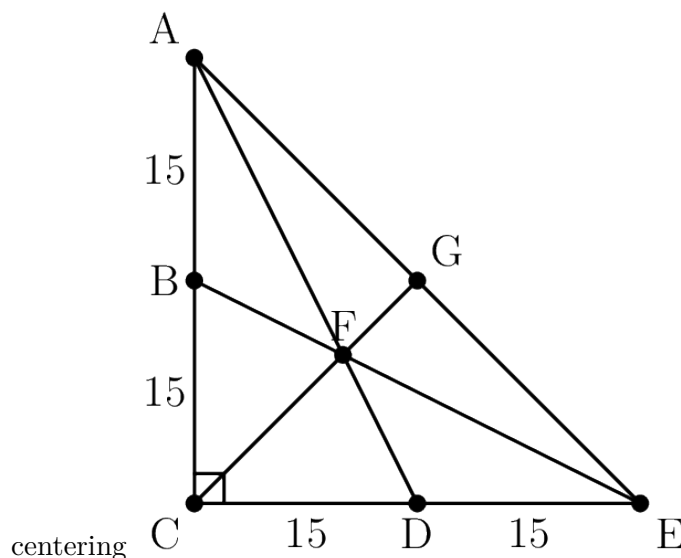
Solution

We begin by noting that $25_b = 2b + 5$ and $52_b = 5b + 2$. The problem tells us that $2 * 25_b = 52_b$, so $2(2b + 5) = 5b + 2$. Solving for b yields the answer **(B)** 8.

16. (AHSME 1965) Let line AC be perpendicular to line CE . Connect A to D , the midpoint of CE , and connect E to B , the midpoint of AC . If AD and EB intersect in point F , and $\overline{BC} = \overline{CD} = 15$ inches, then the area of triangle DFE , in square inches, is:
 (A) 50 (B) $50\sqrt{2}$ (C) 75 (D) $\frac{15}{2}\sqrt{105}$ (E) 100

Answer: C

Solution



Draw $\overline{AE}$, as seen in the diagram. From the problem, we know that $\overline{EB}$ and $\overline{AD}$ are **medians** of $\triangle ACE$. Let G be the midpoint of $\overline{AE}$. Then, $\overline{CG}$ is also a median of $\triangle ACE$, and it goes through $\triangle ACE$'s **centroid**, F . Because medians divide their triangle into 6 smaller triangles of equal area, we know that $[\triangle DFE] = \frac{1}{6}[\triangle ACE]$. Because $[\triangle ACE] = \frac{1}{2} * (15 + 15)^2 = \frac{900}{2} = 450$, $[\triangle DFE] = \frac{450}{6} = 75$. Thus, our answer is **(C) 75**.

17. (AHSME 1965) Given the true statement: The picnic on Sunday will not be held only if the weather is not fair. We can then conclude that:
- (A) If the picnic is held, Sunday's weather is undoubtedly fair.
 - (B) If the picnic is not held, Sunday's weather is possibly unfair.
 - (C) If it is not fair Sunday, the picnic will not be held.
 - (D) If it is fair Sunday, the picnic may be held.
 - (E) If it is fair Sunday, the picnic must be held.

Answer: E

Solution

The given statement tells us that the weather being not fair is the only thing that can cancel the picnic on Sunday. It does not say that the weather being not fair necessarily cancels the picnic, so choice (A) is out. If the weather is fair, however, nothing can stop the picnic from happening, so choice **(E)** is correct.

Choice (B) is incorrect, because if the picnic is cancelled, then the weather *must* be unfair, because the weather is the only thing that can stop the picnic.

Choice (C) is incorrect, because it is the **contrapositive** of choice (A), so (A) and (C) are logically equivalent.

Choice (D) is incorrect, because the picnic *must* be held if foul weather is not there to prevent it.

18. (AHSME 1965) If $1 - y$ is used as an approximation to the value of $\frac{1}{1+y}$, $|y| < 1$, the ratio of the error made to the correct value is:

- (A) y (B) y^2 (C) $\frac{1}{1+y}$ (D) $\frac{y}{1+y}$ (E) $\frac{y^2}{1+y}$

Answer: B

Solution

The error made in this approximation is $\frac{1}{1+y} - (1-y)$, and the correct value is $\frac{1}{1+y}$. Taking the ratio of these two values, we have:

$$\begin{aligned} \frac{\frac{1}{1+y} - (1-y)}{\frac{1}{1+y}} &= \frac{[\frac{1}{1+y} - (1-y)][1+y]}{[\frac{1}{1+y}][1+y]} \\ &= \frac{1 - (1-y^2)}{1} \\ &= y^2 \end{aligned}$$

Thus, our answer is (B) y^2

19. (AHSME 1965) If $x^4 + 4x^3 + 6px^2 + 4qx + r$ is exactly divisible by $x^3 + 3x^2 + 9x + 3$, the value of $(p+q)r$ is:

- (A) -18 (B) 12 (C) 15 (D) 27 (E) 45

Answer: C

Solution 1

Let $f(x) = x^3 + 3x^2 + 9x + 3$ and $g(x) = x^4 + 4x^3 + 6px^2 + 4qx + r$.

Let 3 roots of $f(x)$ be r_1, r_2 and r_3 . As $f(x)|g(x)$, 3 roots of 4 roots of $g(x)$ will be same as roots of $f(x)$. Let the 4th root of $g(x)$ be r_4 . By **Vieta's Formulas**

In $f(x)$

$$r_1 + r_2 + r_3 = -3$$

$$r_1r_2 + r_2r_3 + r_1r_3 = 9$$

$$r_1r_2r_3 = -3$$

In $g(x)$

$$r_1 + r_2 + r_3 + r_4 = -4$$

$$\Rightarrow r_4 = -1$$

$$r_1r_2 + r_1r_3 + r_1r_4 + r_2r_3 + r_2r_4 + r_3r_4 = 6p$$

$$\Rightarrow r_1r_2 + r_1r_3 + r_2r_3 + r_4(r_1 + r_2 + r_3) = 6p$$

$$\Rightarrow 9 + (-1)(-3) = 6p$$

$$\Rightarrow p = 2$$

$$r_1r_2r_3 + r_1r_3r_4 + r_1r_2r_4 + r_2r_3r_4 = -4q$$

$$\Rightarrow -3 + r_4(r_1r_3 + r_1r_2 + r_2r_3) = -4q$$

$$\Rightarrow -3 + (-1)(9) = -4q$$

$$\Rightarrow q = 3$$

$$r_1r_2r_3r_4 = r$$

$$\Rightarrow (-3)(-1) = r$$

$$\Rightarrow r = 3$$

$$\text{so } (p + q)r = \boxed{\text{(C) } 15}$$

By Ahmed_Ashhab

Solution 2

Notice that to obtain the x^4 term one must multiply $x^4 + 4x^3 + 6px^2 + 4qx + r$ by some linear function of the form $x - a$. Looking at the x^3 term, it is clear that a must equal 1. Therefore by multiplying $x^4 + 4x^3 + 6px^2 + 4qx + r$ by $x + 1$, the product will be $x^4 + 4x^3 + 12x^2 + 12x + 3$. Therefore $p = 2$, $q = 3$, $r = 3$. Thus $(2 + 3)3 = \boxed{\text{(C) } 15}$

20. (AHSME 1965) For every n the sum of n terms of an arithmetic progression is $2n + 3n^2$. The r th term is:

(A) $3r^2$ (B) $3r^2 + 2r$ (C) $6r - 1$ (D) $5r + 5$ (E) $6r + 2$

Answer: C

Solution

In order to calculate the r th term of this arithmetic sequence, we can subtract the sum of the first $r - 1$ terms from the sum of the first r terms of the sequence. Plugging in r and $r - 1$ as values of n in the given expression and subtracting yields

$$(3r^2 + 2r) - (3r^2 - 4r + 1).$$

Simplifying gives us the final answer of $\boxed{\text{(C) } 6r - 1}$.

21. (AHSME 1965) It is possible to choose $x > \frac{2}{3}$ in such a way that the value of $\log_{10}(x^2 + 3) - 2\log_{10} x$ is

(A) negative (B) zero (C) one
(D) smaller than any positive number that might be specified
(E) greater than any positive number that might be specified

Answer: D

Solution

By the rules of **logarithms**, $\log_{10}(x^2 + 3) - 2\log_{10} x = \log_{10}\left(\frac{x^2 + 3}{x^2}\right) = \log_{10}\left(1 + \frac{3}{x^2}\right)$. As x goes to infinity, $1 + \frac{3}{x^2}$ gets arbitrarily close to 1 (without ever reaching it), so $\log_{10}\left(1 + \frac{3}{x^2}\right)$ gets arbitrarily close to $\log_{10}(1) = 0$ (without ever reaching it). Furthermore, because $1 + \frac{3}{x^2} > 1$, $\log\left(1 + \frac{3}{x^2}\right)$ is never negative. Thus, we can choose a real $x > \frac{2}{3}$ such that the given expression is

$\boxed{\text{(D) smaller than any positive number that might be specified}}$.

22. (AHSME 1965) If $a_2 \neq 0$ and r and s are the roots of $a_0 + a_1x + a_2x^2 = 0$, then the equality $a_0 + a_1x + a_2x^2 = a_0\left(1 - \frac{x}{r}\right)\left(1 - \frac{x}{s}\right)$ holds:

(A) for all values of x , $a_0 \neq 0$ (B) for all values of x
(C) only when $x = 0$ (D) only when $x = r$ or $x = s$
(E) only when $x = r$ or $x = s$, $a_0 \neq 0$

Answer: A

Solution

a_0 cannot be 0, because $a_2 \neq 0$, so the polynomial can take non-zero values when $a_0 = 0$ and thereby not satisfy the equation. Expanding $a_0\left(1 - \frac{x}{r}\right)\left(1 - \frac{x}{s}\right)$ gives

us $a_0(1 - \frac{x}{r} - \frac{x}{s} + \frac{x^2}{rs})$. Using **Vieta's formulas**, we see that $rs = \frac{a_0}{a_2}$, so $a_0 = a_2rs$. Plugging this into the expanded right hand side of the given equation, we see that that side equals $a_0 - a_2(r + s)x + a_2x^2$. Using Vieta's formulas again, we equate this expression to $a_0 + a_1x + a_2x^2$, which is the left hand side of the given equation. Thus, as long as $a_0 \neq 0$, the equation holds for all values of x . This fact corresponds with answer choice **(A)**.

23. (AHSME 1965) If we write $|x^2 - 4| < N$ for all x such that $|x - 2| < 0.01$, the smallest value we can use for N is:
(A) .0301 **(B)** .0349 **(C)** .0399 **(D)** .0401 **(E)** .0499

Answer: D

Solution

There are two cases: (i) $x - 2 > 0$ and (ii) $x - 2 < 0$. In case (i), $0 < x - 2 < 0.01$, so $4 < x + 2 < 4.01$. In case (ii), $-0.01 < x - 2 < 0$, so $3.99 < x + 2 < 4$. Because $(x + 2)$ in case (i) can take a larger value, we use it to determine the upper bound for $|x^2 - 4|$ under the given restrictions. Now, we can see that $|x^2 - 4| = |x - 2||x + 2| < 0.01 * 4.01 = \boxed{0.0401}$, which is answer choice **(D)**.

24. (AHSME 1965) Given the sequence $10^{\frac{1}{11}}, 10^{\frac{2}{11}}, 10^{\frac{3}{11}}, \dots, 10^{\frac{n}{11}}$, the smallest value of n such that the product of the first n members of this sequence exceeds 100000 is:
(A) 7 **(B)** 8 **(C)** 9 **(D)** 10 **(E)** 11

Answer: E

Solution

Note that the given sequence is a **geometric sequence** with a common ratio $10^{\frac{1}{11}}$. Let the product of the first n terms of the sequence be denoted P_n . It is a consequence of the laws of exponents that $P_1 = (10^{\frac{1}{11}})^1$, $P_2 = (10^{\frac{1}{11}})^{1+2}$, and, in general, $P_n = (10^{\frac{1}{11}})^{T_n}$, where T_n denotes the n th **triangular number**. Setting P_n equal to 100,000, we see that:

$$\begin{aligned} (10^{\frac{1}{11}})^{\frac{n(n+1)}{2}} &= 10^5 \\ \frac{1}{11} * \frac{n(n+1)}{2} &= 5 \\ n(n+1) &= 110 \\ n^2 + n - 110 &= 0 \\ (n-10)(n+11) &= 0 \end{aligned}$$

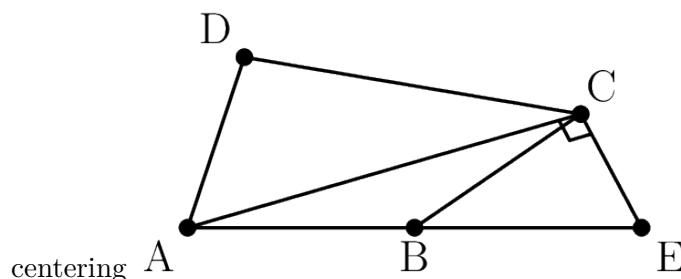
Because n must be positive, we are left with $n = 10$. Given this information, choice (D) may seem appealing. Do not be fooled. The product of the first n terms is *exactly* 100,000, but the problem asks for the smallest n such that P_n *exceeds* 100,000. Thus, the minimum value of n which satisfies the problem is **(E) 11**.

25. (AHSME 1965) Let $ABCD$ be a quadrilateral with AB extended to E so that $\overline{AB} = \overline{BE}$. Lines AC and CE are drawn to form $\angle ACE$. For this angle to be a right angle it is necessary that quadrilateral $ABCD$ have:

- (A) all angles equal (B) all sides equal
 (C) two pairs of equal sides (D) one pair of equal sides
 (E) one pair of equal angles

Answer: D

Solution



Because $\triangle ACE$ is right, the midpoint of its hypotenuse (namely, B) is its **orthocenter**. Thus, $AB = BC$, and so two side lengths of quadrilateral $ABCD$ are equal. The placement of D is irrelevant. Thus, our answer is **(D) one pair of equal sides**.

26. (AHSME 1965) For the numbers a, b, c, d, e define m to be the arithmetic mean of all five numbers; k to be the arithmetic mean of a and b ; l to be the arithmetic mean of c, d , and e ; and p to be the arithmetic mean of k and l . Then, no matter how a, b, c, d , and e are chosen, we shall always have:

- (A) $m = p$ (B) $m \geq p$ (C) $m > p$
 (D) $m < p$ (E) none of these

Answer: E

Solution

We shall begin by eliminating some options through counterexamples. If $a = b = c = d = e$, then $m = k = l = a$, so $p = a$, and $m = p$. Answers (C) and (D) do not allow for $m = p$, so they can be eliminated. If we set $a = 60$ and $b = c = d = e = 0$, then $m = 12$, $k = 30$, $l = 0$, and $p = 15$. Here, $m < p$, so we can throw out options (A) and (B) as well. Now, we are left with only option **(E) none of these**.

27. (AHSME 1965) When $y^2 + my + 2$ is divided by $y - 1$ the quotient is $f(y)$ and the remainder is R_1 . When $y^2 + my + 2$ is divided by $y + 1$ the quotient is $g(y)$ and the remainder is R_2 . If $R_1 = R_2$ then m is:

- (A) 0 (B) 1 (C) 2 (D) -1 (E) an undetermined constant

Answer: A

Solution

$$\text{Let } h(y) = y^2 + my + 2$$

$$h(y) = y^2 + my + 2 = f(y)(y - 1) + R_1$$

$$h(y) = y^2 + my + 2 = g(y)(y + 1) + R_2$$

$$h(1) = 3 + m = R_1$$

$$h(-1) = 3 - m = R_2$$

$$m = 0$$

Thus, the answer is **(A) 0**.

Solution 2

As in the other solution, let $h(y) = y^2 + my + 2$. By dividing $h(y)$ by $(y - 1)$ (through either **long** or **synthetic** division), we get a remainder $R_1 = m + 3$. Similarly, dividing $h(y)$ by $(y + 1)$ yields the remainder $R_2 = 3 - m$. Setting $R_1 = R_2$, we see that $m + 3 = 3 - m$, which only holds when $m = 0$, which corresponds to answer choice **(A)**.

28. (AHSME 1965) An escalator (moving staircase) of n uniform steps visible at all times descends at constant speed. Two boys, A and Z , walk down the escalator steadily as it moves, A negotiating twice as many escalator steps per minute as Z . A reaches the bottom after taking 27 steps while Z reaches the bottom after taking 18 steps. Then n is:

(A) 63 (B) 54 (C) 45 (D) 36 (E) 30

Answer: B

Solution

If we let Z 's speed be z steps/minute, then A 's speed is $2z$ steps/minute. Let t_a be the time A spent on the escalator, and let t_z be the time Z spent on the escalator. Then, we know that A walked down $2zt_a = 27$ steps, and Z walked down $zt_z = 18$ steps. Dividing the first equation by the second, we see that:

$$\begin{aligned}\frac{2zt_a}{zt_z} &= \frac{27}{18} \\ \frac{2t_a}{t_z} &= \frac{3}{2} \\ t_a &= \frac{3}{4}t_z\end{aligned}$$

Thus, because A was on the escalator for $\frac{3}{4}$ as long as Z was, A only gained $\frac{3}{4}$ as many "free" steps (i.e. steps that do not have to be taken because the escalator is moving down). We know that A gained $(n - 27)$ free steps, and Z gained $(n - 18)$ free steps. Thus we have the following equation: $n - 27 = \frac{3}{4}(n - 18)$. Solving for n gives us **(B) 54**.

29. (AHSME 1965) Of 28 students taking at least one subject the number taking Mathematics and English only equals the number taking Mathematics only. No student takes English only or History only, and six students take Mathematics and History, but not English. The number taking English and History only is five times the number taking all three subjects. If the number taking all three subjects is even and non-zero, the number taking English and Mathematics only is:

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

Answer: A

Solution

The number of students taking all three subjects cannot be greater than or equal to 4, because then there would be 20 or more students taking English and History only, and we know that 6 students are taking Mathematics and History only. These three values sum to a number greater than 28, the total number of students. Thus,

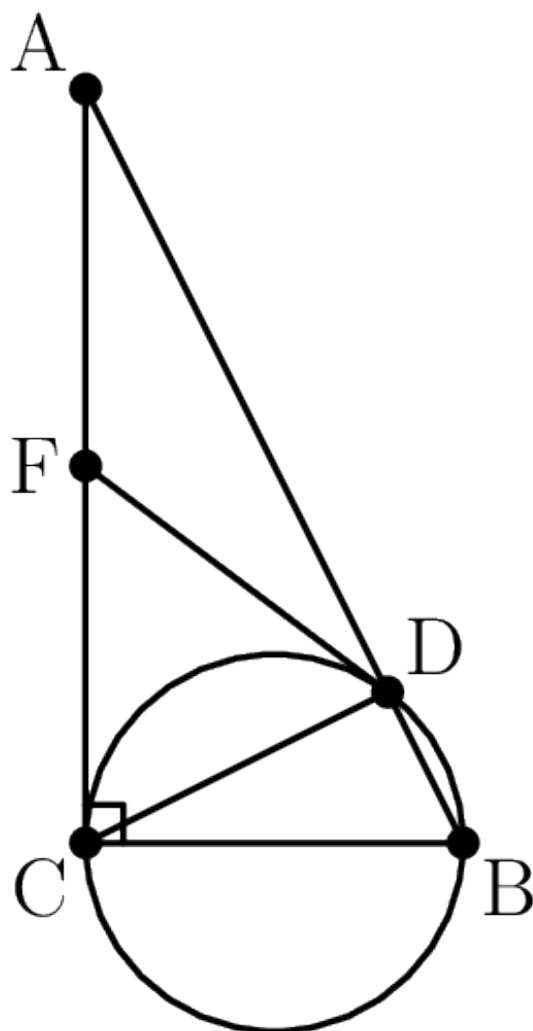
because we know that the number of students taking all three subjects is a non-zero even number, there must be 2 of them. Thus, 10 people must be taking only English and History. With the knowledge of 6 students taking only Mathematics and History, we know the classes of 18 students so far, so we have 10 students left who are either taking Mathematics and English only or Mathematics only. Because these two categories have an equal number of students, the number of students taking English and Mathematics only is **(A) 5**.

30. (AHSME 1965) Let BC of right triangle ABC be the diameter of a circle intersecting hypotenuse AB in D . At D a tangent is drawn cutting leg CA in F . This information is not sufficient to prove that

- (A) DF bisects CA (B) DF bisects $\angle CDA$
 (C) $DF = FA$ (D) $\angle A = \angle BCD$ (E) $\angle CFD = 2\angle A$

Answer: B

Solution 1



centering

We will prove every result except for **(B)**.

By **Thales' Theorem**, $\angle CDB = 90^\circ$ and so $\angle CDA = 90^\circ$. FC and FD are both tangents to the same circle, and hence equal. Let $\angle CFD = \alpha$. Then

$\angle FDC = \frac{180^\circ - \alpha}{2}$, and so $\angle FDA = \frac{\alpha}{2}$. We also have $\angle AFD = 180^\circ - \alpha$, which implies $\angle FAD = \frac{\alpha}{2}$. This means that $CF = DF = FA$, so DF indeed bisects CA . We also know that $\angle BCD = 90 - \frac{180^\circ - \alpha}{2} = \frac{\alpha}{2}$, hence $\angle A = \angle BCD$. And $\angle CFD = 2\angle A$ as $\alpha = \frac{\alpha}{2} \times 2$.

Since all of the results except for B are true, our answer is **B**.

Solution 2

It's easy to verify that $\angle CDA$ always equals 90° . Since $\angle CDF$ changes depending on the sidelengths of the triangle, we cannot be certain that $\angle CDF = 45^\circ$. Hence our answer is **B**.

31. (AHSME 1965) The number of real values of x satisfying the equality $(\log_a x)(\log_b x) = \log_a b$, where $a > 0, b > 0, a \neq 1, b \neq 1$, is:
 (A) 0 (B) 1 (C) 2 (D) a finite integer greater than 2 (E) not finite

Answer: C

Solution

From the properties of **logarithms**, we know that $(\log_a x)(\log_x b) = \log_a b$, and $\log_x b = \frac{1}{\log_b x}$. Thus, to satisfy the given equation, we are looking for values of x when $\log_b x = \frac{1}{\log_b x}$, or when $\log_b x = \pm 1$. This happens either when $x = b$ or $x = \frac{1}{b}$. Thus we have 2 solutions, consistent with answer choice **(C)**.

32. (AHSME 1965) An article costing C dollars is sold for 100 *at a loss of x percent of the selling price. It is then sold for S' dollars, then x is:*
 (A) undetermined (B) $\frac{80}{9}$ (C) 10 (D) $\frac{95}{9}$ (E) $\frac{100}{9}$

Answer: C

Solution

The magnitude of the loss after the first sale is $C - 100$, which equals $x\%$ of the selling price, \$100. Thus, $C - 100 = 100 * \frac{x}{100}$, and so $C = x + 100$. The profit made after the second sale, $S' - 100$, is x . Because $x \geq 0$, $x = 10$, and so we choose answer **(C) 10**.

33. (AHSME 1965) If the number $15!$, that is, $15 \cdot 14 \cdot 13 \dots 1$, ends with k zeros when given to the base 12 and ends with h zeros when given to the base 10, then $k + h$ equals:
 (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

Answer: D

Solution

We can use **Legendre's Formula** to find the number of 0s in base 10

$$\lfloor \frac{15}{5} \rfloor + \lfloor \frac{15}{25} \rfloor = 3$$

So $h = 3$. Likewise, we are looking for the number of 2^2 s and 3s that divide $15!$, so we use Legendre's again.

$$\lfloor \frac{15}{2} \rfloor + \lfloor \frac{15}{4} \rfloor + \lfloor \frac{15}{8} \rfloor = 7 + 3 + 1 = 11$$

$$\lfloor \frac{15}{3} \rfloor + \lfloor \frac{15}{9} \rfloor = 5 + 1 = 6$$

Thus, $3^6 | 15!$ and $2^{11} | 15! \Rightarrow (2^2)^5 | 15!$ So $k = 5$, and $5 + 3 = 8$, which corresponds to answer **(D)**.

34. (AHSME 1965) For $x \geq 0$ the smallest value of $\frac{4x^2+8x+13}{6(1+x)}$ is:

- (A) 1 (B) 2 (C) $\frac{25}{12}$ (D) $\frac{13}{6}$ (E) $\frac{34}{5}$

Answer: B

Solution 1

To begin, let's denote the equation, $\frac{4x^2+8x+13}{6(1+x)}$ as $f(x)$. Let's notice that:

$$\begin{aligned} f(x) &= \frac{4x^2 + 8x + 13}{6(1+x)} \\ &= \frac{4(x^2 + 2x) + 13}{6(x+1)} \\ &= \frac{4(x^2 + 2x + 1 - 1) + 13}{6(x+1)} \\ &= \frac{4(x+1)^2 + 9}{6(x+1)} \\ &= \frac{4(x+1)^2}{6(x+1)} + \frac{9}{6(1+x)} \\ &= \frac{2(x+1)}{3} + \frac{3}{2(x+1)} \end{aligned}$$

After this simplification, we may notice that we may use calculus, or the **AM-GM inequality** to finish this problem because $x \geq 0$, which implies that both $\frac{2(x+1)}{3}$ and $\frac{3}{2(x+1)}$ are greater than zero. Continuing with AM-GM:

$$\frac{\frac{2(x+1)}{3} + \frac{3}{2(x+1)}}{2} \geq \sqrt{\frac{2(x+1)}{3} \cdot \frac{3}{2(x+1)}}$$

$$f(x) \geq 2$$

Therefore, $f(x) = \frac{4x^2+8x+13}{6(1+x)} \geq 2$, **(B)**

(With equality when $\frac{2(x+1)}{3} = \frac{3}{2(x+1)}$, or $x = \frac{1}{2}$)

Solution 2 (Calculus)

Let $f(x) = \frac{4x^2+8x+13}{6(1+x)}$. Take the derivative of $f(x)$ using the quotient rule.

$$\begin{aligned} f(x) &= \frac{4x^2 + 8x + 13}{6(1+x)} \\ f'(x) &= \frac{1}{6} * \frac{(4x^2 + 8x + 13)'(1+x) - (4x^2 + 8x + 13)(1+x)'}{(1+x)^2} \\ &= \frac{(8x + 8)(1+x) - (4x^2 + 8x + 13)(1)}{6(1+x)^2} \\ &= \frac{4x^2 + 8x - 5}{6(1+x)^2} \end{aligned}$$

Next, set the numerator equal to zero to find the x -value of the minimum:

$$\begin{aligned} 4x^2 + 8x - 5 &= 0 \\ (2x + 5)(2x - 1) &= 0 \end{aligned}$$

From the problem, we know that $x \geq 0$, so we are left with $x = \frac{1}{2}$. Plugging $x = \frac{1}{2}$ into $f(x)$, we get:

$$\begin{aligned} f\left(\frac{1}{2}\right) &= \frac{4\left(\frac{1}{2}\right)^2 + 8\left(\frac{1}{2}\right) + 13}{6\left(1 + \left(\frac{1}{2}\right)\right)} \\ &= \frac{1 + 4 + 13}{6\left(\frac{3}{2}\right)} \\ &= \frac{18}{9} \\ &= 2 \end{aligned}$$

Thus, our answer is (B) 2.

Solution 3 (answer choices, no AM-GM or calculus)

We go from A through E and we look to find the smallest value so that $x \geq 0$, so we start from A:

$$\frac{4x^2 + 8x + 13}{6x + 6} = 1$$

$$4x^2 + 8x + 13 = 6x + 6$$

$$4x^2 + 2x + 7 = 0$$

However by the quadratic formula there are no real solutions of x , so x cannot be greater than 0. We move on to B:

$$\frac{4x^2 + 8x + 13}{6x + 6} = 2$$

$$4x^2 + 8x + 13 = 12x + 12$$

$$4x^2 - 4x + 1 = 0$$

$$(2x - 1)^2 = 0$$

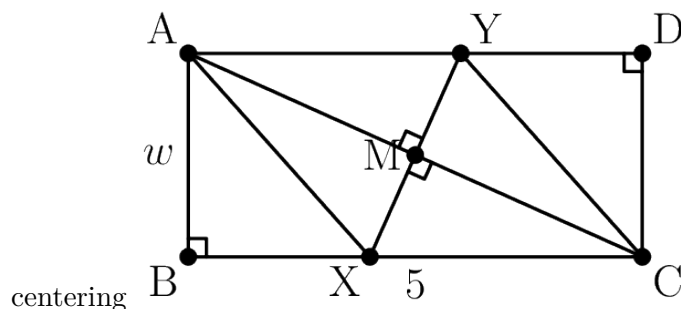
There is one solution: $x = \frac{1}{2}$, which is greater than 0, so 2 works as a value. Since all the other options are bigger than 2 or invalid, the answer must be **B**

35. (AHSME 1965) The length of a rectangle is 5 inches and its width is less than 4 inches. The rectangle is folded so that two diagonally opposite vertices coincide. If the length of the crease is $\sqrt{6}$, then the width is:

(A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) 2 (D) $\sqrt{5}$ (E) $\sqrt{\frac{11}{2}}$

Answer: D

Solution



Let the rectangle be $ABCD$ with $AB = CD = w$ and $AD = BC = 5$, as in the diagram. We desire a line such that reflecting point C across that line yields point A . For this to happen, the line must be perpendicular to the diagonal $\overline{AC}$, and it must go through the midpoint of $\overline{AC}$ (let it be point M). Let the intersection of this line with $\overline{BC}$ be point X and with $\overline{AD}$ be point Y . From the problem, we know that $XY = \sqrt{6}$. By **HL congruence**, $\triangle AMY \cong \triangle CMX$, so $AM = CM = x$, where x is some number. Furthermore, $XM = MY = \frac{\sqrt{6}}{2}$. By **AA similarity**, $\triangle MXC \sim \triangle BAC$, so $\frac{MX}{MC} = \frac{BA}{BC}$. $MX = \frac{\sqrt{6}}{2}$, $MC = x$, $BA = w$, and $BC = 5$, so we can rewrite this proportion to solve for x in terms of w :

$$\begin{aligned} \frac{\sqrt{6}/2}{x} &= \frac{w}{5} \\ \frac{5\sqrt{6}}{2} &= xw \\ x &= \frac{5\sqrt{6}}{2w} \end{aligned}$$

By the **Pythagorean Theorem** on $\triangle ABC$, we know that $w^2 + 25 = 4x^2$, and we can plug in our new expression for x into this equation to solve for w :

$$w^2 + 25 = 4\left(\frac{5\sqrt{6}}{2w}\right)^2$$

$$\begin{aligned}
 w^2 + 25 &= \frac{25 * 6}{w^2} \\
 w^4 + 25w^2 - 150 &= 0 \\
 (w^2 - 5)(w^2 + 30) &= 0
 \end{aligned}$$

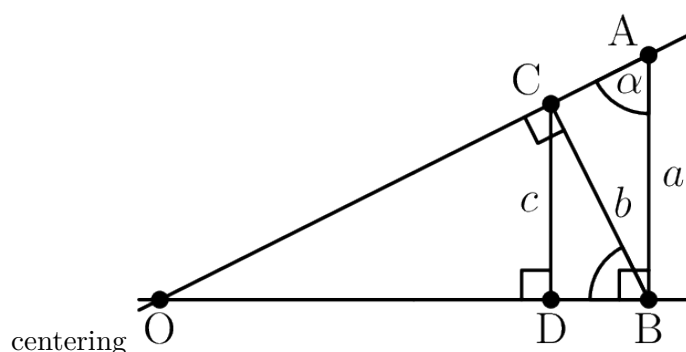
Because $w > 0$, $w^2 = 5$, and so $w = \sqrt{5}$, which corresponds to answer choice **(D)**.

36. (AHSME 1965) Given distinct straight lines OA and OB . From a point in OA a perpendicular is drawn to OB ; from the foot of this perpendicular a line is drawn perpendicular to OA . From the foot of this second perpendicular a line is drawn perpendicular to OB ; and so on indefinitely. The lengths of the first and second perpendiculars are a and b , respectively. Then the sum of the lengths of the perpendiculars approaches a limit as the number of perpendiculars grows beyond all bounds. This limit is:

(A) $\frac{b}{a-b}$ (B) $\frac{a}{a-b}$ (C) $\frac{ab}{a-b}$ (D) $\frac{b^2}{a-b}$ (E) $\frac{a^2}{a-b}$

Answer: E

Solution 1



For simplicity, let the first perpendicular from $\overrightarrow{OA}$ to $\overrightarrow{OB}$ be $\overline{AB}$, and let the second perpendicular have foot C on $\overrightarrow{OA}$. Further, let the perpendicular from C to $\overrightarrow{OB}$ have foot D and length c , as in the diagram. Also, let $\angle OAB = \alpha$. From the problem, we have $AB = a$ and $BC = b$. By **AA similarity**, we have $\triangle OCB \sim \triangle OBA$, so $\angle CBO = \alpha$ as well. In $\triangle ABC$, we see that $\sin \alpha = \frac{b}{a}$, and, in $\triangle CDB$, $\sin \alpha = \frac{c}{b}$. Equating these two expressions for $\sin \alpha$, we get that $\frac{b}{a} = \frac{c}{b}$, or, because $a, b, c > 0$, $b = \sqrt{ac}$. Thus, b is the **geometric mean** of a and c . Note that if we remove the first perpendicular (i.e. the one with length a), we are left with a smaller version of the original problem, which will have the same equation for the limit (but this time expressed in terms of b and c rather than a and b). Thus, if we let the length of the fourth perpendicular be d , then c will equal the geometric mean of b and d , and so on for the infinitude of perpendiculars. Thus, because the length of a given perpendicular (except the first one) is the geometric mean of the two adjacent perpendiculars, the lengths of the perpendiculars form a **geometric sequence**. Because the sequence's first two terms are a and b , it has common ratio $\frac{b}{a}$. Because $b < a$, the common ratio is positive and less than 1, so the sequence's infinite **geometric series** converges. This infinite sum is given by

$$\frac{a}{1 - \frac{b}{a}} = \frac{a^2}{a - b}, \text{ which is answer choice } \mathbf{(E)}.$$

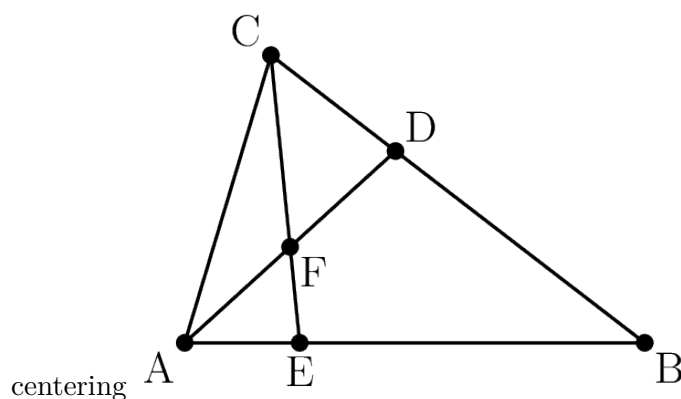
Solution 2 (Answer choices, intuition)

Let a and b be measured with some units of length (say, meters). The limit of the sum of the lengths of the perpendiculars would, then, be measured in meters as well. Performing **dimensional analysis** on each of the answer choices, we can eliminate options (A) and (B), because they have units $\frac{\text{meters}}{\text{meters}}$, so they are dimensionless. Unfortunately, our other three options all have units of meters. Now, we have a $\frac{1}{3}$ chance of guessing the answer correctly, but we can go further. Think of what happens as $\angle BOA$ approaches 0° . Then, a approaches 0, and the perpendiculars have nearly no space to "bounce between" the two lines, and so they *likely* have zero total length. Likewise, if we think of what happens as $\angle BOA$ approaches 90° , all of the perpendiculars except the first one (with length a) go to zero. From this intuition, one would think that a would have a larger impact upon the total length of the perpendiculars. Out of the three answer choices left, this conjecture is only consistent with choice **(E)** $\frac{a^2}{a-b}$. To reinforce this decision, think about what happens as $\angle BOA$ approaches 90° again. We would *expect* the sum to approach a and b to approach 0. Plugging in $b = 0$ into the expression in choice (E), we get $\frac{a^2}{a} = a$, which is what we expected. On the other hand, options (C) and (D) equate to 0, which is clearly false.

37. (AHSME 1965) Point E is selected on side AB of $\triangle ABC$ in such a way that $AE : EB = 1 : 3$ and point D is selected on side BC such that $CD : DB = 1 : 2$. The point of intersection of AD and CE is F . Then $\frac{EF}{FC} + \frac{AF}{FD}$ is:
- (A) $\frac{4}{5}$ (B) $\frac{5}{4}$ (C) $\frac{3}{2}$ (D) 2 (E) $\frac{5}{2}$

Answer: C

Solution



We use **mass points** for this problem. Let m_A denote the mass of point A . Rewrite the expression we are finding as

$$\frac{EF}{FC} + \frac{AF}{FD} = \frac{FE}{FC} + \frac{FA}{FD} = \frac{m_C}{m_E} + \frac{m_D}{m_A}$$

Now, let $m_C = 2$. We then have $2 \cdot 1 = m_B \cdot 2$, so $m_B = 1$, and $m_D = 2 + 1 = 3$. We can let $m_A = 3$. We have $m_E = m_A + m_B = 3 + 1 = 4$. From here, substitute the respective values to get

$$\frac{m_C}{m_E} + \frac{m_D}{m_A} = \frac{2}{4} + \frac{3}{3} = \frac{1}{2} + 1 = \frac{3}{2}$$

This answer corresponds to (C).

38. (*AHSME 1965*) A takes m times as long to do a piece of work as B and C together; B takes n times as long as C and A together; and C takes x times as long as A and B together. Then x , in terms of m and n , is:

(A) $\frac{2mn}{m+n}$ (B) $\frac{1}{2(m+n)}$ (C) $\frac{1}{m+n-mn}$ (D) $\frac{1-mn}{m+n+2mn}$ (E) $\frac{m+n+2}{mn-1}$

Answer: E

Solution 1

Let a , b , and c be the speeds at which A , B and C work, respectively. Also, let the piece of work be worth one unit of work. Then, using the information from the problem along with basic **rate** formulas, we obtain the following equations:

$$\begin{aligned}\frac{1}{a} &= m * \frac{1}{b+c} \\ \frac{1}{b} &= n * \frac{1}{a+c} \\ \frac{1}{c} &= x * \frac{1}{a+b}\end{aligned}$$

These equations can be rearranged into the following:

$$\begin{aligned}\text{(i)} \quad ma &= b+c \\ \text{(ii)} \quad nb &= a+c \\ \text{(iii)} \quad xc &= a+b\end{aligned}$$

Solving for a in equation (i) gives us $a = \frac{b+c}{m}$. Substituting this expression for a into equation (ii) yields:

$$\begin{aligned}nb &= \frac{b+c}{m} + c \\ mnb &= b+c+mc \\ (mn-1)b &= (m+1)c \\ b &= \frac{(m+1)c}{mn-1}\end{aligned}$$

Finally, substituting our expressions for a and b into equation (iii) yields our final answer:

$$\begin{aligned}xc &= \frac{b+c}{m} + \frac{(m+1)c}{mn-1} \\ &= \frac{\frac{(m+1)c}{mn-1} + c}{m} + \frac{(m+1)c}{mn-1} \\ &= \frac{c}{m} \left(\frac{m+1+mn-1}{mn-1} + \frac{m^2+m}{mn-1} \right) \\ &= \frac{c}{m} \left(\frac{m^2+mn+2m}{mn-1} \right) \\ &= c \left(\frac{m+n+2}{mn-1} \right)\end{aligned}$$

Thus, $x = \boxed{\text{(E)} \frac{m+n+2}{mn-1}}.$

Solution 2 (Answer choices)

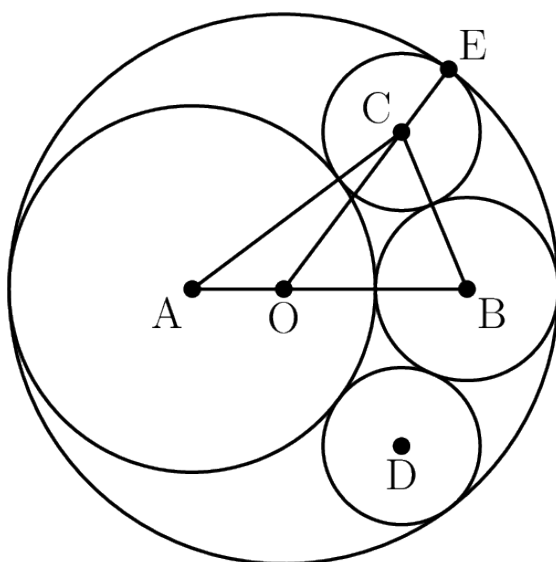
If we let A , B , and C work at the same speed, then it is clear that $m = n = x = 2$. After plugging in $m = n = 2$ into all of the answer choices, we see that the only two choices which give a value $x = 2$ are choices (A) and (E). Now suppose that A and B work at the same speed, but C does no work at all. Then, $m = n = 1$, but x is undefined. After plugging in $m = n = 1$ into choices (A) and (E), we see that the only choice which is undefined is choice $\boxed{\text{(E)} \frac{m+n+2}{mn-1}}.$

39. (AHSME 1965) A foreman noticed an inspector checking a 3"-hole with a 2"-plug and a 1"-plug and suggested that two more gauges be inserted to be sure that the fit was snug. If the new gauges are alike, then the diameter, d , of each, to the nearest hundredth of an inch, is:

(A) .87 (B) .86 (C) .83 (D) .75 (E) .71

Answer: B

Solution 1



centering

Let the center of the 3" circle be O , that of the 2" circle be A , that of the 1" circle be B , and those of the circles of unknown radius (let their radii have length r) be C and D , as in the diagram. Also, in this problem, no two circles share a center, so let the circles be named by their corresponding centers (so, the 3" circle is circle O , etc.). Extend $\overline{OC}$ past C to intersect circle O at point E . Because circle O has radius $\frac{3}{2}$ and circle A has radius 1, $OA = \frac{3}{2} - 1 = \frac{1}{2}$. Likewise, because B has radius $\frac{1}{2}$, $OB = \frac{3}{2} - \frac{1}{2} = 1$. Thus, $AB = \frac{3}{2}$. Furthermore, because the line connecting the centers of two tangent circles goes through their point of tangency, $AC = r + 1$ and $BC = r + \frac{1}{2}$. Because $OE = \frac{3}{2}$ and $CE = r$, $OC = \frac{3}{2} - r$. With this information, we can now apply **Stewart's Theorem** to $\triangle ABC$ to solve for r :

$$OB \cdot OA \cdot AB + OC^2 \cdot AB = AC^2 \cdot OB + BC^2 \cdot OA$$

$$4[(1)(\frac{1}{2})(\frac{3}{2}) + (\frac{3}{2} - r)^2(\frac{3}{2})] = 4[(r + 1)^2(1) + (r + \frac{1}{2})^2(\frac{1}{2})]$$

$$\begin{aligned}
3 + 6\left(\frac{9}{4} - 3r + r^2\right) &= 4(r^2 + 2r + 1) + 2\left(r^2 + r + \frac{1}{4}\right) \\
3 + \frac{27}{2} - 18r + 6r^2 &= 4r^2 + 8r + 4 + 2r^2 + 2r + \frac{1}{2} \\
3 + \frac{27-1}{2} - 4 + 6r^2 - 6r^2 &= 18r + 8r + 2r \\
3 + 13 - 4 &= 28r \\
28r &= 12 \\
r &= \frac{3}{7}
\end{aligned}$$

Because the question asks for the diameter of the circle, we calculate $2r = \frac{6}{7} \approx$

(B) 0.86.

Solution 2 (Overkill?)

This is Descartes' formula for four mutually tangent circles with radii r_1, r_2, r_3, r_4

Let k_i denote $\frac{1}{r_i}$. This is the curvature.

Note: A negative radius means that the other circles are internally tangent to it.

Then,

$$(k_1 + k_2 + k_3 + k_4)^2 = 2(k_1^2 + k_2^2 + k_3^2 + k_4^2)$$

First, scale the problem up by two, so the radii are 1", 2", 3", and d"

Next, let k be $\frac{1}{d}$

Using the formula, we get:

$$\left(\frac{7}{6} + k\right)^2 = 2\left(\frac{49}{36} + k^2\right)$$

$$\frac{49}{36} + \frac{7}{3}k + k^2 = \frac{98}{36} + 2k^2$$

(You'll see why I didn't simplify $\frac{98}{36}$)

$$k^2 - \frac{7}{3} - \frac{49}{36} = 0$$

Using the quadratic formula:

$$k = \frac{\frac{7}{3} \pm \sqrt{\frac{49}{9} - 4 \cdot 1 \cdot \frac{49}{36}}}{2}$$

$$k = \frac{\frac{7}{3} \pm \sqrt{\frac{49}{9} - \frac{49}{9}}}{2}$$

$$k = \frac{\frac{7}{3} \pm \sqrt{0}}{2}$$

$$k = \frac{\frac{7}{3} \pm 0}{2}$$

$$k = \frac{7}{6}$$

$$k = \frac{7}{6}$$

So the diameter $d = \frac{6}{7}$, which is about **(B)**.86

40. (AHSME 1965) Let n be the number of integer values of x such that $P = x^4 + 6x^3 + 11x^2 + 3x + 31$ is the square of an integer. Then n is:

(A) 4 (B) 3 (C) 2 (D) 1 (E) 0

Answer: D

Solution

First, we wish to factor P into a more manageable form. From the beginning of P , we notice $x^4 + 6x^3$, which gives us the idea to use $(x^2 + 3x)^2 = x^4 + 6x^3 + 9x^2$.

This gives us

$$P = (x^2 + 3x)^2 + 2x^2 + 3x + 31$$

This is not useful, but it gives us a place to start from.

We can then try $(x^2 + 3x + 1) = x^4 + 6x^3 + 11x^2 + 6x + 1$.

$$P = (x^2 + 3x + 1)^2 - 3(x - 10)$$

This is much more useful, as it moves all non-linear terms inside of a squared expression.

We can then say $P = (x^2 + 3x + 1)^2 - 3(x - 10) = a^2$, where a^2 is the square of an integer mentioned on the problem. Right from here, we can set $x = 10$, which cancels out the $3(x - 10)$, giving $(100 + 30 + 1)^2 = a^2$. This gives us one solution, $x = 10$, $a = 131$.

We can then rearrange the expression, giving us $(x^2 + 3x + 1)^2 - a^2 = 3(x - 10)$. Factoring using difference of squares, we obtain

$$(x^2 + 3x + 1 + |a|)(x^2 + 3x + 1 - |a|) = 3(x - 10)$$

We can then state that when x is greater than 3 and less than -9 , $x^2 + 3x + 1$ will be greater than $|3x - 10|$. This is obtained by setting $x^2 + 3x + 1 > |3x - 10|$ and then solving the inequality. We can then conclude that $x^2 + 3x + 1 + |a| > |3x - 10|$.

Next, we claim that $x^2 + 3x + 1 - |a| \geq 1$ or $x^2 + 3x + 1 - |a| \leq -1$ when $x \neq 10$. We can prove this by first noting that since x and a are integers, $x^2 + 3x + 1 - |a|$ is an integer. Next, we shall assume that $x^2 + 3x + 1 - |a| = 0$. Solving this and plugging back into the original equation, we obtain $(2|a|)(0) = 3(x - 10)$. Solving we obtain $x = 10$, which is a contradiction to $x \neq 10$. Therefore, $x^2 + 3x + 1 - |a| \neq 0$ and $x^2 + 3x + 1 - |a| \geq 1$ or $x^2 + 3x + 1 - |a| \leq -1$.

Finally, we can go back to the equation

$$(x^2 + 3x + 1 + |a|)(x^2 + 3x + 1 - |a|) = 3(x - 10)$$

We note that since $(x^2 + 3x + 1 + |a|)$ is larger than $3(x - 10)$, in order for there to be solutions, $(x^2 + 3x + 1 - |a|)$ must be in the range $(-1, 1)$. However, this contradicts what was proven earlier, so when $x \neq 10$ and $x < -9$ or $x > 3$, there are no solutions for x .

Now, all that remains to be checked are values of x between -9 and 3 . Using brute force and checking each value individually, we can assert that there are no such solutions for x , leaving us with only 1 solution, $x = 10$. Therefore, the answer is

(D)

Solution by treetor10145

§2 AHSME 1966

1. (AHSME 1966) Given that the ratio of $3x - 4$ to $y + 15$ is constant, and $y = 3$ when $x = 2$, then, when $y = 12$, x equals:

(A) $\frac{1}{8}$ (B) $\frac{7}{3}$ (C) $\frac{7}{8}$ (D) $\frac{7}{2}$ (E) 8

Answer: B

Solution

Let k be the constant ratio. Then $k = \frac{3(2)-4}{(3)+15} = \frac{1}{9} = \frac{3x-4}{(12)+15}$. Solving gives $3x - 4 = 3 \Rightarrow x = \frac{7}{3} \Rightarrow$ (B).

2. (AHSME 1966) When the base of a triangle is increased 10% and the altitude to this base is decreased 10%, the change in area is
 (A) 1% increase (B) $\frac{1}{2}$ % increase (C) 0% (D) $\frac{1}{2}$ % decrease (E) 1% decrease

Answer: E

3. (AHSME 1966) If the arithmetic mean of two numbers is 6 and their geometric mean is 10, then an equation with the given two numbers as roots is:
 (A) $x^2+12x+100=0$ (B) $x^2+6x+100=0$ (C) $x^2-12x-10=0$ (D) $x^2-12x+100=0$ (E) $x^2-6x+100=0$

Answer: D

Solution

Let the numbers be η and ζ .

$$\frac{\eta + \zeta}{2} = 6 \Rightarrow \eta + \zeta = 12$$

.

$$\sqrt{\eta\zeta} = 10 \Rightarrow \eta\zeta = 100$$

.

The **monic quadratic** with roots η and ζ is $x^2 - (\eta + \zeta)x + \eta\zeta$. Therefore, an equation with η and ζ as roots is $x^2 - 12x + 100 = 0 \Rightarrow$ (D)

4. (AHSME 1966) Circle I is circumscribed about a given square and circle II is inscribed in the given square. If r is the ratio of the area of circle I to that of circle II, then r equals:
 (A) $\sqrt{2}$ (B) 2 (C) $\sqrt{3}$ (D) $2\sqrt{2}$ (E) $2\sqrt{3}$

Answer: B

Solution

Make half of the square's side x . Now the radius of the smaller circle is x , so its area is πx^2 . Now find the diameter of the bigger circle. Since half of the square's side is x , the full side is $2x$. Using the Pythagorean theorem, you get the diagonal to be $2\sqrt{2}x$. Half of that is the radius, or $x\sqrt{2}$. Using the same equation as before, you get the area of the larger circle to be $2x^2\pi$. Putting one over the other and dividing, you get two as the answer: or $\boxed{(B)}$.

5. (AHSME 1966) The number of values of x satisfying the equation

$$\frac{2x^2-10x}{x^2-5x} = x - 3$$

is:

- (A) zero (B) one (C) two (D) three (E) an integer greater than 3

Answer: A

Solution

Since $x^2 - 5x$ is in the denominator, $x \neq 0, 5$. Simplifying,

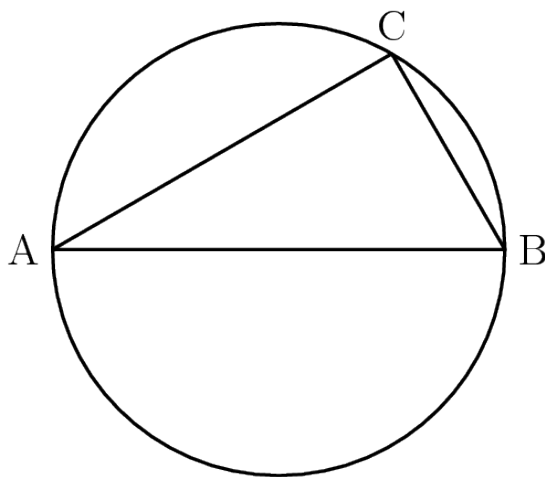
$$2 \left(\frac{x^2 - 5}{x^2 - 5} \right) = 2 = x - 3$$

Thus $x = 5$, which isn't in the **domain** of the equation. Thus there are no values of $x \Rightarrow$ (A).

6. (AHSME 1966) AB is the diameter of a circle centered at O . C is a point on the circle such that angle BOC is 60° . If the diameter of the circle is 5 inches, the length of chord AC , expressed in inches, is:

(A) 3 (B) $\frac{5\sqrt{2}}{2}$ (C) $\frac{5\sqrt{3}}{2}$ (D) $3\sqrt{3}$ (E) none of these

Answer: C

Solution

centering

We see that $\angle A$ is half the measure of $\angle BOC$, so $\angle A = 30^\circ$. That makes ABC a $30 - 60 - 90$ triangle and sidelength $\overline{AC}$ equal to $\frac{5\sqrt{3}}{2}$. C

7. (AHSME 1966) Let $\frac{35x-29}{x^2-3x+2} = \frac{N_1}{x-1} + \frac{N_2}{x-2}$ be an identity in x . The numerical value of N_1N_2 is:

(A) -246 (B) -210 (C) -29 (D) 210 (E) 246

Answer: A

Solution

$$\frac{N_1}{x-1} + \frac{N_2}{x-2} = \frac{N_1(x-2) + N_2(x-1)}{(x-1)(x-2)} = \frac{(N_1 + N_2)x - (2N_1 + N_2)}{x^2 - 3x + 2}$$

Comparing **coefficients**, we have the system of two equations

$$\begin{aligned} N_1 + N_2 &= 35 \\ N_1 + 2N_2 &= 29 \end{aligned}$$

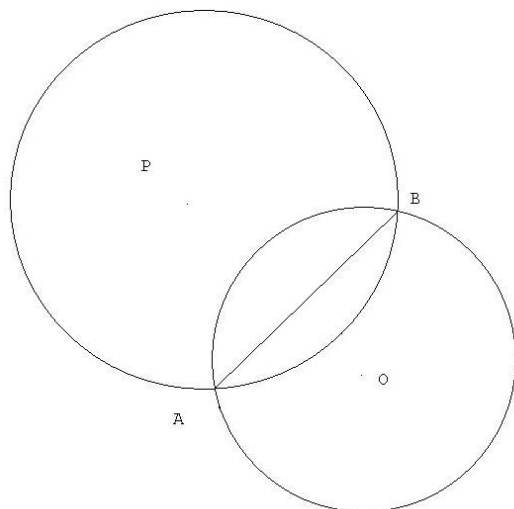
Subtracting the first equation from the second yields $N_2 = -6$. Substituting yields $N_1 = 41$, and their product is $-246 \Rightarrow$ (A).

8. (AHSME 1966) The length of the common chord of two intersecting circles is 16 feet. If the radii are 10 feet and 17 feet, a possible value for the distance between the centers of the circles, expressed in feet, is:

(A) 27 (B) 21 (C) $\sqrt{389}$ (D) 15 (E) undetermined

Answer: B

Solution



centering

Let O be the center of the circle of radius 10 and P be the center of the circle of radius 17. Chord $\overline{AB} = 16$ feet.

$\overline{OA} = \overline{OB} = 10$ feet, since they are radii of the same circle. Hence, $\triangle OAB$ is isosceles with base AB . The height of $\triangle OAB$ from O to AB is $\sqrt{\overline{OB}^2 - (\frac{\overline{AB}}{2})^2} = \sqrt{10^2 - (\frac{16}{2})^2} = \sqrt{100 - 8^2} = \sqrt{100 - 64} = \sqrt{36} = 6$

Similarly, $\overline{PA} = \overline{PB} = 17$. Therefore, $\triangle PAB$ is also isosceles with base AB . The height of the triangle from P to AB is $\sqrt{\overline{PB}^2 - (\frac{\overline{AB}}{2})^2} = \sqrt{17^2 - (\frac{16}{2})^2} = \sqrt{289 - 8^2} = \sqrt{289 - 64} = \sqrt{225} = 15$

The distance between the centers of the circles (points P and O) is the sum of the heights of $\triangle OAB$ and $\triangle PAB$, which is $6 + 15 = 21 \Rightarrow$ **(B)**

9. (AHSME 1966) If $x = (\log_8 2)^{(\log_2 8)}$, then $\log_3 x$ equals:

(A) -3 (B) $-\frac{1}{3}$ (C) $\frac{1}{3}$ (D) 3 (E) 9

Answer: A

Solution

By definition, $\log_8 2 = \frac{1}{3}$ and $\log_2 8 = 3$, so $\log_3 \left(\frac{1}{3}\right)^3 = \log_3 3^{-3} = -3 \Rightarrow$ (A).

10. (AHSME 1966) If the sum of two numbers is 1 and their product is 1, then the sum of their cubes is:

(A) 2 (B) $-2 - \frac{3i\sqrt{3}}{4}$ (C) 0 (D) $-\frac{3i\sqrt{3}}{4}$ (E) -2

Answer: E

Solution

Let the two numbers be a, b ; then $a^3 + b^3 = (a+b)(a^2 - ab + b^2) = (a+b)[(a+b)^2 - 3ab] = (1)[1^2 - 3 \cdot 1] = -2 \Rightarrow (E)$.

11. (AHSME 1966) The sides of triangle BAC are in the ratio $2 : 3 : 4$. BD is the angle-bisector drawn to the shortest side AC , dividing it into segments AD and CD . If the length of AC is 10, then the length of the longer segment of AC is:

(A) $3\frac{1}{2}$ (B) 5 (C) $5\frac{5}{7}$ (D) 6 (E) $7\frac{1}{2}$

Answer: C

Solution

By the **Angle Bisector Theorem**, we have $\frac{BA}{AD} = \frac{BC}{CD}$ which implies $\frac{AD}{DC} = \frac{BA}{BC} = \frac{3}{4}$. So $AC = 10 = AD + DC = \frac{7}{4}DC$, and thus $DC = \frac{40}{7} = 5\frac{5}{7}$. Hence C

12. (AHSME 1966) The number of real values of x that satisfy the equation

$$(2^{6x+3})(4^{3x+6}) = 8^{4x+5}$$

is:

(A) zero (B) one (C) two (D) three (E) greater than 3

Answer: E

Solution

We know that $2^{6x+3} \cdot 4^{3x+6} = 2^{6x+3} \cdot (2^2)^{3x+6} = 2^{6x+3} \cdot 2^{6x+12} = 2^{12x+15}$. We also know that $8^{4x+5} = (2^3)^{4x+5} = 2^{12x+15}$. There are infinite solutions to the equation $2^{12x+15} = 2^{12x+15}$, so the answer is (E) greater than 3.

13. (AHSME 1966) The number of points with positive rational coordinates selected from the set of points in the xy -plane such that $x + y \leq 5$, is:

(A) 9 (B) 10 (C) 14 (D) 15 (E) infinite

Answer: E

Solution

E

14. (AHSME 1966) The length of rectangle $ABCD$ is 5 inches and its width is 3 inches. Diagonal AC is divided into three equal segments by points E and F . The area of triangle BEF , expressed in square inches, is:

(A) $\frac{3}{2}$ (B) $\frac{5}{3}$ (C) $\frac{5}{2}$ (D) $\frac{1}{3}\sqrt{34}$ (E) $\frac{1}{3}\sqrt{68}$

Answer: C

Solution

Draw the rectangle $ABCD$ with $AB = 5$ and $AD = 3$. We created our diagonal, AC and use the Pythagorean Theorem to find the length of AC , which is $\sqrt{34}$. Since EF breaks the diagonal into 3 equal parts, the length of EF is $\frac{\sqrt{34}}{3}$. The only other thing we need is the height of BEF . Realize that the height of BEF is also the height of right triangle ABC using AC as the base. The area of ABC is $\frac{15}{2}$ (using the side lengths of the rectangle). The height of ABC to base AC is $\frac{15}{2}$ divided by $\sqrt{34} \cdot 2$ (remember, we multiply by 2 because we are finding the height from the area of a triangle which is $\frac{bh}{2}$). That simplifies to $\frac{15}{\sqrt{34}}$ which equal to $\frac{15\sqrt{34}}{34}$. Now doing all the arithmetic, $\frac{\sqrt{34}}{3} \cdot \frac{15\sqrt{34}}{34} = \frac{5 \cdot 3}{3 \cdot 2} = \frac{5}{2}$.

Solution 2

Draw rectangle $ABCD$ with $AB = 5$ and $BC = 3$. Next draw a height from vertex B to diagonal AC intersecting at G and similarly from D to AC intersecting at J . Let's call this height h and we can notice that $BG = DJ = h$. Since the points E and F are trisecting diagonal AC the bases and heights of triangles BEA , BEF , BEC , DEA , DEF , and DEC have the same areas. Hence the area of one of these triangles such as BEF is $\frac{5 \cdot 3}{2} = \frac{5}{2}$. So our answer is **C**.

15. (AHSME 1966) If $x - y > x$ and $x + y < y$, then

(A) $y < x$ (B) $x < y$ (C) $x < y < 0$ (D) $x < 0, y < 0$ (E) $x < 0, y > 0$

Answer: D

Solution

From $x - y > x$, we get that $-y > 0 \implies y < 0$. From $x + y < y$, we get that $x < 0$. So, our final answer is **D**.

16. (AHSME 1966) If $\frac{4^x}{2^{x+y}} = 8$ and $\frac{9^{x+y}}{3^{5y}} = 243$, x and y real numbers, then xy equals:

(A) $\frac{12}{5}$ (B) 4 (C) 6 (D) 12 (E) -4

Answer: B

Solution

$\frac{4^x}{2^{x+y}} = 8 \implies 4^x = 2^{x+y+3} \implies 2x = x + y + 3 \implies x = y + 3$. $\frac{9^{x+y}}{3^{5y}} = 243 \implies 9^{x+y} = 3^{5y+5} \implies 2x + 2y = 5y + 5 \implies 2x = 3y + 5$. So, $2y + 6 = 3y + 5 \implies y = 1 \implies x = 4$. Therefore, $xy = 4$ or **B**.

17. (AHSME 1966) The number of distinct points common to the curves $x^2 + 4y^2 = 1$ and $4x^2 + y^2 = 4$ is:

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Answer: C

Solution

Let $a = x^2$ and $b = y^2$. We now have a system of 2 linear equations: $a + 4b = 1$
 $4a + b = 4$ Multiplying the first equation by 4 and then subtracting the second equation from the first one, we get: $15b = 0$

$b = 0$ Now, we can substitute b to solve for a : $4a + 0 = 4$

$a = 1$ Now note that $x^2 = 1 \rightarrow x = \pm 1$ and $y^2 = 0 \rightarrow y = 0$, so the solutions are $(1, 0), (-1, 0)$. There are two solutions, meaning that the answer is **C**.

18. (AHSME 1966) In a given arithmetic sequence the first term is 2, the last term is 29, and the sum of all the terms is 155. The common difference is:

(A) 3 (B) 2 (C) $\frac{27}{19}$ (D) $\frac{13}{9}$ (E) $\frac{23}{38}$

Answer: A

Solution

A

19. (AHSME 1966) Let s_1 be the sum of the first n terms of the arithmetic sequence 8, 12, $\dots$ and let s_2 be the sum of the first n terms of the arithmetic sequence 17, 19, $\dots$. Assume $n \neq 0$. Then $s_1 = s_2$ for:

(A) no value of n (B) one value of n (C) two values of n (D) four values of n (E) more than four values of n

Answer: B

Solution

B

20. (AHSME 1966) The negation of the proposition "For all pairs of real numbers a, b , if $a = 0$, then $ab = 0$ " is: There are real numbers a, b such that

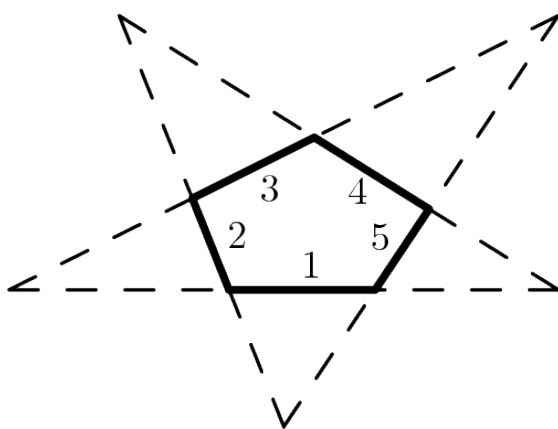
(A) $a \neq 0$ and $ab \neq 0$ (B) $a \neq 0$ and $ab = 0$ (C) $a = 0$ and $ab \neq 0$
 (D) $ab \neq 0$ and $a \neq 0$ (E) $ab = 0$ and $a \neq 0$

Answer: C

Solution

C

21. (AHSME 1966)



centering

An " n -pointed star" is formed as follows: the sides of a convex polygon are numbered consecutively $1, 2, \dots, k, \dots, n$, $n \geq 5$; for all n values of k , sides k and $k + 2$ are non-parallel, sides $n + 1$ and $n + 2$ being respectively identical with sides 1 and 2; prolong the n pairs of sides numbered k and $k + 2$ until they meet. (A figure is shown for the case $n = 5$).

Let S be the degree-sum of the interior angles at the n points of the star; then S equals:

(A) 180 (B) 360 (C) $180(n + 2)$ (D) $180(n - 2)$ (E) $180(n - 4)$

Answer: E

Solution

E

22. (AHSME 1966) Consider the statements: (I) $\sqrt{a^2 + b^2} = 0$, (II) $\sqrt{a^2 + b^2} = ab$, (III) $\sqrt{a^2 + b^2} = a + b$, (IV) $\sqrt{a^2 + b^2} = a - b$, where we allow a and b to be real or complex numbers. Those statements for which there exist solutions other than $a = 0$ and $b = 0$, are:

(A) (I),(II),(III),(IV) (B) (II),(III),(IV) only (C) (I),(III),(IV) only (D) (III),(IV) only (E) (I)

Answer: A

Solution

We are given the following statements:

- a) $\sqrt{a^2 + b^2} = 0$
 b) $\sqrt{a^2 + b^2} = ab$
 c) $\sqrt{a^2 + b^2} = a + b$
 d) $\sqrt{a^2 + b^2} = a - b$

We are asked to find the solutions where these statements hold, excluding the trivial solution $a = 0$ and $b = 0$.

Statement (I): $\sqrt{a^2 + b^2} = 0$

Squaring both sides:

$$a^2 + b^2 = 0$$

For real or complex numbers, $a^2 + b^2 = 0$ implies that both $a = 0$ and $b = 0$ because the sum of squares of two real or complex numbers is 0 only if both are 0.

Thus, the only solution is $a = 0$ and $b = 0$.

Conclusion for Statement (I): There are no solutions other than $a = 0$ and $b = 0$.

Statement (II): $\sqrt{a^2 + b^2} = ab$

Squaring both sides:

$$a^2 + b^2 = (ab)^2 = a^2b^2$$

Rearranging:

$$a^2 + b^2 - a^2b^2 = 0$$

Factoring:

$$a^2(1 - b^2) + b^2 = 0$$

Testing for specific solutions:

- If $a = 1$ and $b = 1$, we get:

$$\sqrt{1^2 + 1^2} = 1 \cdot 1 \quad \Rightarrow \quad \sqrt{2} = 1$$

which is false. Thus, there are no simple real solutions.

- For complex solutions, trying specific cases like $a = i$ and $b = 1$ also results in contradictions.

Thus, this equation has no nonzero solutions.

Conclusion for Statement (II): No solutions other than $a = 0$ and $b = 0$.

Statement (III): $\sqrt{a^2 + b^2} = a + b$

Squaring both sides:

$$a^2 + b^2 = (a + b)^2$$

Expanding the right-hand side:

$$a^2 + b^2 = a^2 + 2ab + b^2$$

Canceling $a^2 + b^2$ from both sides:

$$0 = 2ab$$

Thus:

$$ab = 0$$

This implies that either $a = 0$ or $b = 0$. Therefore, there are solutions other than $a = 0$ and $b = 0$, such as $a = 0$ and $b \neq 0$, or $a \neq 0$ and $b = 0$.

Conclusion for Statement (III): There are solutions other than $a = 0$ and $b = 0$.

Statement (IV): $\sqrt{a^2 + b^2} = a - b$

Squaring both sides:

$$a^2 + b^2 = (a - b)^2$$

Expanding the right-hand side:

$$a^2 + b^2 = a^2 - 2ab + b^2$$

Canceling $a^2 + b^2$ from both sides:

$$0 = -2ab$$

Thus:

$$ab = 0$$

This implies that either $a = 0$ or $b = 0$. Therefore, there are solutions other than $a = 0$ and $b = 0$, such as $a = 0$ and $b \neq 0$, or $a \neq 0$ and $b = 0$.

Conclusion for Statement (IV): There are solutions other than $a = 0$ and $b = 0$.

Final Conclusion:

The statements that have solutions other than $a = 0$ and $b = 0$ are (III) and (IV).

Thus, the correct answer is:

(A) (I), (II), (III), (IV)

23. (AHSME 1966) If x is real and $4y^2 + 4xy + x + 6 = 0$, then the complete set of values of x for which y is real, is:

- (A) $x \leq -2$ or $x \geq 3$ (B) $x \leq 2$ or $x \geq 3$ (C) $x \leq -3$ or $x \geq 2$
 (D) $-3 \leq x \leq 2$ (E) $-2 \leq x \leq 3$

Answer: A

Solution

We treat the equation as a quadratic equation in y for which the discriminant

$$D = 16x^2 - 16(x + 6) = 16(x^2 - x - 6) = 16(x - 3)(x + 2)$$

For y to be real $D \geq 0$. This inequality is satisfied when $x \leq -2$ or $x \geq 3$ or $\boxed{A}$

24. (AHSME 1966) If $\log_M N = \log_N M$, $M \neq N$, $MN > 0$, $M \neq 1$, $N \neq 1$, then MN equals:

(A) $\frac{1}{2}$ (B) 1 (C) 2 (D) 10
(E) a number greater than 2 and less than 10

Answer: B

Solution

If we change the base of $\log_M N$ to base N , we get $\frac{1}{\log_N M} = \log_N M$. Multiplying both sides by $\log_N M$, we get $\log_N M^2 = 1$. Since $N \neq M$, $\log_N M = -1$. So $N^{-1} = M \rightarrow \frac{1}{N} = M \rightarrow MN = 1$. So the answer is $\boxed{B}$.

25. (AHSME 1966) If $F(n+1) = \frac{2F(n)+1}{2}$ for $n = 1, 2, \dots$ and $F(1) = 2$, then $F(101)$ equals:

(A) 49 (B) 50 (C) 51 (D) 52 (E) 53

Answer: D

Solution

Notice that $\frac{2F(n)+1}{2} = F(n) + \frac{1}{2}$.

This means that for every single increment n goes up from 1, $F(n)$ will increase by $\frac{1}{2}$. Since 101 is 100 increments from 1, $F(n)$ will increase $\frac{1}{2} \times 100 = 50$.

Since $F(1) = 2$, $F(101)$ will equal $2 + 50 = \boxed{(D) 52}$.

Solution by davidaops.

26. (AHSME 1966) Let m be a positive integer and let the lines $13x + 11y = 700$ and $y = mx - 1$ intersect in a point whose coordinates are integers. Then m can be:

(A) 4 only (B) 5 only (C) 6 only (D) 7 only
(E) one of the integers 4,5,6,7 and one other positive integer

Answer: C

Solution

Substitute the second equation into the first one, we have $13x + 11mx - 11 = 700$.

So $(13 + 11m)x = 711$. So $13 + 11m$ is a factor of 711. $711 = 3 * 3 * 79$, so the factors of 711 are: 1, 3, 9, 79, 237, 711.

Clearly, because $m \geq 1$, so $13 + 11m \geq 24$. So we only need to check whether m is an integer when $11m + 13 = 79, 237, 711$.

When $11m + 13 = 79$, $m = 6$.

Checking the other two choices, m doesn't yield to be an integer. So $m = 6$ is the only option. Select $\boxed{C}$.

27. (*AHSME 1966*) At his usual rate a man rows 15 miles downstream in five hours less time than it takes him to return. If he doubles his usual rate, the time downstream is only one hour less than the time upstream. In miles per hour, the rate of the stream's current is:

(A) 2 (B) $\frac{5}{2}$ (C) 3 (D) $\frac{7}{2}$ (E) 4

Answer: A

Solution

Let the speed of rowing in still water and water speed be r, w .

Then:

$$\frac{15}{r-w} - \frac{15}{r+w} = 5.$$

$$\frac{15}{2r-w} - \frac{15}{2r+w} = 1.$$

Solving (remember: both variables must be positive!), we have $r = 4, w = 2$. Select

A.

28. (*AHSME 1966*) Five points O, A, B, C, D are taken in order on a straight line with distances $OA = a, OB = b, OC = c$, and $OD = d$. P is a point on the line between B and C and such that $AP : PD = BP : PC$. Then OP equals:

(A) $\frac{b^2-bc}{a-b+c-d}$ (B) $\frac{ac-bd}{a-b+c-d}$
 (C) $-\frac{bd+ac}{a-b+c-d}$ (D) $\frac{bc+ad}{a+b+c+d}$ (E) $\frac{ac-bd}{a+b+c+d}$

Answer: B

Solution

B

29. (*AHSME 1966*) The number of positive integers less than 1000 divisible by neither 5 nor 7 is:

(A) 688 (B) 686 (C) 684 (D) 658 (E) 630

Answer: B

Solution

The number of positive integers under 1000 that are divisible by 5 is $\lfloor \frac{999}{5} \rfloor = 199$. The number of positive integers under 1000 that are divisible by 7 is $\lfloor \frac{1000}{7} \rfloor = 142$. Adding them together, we get 341. However, we have over counted the numbers which are divisible by 35. There are $\lfloor \frac{999}{35} \rfloor = 28$ of these. So, the number of positive integers divisible by 7 or 5 under 1000 is $341 - 28 = 313$. We can conclude that the number of positive integers divisible by neither 5 or 7 is $999 - 313 = 686$ or answer choice B.

30. (*AHSME 1966*) If three of the roots of $x^4 + ax^2 + bx + c = 0$ are 1, 2, and 3 then the value of $a + c$ is:

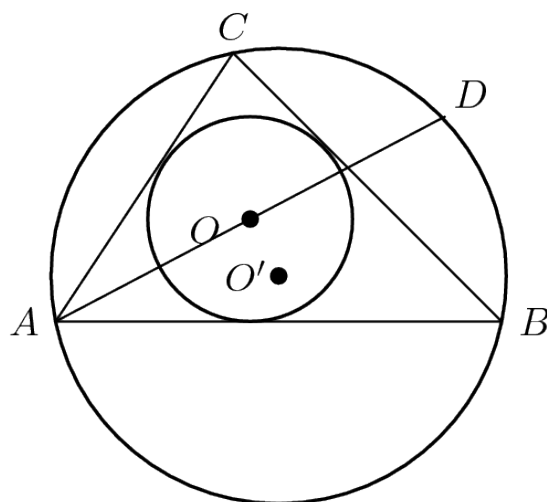
(A) 35 (B) 24 (C) -12 (D) -61 (E) -63

Answer: D

Solution

Since this is a quartic equation, there are going to be 4 solutions. By Vieta's formulas, since the x^3 term is 0, the sum of the roots is also 0. Therefore, the 4th root of this polynomial is -6 . Lastly, by Vieta's Formulas, $a + c = -(1 \cdot 2 + 1 \cdot 3 + 1 \cdot -6 + 2 \cdot 3 + 2 \cdot -6 + 3 \cdot -6) + 1 \cdot 2 \cdot 3 \cdot -6 = -25 - 36 = -61 =$ D

31. (AHSME 1966)



centering

Triangle ABC is inscribed in a circle with center O' . A circle with center O is inscribed in triangle ABC . AO is drawn, and extended to intersect the larger circle in D . Then we must have:

- (A) $CD = BD = O'D$ (B) $AO = CO = OD$ (C) $CD = CO = BD$
 (D) $CD = OD = BD$ (E) $O'B = O'C = OD$

Answer: D**Solution**

We will prove that $\triangle DOB$ and $\triangle COD$ is isosceles, meaning that $CD = OD = BD$ and hence **D**.

Let $\angle A = 2\alpha$ and $\angle B = 2\beta$. Since the incentre of a triangle is the intersection of its angle bisectors, $\angle OAB = \alpha$ and $\angle ABO = \beta$. Hence $\angle DOB = \alpha + \beta$. Since quadrilateral $ABCD$ is cyclic, $\angle CAD = \alpha = \angle CBD$. So $\angle OBD = \angle OBC + \angle CBD = \alpha + \beta = \angle DOB$. This means that $\triangle DOB$ is isosceles, and hence $BD = OD$.

Now let $\angle C = 2\gamma$ which means $\angle ACO = \angle COD = \gamma$. Since $ABCD$ is cyclic, $\angle DAB = \alpha = \angle DCB$. Also, $\angle DAC = \alpha$ so $\angle DOC = \alpha + \gamma$. Thus, $\angle OCD = \angle OCB + \angle BCD = \gamma + \alpha = \angle DOC$ which means $\triangle COD$ is isosceles, and hence $CD = OD = BD$.

Thus our answer is **D**.

32. (AHSME 1966) Let M be the midpoint of side AB of triangle ABC . Let P be a point on AB between A and M , and let MD be drawn parallel to PC and intersecting BC at D . If the ratio of the area of triangle BPD to that of triangle ABC is denoted by r , then

- (A) $\frac{1}{2} < r < 1$, depending upon the position of P
 (B) $r = \frac{1}{2}$, independent of the position of P
 (C) $\frac{1}{2} \leq r < 1$, depending upon the position of P
 (D) $\frac{1}{3} < r < \frac{2}{3}$, depending upon the position of P
 (E) $r = \frac{1}{3}$, independent of the position of P

Answer: B

Solution

Both triangles are similar by some ratio r . Place point P on A . Then, line MD intersects BC at midpoint D , and the ratio of the two triangles is $\frac{1}{2}$. Place point P on M , and then the same thing occurs, line MD intersects BC at midpoint D , and the ratio is $\frac{1}{2}$. Now as P moves between AM , D follows it to maintain parallelism. However, the only exception is when P reaches past the midpoint of AM , and going towards A , when D moves back toward the midpoint of BC to maintain parallelism. Because both triangles are always similar at the extrema points, that are, A and M , as P moves around, the value of r won't change as it is essentially the same as the cevian MD of triangle BPD moving along the base BP . So essentially, that the effect of P moving along line AM is essentially just moving the cevian MD , as it changes as P moves, along the base. However, because the two triangles are similar, moving the cevian MD is the same as moving the cevian CP , and they maintain the same ratio. At the extrema's, these values of r were $\frac{1}{2}$, so as P moves, the value of r becomes unchanged and independent of that from the position of P . Thus our answer is **B**.

33. (AHSME 1966) If $ab \neq 0$ and $|a| \neq |b|$, the number of distinct values of x satisfying the equation

$$\frac{x-a}{b} + \frac{x-b}{a} = \frac{b}{x-a} + \frac{a}{x-b},$$

is:

- (A) zero (B) one (C) two (D) three (E) four

Answer: D

Solution

Let $m = \frac{x-a}{b}$ and $n = \frac{x-b}{a}$ then we have

$$m + n = \frac{1}{m} + \frac{1}{n}$$

$$m + n = \frac{m+n}{mn}$$

Notice that the equation is possible iff $m + n = 0$ or $mn = 1$.

If $m + n = 0$ then

$$\frac{x-a}{b} + \frac{x-b}{a} = 0$$

$$\frac{x-a}{b} = \frac{b-x}{a}$$

$$x = \frac{a^2 + b^2}{a + b}$$

Which yields 1 solution for x .

If $mn = 0$ then

$$\left(\frac{x-a}{b}\right)\left(\frac{x-b}{a}\right) = 1$$

$$x^2 - (a+b)x = 0$$

Solving the quadratic gets another 2 solutions for x .

Thus there are **(D) three** solutions in total.

34. (*AHSME 1966*) Let r be the speed in miles per hour at which a wheel, 11 feet in circumference, travels. If the time for a complete rotation of the wheel is shortened by $\frac{1}{4}$ of a second, the speed r is increased by 5 miles per hour. Then r is:
 (A) 9 (B) 10 (C) $10\frac{1}{2}$ (D) 11 (E) 12

Answer: B

Solution

The circumference of the wheel is $\frac{11}{5280}$ miles. Let the time for the rotation in seconds be t . So $rt = \frac{11}{5280} * 3600$. We also know reducing the time by $\frac{1}{4}$ of a second makes r increase by 5. So $(r + 5)(t - \frac{1}{4}) = \frac{11}{5280} * 3600$. Solving for r we get $r = 10$. So our answer is (B) 10.

35. (*AHSME 1966*) Let O be an interior point of triangle ABC , and let $s_1 = OA + OB + OC$. If $s_2 = AB + BC + CA$, then
 (A) for every triangle $s_2 > 2s_1, s_1 \leq s_2$
 (B) for every triangle $s_2 > 2s_1, s_1 < s_2$
 (C) for every triangle $s_1 > \frac{1}{2}s_2, s_1 < s_2$
 (D) for every triangle $s_2 \geq 2s_1, s_1 \leq s_2$
 (E) neither (A) nor (B) nor (C) nor (D) applies to every triangle

Answer: C

Solution

By the Triangle Inequality, we see that $s_1 > \frac{1}{2}s_2$, therefore C. -Dark_Lord

36. (*AHSME 1966*) Let $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \cdots + a_{2n}x^{2n}$ be an identity in x . If we let $s = a_0 + a_2 + a_4 + \cdots + a_{2n}$, then s equals:
 (A) 2^n (B) $2^n + 1$ (C) $\frac{3^n - 1}{2}$ (D) $\frac{3^n}{2}$ (E) $\frac{3^n + 1}{2}$

Answer: E

Solution 1

Let $f(x) = (1 + x + x^2)^n$ then we have

$$f(1) = a_0 + a_1 + a_2 + \dots + a_{2n} = (1 + 1 + 1)^n = 3^n$$

$$f(-1) = a_0 - a_1 + a_2 - \dots + a_{2n} = (1 - 1 + 1)^n = 1$$

Adding yields

$$f(1) + f(-1) = 2(a_0 + a_2 + a_4 + \dots + a_{2n}) = 3^n + 1$$

Thus $s = \frac{3^n + 1}{2}$, or (E).

37. (*AHSME 1966*) Three men, Alpha, Beta, and Gamma, working together, do a job in 6 hours less time than Alpha alone, in 1 hour less time than Beta alone, and in one-half the time needed by Gamma when working alone. Let h be the number of hours needed by Alpha and Beta, working together, to do the job. Then h equals:
 (A) $\frac{5}{2}$ (B) $\frac{3}{2}$ (C) $\frac{4}{3}$ (D) $\frac{5}{4}$ (E) $\frac{3}{4}$

Answer: C

Solution

Let A, B, C denote the number of hours needed by Alpha, Beta, Gamma, respectively. We also have their respective efficiency $\frac{1}{A}$, $\frac{1}{B}$, and $\frac{1}{C}$. Thus we get the equations

$$\frac{1}{A} + \frac{1}{B} + \frac{1}{C} = \frac{1}{A-6}$$

$$\frac{1}{A} + \frac{1}{B} + \frac{1}{C} = \frac{1}{B-1}$$

$$\frac{1}{A} + \frac{1}{B} + \frac{1}{C} = \frac{2}{C} \implies \frac{1}{A} + \frac{1}{B} = \frac{1}{C}$$

Equating the first 2 equations gets

$$\frac{1}{A-6} = \frac{1}{B-1} \implies A = B + 5$$

Substituting the new relation along with the third equation into the first equation gets

$$\frac{2}{A} + \frac{2}{A-5} = \frac{1}{A-6}$$

Solving the quadratic gets $A = 3$ or $\frac{20}{3}$. Since $B = A - 5 > 0$, $A = \frac{20}{3}$ is the only legit solution.

Thus $B = \frac{5}{3}$ and $h = \frac{1}{\frac{1}{A} + \frac{1}{B}} = \frac{4}{3}$, **(C)**.

38. (AHSME 1966) In triangle ABC the medians AM and CN to sides BC and AB , respectively, intersect in point O . P is the midpoint of side AC , and MP intersects CN in Q . If the area of triangle OMQ is n , then the area of triangle ABC is:

(A) $16n$ (B) $18n$ (C) $21n$ (D) $24n$ (E) $27n$

Answer: D

Solution

Construct triangle $\triangle ABC$ with points M, N, P being the midpoints of sides $\overline{CB}, \overline{AB}, \overline{AC}$, respectively. Proceed by drawing all medians. Then draw all medians (so draw $\overline{AM}, \overline{BP}, \overline{CN}$). Next, draw line $\overline{PM}$ and label $\overline{PM}$'s intersection with $\overline{CN}$ as the point Q . From the problem, the area of $\triangle QMO$ is n , but by vertical angles we know that $\angle QOM = \angle AOM$. Furthermore, since line $\overline{PM}$ is drawn from the midpoint of $\overline{AC}$ to the midpoint of $\overline{CB}$, we know that $\overline{PM}$ is parallel to $\overline{AB}$ (via SAS similarity on triangles PCM and ABC). From these parallel lines we know that $\angle PMO = \angle OAB$ which indicates that $\triangle QOM \sim \triangle ANO$. The linear ratio from $\triangle QOM$ to $\triangle ANO$ is 1:2 because line segment $\overline{OM}$ is one half of line segment $\overline{AO}$ since $\overline{AO}$ and $\overline{OM}$ make up the median $\overline{AM}$. Thus the area ratio is 1:4. So $\triangle ANO$ has area $4n$. Since $\triangle ONB$ has the same height and base as $\triangle ANO$ we know that the area of $\triangle AOB = 8n$. The medians form 3 triangles each with area $1/3$ of the total triangle (these triangles are $\triangle AOB, \triangle COB, \triangle COA$). Thus since $\triangle AOB = 4n \implies \triangle ABC = 24n$ **(D)**.
multiply by 3

- LJ

39. (AHSME 1966) In base R_1 the expanded fraction F_1 becomes $.373737\cdots$, and the expanded fraction F_2 becomes $.737373\cdots$. In base R_2 fraction F_1 , when expanded, becomes $.252525\cdots$, while the fraction F_2 becomes $.525252\cdots$. The sum of R_1 and R_2 , each written in the base ten, is:

- (A) 24 (B) 22 (C) 21 (D) 20 (E) 19

Answer: E

Solution

First, let's write F_1 as a proper fraction in base R_1 . To do that, note that: $F_1 = 0.373737 \dots$. Multiplying this equation on both sides R_1^2 , we get: $R_1^2 F_1 = 37.373737 \dots$. Subtracting the first equation from the second one, we get: $R_1^2 F_1 - F_1 = 37$

$$F_1(R_1^2 - 1) = 37$$

$$F_1 = \frac{3R_1+7}{R_1^2-1}$$

Using a very similar method as above, we can see that: $F_1 = \frac{3R_1+7}{R_1^2-1} =$

$$\frac{2R_2+5}{R_2^2-1} \text{ and } F_2 = \frac{7R_1+3}{R_1^2-1} = \frac{5R_2+2}{R_2^2-1}. \text{ Dividing the 2 equations to get some potential}$$

solutions, we get (note that we don't have to worry about division by zero because

$$R_1, R_2 > 7): \frac{3R_1+7}{7R_1+3} = \frac{2R_2+5}{5R_2+2}$$

$$15R_1R_2 + 6R_1 + 35R_2 + 14 = 14R_1R_2 + 35R_1 + 6R_2 + 15$$

$$R_1R_2 - 29R_1 + 29R_2 = 1$$

$(R_1 + 29)(R_2 - 29) = -840$ Since non-integer bases are rarely used, we can try to

assume that the solution is positive integers and see if we get any solutions. Notice

that $(R_2 - 29)$ has to be a negative factor of 840. We need to plug in values of

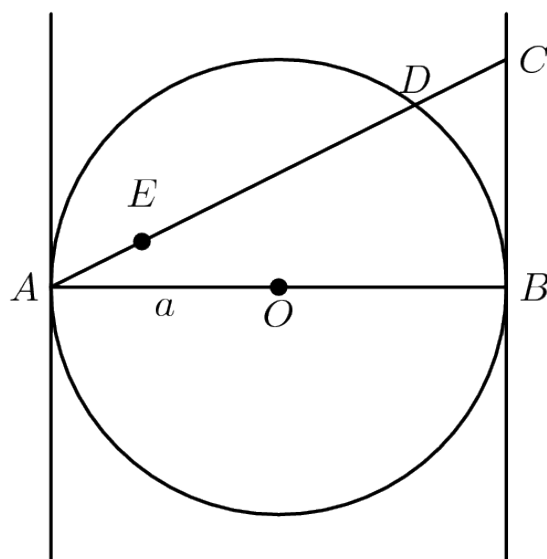
$R_2 > 7$. 840 divides -21, so we plug in 8 to check. Luckily, when $R_2 = 8$, we see

that $R_1 = 11$, and furthermore, $11+8=19$ is one of the answers. We can quickly

test it into the original equations $F_1 = \frac{3R_1+7}{R_1^2-1} = \frac{2R_2+5}{R_2^2-1}$ and $F_2 = \frac{7R_1+3}{R_1^2-1} = \frac{5R_2+2}{R_2^2-1}$,

and we see they indeed work. Therefore the answer is (E)19

40. (AHSME 1966)



centering

In this figure AB is a diameter of a circle, centered at O , with radius a . A chord AD is drawn and extended to meet the tangent to the circle at B in point C . Point E is taken on AC so the $AE = DC$. Denoting the distances of E from the tangent through A and from the diameter AB by x and y , respectively, we can deduce the relation:

$$(A) y^2 = \frac{x^3}{2a-x} \quad (B) y^2 = \frac{x^3}{2a+x} \quad (C) y^4 = \frac{x^2}{2a-x}$$

$$(D) x^2 = \frac{y^2}{2a-x} \quad (E) x^2 = \frac{y^2}{2a+x}$$

Answer: A

Solution

A

You can troll this problem completely by taking $x=y=r$. Orion2010

§3 AHSME 1967

1. (*AHSME 1967*) The three-digit number $2a3$ is added to the number 326 to give the three-digit number $5b9$. If $5b9$ is divisible by 9, then $a + b$ equals
 (A) 2 (B) 4 (C) 6 (D) 8 (E) 9

Answer: C**Solution**

If $5b9$ is divisible by 9, this must mean that $5 + b + 9$ is a multiple of 9. So,

$$5 + b + 9 = 9, 18, 27, 36 \dots$$

Because $5 + 9 = 14$ and b is in between 0 and 9,

$$5 + b + 9 = 18$$

$$b = 4$$

The question states that

$$2a3 + 326 = 549$$

so

$$2a3 = 549 - 326$$

$$2a3 = 223$$

$$a = 2$$

$$a + b = 6$$

which is answer choice C.

2. (*AHSME 1967*) An equivalent of the expression

$$\left(\frac{x^2+1}{x}\right)\left(\frac{y^2+1}{y}\right) + \left(\frac{x^2-1}{y}\right)\left(\frac{y^2-1}{x}\right), \quad xy \neq 0,$$

is:

- (A) 1 (B) $2xy$ (C) $2x^2y^2 + 2$ (D) $2xy + \frac{2}{xy}$ (E) $\frac{2x}{y} + \frac{2y}{x}$

Answer: D**Solution 1**

$$\begin{aligned} & \left(\frac{x^2+1}{x}\right)\left(\frac{y^2+1}{y}\right) + \left(\frac{x^2-1}{y}\right)\left(\frac{y^2-1}{x}\right) \\ &= \frac{(x^2+1)(y^2+1)}{xy} + \frac{(x^2-1)(y^2-1)}{xy} \\ &= \frac{(x^2y^2 + x^2 + y^2 + 1) + (x^2y^2 - x^2 - y^2 + 1)}{xy} \end{aligned}$$

$$\frac{x^2y^2 + x^2 + y^2 + 1 + x^2y^2 - x^2 - y^2 + 1}{xy}$$

$$\frac{2x^2y^2 + 2}{xy}$$

$$\frac{2x^2y^2}{xy} + \frac{2}{xy}$$

$$\frac{2xy * xy}{xy} + \frac{2}{xy}$$

$$2xy + \frac{2}{xy}$$

The answer is $\boxed{D}$

Solution 2 (Answer Choices)

The denominator before simplification is xy . As the constant term will not be divisible by the denominator, we inspect that the fraction with only the constant term in the numerator is $\frac{(1 \cdot 1) + (-1 \cdot -1)}{xy}$. The only answer choice that matches this is $\boxed{D}$. megaboy6679

3. (AHSME 1967) The side of an equilateral triangle is s . A circle is inscribed in the triangle and a square is inscribed in the circle. The area of the square is:
- (A) $\frac{s^2}{24}$ (B) $\frac{s^2}{6}$ (C) $\frac{s^2\sqrt{2}}{6}$ (D) $\frac{s^2\sqrt{3}}{6}$ (E) $\frac{s^2}{3}$

Answer: B

Solution

The radius of our circle is $\frac{\sqrt{3}}{6}s$. This means the square has side length $\frac{\sqrt{6}}{6}s$ which has area of $\frac{s^2}{6} = \boxed{B}$.

4. (AHSME 1967) Given $\frac{\log a}{p} = \frac{\log b}{q} = \frac{\log c}{r} = \log x$, all logarithms to the same base and $x \neq 1$. If $\frac{b^2}{ac} = x^y$, then y is:
- (A) $\frac{q^2}{p+r}$ (B) $\frac{p+r}{2q}$ (C) $2q - p - r$ (D) $2q - pr$ (E) $q^2 - pr$

Answer: C

Solution

We are given:

$$\frac{b^2}{ac} = x^y$$

Taking the logarithm on both sides:

$$\log \left(\frac{b^2}{ac} \right) = \log x^y$$

Using the properties of logarithms:

$$2 \log b - \log a - \log c = y \log x$$

Substituting the values given in the problem statement:

$$2q \log x - p \log x - r \log x = y \log x$$

Since $x \neq 1$, dividing each side by $\log x$ we get:

$$y = \boxed{(C) \ 2q - p - r}$$

5. (AHSME 1967) A triangle is circumscribed about a circle of radius r inches. If the perimeter of the triangle is P inches and the area is K square inches, then $\frac{P}{K}$ is:

(A) independent of the value of r (B) $\frac{\sqrt{2}}{r}$ (C) $\frac{2}{\sqrt{r}}$ (D) $\frac{2}{r}$ (E) $\frac{r}{2}$

Answer: D

Solution

The area K of the triangle can be expressed in terms of its inradius r and its semiperimeter s as:

$$K = r \times s = r \times \frac{P}{2}$$

$$\text{So, } \frac{P}{K} = \boxed{\text{(C)} \frac{2}{r}}.$$

6. (AHSME 1967) If $f(x) = 4^x$ then $f(x+1) - f(x)$ equals:

(A) 4 (B) $f(x)$ (C) $2f(x)$ (D) $3f(x)$ (E) $4f(x)$

Answer: D

Solution

The desired expression is equal to $4^{x+1} - 4^x$. Using the fact that $4^{x+1} = 4^x * 4$, we see that the answer is $3 * 4^x$. D

7. (AHSME 1967) If $\frac{a}{b} < -\frac{c}{d}$ where a , b , c , and d are real numbers and $bd \neq 0$, then:

(A) a must be negative (B) a must be positive (C) a must not be zero (D) a can be negative
(E) a can be positive, negative, or zero

Answer: E

Solution

Notice that b and d are irrelevant to the problem. Also notice that the negative sign is irrelevant since c can be any real number. So the question is equivalent to asking what values a can take on given the inequality $a < c$. The answer, of course, is that a could be anything, or (E) a can be positive, negative, or zero

8. (AHSME 1967) To m ounces of a $m\%$ solution of acid, x ounces of water are added to yield a $(m-10)\%$ solution. If $m > 25$, then x is

(A) $\frac{10m}{m-10}$ (B) $\frac{5m}{m-10}$ (C) $\frac{m}{m-10}$ (D) $\frac{5m}{m-20}$
(E) not determined by the given information

Answer: A

Solution

A

9. (AHSME 1967) Let K , in square units, be the area of a trapezoid such that the shorter base, the altitude, and the longer base, in that order, are in arithmetic progression. Then:

(A) K must be an integer (B) K must be a rational fraction
(C) K must be an irrational number (D) K must be an integer or a rational fraction
(E) taken alone neither (A) nor (B) nor (C) nor (D) is true

Answer: E

Solution

From the problem we can set the altitude equal to a , the shorter base equal to $a - d$, and the longer base equal to $a + d$. By the formula for the area of a trapezoid, we have $K = a^2$. However, since a can equal any real number $(3, 2.7, \pi)$, none of the statements A, B, C, D need to be true, so the answer is E.

10. (AHSME 1967) If $\frac{a}{10^x-1} + \frac{b}{10^x+2} = \frac{2 \cdot 10^x + 3}{(10^x-1)(10^x+2)}$ is an identity for positive rational values of x , then the value of $a - b$ is:

(A) $\frac{4}{3}$ (B) $\frac{5}{3}$ (C) 2 (D) $\frac{11}{4}$ (E) 3

Answer: A

Solution

Given the equation:

$$\frac{a}{10^x-1} + \frac{b}{10^x+2} = \frac{2 \cdot 10^x + 3}{(10^x-1)(10^x+2)}$$

Let's simplify by letting $y = 10^x$. The equation becomes:

$$\frac{a}{y-1} + \frac{b}{y+2} = \frac{2y+3}{(y-1)(y+2)}$$

Multiplying each term by the common denominator $(y-1)(y+2)$, we obtain:

$$a(y+2) + b(y-1) = 2y+3$$

For the equation to be an identity, the coefficients of like terms on both sides must be equal. Equating the coefficients of y and the constant terms, we get the system of equations:

$$\begin{aligned} a + b &= 2 && \text{(from coefficients of } y) \\ 2a - b &= 3 && \text{(from constant terms)} \end{aligned}$$

Solving this system, we find:

$$a = \frac{5}{3} \quad \text{and} \quad b = \frac{1}{3}$$

Thus, the difference is:

$$a - b = \boxed{\text{(A)} \frac{4}{3}}$$

11. (AHSME 1967) If the perimeter of rectangle $ABCD$ is 20 inches, the least value of diagonal $\overline{AC}$, in inches, is:

(A) 0 (B) $\sqrt{50}$ (C) 10 (D) $\sqrt{200}$ (E) none of these

Answer: B

Solution

For rectangle $ABCD$ with perimeter 20, the diagonal AC is given by:

$$AC = \sqrt{l^2 + w^2}$$

To minimize AC , l and w should be equal (i.e., the rectangle is a square). Thus, $l = w = 5$. So, the minimum AC is:

$$AC = \sqrt{5^2 + 5^2} = \boxed{\text{(B)} \sqrt{50}}$$

proloto

12. (AHSME 1967) If the (convex) area bounded by the x-axis and the lines $y = mx + 4$, $x = 1$, and $x = 4$ is 7, then m equals:

(A) $-\frac{1}{2}$ (B) $-\frac{2}{3}$ (C) $-\frac{3}{2}$ (D) -2 (E) none of these

Answer: B

13. (AHSME 1967) A triangle ABC is to be constructed given a side a (opposite angle A), angle B , and h_c , the altitude from C . If N is the number of noncongruent solutions, then N

(A) is 1 (B) is 2 (C) must be zero (D) must be infinite (E) must be zero or infinite

Answer: E

14. (AHSME 1967) Let $f(t) = \frac{t}{1-t}$, $t \neq 1$. If $y = f(x)$, then x can be expressed as

(A) $f\left(\frac{1}{y}\right)$ (B) $-f(y)$ (C) $-f(-y)$ (D) $f(-y)$ (E) $f(y)$

Answer: C

Solution

Since we know that $y = f(x)$, we can solve for y in terms of x . This gives us

$$y = \frac{x}{1-x}$$

$$\Rightarrow y(1-x) = x$$

$$\Rightarrow y - yx = x$$

$$\Rightarrow y = yx + x$$

$$\Rightarrow y = x(y+1)$$

$$\Rightarrow x = \frac{y}{y+1}$$

Therefore, we want to find the function with y that outputs $\frac{y}{y+1}$. Listing out the possible outputs from each of the given functions we get $f\left(\frac{1}{y}\right) = \frac{1}{y-1}$

$$f(y) = \frac{1}{1-y}$$

$$f(-y) = \frac{-y}{y+1}$$

$$-f(y) = \frac{y}{y-1}$$

$$-f(-y) = \frac{y}{y+1}$$

Since $-f(-y) = \frac{y}{y+1} = x$ the answer must be $\boxed{\text{C}}$.

15. (AHSME 1967) The difference in the areas of two similar triangles is 18 square feet, and the ratio of the larger area to the smaller is the square of an integer. The area of the smaller triangle, in square feet, is an integer, and one of its sides is 3 feet. The corresponding side of the larger triangle, in feet, is:

(A) 12 (B) 9 (C) $6\sqrt{2}$ (D) 6 (E) $3\sqrt{2}$

Answer: D

Solution

If the areas of the two triangles are A_1 and A_2 with $A_1 > A_2$, we are given that $A_1 - A_2 = 18$ and $\frac{A_1}{A_2} = k^2$

Plugging in the second equation into the first leads to $(k^2 - 1)A_2 = 18$.

If A_2 is an integer, it can only be a factor of 18 - namely 1, 2, 3, 6, 9, 18.

Since $A_1 = A_2 + 18$, this would lead to $A_1 = 19, 20, 21, 24, 27, 36$.

This would lead to $\frac{A_1}{A_2} = 19, 10, 7, 4, 3, 2$, and the only square is 4.

Thus, the ratio of the areas is 4, and hence the ratio of the sides is $\sqrt{4} = 2$. The corresponding side has a length of $3 \cdot 2 = 6$, which is option **[D]**.

16. (AHSME 1967) Let the product $(12)(15)(16)$, each factor written in base b , equal 3146 in base b . Let $s = 12 + 15 + 16$, each term expressed in base b . Then s , in base b , is

(A) 43 (B) 44 (C) 45 (D) 46 (E) 47

Answer: B

Solution

Converting everything into base 10, we have $(b+2)(b+5)(b+6) = 3b^3 + b^2 + 4b + 6$. Looking ahead, the constant term of the polynomial will be $2 \cdot 5 \cdot 6 - 6 = 54$. By the Rational Root Theorem, the only possible integer roots are 1, 2, 3, 6, 9, 18, 27, 54. Bases 1, 2, 3, 6 do not have a 6 as a digit. Testing $b = 9$ gives a solution that works.

Therefore, we are working in base 9. Adding the units place in base 9, $2+5+6 = 14_9$, so we carry the 1 to get a total of 44_9 , which is option **[B]**.

17. (AHSME 1967) If r_1 and r_2 are the distinct real roots of $x^2 + px + 8 = 0$, then it must follow that:

(A) $|r_1 + r_2| > 4\sqrt{2}$ (B) $|r_1| > 3$ or $|r_2| > 3$
 (C) $|r_1| > 2$ and $|r_2| > 2$ (D) $r_1 < 0$ and $r_2 < 0$ (E) $|r_1 + r_2| < 4\sqrt{2}$

Answer: A

Solution

We are given that the roots are real, so the discriminant is positive, which means $p^2 - 4(8)(1) > 0$. This leads to $|p| > 4\sqrt{2}$. By Vieta, the sum of the roots is $-p$, so we have $|-(r_1 + r_2)| \geq 4\sqrt{2}$, or $|r_1 + r_2| > 4\sqrt{2}$, which is option **[A]**.

18. (AHSME 1967) If $x^2 - 5x + 6 < 0$ and $P = x^2 + 5x + 6$ then

(A) P can take any real value (B) $20 < P < 30$
 (C) $0 < P < 20$ (D) $P < 0$ (E) $P > 30$

Answer: B

Solution

We are given that $x^2 - 5x + 6 < 0$, which, when factored, gives $(x-2)(x-3) < 0$. This has a solution of $2 < x < 3$, because the original quadratic is U-shaped, and thus dips below the x-axis between the roots.

Since $x^2 + 5x + 6$ has a vertex minimum at $x = -\frac{5}{2}$, so it is increasing on the interval $[2, 3]$. Thus, evaluating P at $x = 2$ and $x = 3$ will give our bounds, and doing so gives $20 < P < 30$, or **[B]**.

19. (AHSME 1967) The area of a rectangle remains unchanged when it is made $2\frac{1}{2}$ inches longer and $\frac{2}{3}$ inch narrower, or when it is made $2\frac{1}{2}$ inches shorter and $\frac{4}{3}$ inch wider. Its area, in square inches, is:

(A) 30 (B) $\frac{80}{3}$ (C) 24 (D) $\frac{45}{2}$ (E) 20

Answer: E

Solution

We are given $xy = (x + \frac{5}{2})(y - \frac{2}{3}) = (x - \frac{5}{2})(y + \frac{4}{3})$

FOILING each side gives:

$$xy = xy - \frac{2}{3}x + \frac{5}{2}y - \frac{5}{3} = xy + \frac{4}{3}x - \frac{5}{2}y - \frac{10}{3}$$

Taking the last two parts and moving everything to the left gives:

$$-2x + 5y + \frac{5}{3} = 0$$

Taking the first two parts and multiplying by 3 gives $-2x + \frac{15}{2}y - 5 = 0$

Solving both equations for $-2x$ and setting them equal to each other gives $5y + \frac{5}{3} = \frac{15}{2}y - 5$, which leads to $y = \frac{8}{3}$

Plugging that in to $-2x + 5y + \frac{5}{3} = 0$ gives $x = \frac{15}{2}$.

The area of the rectangle is $xy = 20$, or **E**.

20. (AHSME 1967) A circle is inscribed in a square of side m , then a square is inscribed in that circle, then a circle is inscribed in the latter square, and so on. If S_n is the sum of the areas of the first n circles so inscribed, then, as n grows beyond all bounds, S_n approaches:

(A) $\frac{\pi m^2}{2}$ (B) $\frac{3\pi m^2}{8}$ (C) $\frac{\pi m^2}{3}$ (D) $\frac{\pi m^2}{4}$ (E) $\frac{\pi m^2}{8}$

Answer: A

Solution

All answers correctly grow as m^2 , so we let $m = 1$.

The radius of the first circle is $\frac{1}{2}$, so its area is $\frac{\pi}{4}$.

The diagonal of the second square is the diameter of the first circle, which is 1. Therefore, the side length of the square is $\frac{\sqrt{2}}{2}$.

Now we note that the picture is self-similar; if we erase the outer square, erase the outer circle, rotate the picture, and dilate the picture from the center in both the x- and y-directions by an equal scaling factor, we will get the original picture. Therefore, the side lengths of successive squares form a geometric sequence with common ratio R , as do the radii of the circles. On the other hand, the areas of the squares (and areas of the circles) form a geometric sequence with common ratio R^2 .

Since the first square has side 1 and the second square has side $\frac{\sqrt{2}}{2}$, we know $R = \frac{\sqrt{2}}{2}$, and $R^2 = \frac{1}{2}$.

Since the area of the first circle is $\frac{\pi}{4}$, and the common ratio of areas is $\frac{1}{2}$, the sum of all the areas of the circles is the sum of the infinite geometric sequence, which is $\frac{A_1}{1-r} = \frac{\frac{\pi}{4}}{1-\frac{1}{2}} = \frac{\pi}{2}$. This is for side length $m = 1$, and as noted before, there should be an m^2 factor in addition to this number to generalize it from the unit square. This gives answer **A**.

21. (AHSME 1967) In right triangle ABC the hypotenuse $\overline{AB} = 5$ and leg $\overline{AC} = 3$. The bisector of angle A meets the opposite side in A_1 . A second right triangle PQR is then constructed with hypotenuse $\overline{PQ} = A_1B$ and leg $\overline{PR} = A_1C$. If the bisector of angle P meets the opposite side in P_1 , the length of PP_1 is:

(A) $\frac{3\sqrt{6}}{4}$ (B) $\frac{3\sqrt{5}}{4}$ (C) $\frac{3\sqrt{3}}{4}$ (D) $\frac{3\sqrt{2}}{2}$ (E) $\frac{15\sqrt{2}}{16}$

Answer: B

Solution

B

22. (AHSME 1967) For natural numbers, when P is divided by D , the quotient is Q and the remainder is R . When Q is divided by D' , the quotient is Q' and the remainder is R' . Then, when P is divided by DD' , the remainder is:

(A) $R + R'D$ (B) $R' + RD$ (C) RR' (D) R (E) R'

Answer: A

Solution

We are given $P = QD + R$ and $Q = D'Q' + R'$.

Plugging the second equation into the first yields:

$$P = (D'Q' + R')D + R$$

$$P = (DD')Q' + (R'D + R)$$

If we divide P by DD' , the quotient would be Q' , and the remainder would be $R'D + R$, which is option A.

23. (AHSME 1967) If x is real and positive and grows beyond all bounds, then $\log_3(6x - 5) - \log_3(2x + 1)$ approaches:

(A) 0 (B) 1 (C) 3 (D) 4 (E) no finite number

Answer: B

Solution

Since $\log_b x - \log_b y = \log_b \frac{x}{y}$, the expression is equal to $\log_3 \frac{6x-5}{2x+1}$.

The expression $\frac{6x-5}{2x+1}$ is equal to $3 - \frac{8}{2x+1}$. As x gets large, the second term approaches 0, and thus $\frac{6x-5}{2x+1}$ approaches 3. Thus, the expression approaches $\log_3 3$, which is 1.

Alternately, we divide the numerator and denominator of $\frac{6x-5}{2x+1}$ by x to get $\frac{6-\frac{5}{x}}{2+\frac{1}{x}}$.

As x grows large, both fractions approach 0, leaving $\frac{6}{2} = 3$, and so the expression approaches $\log_3 3 = 1$.

With either reasoning, the answer is B

24. (AHSME 1967) The number of solution-pairs in the positive integers of the equation $3x + 5y = 501$ is:

(A) 33 (B) 34 (C) 35 (D) 100 (E) none of these

Answer: A

Solution

We have $y = \frac{501-3x}{5}$. Thus, $501 - 3x$ must be a positive multiple of 5. If $x = 2$, we find our first positive multiple of 5. From there, we note that $x = 2 + 5k$ will always return a multiple of 5 for $501 - 3x$. Our first solution happens at $k = 0$.

We now want to find the smallest multiple of 5 that will work. If $x = 2 + 5k$, then we have $501 - 3x = 501 - 3(2 + 5k)$, or $495 - 15k$. When $k = 32$, the expression is equal to 15, and when $k = 33$, the expression is equal to 0, which will no longer work.

Thus, all integers from $k = 0$ to $k = 32$ will generate an $x = 2 + 5k$ that will be a positive integer, and which will in turn generate a y that is also a positive integer. So, the answer is $\boxed{\text{A}}$.

25. (AHSME 1967) For every odd number $p > 1$ we have:

(A) $(p-1)^{\frac{1}{2}(p-1)} - 1$ is divisible by $p-2$ (B) $(p-1)^{\frac{1}{2}(p-1)} + 1$ is divisible by p
 (C) $(p-1)^{\frac{1}{2}(p-1)}$ is divisible by p (D) $(p-1)^{\frac{1}{2}(p-1)} + 1$ is divisible by $p+1$
 (E) $(p-1)^{\frac{1}{2}(p-1)} - 1$ is divisible by $p-1$

Answer: A

Solution

Given that p is odd, $p-1$ must be even, therefore $\frac{1}{2}(p-1)$ must be an integer, which will be denoted as n .

$$(p-1)^n - 1$$

By sum and difference of powers

$$\begin{aligned} &= ((p-1) - 1)((p-1)^{n-1} + \cdots + 1^{n-1}) \\ &= (p-2)((p-1)^{n-1} + \cdots + 1^{n-1}) \end{aligned}$$

$p-2$ divide $(p-1)^{\frac{1}{2}(p-1)} \boxed{\text{A}}$.

26. (AHSME 1967) If one uses only the tabular information $10^3 = 1000$, $10^4 = 10,000$, $2^{10} = 1024$, $2^{11} = 2048$, $2^{12} = 4096$, $2^{13} = 8192$, then the strongest statement one can make for $\log_{10} 2$ is that it lies between:

(A) $\frac{3}{10}$ and $\frac{4}{11}$ (B) $\frac{3}{10}$ and $\frac{4}{12}$ (C) $\frac{3}{10}$ and $\frac{4}{13}$ (D) $\frac{3}{10}$ and $\frac{40}{132}$ (E) $\frac{3}{11}$ and $\frac{40}{132}$

Answer: C

Solution

Since 1024 is greater than 1000.

$$\log(1024) > 3$$

$$10 * \log(2) > 3$$

$$\text{and } \log(2) > 3/10.$$

Similarly, $8192 < 10000$, so $\log(8192) < 4$

$$13 * \log(2) < 4$$

$$\text{and } \log(2) < 4/13$$

Therefore $3/10 < \log 2 < 4/13$ so the answer is $\boxed{\text{C}}$

27. (AHSME 1967) Two candles of the same length are made of different materials so that one burns out completely at a uniform rate in 3 hours and the other in 4 hours. At what time P.M. should the candles be lighted so that, at 4 P.M., one stub is twice the length of the other?

(A) 1 : 24 (B) 1 : 28 (C) 1 : 36 (D) 1 : 40 (E) 1 : 48

Answer: C

Solution

If the candles both have length ℓ , then the candle that burns in 3 hours has a stub of ℓ at 0 minutes, and a stub of 0 at 180 minutes. Since the candle burns at a constant rate (i.e. linearly), the stub length of this candle t minutes after being lit is $f(t) = \frac{\ell}{180}(180 - t)$, since $f(0) = \ell$ and $f(180) = 0$.

Similarly, for the second candle that burns out in 240 minutes, $g(t) = \frac{\ell}{240}(240 - t)$. Since the first candle burns out faster, if the two candles are lighted simultaneously, it will always have a shorter stub. The problem asks for when $g(t) = 2f(t)$. Solving this equation gives:

$$\frac{\ell}{240}(240 - t) = 2 \cdot \frac{\ell}{180}(180 - t)$$

$$240 - t = \frac{480}{180}(180 - t)$$

$$240 - t = 480 - \frac{480}{180}t$$

$$\frac{8}{3}t - t = 480 - 240$$

$$t = \frac{3}{5} \cdot 240$$

$$t = 144$$

So, the second candle will have a stub twice as big as the first candle 144 minutes after they are both lit. If we want this to happen at 4 PM, the candles have to be lit 144 minutes earlier, or 2 hours and 24 minutes earlier. This is at 1:36 PM, which is option **C**

28. (AHSME 1967) Given the two hypotheses: I Some Mems are not Ens and II No Ens are Veens. If "some" means "at least one," we can conclude that:

(A) Some Mems are not Veens (B) Some Veens are not Mems
 (C) No Mem is a Vee (D) Some Mems are Veens
 (E) Neither (A) nor (B) nor (C) nor (D) is deducible from the given statements

Answer: E

Solution

E

29. (AHSME 1967) $\overline{AB}$ is a diameter of a circle. Tangents $\overline{AD}$ and $\overline{BC}$ are drawn so that $\overline{AC}$ and $\overline{BD}$ intersect in a point on the circle. If $\overline{AD} = a$ and $\overline{BC} = b$, $a \neq b$, the diameter of the circle is:

(A) $|a - b|$ (B) $\frac{1}{2}(a + b)$ (C) $\sqrt{ab}$ (D) $\frac{ab}{a+b}$ (E) $\frac{1}{2} \frac{ab}{a+b}$

Answer: C

Solution

C

30. (AHSME 1967) A dealer bought n radios for d dollars, d a positive integer. He contributed two radios to a community bazaar at half their cost. The rest he sold at a profit of 8 on each radio sold. If the overall profit was 72, then the least possible value of n for the given information is:

(A) 18 (B) 16 (C) 15 (D) 12 (E) 11

Answer: D

Solution

The first 2 radios are sold for $\frac{d}{2}$ dollars each, for income of $2 \cdot \frac{d}{2}$, or simply d .

The remaining $n - 2$ radios are sold for $d + 8$ dollars each, for income of $(n - 2)(d + 8)$, which expands to $nd + 8n - 2d - 16$.

The amount invested in buying n radios for d dollars is nd .

Thus, the total profit is $(d) + (nd - 2d + 8n - 16) - nd$, which simplifies to $8n - d - 16$.

Setting this equal to 72 gives $8n - d - 16 = 72$, or $n = 11 + \frac{d}{8}$.

Since n must be an integer, d must be a multiple of 8. If we want to minimize n , we want to minimize d . Since $d > 0$, setting $d = 8$ yields $n = 12$, giving the answer of **D**

31. (AHSME 1967) Let $D = a^2 + b^2 + c^2$, where a, b , are consecutive integers and $c = ab$. Then $\sqrt{D}$ is:
- (A) always an even integer (B) sometimes an odd integer, sometimes not
 (C) always an odd integer (D) sometimes rational, sometimes not
 (E) always irrational

Answer: C

Solution

Let $a = x, b = x + 1, c = x(x + 1)$. Then $D = x^2 + (x + 1)^2 + x^2(x + 1)^2$, which simplifies to $x^4 + 2x^3 + 3x^2 + 2x + 1$.

From the options, we want to test if D is always a perfect square. Because the polynomial expression for D is quartic, if it is a perfect square, it would be the square of a quadratic expression. Thus, D could be written in the form $(Ax^2 + Bx + C)^2$ for some (A, B, C) .

Setting $(Ax^2 + Bx + C)^2 = x^4 + 2x^3 + 3x^2 + 2x + 1$, we can compare coefficients. From the x^4 coefficient, we get $A = \pm 1$. Note that if $(Ax^2 + Bx + C)^2$ works, so does $(-Ax^2 - Bx - C)^2$, so we can arbitrarily pick $A = 1$.

We now have $(x^2 + Bx + C)^2 = x^4 + 2x^3 + 3x^2 + 2x + 1$. Setting the cubic terms equal gives $2B = 2$, or $B = 1$. This leaves $(x^2 + x + C)^2 = x^4 + 2x^3 + 3x^2 + 2x + 1$. We can quickly inspect the constant term to determine that $C = \pm 1$. We reject $C = -1$, since the quadratic and linear terms won't match up, which leaves $(x^2 + x + 1)^2$ as the only possibility - and, in fact, it works.

Thus, D is always the square of an integer - namely $x^2 + x + 1$. This in turn means that $\sqrt{D}$ is always rational, which leaves choices A, B, C as the only possible correct answers.

The question now is whether $\sqrt{D}$, or $x^2 + x + 1$, is odd, even, or could be both. We have two cases for x :

If $x \equiv 0 \pmod{2}$, then $x^2 \equiv 0^2 \pmod{2}$. This means $x^2 + x + 1 \equiv 0 + 0 + 1 \equiv 1 \pmod{2}$.

If $x \equiv 1 \pmod{2}$, then $x^2 \equiv 1^2 \pmod{2}$, and $x^2 + x + 1 \equiv 1 + 1 + 1 \equiv 1 \pmod{2}$.

Either way, $x^2 + x + 1$ is an odd integer, and the answer is **C**

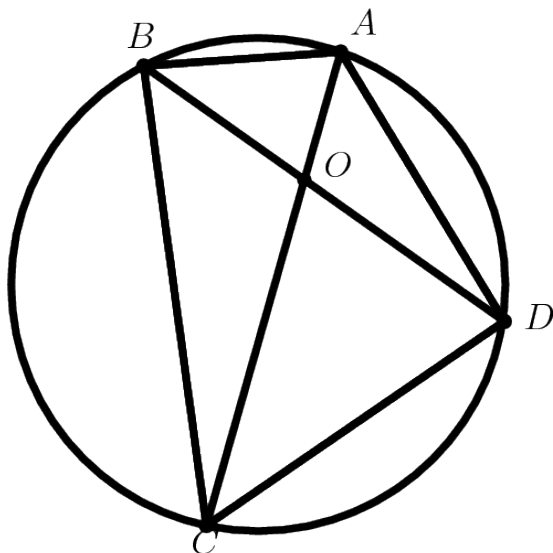
32. (AHSME 1967) In quadrilateral $ABCD$ with diagonals $\overline{AC}$ and $\overline{BD}$ intersecting at O , $\overline{BO} = 4$, $\overline{OD} = 6$, $\overline{AO} = 8$, $\overline{OC} = 3$, and $\overline{AB} = 6$. The length of $\overline{AD}$ is:
- (A) 9 (B) 10 (C) $6\sqrt{3}$ (D) $8\sqrt{2}$ (E) $\sqrt{166}$

Answer: E

Solution 1

After drawing the diagram, we see that we actually have a lot of lengths to work with. Considering triangle ABD, we know values of $AB, BD (BD = BO + OD)$, but we want to find the value of AD . We can apply Stewart's theorem now, letting $m = 4, n = 6, AD = X, AB = 6$, and we have $10 \cdot 6 \cdot 4 + 8 \cdot 8 \cdot 10 = x^2 + 36 \cdot 6$, and we see that $x = \boxed{\text{(E)} \sqrt{166}}$

Solution 2



centering

(Diagram not to scale)

Since $AO \cdot OC = BO \cdot OD$, $ABCD$ is cyclic through power of a point. From the given information, we see that $\triangle AOB \sim \triangle DOC$ and $\triangle BOC \sim \triangle AOD$. Hence, we can find $CD = \frac{9}{2}$ and $AD = 2 \cdot BC$. Letting BC be x , we can use Ptolemy's to get

$$6 \cdot \frac{9}{2} + 2x^2 = 10 \cdot 11 \implies x = \sqrt{\frac{83}{2}}$$

Since we are solving for $AD = 2x = 2 \cdot \sqrt{\frac{83}{2}} = \sqrt{4 \cdot \frac{83}{2}} = \boxed{\text{(E)} \sqrt{166}}$

- PhunsukhWangdu

Solution 3:

The solution says it all. Since $\angle AOD$ is supplementary to $\angle AOB$, $\cos(\angle AOD) = \cos(180^\circ - \angle AOB) = -\cos(\angle AOB)$. The law of cosines on $\triangle AOB$ gives us $\cos(\angle AOB) = \frac{8^2 + 4^2 - 6^2}{(2)(8)(4)} = \frac{11}{16}$. Again, we can use the law of cosines on $\triangle AOD$, which gives us

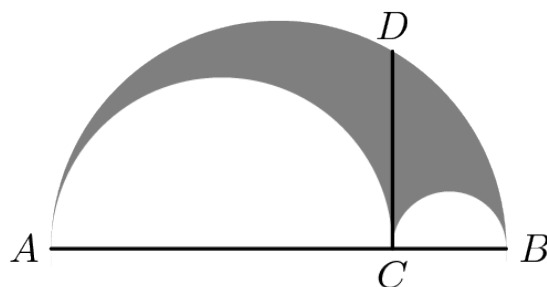
$$\begin{aligned} AD &= \sqrt{8^2 + 6^2 - 2(8)(6)\cos(\angle AOD)} \\ &= \sqrt{8^2 + 6^2 - (2)(8)(6)\cos(\angle 180^\circ - AOB)} \\ &= \sqrt{8^2 + 6^2 + (2)(8)(6)\cos(\angle AOB)} \\ &= \sqrt{8^2 + 6^2 + (2)(8)(6)\left(\frac{11}{16}\right)} \\ &= \sqrt{166} \end{aligned}$$

which gives us **(E)** $\sqrt{166}$

Note that this solution works even if the quadrilateral is not cyclic, and in general, it works if an angle's supplement is known. (Anyone come from aops volume 2 lmao.)

-Tanstorm

33. (AHSME 1967)



centering

In this diagram semi-circles are constructed on diameters $\overline{AB}$, $\overline{AC}$, and $\overline{CB}$, so that they are mutually tangent. If $\overline{CD} \perp \overline{AB}$, then the ratio of the shaded area to the area of a circle with $\overline{CD}$ as radius is:

- (A) 1 : 2 (B) 1 : 3 (C) $\sqrt{3} : 7$ (D) 1 : 4 (E) $\sqrt{2} : 6$

Answer: D

Solution

To make the problem much simpler while staying in the constraints of the problem, position point C halfway between A and B . Then, call $\overline{AC} = \overline{BC} = r$. The area of the shaded region is then

$$\frac{\pi r^2 - \pi(r/2)^2 - \pi(r/2)^2}{2} = \frac{\pi r^2}{4}$$

Because $\overline{CD} = r$ the area of the circle with $\overline{CD}$ as radius is πr^2 . Our ratio is then

$$\frac{\pi r^2}{4} : \pi r^2 = 1 : 4$$

Which corresponds with answer **(D)**

34. (AHSME 1967) Points D , E , F are taken respectively on sides AB , BC , and CA of triangle ABC so that $AD : DB = BE : EC = CF : FA = 1 : n$. The ratio of the area of triangle DEF to that of triangle ABC is:

- (A) $\frac{n^2-n+1}{(n+1)^2}$ (B) $\frac{1}{(n+1)^2}$ (C) $\frac{2n^2}{(n+1)^2}$ (D) $\frac{n^2}{(n+1)^2}$ (E) $\frac{n(n-1)}{n+1}$

Answer: A

Solution

WLOG, let's assume that $\triangle ABC$ is equilateral. Therefore, $[ABC] = \frac{(1+n)^2\sqrt{3}}{4}$ and $[DBE] = [ADF] = [EFC] = n \cdot \sin(60)/2$. Then $[DEF] = \frac{(n^2-n+1)\sqrt{3}}{4}$. Finding the ratio yields **[A]**. -Dark_Lord

35. (AHSME 1967) The roots of $64x^3 - 144x^2 + 92x - 15 = 0$ are in arithmetic progression. The difference between the largest and smallest roots is:

(A) 2 (B) 1 (C) $\frac{1}{2}$ (D) $\frac{3}{8}$ (E) $\frac{1}{4}$

Answer: B

Solution 1

By **Vieta's Formulas**, the sum of the roots is $-\frac{-144}{64} = \frac{9}{4}$. Because the roots are in arithmetic progression, the middle root is the average of the other two roots, and is also the average of all three roots. Therefore, $\frac{9}{3} = \frac{3}{1}$ is the middle root.

The other two roots have a sum of $\frac{9}{4} - \frac{3}{1} = \frac{3}{4}$. By Vieta on the original cubic, the product of all 3 roots is $-\frac{-15}{64} = \frac{15}{64}$, so the product of the remaining two roots is $\frac{\frac{15}{64}}{\frac{3}{1}} = \frac{5}{16}$. If the sum of the two remaining roots is $\frac{3}{2} = \frac{24}{16}$, and the product is $\frac{5}{16}$, the two remaining roots are also the two roots of $16x^2 - 24x + 5 = 0$.

The two remaining roots are thus $x = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$, and they have a difference of $\frac{2\sqrt{B^2 - 4AC}}{2A}$. Plugging in gives $\frac{\sqrt{24^2 - 4(16)(5)}}{16}$, which is equal to 1, which is answer **[B]**.

Solution 2

Because the roots are in arithmetic progression, let the roots be $r - d, r$, and $r + d$ with $d \geq 0$. From **Vieta's Formulas**, we know that $(r - d) + r + (r + d) = \frac{144}{64}$, so $3r = \frac{9}{4}$, and $r = \frac{3}{4}$.

By using Vieta again, we see that $(r - d)(r)(r + d) = \frac{15}{64}$. Substituting $r = \frac{3}{4}$ into this equation, we can now solve for d :

$$\begin{aligned} \left(\frac{3}{4} - d\right)\left(\frac{3}{4}\right)\left(\frac{3}{4} + d\right) &= \frac{15}{64} \\ \frac{3}{4}\left(\frac{9}{16} - d^2\right) &= \frac{15}{64} \\ \frac{9}{16} - d^2 &= \frac{5}{16} \\ d^2 &= \frac{4}{16} = \frac{1}{4} \\ d &= \pm \frac{1}{2} \end{aligned}$$

Because we defined $d \geq 0$, $d = \frac{1}{2}$. Therefore, the difference between the largest and smallest roots, $r + d$ and $r - d$, is $2d = 2 \cdot \frac{1}{2} = \mathbf{(B) 1}$.

36. (AHSME 1967) Given a geometric progression of five terms, each a positive integer less than 100. The sum of the five terms is 211. If S is the sum of those terms in the progression which are squares of integers, then S is:

(A) 0 (B) 91 (C) 133 (D) 195 (E) 211

Answer: C

Solution

Let the first term be a and the common ratio be r , and WLOG let $r \geq 1$. The five terms are a, ar, ar^2, ar^3, ar^4 , and the sum is $a(1 + r + r^2 + r^3 + r^4)$. Clearly r must be rational for all terms to be integers. If r were an integer, it could not be 1, since a would equal $\frac{211}{1+r+r^2+r^3+r^4}$, which is not an integer. In fact, quickly testing $r = 2, 3, 4$ shows that r cannot be an integer.

We now consider non-integers. If $r = \frac{x}{y}$ and $\gcd(x, y) = 1$, then a would have to be divisible by y^4 , since $ar^4 = a\frac{x^4}{y^4}$ is an integer. If $y = 3$, then a would have to be a multiple of 81, which would make the five terms sum to at least $5 \cdot 81$. It only gets worse if $y > 3$. Thus, $y = 2$, and a is a multiple of $2^4 = 16$. Let $a = 16k$.

We now look at the last term, $ar^4 = 16k\frac{x^4}{16} = kx^4$. The smallest allowable values for x , given that x cannot be even, are 3 and 5. If $x = 5$, the last term will be way too big. Thus, $x = 3$ is the only possibility.

We now have a sum of $a(1 + r + r^2 + r^3 + r^4) = 211$, and we know that $r = \frac{3}{2}$ is the only possibility. The terms in the parentheses happen to equal $\frac{211}{16}$ when you plug them in, so $a = 16$.

Thus, the terms are 16, 24, 36, 54, 81, and the first, third, and fifth terms are squares, with a sum of 133, which is answer C.

37. (AHSME 1967) Segments $AD = 10$, $BE = 6$, $CF = 24$ are drawn from the vertices of triangle ABC , each perpendicular to a straight line RS , not intersecting the triangle. Points D, E, F are the intersection points of RS with the perpendiculars. If x is the length of the perpendicular segment GH drawn to RS from the intersection point G of the medians of the triangle, then x is:

(A) $\frac{40}{3}$ (B) 16 (C) $\frac{56}{3}$ (D) $\frac{80}{3}$ (E) undetermined

Answer: A

Solution

A

WLOG let RS be the x-axis, or at least horizontal. The three lengths represent the y-coordinates of points A, B, C . As G is by definition the average of A, B, C , it's coordinates are the average of the coordinates of A, B , and C . Hence, the y-coordinate of G , which is also the distance from G to RS , is the average of the y-coordinates of A, B , and C , or $\frac{10+6+24}{3}$, hence the result.

38. (AHSME 1967) Given a set S consisting of two undefined elements "pib" and "maa", and the four postulates: P_1 : Every pib is a collection of maas, P_2 : Any two distinct pibs have one and only one maa in common, P_3 : Every maa belongs to two and only two pibs, P_4 : There are exactly four pibs. Consider the three theorems: T_1 : There are exactly six maas, T_2 : There are exactly three maas in each pib, T_3 : For each maa there is exactly one other maa not in the same pib with it. The theorems which are deducible from the postulates are:

(A) T_3 only (B) T_2 and T_3 only (C) T_1 and T_2 only
(D) T_1 and T_3 only (E) all

Answer: E

Solution

E

39. (*AHSME 1967*) Given the sets of consecutive integers $\{1\}, \{2, 3\}, \{4, 5, 6\}, \{7, 8, 9, 10\}, \dots$, where each set contains one more element than the preceding one, and where the first element of each set is one more than the last element of the preceding set. Let S_n be the sum of the elements in the n th set. Then S_{21} equals:

(A) 1113 (B) 4641 (C) 5082 (D) 53361 (E) none of these

Answer: B

Solution

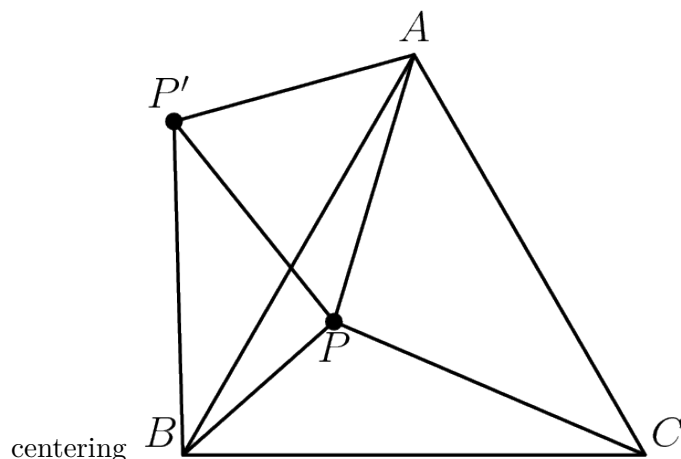
The last element of the 21st set is $\frac{21 \times 22}{2} = 231$, and hence the first element of the 21st set is 211. So $S_{21} = \frac{231 \times 232}{2} - \frac{210 \times 211}{2} = 4641$, hence our answer is B

40. (*AHSME 1967*) Located inside equilateral triangle ABC is a point P such that $PA = 8$, $PB = 6$, and $PC = 10$. To the nearest integer the area of triangle ABC is:

(A) 159 (B) 131 (C) 95 (D) 79 (E) 50

Answer: D

Solution 1



Notice that $6^2 + 8^2 = 10^2$. That makes us want to construct a right triangle.

Rotate $\triangle APC$ 60° about A. Note that $\triangle PAC \cong \triangle P'AB$, so

$$\angle P'AP = \angle PAB + \angle P'AB = \angle PAB + \angle PAC = 60^\circ.$$

Therefore, $\triangle APP'$ is equilateral, so $P'P = 8$, which means $\angle P'PB = 90^\circ$.

Let $\angle BP'P = \alpha$. Notice that $\cos \alpha = \frac{8}{10} = \frac{4}{5}$, and $\sin \alpha = \frac{3}{5}$.

Applying the Law of Cosines to $\triangle APC$ (remembering $\angle APC = \angle AP'B$):

$$\begin{aligned} AC^2 &= 10^2 + 8^2 - 2 \cdot 10 \cdot 8 \cdot \cos(60^\circ + \alpha) \\ &= 164 - 160(\cos 60^\circ \cos \alpha - \sin 60^\circ \sin \alpha) \\ &= 164 - 160\left(\frac{2}{5} - \frac{3\sqrt{3}}{10}\right) \\ &= 164 - 16(4 - 3\sqrt{3}) \\ &= 100 + 48\sqrt{3}. \end{aligned}$$

We want to find the area of $\triangle ABC$, which is

$$AC^2 \cdot \frac{\sqrt{3}}{4} = 25\sqrt{3} + 36 \approx \boxed{(D)79}.$$

Solution 2 (Magic Formula)

Fun formula: Given a point whose distances from the vertices of an equilateral triangle are a , b , and c , the side length of the triangle is:

$$s = \sqrt{\frac{1}{2} \left(a^2 + b^2 + c^2 \pm \sqrt{6(a^2b^2 + b^2c^2 + c^2a^2) - 3(a^4 + b^4 + c^4)} \right)}$$

Given that the area of an equilateral triangle is $\frac{\sqrt{3}}{4}s^2$, the answer is:

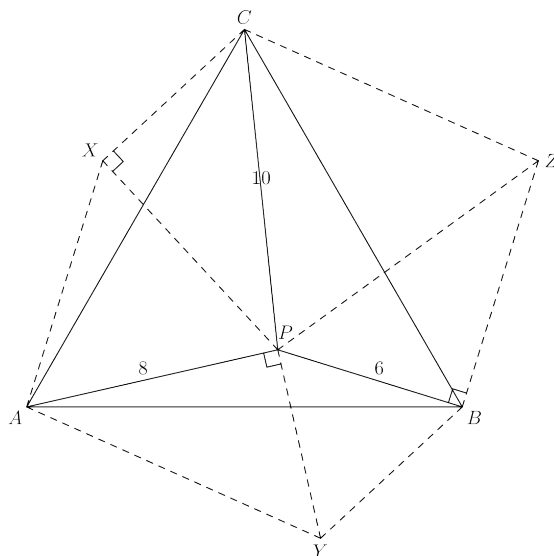
$$\begin{aligned} [ABC] &= \frac{\sqrt{3}}{4} \cdot \frac{1}{2} \left(a^2 + b^2 + c^2 \pm \sqrt{6(a^2b^2 + b^2c^2 + c^2a^2) - 3(a^4 + b^4 + c^4)} \right) \\ &= \frac{\sqrt{3}}{8} \left(200 \pm \sqrt{6 \cdot 16(3^2 4^2 + 4^2 5^2 + 5^2 3^2) - 3 \cdot 16(3^4 + 4^4 + 5^4)} \right) \\ &= \frac{\sqrt{3}}{8} \left(200 \pm \sqrt{96(144 + 400 + 225) - 48(81 + 256 + 625)} \right) \\ &= \frac{\sqrt{3}}{8} \left(200 \pm \sqrt{96 \cdot 769 - 48 \cdot 962} \right) = \frac{\sqrt{3}}{8} \left(200 \pm \sqrt{96 \cdot 769 - 96 \cdot 481} \right) \\ &= \frac{\sqrt{3}}{8} \left(200 \pm \sqrt{96 \cdot 288} \right) = \frac{\sqrt{3}}{8} \left(200 \pm \sqrt{96 \cdot 96 \cdot 3} \right) \\ &= 25\sqrt{3} \pm 36 \approx \{6.5, \text{ or } 78.5\} \end{aligned}$$

6.5 is not a choice, therefore the answer is $\boxed{(D) 79}$.

(Note that the 6.5 answer is actually the solution for when point P is *exterior* to $\triangle ABC$.)

Solution 3

Rotate P and B 60° CCW around A , becoming X and C . Rotate P and C 60° CCW around B , becoming Y and A . Rotate P and A 60° CCW around C , becoming Z and B :



centering

Notice that since $\triangle AXC \cong \triangle APB$, $\triangle BYA \cong \triangle BPC$, and $\triangle CZB \cong \triangle CPA$, then

$$[AYBZCX] = 2 \cdot [ABC]$$

Now the area of the big hexagon is easy to compute since it's comprised of 3 equilateral triangle and 3 right triangles:

$$\begin{aligned} [ABC] &= \frac{1}{2}[AYBZCX] = \frac{1}{2} \left(\underbrace{3 \cdot \frac{1}{2} \cdot 6 \cdot 8}_{3 \text{ right triangles}} + \underbrace{\frac{\sqrt{3}}{4} (6^2 + 8^2 + 10^2)}_{3 \text{ equilateral triangles}} \right) \\ &= \frac{1}{2} \left(72 + \frac{\sqrt{3}}{4} \cdot 200 \right) = 36 + 25\sqrt{3} \\ &\approx \boxed{\text{(D) } 79} \end{aligned}$$

Solution 4 (Answer Choices, Approximation)

Let s be the side length of ABC . Notice that $s \leq 14$ by the triangle inequality. This means that

$$[ABC] \leq \frac{14^2 \sqrt{3}}{2} \approx 84.87.$$

This automatically rules out choices A, B , and C . Now, we will look at if the area is 50. By the equilateral triangle area formula, s would equal $10\sqrt{\frac{2}{\sqrt{3}}} \approx 10.75$. This is very close to 10. If $s = 10$, $\angle APB = 90$ and $\angle APC, \angle BPC < 90$ by the Pythagorean theorem and Pythagorean inequalities. Thus,

$$\angle APB + \angle APC + \angle BPC < 270.$$

$\angle APB + \angle APC + \angle BPC$ needs to be 360, and it probably cannot increase by more than 90 by just adding 0.75 to s (more rigorous proof below) Thus, the only viable answer choice is $\boxed{79}$.

- In fact, for $s < \sqrt{136} \approx 11.66$ ($[ABC] < 34\sqrt{3} \approx 58.88$) $\angle APC, \angle BPC < 90$ still holds. As $\angle APB < 180$,

$$\angle APB + \angle APC + \angle BPC < 360.$$

Thus, we know for sure that the nearest integer to the area cannot be 50.

§4 AHSME 1968

1. (AHSME 1968) Let P units be the increase in circumference of a circle resulting from an increase in π units in the diameter. Then P equals:
(A) $\frac{1}{\pi}$ (B) π (C) $\frac{\pi^2}{2}$ (D) π^2 (E) 2π

Answer: D

Solution

Let d be the diameter of the original circle. If d is increased by π , then the new circumference is $\pi d + \pi^2$. The difference in circumference is therefore $\pi d + \pi^2 - \pi d = \pi^2$

Therefore, the answer is D

Solution by VivekA

2. (*AHSME 1968*) The real value of x such that 64^{x-1} divided by 4^{x-1} equals 256^{2x} is:

(A) $-\frac{2}{3}$ (B) $-\frac{1}{3}$ (C) 0 (D) $\frac{1}{4}$ (E) $\frac{3}{8}$

Answer: B

Solution

Because $64 = 4^3$ and $256 = 4^4$, we can change all of the numbers in the equation to exponents with base 4 and solve the equation:

$$\begin{aligned}\frac{64^{x-1}}{4^{x-1}} &= 256^{2x} \\ \frac{(4^3)^{x-1}}{4^{x-1}} &= (4^4)^{2x} \\ \frac{4^{3x-3}}{4^{x-1}} &= 4^{8x} \\ 4^{2x-2} &= 4^{8x} \\ 2x-2 &= 8x \\ x-1 &= 4x \\ 3x &= -1 \\ x &= \frac{-1}{3}\end{aligned}$$

Thus, our desired answer is (B) -1/3.

3. (*AHSME 1968*) A straight line passing through the point $(0, 4)$ is perpendicular to the line $x - 3y - 7 = 0$. Its equation is:

(A) $y + 3x - 4 = 0$ (B) $y + 3x + 4 = 0$ (C) $y - 3x - 4 = 0$
(D) $3y + x - 12 = 0$ (E) $3y - x - 12 = 0$

Answer: A

Solution

The original line can be adjusted to **standard form**, in which it is $x - 3y = 7$. Because this line is in standard form, its slope is $\frac{-1}{-3} = \frac{1}{3}$. Thus, the slope of the perpendicular line is the negative reciprocal of this number, -3 . Because we know that this new line passes through the point $(0, 4)$, we can describe this line using **point-slope form**, in which it is $y - 4 = -3(x - 0)$, or $y + 3x - 4$, which is answer choice A.

4. (*AHSME 1968*) Define an operation $\star$ for positive real numbers as $a \star b = \frac{ab}{a+b}$. Then $4 \star (4 \star 4)$ equals:

(A) $\frac{3}{4}$ (B) 1 (C) $\frac{4}{3}$ (D) 2 (E) $\frac{16}{3}$

Answer: C

Solution

$$4 \star 4 \star 4 = 4 \star 2 = \frac{4}{3}$$

Therefore, our answer is C

Solution by VivekA

5. (*AHSME 1968*) If $f(n) = \frac{1}{3}n(n+1)(n+2)$, then $f(r) - f(r-1)$ equals:
 (A) $r(r+1)$ (B) $(r+1)(r+2)$ (C) $\frac{1}{3}r(r+1)$
 (D) $\frac{1}{3}(r+1)(r+2)$ (E) $\frac{1}{3}r(r+1)(2r+1)$

Answer: A

Solution

Plugging in the expressions for $f(r)$ and $f(r-1)$, we see that:

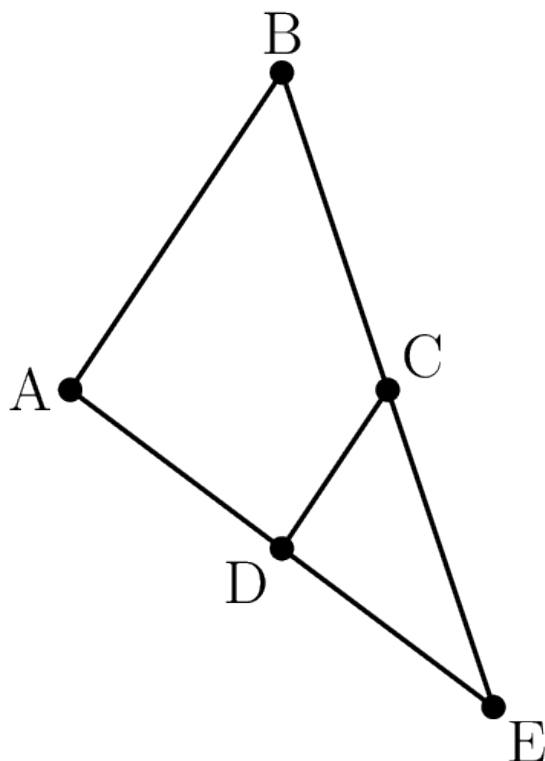
$$\begin{aligned} f(r) - f(r-1) &= \frac{1}{3}r(r+1)(r+2) - \frac{1}{3}(r-1)(r-1+1)(r-1+2) \\ &= \frac{1}{3}[r(r+1)(r+2) - r(r-1)(r+1)] \\ &= \frac{1}{3}[r(r^2 + 3r + 2) - r(r^2 - 1)] \\ &= \frac{1}{3}[r^3 + 3r^2 + 2r - r^3 + r] \\ &= \frac{1}{3}[3r^2 + 3r] \\ &= r^2 + r \\ &= r(r+1), \end{aligned}$$

which is answer choice A.

6. (*AHSME 1968*) Let side AD of convex quadrilateral $ABCD$ be extended through D , and let side BC be extended through C , to meet in point E . Let S be the degree-sum of angles CDE and DCE , and let S' represent the degree-sum of angles BAD and ABC . If $r = S/S'$, then:
 (A) $r = 1$ sometimes, $r > 1$ sometimes
 (B) $r = 1$ sometimes, $r < 1$ sometimes
 (C) $0 < r < 1$ (D) $r > 1$ (E) $r = 1$

Answer: E

Solution



centering

Because $ABCD$ is a convex **quadrilateral**, the sum of its interior angles is 360° . Thus, $S' = \angle BAD + \angle ABC = 360^\circ - (\angle BCD + \angle ADC)$. Furthermore, because $\angle BCD$ and $\angle ADC$ are **supplementary** to $\angle DCE$ and $\angle CDE$, respectively, the four angles sum to $2 \cdot 180^\circ = 360^\circ$, so $\angle BCD + \angle ADC = 360^\circ - (\angle DCE + \angle CDE) = 360^\circ - S$. Plugging this expression for $\angle BCD + \angle ADC$ into the first equation, we see that $S' = 360^\circ - (360^\circ - S) = S$, so $r = \frac{S}{S'} = 1$, which is answer choice **E**.

7. (AHSME 1968) Let O be the intersection point of medians AP and CQ of triangle ABC . If OQ is 3 inches, then OP , in inches, is:

(A) 3 (B) $\frac{9}{2}$ (C) 6 (D) 9 (E) undetermined

Answer: E

Solution

The fact that $OQ = 3$ only tells us that $CQ = 9$. There are infinitely many triangles with a median which has length 9, so we can make no statement about the length of the median AP or the segment OP . Thus, OP is **(E) undetermined**.

8. (AHSME 1968) A positive number is mistakenly divided by 6 instead of being multiplied by 6. Based on the correct answer, the error thus committed, to the nearest percent, is :

(A) 100 (B) 97 (C) 83 (D) 17 (E) 3

Answer: B

Solution

Let n be the original number in question. The number divided by 6 is $\frac{n}{6}$, and the number multiplied by 6 is $6n$. Because $\frac{n}{6} = \frac{1}{36} \cdot 6n$, and $\frac{1}{36} = 0.02\bar{7} \approx 0.03 = 3\%$, the error (which is our deviation from the correct value) is about 97%. Thus, our answer is **B**.

9. (AHSME 1968) The sum of the real values of x satisfying the equality $|x + 2| = 2|x - 2|$ is:

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) 6 (D) $6\frac{1}{3}$ (E) $6\frac{2}{3}$

Answer: E

Solution

There are three cases:

- (i) $(x + 2), (x - 2) \geq 0$
 (ii) $x + 2 \geq 0, x - 2 < 0$
 (iii) $(x + 2), (x - 2) < 0$

In case (i), we solve the equation $x + 2 = 2(x - 2)$ to get $x = 6$. In case (ii), we solve the equation $x + 2 = 2(2 - x)$ to get $x = \frac{2}{3}$. Case (iii) simply produces an equation which is that of case (i) multiplied on both sides by -1 , so it yields the same solutions and thus may be discarded. Thus, if we add the real values of x which satisfy the original equation, we get $6 + \frac{2}{3} = 6\frac{2}{3}$, or answer choice E.

10. (AHSME 1968) Assume that, for a certain school, it is true that

I: Some students are not honest. II: All fraternity members are honest.

A necessary conclusion is:

- (A) Some students are fraternity members.
 (B) Some fraternity member are not students.
 (C) Some students are not fraternity members.
 (D) No fraternity member is a student.
 (E) No student is a fraternity member.

Answer: C

Solution

If some students are dishonest, we know that they must not be fraternity members, because if they were members, then they would be honest. This conclusion aligns with answer choice C.

Choice (A) is incorrect, because if there were no fraternity members, proposition II would be **vacuously true**, and dishonest students are still allowed to exist within the rules of the problem, satisfying proposition I.

Choice (B) is incorrect, because dishonest students can still exist (proposition I) while some other students (the fraternity members, maybe some others) are honest (proposition II).

Choice (D) is incorrect by the same reasoning used against choice (B).

Choice (E) is incorrect, because it is the **contrapositive** of choice (D), so the two are logically equivalent. (Choice (D) could be restated as "if someone is a fraternity member, that someone is not a student," and choice (E) could be restated as "if someone is a student, that someone is not a fraternity member.")

11. (AHSME 1968) If an arc of 60° on circle I has the same length as an arc of 45° on circle II , the ratio of the area of circle I to that of circle II is:

(A) 16 : 9 (B) 9 : 16 (C) 4 : 3 (D) 3 : 4 (E) none of these

Answer: B

Solution

Let the length of the two arcs be denoted x . First, we acknowledge that the length of an arc of a circle is given by the measure of that arc's central angle in radians times the radius of that circle. Note that the central angle of circle I measures $\frac{\pi}{3}$ radians, and the central angle of circle II measures $\frac{\pi}{4}$ radians. Let the radius of circle I be r_1 and the radius of circle II be r_2 . Then, we know that the arc length $x = \frac{\pi}{3}r_1 = \frac{\pi}{4}r_2$, and so $\frac{r_1}{r_2} = \frac{3}{4}$. From this, we see that $\frac{r_1^2}{r_2^2} = \frac{\pi r_1^2}{\pi r_2^2} = \frac{[I]}{[II]} = \frac{9}{16}$. Thus, our answer is (B) 9:16.

12. (AHSME 1968) A circle passes through the vertices of a triangle with side-lengths $7\frac{1}{2}, 10, 12\frac{1}{2}$. The radius of the circle is:

(A) $\frac{15}{4}$ (B) 5 (C) $\frac{25}{4}$ (D) $\frac{35}{4}$ (E) $\frac{15\sqrt{2}}{2}$

Answer: C

Solution

The triangle that goes through all the vertices of the triangle is the circumcircle of the triangle. $(7\frac{1}{2})^2 + 10^2 = (12\frac{1}{2})^2$, so the triangle is a right triangle. The radius of a circumcircle of a right triangle is half the hypotenuse. $\frac{1}{2} \cdot \frac{25}{2} = \frac{25}{4} \implies$ C

13. (AHSME 1968) If m and n are the roots of $x^2 + mx + n = 0, m \neq 0, n \neq 0$, then the sum of the roots is:

(A) $-\frac{1}{2}$ (B) -1 (C) $\frac{1}{2}$ (D) 1 (E) undetermined

Answer: B

Solution

By Vieta's Theorem, $mn = n$ and $-(m+n) = m$. Dividing the first equation by n gives $m = 1$. Multiplying the 2nd by -1 gives $m+n = -m$. The RHS is -1 , so the answer is B

14. (AHSME 1968) If x and y are non-zero numbers such that $x = 1 + \frac{1}{y}$ and $y = 1 + \frac{1}{x}$, then y equals

(A) $x - 1$ (B) $1 - x$ (C) $1 + x$ (D) $-x$ (E) x

Answer: E

Solution 1

We see after multiplying the first equation by y , that

$$xy = y + 1.$$

Similarly, we see that after multiplying the second equation by x , we get that

$$xy = x + 1.$$

Thus $x + 1 = y + 1 \implies x = y$, giving us our final answer of E.

Solution 2

We can first combine the fraction on the second equation, which is

$$y = \frac{x+1}{x}.$$

Then, substituting the value of x from the first equation, we can get

$$y = \frac{1 + \frac{1}{y} + 1}{1 + \frac{1}{y}}.$$

Then, multiplying the denominator on both sides, we get

$y + 1 = 2 + \frac{1}{y} \implies y = 1 + \frac{1}{y}$ which is equivalent to the value of x that was originally provided, giving the final answer of **E**.

15. (AHSME 1968) Let P be the product of any three consecutive positive odd integers. The largest integer dividing all such P is:

(A) 15 (B) 6 (C) 5 (D) 3 (E) 1

Answer: D

Solution

Product P can be written as $2n - 1, 2n + 1, 2n + 3$. Because P is defined as a "3 consecutive odd integer" product implies that P must be divisible by at least 3. A is ruled out because factors of 5 only arise every 5 terms, if we were to take the 3 terms in the middle of the factors of 5 we wouldn't have a factor of 5. Obviously B is impossible because we are multiplying odd numbers and 2 would never become one of our prime factors. C is ruled out with the same logic as A. Lastly E is ruled out because we have already proved that 3 is possible, and the question asks for greatest possible. **D**

- In first step we could also apply modular arithmetic and get the same answer.

16. (AHSME 1968) If x is such that $\frac{1}{x} < 2$ and $\frac{1}{x} > -3$, then:

(A) $-\frac{1}{3} < x < \frac{1}{2}$ (B) $-\frac{1}{2} < x < 3$ (C) $x > \frac{1}{2}$
 (D) $x > \frac{1}{2}$ or $-\frac{1}{3} < x < 0$ (E) $x > \frac{1}{2}$ or $x < -\frac{1}{3}$

Answer: E

Solution

Because $\frac{1}{x} < 2$, $x > \frac{1}{2}$ when x is positive. Because $\frac{1}{x} > -3$, $x < -\frac{1}{3}$ when x is negative. x can never be 0, so these two inequalities cover all cases for the value of x . Thus, our answer is **E**.

17. (AHSME 1968) Let $f(n) = \frac{x_1 + x_2 + \dots + x_n}{n}$, where n is a positive integer. If $x_k = (-1)^k, k = 1, 2, \dots, n$, the set of possible values of $f(n)$ is:

(A) $\{0\}$ (B) $\{\frac{1}{n}\}$ (C) $\{0, -\frac{1}{n}\}$ (D) $\{0, \frac{1}{n}\}$ (E) $\{1, \frac{1}{n}\}$

Answer: C

Solution

Because we start with $x_1 = -1$, and the terms x_k alternate between 1 and -1 , there is either one more -1 than the number of 1s (when n is odd), or there are an equal number of 1s and -1 s (when n is even). In the former case, $f(n) = \frac{-1}{n}$, and, in the latter case, $f(n) = 0$. This is only consistent with answer **C**.

18. (AHSME 1968) Side AB of triangle ABC has length 8 inches. Line DEF is drawn parallel to AB so that D is on segment AC , and E is on segment BC . Line AE extended bisects angle FEC . If DE has length 5 inches, then the length of CE , in inches, is:

(A) $\frac{51}{4}$ (B) 13 (C) $\frac{53}{4}$ (D) $\frac{40}{3}$ (E) $\frac{27}{2}$

Answer: D

Solution

Draw a line passing through A and parallel to BC . Let $\angle FEC = 2n$. By alternate-interior-angles or whatever, $\angle BAE = n$, so BAE is an isosceles triangle, and it follows that $BE = 8$. $\triangle ABC \sim \triangle DEC$. Let $CE = x$. We have

$$\frac{8}{8+x} = \frac{5}{x} \Rightarrow 40 + 5x = 8x \Rightarrow x = CE = \boxed{\frac{40}{3}}.$$

Thus, we choose answer $\boxed{D}$.

19. (AHSME 1968) Let n be the number of ways 10 dollars can be changed into dimes and quarters, with at least one of each coin being used. Then n equals:

(A) 40 (B) 38 (C) 21 (D) 20 (E) 19

Answer: E

Solution

Because we have to use at least one dime and one quarter, this part of the selection is forced. Thus, the number of ways to partition \$10 in the desired manner is equal to the number of ways to partition $\$10 - 10cent - 25cent = \9.65 into quarters and dimes without restriction. However, \$9.65 is not a multiple of 25cent, so we must use dimes until we reach a multiple of 25cent. We first reach such a multiple at \$9.25, and, because we have had no freedom up to this point, the number of ways to partition \$9.25 into dimes and quarters will yield our desired answer. \$9.25 can be partitioned into 37 quarters, and we can replace quarters 2 at a time with 5 dimes to partition it differently. We can replace anywhere from 0 to 18 pairs of quarters with dimes in this manner, giving us $\boxed{19}$ different partitions, which is answer choice $\boxed{E}$.

20. (AHSME 1968) The measures of the interior angles of a convex polygon of n sides are in arithmetic progression. If the common difference is 5° and the largest angle is 160° , then n equals:

(A) 9 (B) 10 (C) 12 (D) 16 (E) 32

Answer: A

Solution

The formula for the sum of the angles in any polygon is $180(n-2)$. Because this particular polygon is convex and has its angles in an arithmetic sequence with its largest angle being 160, we can find the sum of the angles.

$$a_n = 160$$

$$a_1 = 160 - 5(n-1)$$

Plugging this into the formula for finding the sum of an arithmetic sequence...

$$n \left(\frac{160+160-5(n-1)}{2} \right) = 180(n-2).$$

Simplifying, we get $n^2 + 7n - 144$.

Since we want the positive solution to the quadratic, we can easily factor and find the answer is $n = \boxed{9}$.

Hence the answer is $\boxed{A}$

21. (AHSME 1968) If $S = 1! + 2! + 3! + \cdots + 99!$, then the units' digit in the value of S is:

(A) 9 (B) 8 (C) 5 (D) 3 (E) 0

Answer: D

Solution

Note that every factorial after $5!$ has a unit digit of 0, meaning that we can disregard them. Thus, we only need to find the units digit of $1! + 2! + 3! + 4!$, and as $1! + 2! + 3! + 4! \equiv 3 \pmod{10}$, which means that the unit digit is 3, we have our answer of D as desired.

22. (AHSME 1968) A segment of length 1 is divided into four segments. Then there exists a quadrilateral with the four segments as sides if and only if each segment is:

(A) equal to $\frac{1}{4}$
 (B) equal to or greater than $\frac{1}{8}$ and less than $\frac{1}{2}$
 (C) greater than $\frac{1}{8}$ and less than $\frac{1}{2}$
 (D) equal to or greater than $\frac{1}{8}$ and less than $\frac{1}{4}$
 (E) less than $\frac{1}{2}$

Answer: E

Solution

As a consequence of the **Triangle Inequality**, we can form a quadrilateral from the four segments iff the sum of any three sides is greater than the sum of the fourth side. Thus, no segment can have a length $\geq \frac{1}{2}$. All other segment lengths are valid, so long as the inequality is satisfied. Thus, our answer is (E) less than $1/2$.

23. (AHSME 1968) If all the logarithms are real numbers, the equality $\log(x+3) + \log(x-1) = \log(x^2 - 2x - 3)$ is satisfied for:

(A) all real values of x
 (B) no real values of x
 (C) all real values of x except $x = 0$
 (D) no real values of x except $x = 0$
 (E) all real values of x except $x = 1$

Answer: B

Solution

From the given we have

$$\log(x+3) + \log(x-1) = \log(x^2 - 2x - 3)$$

$$\log(x^2 + 2x - 3) = \log(x^2 - 2x - 3)$$

$$x^2 + 2x - 3 = x^2 - 2x - 3$$

$$x = 0$$

However substituting into $\log(x-1)$ gets a negative argument, which is impossible B.

24. (AHSME 1968) A painting 18" X 24" is to be placed into a wooden frame with the longer dimension vertical. The wood at the top and bottom is twice as wide as the

wood on the sides. If the frame area equals that of the painting itself, the ratio of the smaller to the larger dimension of the framed painting is:

- (A) 1 : 3 (B) 1 : 2 (C) 2 : 3 (D) 3 : 4 (E) 1 : 1

Answer: C

Solution

Let the width of the frame on the sides to be x .

Then, the width of the frame on the top and bottom is $2x$.

The area of the frame is then $x \cdot 2x - 18 \cdot 24$

Setting the area of the frame equal to the area of the picture,

$$(2x + 18)(4x + 24) - 18 \cdot 24 = 18 \cdot 24$$

Solving,

$$8x^2 + 120x + 432 = 864$$

$$x^2 + 15x - 54 = 0$$

$$(x + 18)(x - 3) = 0 \Rightarrow x = 3$$

Therefore, the ratio of the smaller side to the larger side is $\frac{18+2 \cdot 3}{24+4 \cdot 3} = \frac{24}{36} = \boxed{2 : 3(C)}$.

25. (AHSME 1968) Ace runs with constant speed and Flash runs x times as fast, $x > 1$. Flash gives Ace a head start of y yards, and, at a given signal, they start off in the same direction. Then the number of yards Flash must run to catch Ace is:

- (A) xy (B) $\frac{y}{x+y}$ (C) $\frac{xy}{x-1}$ (D) $\frac{x+y}{x+1}$ (E) $\frac{x+y}{x-1}$

Answer: C

Solution

Let k denotes the distance Ace needs to run after the y yard. Since the distance they run with same amount of time is proportional to their speed, we have

$$\frac{1}{x} = \frac{k}{y+k}$$

$$k = \frac{y}{x-1}$$

Thus the total distance ran by Flash is

$$y+k = y + \frac{y}{x-1} = \frac{xy}{x-1} \boxed{C}$$

26. (AHSME 1968) Let $S = 2 + 4 + 6 + \cdots + 2N$, where N is the smallest positive integer such that $S > 1,000,000$. Then the sum of the digits of N is:

- (A) 27 (B) 12 (C) 6 (D) 2 (E) 1

Answer: E

Solution

Note that $S = 2(1 + 2 + 3 + \cdots + N) = N(N+1)$. It follows that $N = 1000$, so the sum of the digits of N is $\boxed{E}$.

FrostFox

27. (AHSME 1968) Let $S_n = 1 - 2 + 3 - 4 + \cdots + (-1)^{n-1}n$, where $n = 1, 2, \dots$. Then $S_{17} + S_{33} + S_{50}$ equals:

(A) 0 (B) 1 (C) 2 (D) -1 (E) -2

Answer: B

Solution

If n is even, S_n is negative $n/2$. If n is odd, then S_n is $(n+1)/2$. (These can be found using simple calculations.) Therefore, we know $S_{17} + S_{33} + S_{50} = 9 + 17 - 25$, which is $\boxed{\text{B}}$.

Solution By FrostFox

28. (AHSME 1968) If the arithmetic mean of a and b is double their geometric mean, with $a > b > 0$, then a possible value for the ratio a/b , to the nearest integer, is:

(A) 5 (B) 8 (C) 11 (D) 14 (E) none of these

Answer: D

Solution

$\boxed{\text{D}}$

$$\frac{a+b}{2} = 2 \cdot \sqrt{ab}$$

$$\frac{a}{b} + 1 = 4 \cdot \sqrt{\frac{a}{b}}$$

setting $x = \sqrt{\frac{a}{b}}$ we get a quadratic equation $x^2 + 1 = 4x$ with solutions $x = \frac{4 \pm \sqrt{16-4}}{2}$

$$x^2 = \frac{a}{b} = (4+3) + 4\sqrt{3} = 13.8 = 14.$$

29. (AHSME 1968) Given the three numbers $x, y = x^x, z = x^{x^x}$ with $.9 < x < 1.0$. Arranged in order of increasing magnitude, they are:

(A) x, z, y (B) x, y, z (C) y, x, z (D) y, z, x (E) z, x, y

Answer: A

Solution 1

Seeing that we need to compare values with exponents, we think **logarithms**. Taking the logarithm base x of each term, we obtain 1, x , and x^x . Because $0 < x < 1$, $f(n) = \log_x(n)$ is **monotonically decreasing**, so the order of terms by magnitude in our new set of numbers will be reversed compared to the original set (i.e. if $a < b < c$, then $\log_x(a) > \log_x(b) > \log_x(c)$). However, the order of this set will be reversed again (back to the order of the original set) when we take the logarithm base x a second time. After doing this operation, we find the values 0, 1, and x , which correspond to x , y , and z , respectively. Because $0.9 < x < 1$, $0 < x < 1$, and so, by the correspondence detailed above, $x < z < y$, which yields us answer choice

$\boxed{\text{A}}$.

Solution 2

Because $0 < x < 1$, taking x to the x th power will bring it closer to 1, thereby raising its value. Because we have established that $x < x^x$, and $f(n) = x^n$ is a **monotonically decreasing** function, we know that $x^x > x^{x^x}$. However, because $x^x < 1$, x^{x^x} , compared to x , will be closer to (but still less than) 1. Thus, $x^{x^x} > x$. Putting this all together, we see that $x < x^{x^x} < x^x$, or $x < z < y$, which is answer choice $\boxed{\text{A}}$.

30. (AHSME 1968) Convex polygons P_1 and P_2 are drawn in the same plane with n_1 and n_2 sides, respectively, $n_1 \leq n_2$. If P_1 and P_2 do not have any line segment in common, then the maximum number of intersections of P_1 and P_2 is:

(A) $2n_1$ (B) $2n_2$ (C) n_1n_2 (D) $n_1 + n_2$ (E) none of these

Answer: A

Solution 1

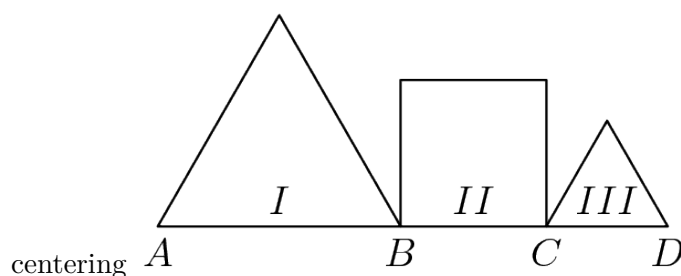
Notice how P_2 can pass through each line segment of P_1 at most twice. To have more than two intersections, the line passing through P_1 would have a zigzag shape which is impossible for convex polygons. Therefore, the intersections does not depend on P_2 and the answer is A

Solution 2

Try to get the answer experimentally. Draw two of the simplest shapes: a square and a triangle and maximize the number of intersections. You will discover it is 6, and the only expression provided that will give 6 is A

Note : this solution isn't risk free, as the option of "None of These" is given as well.

31. (AHSME 1968)



In this diagram, not drawn to scale, Figures I and III are equilateral triangular regions with respective areas of $32\sqrt{3}$ and $8\sqrt{3}$ square inches. Figure II is a square region with area 32 square inches. Let the length of segment AD be decreased by $12\frac{1}{2}\%$ of itself, while the lengths of AB and CD remain unchanged. The percent decrease in the area of the square is:

(A) $12\frac{1}{2}$ (B) 25 (C) 50 (D) 75 (E) $87\frac{1}{2}$

Answer: D

Solution

Given an equilateral triangle with side length s , the area is given by $\frac{s^2\sqrt{3}}{4}$. Setting this equation equal to the area of triangle I , $32\sqrt{3}$, we find that $s = 8\sqrt{2}$. Because triangle III is also equilateral, it is similar to triangle I , and, because it has a quarter of the area of I , it has $\sqrt{\frac{1}{4}} = \frac{1}{2}$ of the side length. Thus, its sides have a length of $4\sqrt{2}$. Square II initially has an area of 32, so its side length starts at $\sqrt{32} = 4\sqrt{2}$. The initial length AD , therefore, is $16\sqrt{2}$. Because AD decreases by 12.5% , it becomes $\frac{7}{8}$ of its initial value, which is $14\sqrt{2}$. Because the sides of the triangles remain unchanged, this decrease of $2\sqrt{2}$ must come from the side length of the square. Thus, the square's final side is $2\sqrt{2}$, which gives an area of 8 square inches. 32 to 8 is a decrease of 75% . Therefore, our answer is (D).

32. (AHSME 1968) A and B move uniformly along two straight paths intersecting at right angles in point O . When A is at O , B is 500 yards short of O . In two minutes they are equidistant from O , and in 8 minutes more they are again equidistant from O . Then the ratio of A 's speed to B 's speed is:

(A) 4 : 5 (B) 5 : 6 (C) 2 : 3 (D) 5 : 8 (E) 1 : 2

Answer: C

Solution

Let the speed of A be a and the speed of B be b . The first time that A and B will be equidistant from O , B will have not yet reached O . Thus, after two minutes, B 's distance from O will be $500 - 2b$, and A 's distance from O will be $2a$. Setting these expressions equal to each other and dividing by 2, we see that $a = 250 - b$.

After another eight minutes (or after a total of ten minutes since A was at O), A and B will again be equidistant from O , but this time B will have passed O . The distance A will be from O is $10a$, and the distance B will be from O is $10b - 500$. Setting these expressions equal to each other and dividing by 10, we see that $a = b - 50$.

Adding the two equations that we have obtained above, we see that $2a = 250 - b + b - 50$, and so $a = 100$. Substituting this value of a into the second equation, we see that $100 = b - 50$, or $b = 150$. Then, $\frac{a}{b} = \frac{100}{150} = \frac{2}{3}$, so the ratio of A 's speed to that of B is (C) 2:3.

33. (AHSME 1968) A number N has three digits when expressed in base 7. When N is expressed in base 9 the digits are reversed. Then the middle digit is:

(A) 0 (B) 1 (C) 3 (D) 4 (E) 5

Answer: A

Solution

Call the number $\overline{abc}$ in base 7.

Then, $49a + 7b + c = 81c + 9b + a$. (Breaking down the number in base-form).

After combining like terms and moving the variables around, $48a = 2b + 80c$, $b = 40c - 24a = 8(5c - 2a)$. This shows that b is a multiple of 8 (we only have to find the middle digit under *one* of the bases). Thus, $b = 0$ (since $8 > 6$, the largest digit in base 7).

Select A as our answer.

34. (AHSME 1968) With 400 members voting the House of Representatives defeated a bill. A re-vote, with the same members voting, resulted in the passage of the bill by twice the margin by which it was originally defeated. The number voting for the bill on the revote was $\frac{12}{11}$ of the number voting against it originally. How many more members voted for the bill the second time than voted for it the first time?

(A) 75 (B) 60 (C) 50 (D) 45 (E) 20

Answer: B

Solution

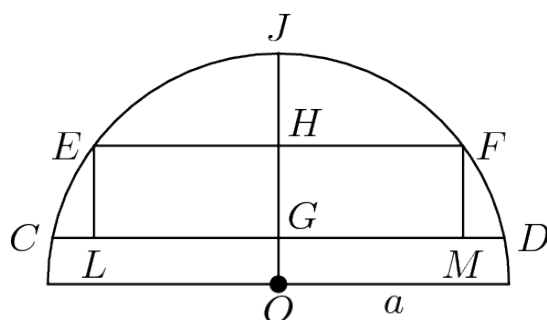
Let the number of votes for the bill the first time be a and the number of votes for the bill the second time be b . Then, from the problem, we know that $b = \frac{12}{11}(400 - a)$. Furthermore, the original vote's margin of defeat was $(400 - a) - a = 400 - 2a$,

and the revote's margin of victory was $b - (400 - b) = 2b - 400$. Thus, from the problem, we know that $2(400 - 2a) = 2b - 400$. Substituting the value of b we obtained earlier into this equation, we see that:

$$\begin{aligned}
 2(400 - 2a) &= 2 * \frac{12}{11}(400 - a) - 400 \\
 400 - 2a &= \frac{400 * 12}{11} - \frac{12a}{11} - 200 \\
 200 - a &= \frac{400 * 6}{11} - \frac{6a}{11} - 100 \\
 300 - \frac{2400}{11} &= \frac{5a}{11} \\
 5a &= 3300 - 2400 \\
 5a &= 900 \\
 a &= 180
 \end{aligned}$$

From this fact, we can conclude that $c = \frac{12}{11} * (400 - 180) = 240$, so $c - a = \boxed{60}$, which is answer choice **B**.

35. (AHSME 1968)



centering

In this diagram the center of the circle is O , the radius is a inches, chord EF is parallel to chord CD . O, G, H, J are collinear, and G is the midpoint of CD . Let K (sq. in.) represent the area of trapezoid $CDFE$ and let R (sq. in.) represent the area of rectangle $ELMF$. Then, as CD and EF are translated upward so that OG increases toward the value a , while JH always equals HG , the ratio $K : R$ becomes arbitrarily close to:

- (A) 0 (B) 1 (C) $\sqrt{2}$ (D) $\frac{1}{\sqrt{2}} + \frac{1}{2}$ (E) $\frac{1}{\sqrt{2}} + 1$

Answer: D

Solution

Let $OG = a - 2h$, where $h = JH = HG$. Since the areas of rectangle $EHGL$ and trapezoid $EHGC$ are both half of rectangle $LMFE$ and trapezoid $EFDC$, respectively, the ratios between their areas will remain the same, so let us consider rectangle $EHGL$ and trapezoid $EHGC$.

Draw radii OC and OE , both of which obviously have length a . By the Pythagorean Theorem, the length of EH is $\sqrt{a^2 - (OG + h)^2}$, and the length of CG is $\sqrt{a^2 - OG^2}$. It follows that the area of rectangle $EHGL$ is

$$EH \cdot HG = h\sqrt{a^2 - (OG + h)^2}$$

while the area of trapezoid $EHGC$ is

$$\frac{HG}{2}(EH + CG) = \frac{h}{2} \left(\sqrt{a^2 - (OG + h)^2} + \sqrt{a^2 - OG^2} \right).$$

Now, we want to find the limit, as OG approaches a , of $\frac{K}{R}$. Note that this is equivalent to finding the same limit as h approaches 0. Substituting $a - 2h$ into OG yields that trapezoid $EHGC$ has area

$$\frac{h}{2} \left(\sqrt{a^2 - (a - 2h + h)^2} + \sqrt{a^2 - (a - 2h)^2} \right) = \frac{h}{2} \left(\sqrt{2ah - h^2} + \sqrt{4ah - 4h^2} \right)$$

and that rectangle $EHGL$ has area

$$h\sqrt{a^2 - (a - 2h + h)^2} = h \left(\sqrt{2ah - h^2} \right).$$

Our answer thus becomes

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{\frac{h}{2}(\sqrt{2ah - h^2} + \sqrt{4ah - 4h^2})}{h(\sqrt{2ah - h^2})} &= \lim_{h \rightarrow 0} \left[\frac{1}{2} \cdot \frac{\sqrt{h}(\sqrt{2a - h} + 2\sqrt{a - h})}{\sqrt{h}(\sqrt{2a - h})} \right] \\ &\Rightarrow \frac{1}{2} \cdot \frac{\sqrt{2a} + 2\sqrt{a}}{\sqrt{2a}} = \frac{1}{2} \left(1 + \frac{2}{\sqrt{2}} \right) = \boxed{\text{(D)} \frac{1}{2} + \frac{1}{\sqrt{2}}}. \end{aligned}$$

§5 AHSME 1969

1. (AHSME 1969) When x is added to both the numerator and denominator of the fraction $\frac{a}{b}$, $a \neq b$, $b \neq 0$, the value of the fraction is changed to $\frac{c}{d}$. Then x equals:
 (A) $\frac{1}{c-d}$ (B) $\frac{ad-bc}{c-d}$ (C) $\frac{ad-bc}{c+d}$ (D) $\frac{bc-ad}{c-d}$ (E) $\frac{bc+ad}{c-d}$

Answer: B

Solution

$$\frac{a+x}{b+x} = \frac{c}{d},$$

$$bc + cx = ad + dx,$$

$$(c - d)x = ad - bc,$$

$$x = \frac{ad-bc}{c-d}. \text{ The answer is } \boxed{\text{B}}.$$

2. (AHSME 1969) If an item is sold for x dollars, there is a loss of 15% based on the cost. If, however, the same item is sold for y dollars, there is a profit of 15% based on the cost. The ratio of $y : x$ is:

(A) $23 : 17$ (B) $17y : 23$ (C) $23x : 17$
 (D) dependent upon the cost (E) none of these.

Answer: A

Solution

Let c be the cost. Selling the item for x dollars equates to losing 15% of c , so $x = .85c$. Selling the item for y dollars equates to profiting by 15% of c , so $y = 1.15c$. Therefore $\frac{y}{x} = \frac{1.15c}{.85c} = \frac{23}{17}$. The answer is **(A)**.

3. (AHSME 1969) If N , written in base 2, is 11000, the integer immediately preceding N , written in base 2, is:

(A) 10001 (B) 10010 (C) 10011 (D) 10110 (E) 10111

Answer: E

Solution

$11000_2 - 1_2 = 10111_2$. **(E)**

4. (AHSME 1969) Let a binary operation $\star$ on ordered pairs of integers be defined by $(a, b) \star (c, d) = (a - c, b + d)$. Then, if $(3, 3) \star (0, 0)$ and $(x, y) \star (3, 2)$ represent identical pairs, x equals:

(A) -3 (B) 0 (C) 2 (D) 3 (E) 6

Answer: E

Solution

Performing the operation based on the definition, $(3, 3) \star (0, 0) = (3, 3)$ and $(x, y) \star (3, 2) = (x - 3, y + 2)$. Because the outputs are identical pairs, they must equal each other, so $3 = x - 3$. Solving for x yields $x = 6$, which is answer choice **(E)**.

5. (AHSME 1969) If a number N , $N \neq 0$, diminished by four times its reciprocal, equals a given real constant R , then, for this given R , the sum of all such possible values of N is

(A) $\frac{1}{R}$ (B) R (C) 4 (D) $\frac{1}{4}$ (E) $-R$

Answer: B

Solution

Write an equation from the given information.

$$N - \frac{4}{N} = R$$

$$N^2 - 4 = RN$$

$$N^2 - RN - 4 = 0$$

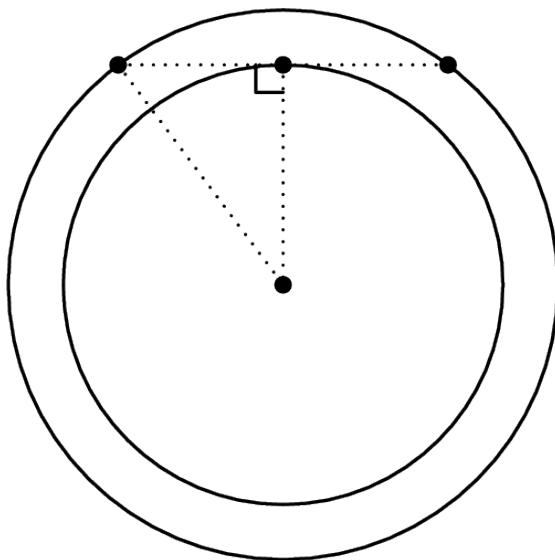
By **Vieta's Formulas**, the sum of all possible values of N for a given R is R , so the answer is **(B)**.

6. (AHSME 1969) The area of the ring between two concentric circles is $12\frac{1}{2}\pi$ square inches. The length of a chord of the larger circle tangent to the smaller circle, in inches, is:

(A) $\frac{5}{\sqrt{2}}$ (B) 5 (C) $5\sqrt{2}$ (D) 10 (E) $10\sqrt{2}$

Answer: C

Solution



centering

Let a be radius of larger circle, and b be radius of smaller circle. The area of the ring can be determined by subtracting the area of the smaller circle from the area of the larger circle, so

$$\begin{aligned}\pi a^2 - \pi b^2 &= \frac{25\pi}{2} \\ a^2 - b^2 &= \frac{25}{2}\end{aligned}$$

From the diagram, by using the **Pythagorean Theorem**, half of the chord is $\frac{5}{\sqrt{2}} = \frac{5\sqrt{2}}{2}$ units long, so the the length of the entire chord is **(C) $5\sqrt{2}$** .

7. (AHSME 1969) If the points $(1, y_1)$ and $(-1, y_2)$ lie on the graph of $y = ax^2 + bx + c$, and $y_1 - y_2 = -6$, then b equals:

(A) -3 (B) 0 (C) 3 (D) $\sqrt{ac}$ (E) $\frac{a+c}{2}$

Answer: A

Solution

Because the two points are on the quadratic, $y_1 = a + b + c$ and $y_2 = a - b + c$. Because $y_1 - y_2 = -6$,

$$(a + b + c) - (a - b + c) = -6$$

$$2b = -6$$

$$b = -3$$

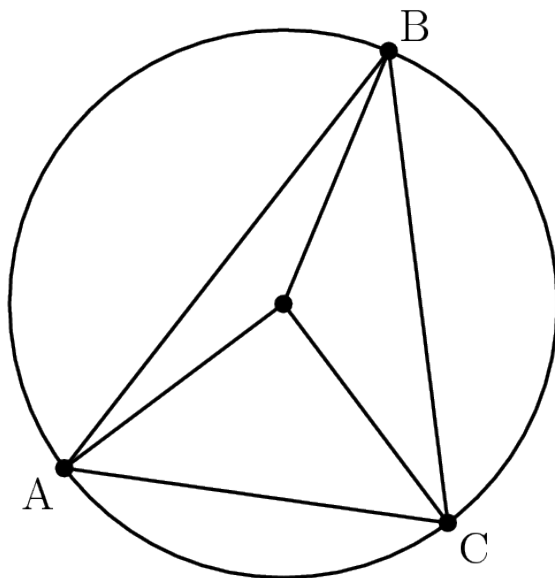
The answer is **(A)**.

8. (AHSME 1969) Triangle ABC is inscribed in a circle. The measure of the non-overlapping minor arcs AB , BC and CA are, respectively, $x + 75^\circ$, $2x + 25^\circ$, $3x - 22^\circ$. Then one interior angle of the triangle is:

(A) $57\frac{1}{2}^\circ$ (B) 59° (C) 60° (D) 61° (E) 122°

Answer: D

Solution



centering

Because the triangle is inscribed, the sum of the minor arcs equals 360° . Thus,

$$x + 75 + 2x + 25 + 3x - 22 = 360$$

$$6x + 78 = 360$$

Solving this yields $x = 47$, so the inscribed angles are 122° , 99° , and 119° . Noting that an angle of $\triangle ABC$ is half of its corresponding inscribed angle, so the angles of $\triangle ABC$ are 59.5° , 49.5° , and **(D) 61°** .

9. (AHSME 1969) The arithmetic mean (ordinary average) of the fifty-two successive positive integers beginning at 2 is:

(A) 27 (B) $27\frac{1}{4}$ (C) $27\frac{1}{2}$ (D) 28 (E) $28\frac{1}{2}$

Answer: C

Solution

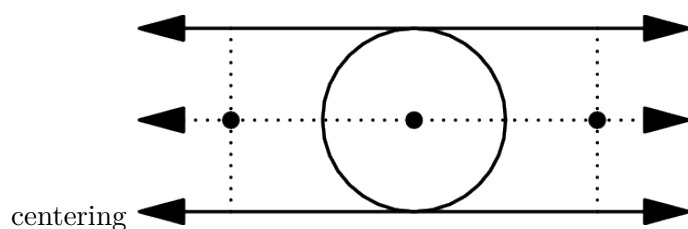
To solve the problem, find the sum of the first 52 terms of an **arithmetic sequence** with first term 2 and common difference 1 and divide that by 52. The 52nd term of the sequence is $2 + 51 = 53$, so the sum of the first 52 terms of the sequence is $\frac{52(2+53)}{2} = 1430$. Thus, the **arithmetic mean** is $\frac{1430}{52} = \frac{55}{2} = \mathbf{(C) \ 27\frac{1}{2}}$.

10. (AHSME 1969) The number of points equidistant from a circle and two parallel tangents to the circle is:

(A) 0 (B) 2 (C) 3 (D) 4 (E) ∞

Answer: C

Solution



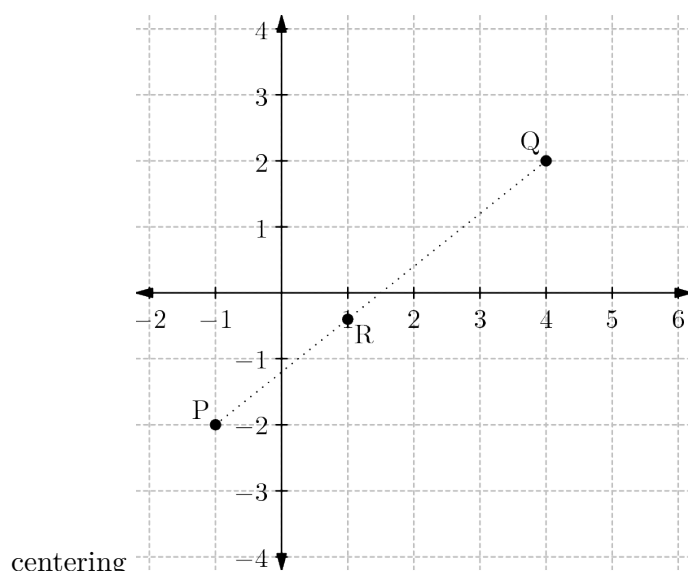
The distance between the two parallel tangents is the length of the circle's diameter, so the distance from a point that satisfies the conditions and the two tangents is the length of the circle's radius. From the diagram, there are **(C) 3** points that satisfies the conditions.

11. (AHSME 1969) Given points $P(-1, -2)$ and $Q(4, 2)$ in the xy -plane; point $R(1, m)$ is taken so that $PR + RQ$ is a minimum. Then m equals:

(A) $-\frac{3}{5}$ (B) $-\frac{2}{5}$ (C) $-\frac{1}{5}$ (D) $\frac{1}{5}$ (E) either $-\frac{1}{5}$ or $\frac{1}{5}$.

Answer: B

Solution



By the **Triangle Inequality**, $PR + QR \geq PQ$, and equality holds if R is on PQ . The equation of the line with P and Q is $y = \frac{4}{5}x - \frac{6}{5}$, so point R is $(1, -\frac{2}{5})$. Thus,

$$m = \textbf{(B)} -\frac{2}{5}.$$

12. (AHSME 1969) Let $F = \frac{6x^2 + 16x + 3m}{6}$ be the square of an expression which is linear in x . Then m has a particular value between:

(A) 3 and 4 (B) 4 and 5 (C) 5 and 6 (D) -4 and -3 (E) -6 and -5

Answer: A

Solution

The quadratic can also be written as

$$x^2 + \frac{8}{3}x + \frac{m}{2}$$

In order for the quadratic to be a square of a linear expression, the constant term must be the square of the x-term divided by 4. Thus,

$$\begin{aligned}\frac{m}{2} &= \left(\frac{8}{3}\right)^2 \cdot \frac{1}{4} \\ \frac{m}{2} &= \frac{16}{9} \\ m &= \frac{32}{9}\end{aligned}$$

The answer is **(A)**.

13. (AHSME 1969) A circle with radius r is contained within the region bounded by a circle with radius R . The area bounded by the larger circle is $\frac{a}{b}$ times the area of the region outside the smaller circle and inside the larger circle. Then $R : r$ equals:
(A) $\sqrt{a} : \sqrt{b}$ (B) $\sqrt{a} : \sqrt{a-b}$ (C) $\sqrt{b} : \sqrt{a-b}$ (D) $a : \sqrt{a-b}$ (E) $b : \sqrt{a-b}$

Answer: B

Solution

The area of the larger circle is πR^2 , and the area of the region outside the smaller circle and inside the larger circle $\pi R^2 - \pi r^2$. Thus,

$$\begin{aligned}\pi R^2 &= \frac{a}{b} \cdot (\pi R^2 - \pi r^2) \\ R^2 &= \frac{a}{b} \cdot R^2 - \frac{a}{b} \cdot r^2 \\ \frac{a}{b} \cdot r^2 &= \left(\frac{a}{b} - 1\right) R^2 \\ \frac{a}{b} \div \frac{a-b}{b} &= \frac{R^2}{r^2} \\ \frac{a}{a-b} &= \frac{R^2}{r^2} \\ \frac{\sqrt{a}}{\sqrt{a-b}} &= \frac{R}{r}\end{aligned}$$

The answer is **(B)**.

14. (AHSME 1969) The complete set of x -values satisfying the inequality $\frac{x^2-4}{x^2-1} > 0$ is the set of all x such that:
(A) $x > 2$ or $x < -2$ or $-1 < x < 1$ (B) $x > 2$ or $x < -2$
(C) $x > 1$ or $x < -2$ (D) $x > 1$ or $x < -1$
(E) x is any real number except 1 or -1

Answer: A

Solution

Factor the **difference of squares**.

$$\frac{(x+2)(x-2)}{(x+1)(x-1)} > 0$$

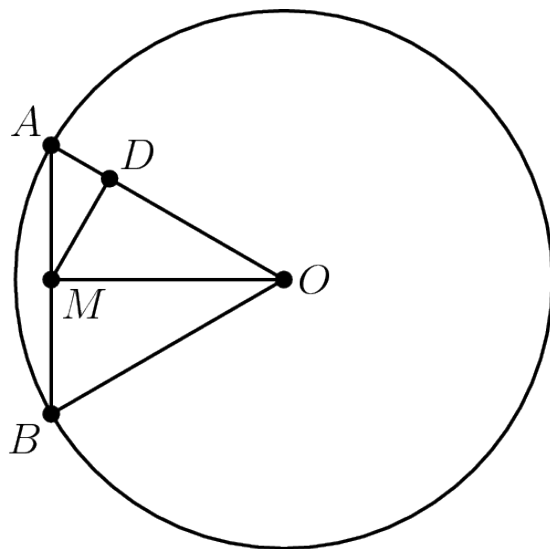
Note that the graph intersects the x-axis at when $x = \pm 2$ or $x = \pm 1$, so check the sign of the result to see if it is positive. After testing, $x < -2$ or $-1 < x < 1$ or $x > 2$, so the answer is **(A)**.

15. (AHSME 1969) In a circle with center O and radius r , chord AB is drawn with length equal to r (units). From O , a perpendicular to AB meets AB at M . From M a perpendicular to OA meets OA at D . In terms of r the area of triangle MDA , in appropriate square units, is:

(A) $\frac{3r^2}{16}$ (B) $\frac{\pi r^2}{16}$ (C) $\frac{\pi r^2 \sqrt{2}}{8}$ (D) $\frac{r^2 \sqrt{3}}{32}$ (E) $\frac{r^2 \sqrt{6}}{48}$

Answer: D

Solution



centering

Because $AO = OB = AB$, $\triangle AOB$ is an equilateral triangle, and $\angle OAB = 60^\circ$. Using 30-60-90 triangles, $AM = \frac{r}{2}$, $AD = \frac{r}{4}$, and $DM = \frac{r\sqrt{3}}{4}$. Thus, the area of

$$\triangle ADM \text{ is } \frac{1}{2} \cdot \frac{r}{4} \cdot \frac{r\sqrt{3}}{4} = \boxed{\text{(D)} \frac{r^2 \sqrt{3}}{32}}.$$

16. (AHSME 1969) When $(a - b)^n$, $n \geq 2$, $ab \neq 0$, is expanded by the binomial theorem, it is found that when $a = kb$, where k is a positive integer, the sum of the second and third terms is zero. Then n equals:
- (A) $\frac{1}{2}k(k - 1)$ (B) $\frac{1}{2}k(k + 1)$ (C) $2k - 1$ (D) $2k$ (E) $2k + 1$

Answer: E

Solution

Since $a = kb$, we can write $(a - b)^n$ as $(kb - b)^n$. Expanding, the second term is $-k^{n-1}b^n \binom{n}{1}$, and the third term is $k^{n-2}b^n \binom{n}{2}$, so we can write the equation

$$-k^{n-1}b^n \binom{n}{1} + k^{n-2}b^n \binom{n}{2} = 0$$

Simplifying and multiplying by two to remove the denominator, we get

$$-2k^{n-1}b^n n + k^{n-2}b^n n(n - 1) = 0$$

Factoring, we get

$$k^{n-2}b^n n(-2k + n - 1) = 0$$

Dividing by $k^{n-2}b^n$ gives

$$n(-2k + n - 1) = 0$$

Since it is given that $n \geq 2$, n cannot equal 0, so we can divide by n , which gives

$$-2k + n - 1 = 0$$

Solving for n gives

$$n = 2k + 1$$

so the answer is **E**.

17. (AHSME 1969) The equation $2^{2x} - 8 \cdot 2^x + 12 = 0$ is satisfied by:

(A) $\log(3)$ (B) $\frac{1}{2}\log(6)$ (C) $1 + \log(\frac{3}{2})$ (D) $1 + \frac{\log(3)}{\log(2)}$ (E) none of these

Answer: D

Solution

Let $2^x = a$. Because $2^{2x} = (2^x)^2$, the given expression can be rewritten as $a^2 - 8a + 12 = 0$. This can be factored as $(a - 6)(a - 2) = 0$, which has solutions $a = 2^x = 6$ and $a = 2^x = 2$. Looking at the answer choices, we see that $x = 1$ is absent. Rewriting $2^x = 6$ as $x = \log_2 6$ and then applying the logarithm addition identity in reverse gives $x = \log_2(2) + \log_2(3) = 1 + \log_2(3)$. Applying the logarithm division identity shows that the answer is **D**.

18. (AHSME 1969) The number of points common to the graphs of $(x - y + 2)(3x + y - 4) = 0$ and $(x + y - 2)(2x - 5y + 7) = 0$ is:

(A) 2 (B) 4 (C) 6 (D) 16 (E) ∞

Answer: B

Solution

By the **Zero Product Property**, $x - y + 2 = 0$ or $3x + y - 4 = 0$ in the first equation and $x + y - 2 = 0$ or $2x - 5y + 7 = 0$ in the second equation. Thus, from the first equation, $y = x + 2$ or $y = -3x + 4$, and from the second equation, $y = -x + 2$ or $y = \frac{2}{5}x + \frac{7}{5}$.

If a point is common to the two graphs, then the point must be in one of the lines in the first equation as well as one of the lines in the second equation. Since the slopes of the lines are different, none of the lines are parallel. Thus, there are $2 \cdot 2 =$ **(B) 4** points of intersection in the two graphs.

19. (AHSME 1969) The number of distinct ordered pairs (x, y) where x and y have positive integral values satisfying the equation $x^4y^4 - 10x^2y^2 + 9 = 0$ is:

(A) 0 (B) 3 (C) 4 (D) 12 (E) ∞

Answer: B

Solution

Let $(xy)^2 = a$. The expression given is equal to $a^2 - 10a + 9 = 0$, which can be factored as $(a - 9)(a - 1) = 0$. Thus, we have $a = (xy)^2 = 9$ and $a = (xy)^2 = 1$. Because x and y are positive, we can eliminate the possibilities where xy is negative. From here it is easy to see that the only integral pairs of x and y are $(3, 1)$, $(1, 3)$, and $(1, 1)$. The answer is **B**.

20. (AHSME 1969) Let P equal the product of 3,659,893,456,789,325,678 and 342,973,489,379,256. The number of digits in P is:

(A) 36 (B) 35 (C) 34 (D) 33 (E) 32

Answer: C

Solution 1

Through inspection, we see that the two digit number $33^2 = 1089 = 4$ digits. Notice that any number that has the form $33abcdefg\dots$ multiplied by another $33qwertyu\dots$ will have its number of digits equal to the sum of the original numbers' digits.

In this case, we see that the first number has 19 digits, and the second number has 15 digits.

Note: this applies for numbers $33 \dots > 99$

Hence, the answer is $19 + 15 = 34$ digits $\implies$ C

Solution 2

We can approximate the product with $10^{18} * 3.6 * 10^{14} * 3.4$. Now observe that $3.6 * 3.4 > 10$, so we can further simplify the product with $10^{18} * 10^{14} * 10 = 10^{33}$ which means the product has 34 (C) digits.

-serpent_121

21. (AHSME 1969) If the graph of $x^2 + y^2 = m$ is tangent to that of $x + y = \sqrt{2m}$, then:
- (A) m must equal $\frac{1}{2}$ (B) m must equal $\frac{1}{\sqrt{2}}$
 (C) m must equal $\sqrt{2}$ (D) m must equal 2
 (E) m may be any non-negative real number

Answer: E

Solution 1

Note that the first equation represents a circle and the second equation represents a line. If a line is tangent to a circle, then it only hits at one point, so there will only be one solution to the **system of equations**.

In the second equation, $y = -x + \sqrt{2m}$. Substitution results in

$$x^2 + x^2 - 2x\sqrt{2m} + 2m = m$$

$$2x^2 - 2x\sqrt{2m} + m = 0$$

In order for the system to have one solution, the **discriminant** must equal 0.

$$4(2m) - 4 \cdot 2 \cdot m = 0$$

$$0 = 0$$

Thus, m can be a non-negative real number, so the answer is (E).

Solution 2

Since $x^2 + y^2 = m$ has **radius** $\sqrt{m}$, and $x + y = \sqrt{2m}$ is a **diagonal line** in the **plane** with **slope** -1 , if the two **curves** are **tangent** then $x + y = \sqrt{2m}$ either has to pass through $(\frac{\sqrt{2m}}{2}, \frac{\sqrt{2m}}{2})$ or $(-\frac{\sqrt{2m}}{2}, -\frac{\sqrt{2m}}{2})$. However, as $\sqrt{2m}$ is **positive**, the **line** can only pass through $(\frac{\sqrt{2m}}{2}, \frac{\sqrt{2m}}{2})$, which it does for all m . Thus the answer is (E).

22. (AHSME 1969) Let K be the measure of the area bounded by the x -axis, the line $x = 8$, and the curve defined by

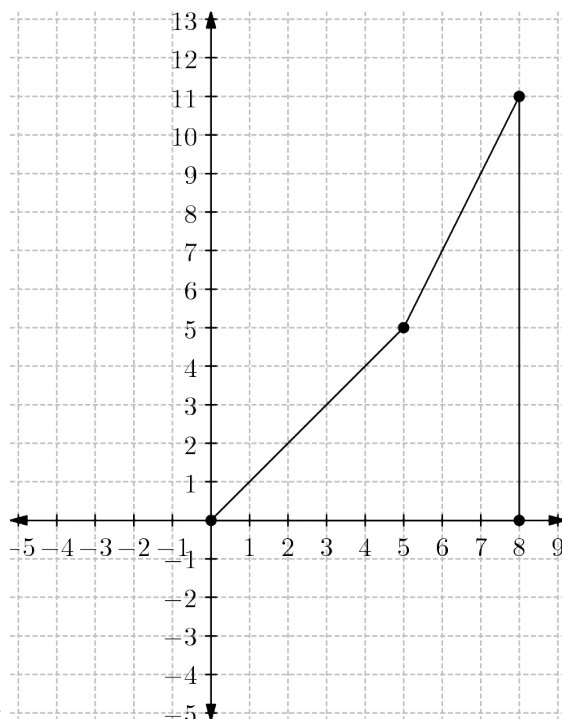
$$f = (x, y) \quad | \quad y = x \text{ when } 0 \leq x \leq 5, y = 2x - 5 \text{ when } 5 \leq x \leq 8.$$

Then K is:

- (A) 21.5 (B) 36.4 (C) 36.5 (D) 44 (E) less than 44 but arbitrarily close to it

Answer: C

Solution



centering

The shape can be divided into a triangle and a trapezoid. For the triangle, the base is 5 and the height is 5, so the area is $\frac{5 \cdot 5}{2} = \frac{25}{2}$. For the trapezoid, the two bases are 5 and 11 and the height is 3, so the area is $\frac{3(5+11)}{2} = 24$. Thus, the total area is $\frac{25}{2} + 24 = \frac{73}{2} = \boxed{\text{(C) } 36.5}$.

23. (AHSME 1969) For any integer $n > 1$, the number of prime numbers greater than $n! + 1$ and less than $n! + n$ is:
- (A) 0 (B) 1
 (C) $\frac{n}{2}$ for n even, $\frac{n+1}{2}$ for n odd
 (D) $n - 1$ (E) n

Answer: A

Solution

Observe that for all $k \in 1 < k < n$, since k divides $n!$, k also divides $n! + k$. Therefore, all numbers a in the range $n! + 1 < a < n! + n$ are composite. Therefore there are 0 primes in that range. $\boxed{\text{A}}$

24. (AHSME 1969) When the natural numbers P and P' , with $P > P'$, are divided by the natural number D , the remainders are R and R' , respectively. When PP' and RR' are divided by D , the remainders are r and r' , respectively. Then:

- (A) $r > r'$ always (B) $r < r'$ always
 (C) $r > r'$ sometimes and $r < r'$ sometimes
 (D) $r > r'$ sometimes and $r = r'$ sometimes
 (E) $r = r'$ always

Answer: E

Solution

The divisors are the same, so take each variable **modulo** D .

$$P \equiv R \pmod{D}$$

$$P' \equiv R' \pmod{D}$$

That means

$$PP' \equiv RR' \pmod{D}$$

Thus, PP' and RR' have the same remainder when divided by D , so the answer is

(E).

25. (AHSME 1969) If it is known that $\log_2(a) + \log_2(b) \geq 6$, then the least value that can be taken on by $a + b$ is:

- (A) $2\sqrt{6}$ (B) 6 (C) $8\sqrt{2}$ (D) 16 (E) none of these

Answer: D

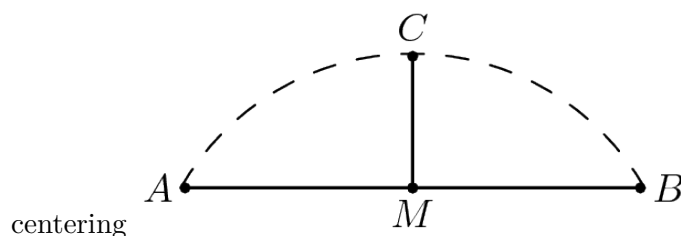
Solution

We use the logarithm property of addition:

$$\begin{aligned} \log_2(a) + \log_2(b) &\geq 6 = \log_2(ab) \geq 6 \\ &\Rightarrow 2^{\log_2(ab)} \geq 2^6 \\ &= ab \geq 64 \end{aligned}$$

Due to the Quadratic **Optimization** or the **AM-GM Inequality**, the least value is obtained when $a = b$. Therefore, $a = b = 8 \Rightarrow a + b = \boxed{(D)16}$

26. (AHSME 1969)



A parabolic arch has a height of 16 inches and a span of 40 inches. The height, in inches, of the arch at the point 5 inches from the center M is:

- (A) 1 (B) 15 (C) $15\frac{1}{3}$ (D) $15\frac{1}{2}$ (E) $15\frac{3}{4}$

Answer: B

Solution

Because the arch has a height of 16 inches, an equation that models the arch is $y = ax^2 + 16$, where x is the horizontal distance from the center and y is the height. The arch has a span of 40 inches, so the arch meets the ground 20 inches from the center. That means $0 = 400a + 16$, so $a = -\frac{1}{25}$.

Thus, the equation that models height based on distance from the center $y = -\frac{1}{25}x^2 + 16$, so the height of the arch 5 inches from the center is $-\frac{1}{25} \cdot 5^2 + 16 = \boxed{\text{(B)} 15}$ inches.

27. (AHSME 1969) A particle moves so that its speed for the second and subsequent miles varies inversely as the integral number of miles already traveled. For each subsequent mile the speed is constant. If the second mile is traversed in 2 hours, then the time, in hours, needed to traverse the n th mile is:

(A) $\frac{2}{n-1}$ (B) $\frac{n-1}{2}$ (C) $\frac{2}{n}$ (D) $2n$ (E) $2(n-1)$

Answer: E

Solution

Let d be the distance already traveled and s_n be the speed for the n^{th} mile. Because the speed for the second and subsequent miles varies inversely as the integral number of miles already traveled,

$$ds_{d+1} = k$$

If the second mile is traversed in 2 hours, then the speed in the second mile (s_2) equals $\frac{1}{2}$ miles per hour. Since one mile has been traveled already, $k = \frac{1}{2}$. Now solve for the speed the n^{th} mile, noting that $n-1$ miles have already been traveled.

$$(n-1)s_{d+1} = \frac{1}{2}$$

$$s_{d+1} = \frac{1}{2(n-1)}$$

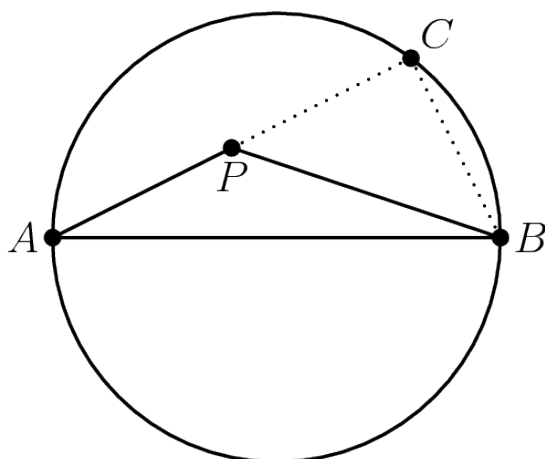
Thus, the time it takes to travel the n^{th} mile is $1 \div \frac{1}{2(n-1)} = \boxed{\text{(E)} 2(n-1)}$ hours.

28. (AHSME 1969) Let n be the number of points P interior to the region bounded by a circle with radius 1, such that the sum of squares of the distances from P to the endpoints of a given diameter is 3. Then n is:

(A) 0 (B) 1 (C) 2 (D) 4 (E) ∞

Answer: E

Solution



centering

Let A and B be points on diameter. Extend AP , and mark intersection with circle as point C .

Because AB is a diameter, $\angle ACB = 90^\circ$. Also, by Exterior Angle Theorem, $\angle ACB + \angle CBP = \angle APB$, so $\angle APB > \angle ACB$, making $\angle APB$ an obtuse angle.

By the **Law of Cosines**, $AP^2 + BP^2 - 2 \cdot AP \cdot BP \cdot \cos \angle APB = 4$. Since $AP^2 + BP^2 = 3$, substitute and simplify to get $\cos \angle APB = -\frac{1}{2 \cdot AP \cdot BP}$. This equation has infinite solutions because for every AP and BP , where $AP + BP \geq 2$ and AP and BP are both less than 2, there can be an obtuse angle that satisfies the equation, so the answer is **(E)**.

Solution 2 (Apollonius Theorem)

Let C_1 be the circle with a centra O and diameter AB . Let P be any random point inside C_1 . Connect OP . We want $AP^2 + BP^2 = 3$ In $\triangle APB$, PO is a median on AB . Using Appolonius Theorem we get:

$$AP^2 + BP^2 = 2(AO^2 + OP^2)$$

$$2(OP^2 + 1) = 3$$

$$2 \cdot OP^2 + 2 = 3$$

$$OP = \frac{1}{\sqrt{2}}$$

We draw a circle C_2 centred at O with a radius of $\frac{1}{\sqrt{2}}$ i.e, OP . Every point on the circle C_2 will satisfy the given conditions. Since there are infinite points on a circle hence there are infinite number of points P that can satisfy the condition.

Hence, the answer is **(E)**.

–Tanstorm

29. (AHSME 1969) If $x = t^{1/(t-1)}$ and $y = t^{t/(t-1)}$, $t > 0, t \neq 1$, a relation between x and y is:

(A) $y^x = x^{1/y}$ (B) $y^{1/x} = x^y$ (C) $y^x = x^y$ (D) $x^x = y^y$ (E) none of these

Answer: C

Solution

Plug in $t = 2$ and test each expression (if the relation works, then both sides are equal). After testing, options A, B, and D are out, but for option C, both sides are equal. To check that C is a valid option, substitute the values of x and y and use exponent properties.

$$x^y = (t^{\frac{1}{t-1}})^{t^{\frac{t}{t-1}}} = t^{\frac{1}{t-1} \cdot t^{\frac{t}{t-1}}}$$

$$y^x = (t^{\frac{t}{t-1}})^{t^{\frac{1}{t-1}}} = t^{\frac{t}{t-1} \cdot t^{\frac{1}{t-1}}}$$

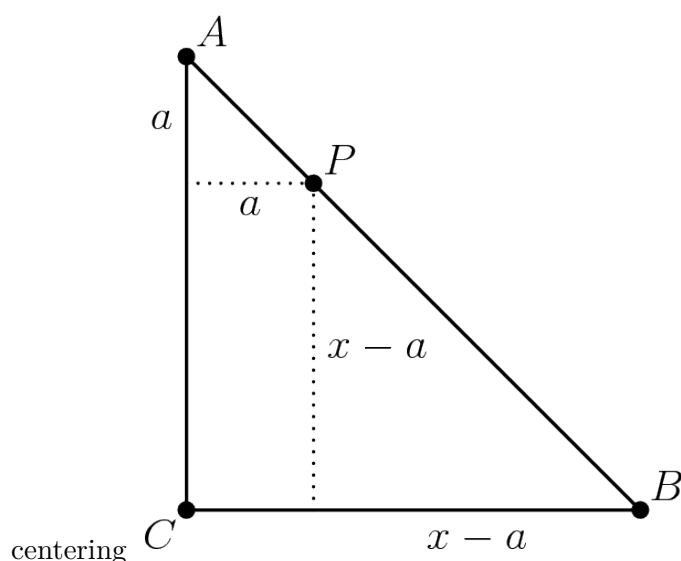
In the second equation, the exponent can be rewritten as $\frac{1}{t-1} \cdot t \cdot t^{\frac{1}{t-1}}$. Since $\frac{t-1}{t-1} = 1$, the exponent can be simplified as $\frac{1}{t-1} \cdot t^{\frac{t-1}{t-1}} \cdot t^{\frac{1}{t-1}} = \frac{1}{t-1} \cdot t^{\frac{t}{t-1}}$. Since the base and exponent of both x^y and y^x are the same, we can confirm that **(C)** $y^x = x^y$.

30. (AHSME 1969) Let P be a point of hypotenuse AB (or its extension) of isosceles right triangle ABC . Let $s = AP^2 + PB^2$. Then:

- (A) $s < 2CP^2$ for a finite number of positions of P
 (B) $s < 2CP^2$ for an infinite number of positions of P
 (C) $s = 2CP^2$ only if P is the midpoint or an endpoint of AB
 (D) $s = 2CP^2$ always
 (E) $s > 2CP^2$ if P is a trisection point of AB

Answer: D

Solution



Consider the case where P is on the hypotenuse of AB . Draw perpendicular lines from P towards the sides. Using the **Pythagorean Theorem**,

$$AP^2 = a^2 + a^2$$

$$BP^2 = (x - a)^2 + (x - a)^2$$

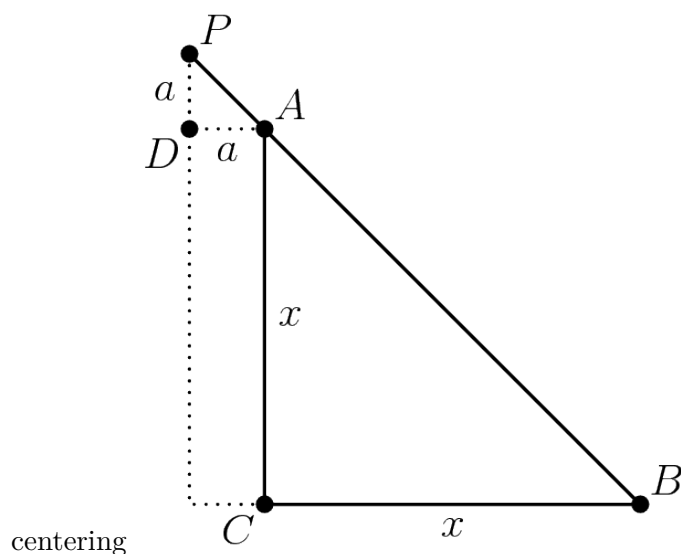
$$CP^2 = a^2 + (x - a)^2$$

This means

$$s = 4a^2 - 4ax + 2x^2$$

$$2 \cdot CP^2 = 4a^2 - 4ax + 2x^2$$

Thus, $s = 2 \cdot CP^2$ when P is on the hypotenuse of AB .



Consider the case where P is on the extension of AB . **WLOG**, let point A be between point P and point B . Extend BC and draw perpendicular line from P . Also, draw point D , where $PD \parallel AC$ and $DA \parallel CB$.

Using the Pythagorean Theorem again,

$$AP^2 = a^2 + a^2$$

$$BP^2 = (a + x)^2 + (a + x)^2$$

$$CP^2 = (a + x)^2 + a^2$$

That means

$$s = 4a^2 + 4ax + 2x^2$$

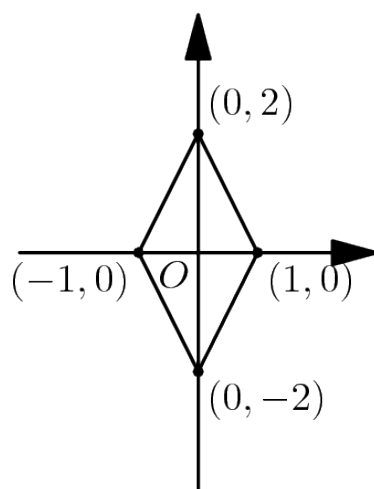
$$2 \cdot CP^2 = 4a^2 + 4ax + 2x^2$$

Thus, $s = 2 \cdot CP^2$ when P is outside the hypotenuse.

In summary, $AP^2 + BP^2 = 2 \cdot CP^2$, so the answer is **(D)**.

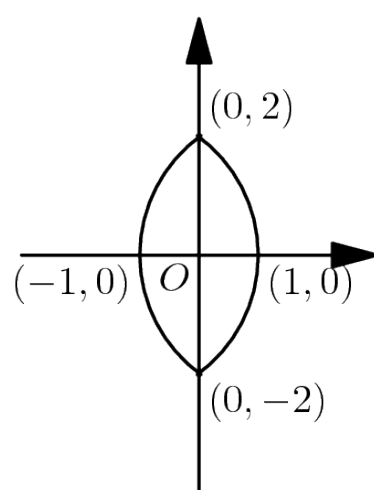
31. (AHSME 1969) Let $OABC$ be a unit square in the xy -plane with $O(0, 0)$, $A(1, 0)$, $B(1, 1)$ and $C(0, 1)$. Let $u = x^2 - y^2$, and $v = 2xy$ be a transformation of the xy -plane into the uv -plane. The transform (or image) of the square is:

(A)



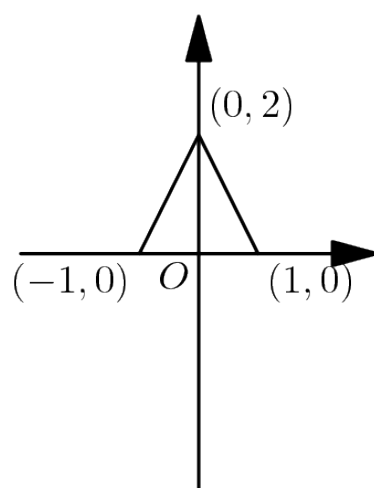
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(B)



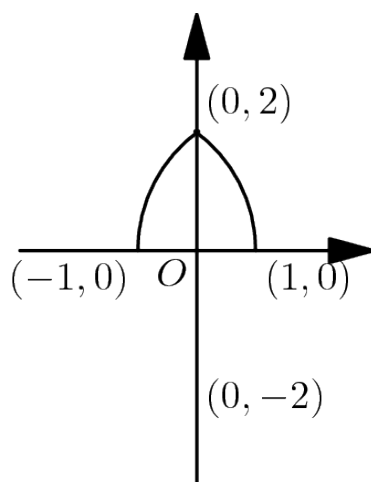
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(C)



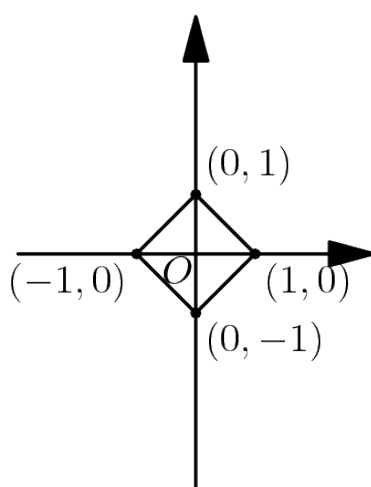
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(D)



centering

(E)



centering

Answer: D**Solution**

Each point on the square can be in the form $(0, y)$, $(1, y)$, $(x, 0)$, and $(x, 1)$, where $0 \leq x, y \leq 1$. Making the appropriate substitutions results in points being $(-y^2, 0)$, $(1 - y^2, 2y)$, $(x^2, 0)$, and $(x^2 - 1, 2x)$ on the uv -plane.

Notice that since $v \geq 0$, none of the points are below the u -axis, so options A, B, and E are out. Since $x = \frac{v}{2}$, $u = (\frac{v}{2})^2 - 1$, so $v = 2\sqrt{u+1}$, where $-1 \leq u \leq 0$. That means some of the lines are not straight, so the answer is **(D)**.

32. (AHSME 1969) Let a sequence $\{u_n\}$ be defined by $u_1 = 5$ and the relationship $u_{n+1} - u_n = 3 + 4(n - 1)$, $n = 1, 2, 3, \dots$. If u_n is expressed as a polynomial in n , the algebraic sum of its coefficients is:

(A) 3 (B) 4 (C) 5 (D) 6 (E) 11

Answer: C**Solution**

Note that the first differences create a linear function, so the sequence u_n is quadratic.

The first three terms of the sequence are 5, 8, and 15. From there, a **system of equations** can be written.

$$a + b + c = 5$$

$$4a + 2b + c = 8$$

$$9a + 3b + c = 15$$

Solve the system to get $a = 2$, $b = -3$, and $c = 6$. The sum of the coefficients is

(C) 5.

Note: Solving the system is extra work, as the answer is described by the first equation.

33. (AHSME 1969) Let S_n and T_n be the respective sums of the first n terms of two arithmetic series. If $S_n : T_n = (7n + 1) : (4n + 27)$ for all n , the ratio of the eleventh term of the first series to the eleventh term of the second series is:

(A) 4 : 3 (B) 3 : 2 (C) 7 : 4 (D) 78 : 71 (E) undetermined

Answer: A

Solution

Let S be the first arithmetic sequence and T be the second arithmetic sequence. If $n = 1$, then $S_1 : T_1 = 8 : 31$. Since S_1 and T_1 are just the first term, the first term of S is $8a$ and the first term of T is $31a$ for some a . If $n = 2$, then $S_2 : T_2 = 15 : 35 = 3 : 7$, so the sum of the first two terms of S is $3b$ and the sum of the first two terms of T is $7b$ for some b . Thus, the second term of S is $3b - 8a$ and the second term of T is $7b - 31a$, so the common difference of S is $3b - 16a$ and the common difference of T is $7b - 62a$.

Thus, using the first terms and common differences, the sum of the first three terms of S equals $\frac{1}{2} \cdot 3(16a + 2(-16a + 3b))$, and the sum of the first three terms of T equals $\frac{1}{2} \cdot 3(62a + 2(-62a + 7b))$. That means

$$\frac{16a + 2(-16a + 3b)}{62a + 2(-62a + 7b)} = \frac{22}{39}$$

$$\frac{6b - 16a}{14b - 62a} = \frac{22}{39}$$

$$\frac{3b - 8a}{7b - 31a} = \frac{22}{39}$$

$$117b - 312a = 154b - 682a$$

$$-37b = -370a$$

$$b = 10a$$

With the substitution, the common difference of S is $14a$, and the common difference of T is $8a$. That means the 11th term of S is $8a + 10(14a) = 148a$, and the 11th term of T is $31a + 10(8a) = 111a$. Thus, the ratio of the eleventh term of the first series to the eleventh term of the second series is $148a : 111a = \mathbf{(A) 4 : 3}$.

Solution 2 (Quick Sol)

Note that the sum of arithmetic sequences can be expressed as a quadratic polynomial of n . From the ratio given in the problem, multiply n to both sides to get quadratic polynomials.

$$S_n = 7n^2 + n, T_n = 4n^2 + 27n$$

From there, the 11th term for S_n and T_n can be calculated.

$$S_{11} - S_{10} = 7 * 11^2 + 11 - (7 * 10^2 + 10) = 148$$

$$T_{11} - T_{10} = 4 * 11^2 + 27 * 11 - (4 * 10^2 + 27 * 10) = 111$$

The ratio is $148 : 111 = \boxed{\text{(A)} 4 : 3}$.

Solution 3 (Quickest Sol)

Since they are arithmetic sequences, we know that the 11th term is the average value of the first 21 terms in each sequence. So the desired ratio of the 11th terms is just $(21 \cdot 7 + 1) : (21 \cdot 4 + 27) = 148 : 111 = \boxed{\text{(A)} 4 : 3}$

-purplepenguin2

Solution 4 (Formal Quick)

Let a , and c be the first term of S_n and T_n respectively and d , b be the common difference of S_n and T_n respectively

$$\frac{S_n}{T_n} = \frac{\frac{n}{2}(2a+(n-1)d)}{\frac{n}{2}(2c+(n-1)b)} = \frac{7n+1}{4n+27}$$

$$\frac{S_n}{T_n} = \frac{(2a+(n-1)d)}{(2c+(n-1)b)}$$

Now major step!!!!

Divide numerator and denominator of R.H.S by 2

$$\frac{S_n}{T_n} = \frac{(a+(\frac{n-1}{2})d)}{(c+(\frac{n-1}{2})b)} = \frac{7n+1}{4n+27}$$

Since we want ratio of 11th terms

$$\frac{n-1}{2} = 10$$

$$n = 21$$

$$\frac{S_{21}}{T_{21}} = \frac{(a+(10)d)}{(c+(10)b)} = \frac{148}{111}$$

which is equal to $\frac{4}{3}$

Actually this solution was given by my maths teacher.

34. (AHSME 1969) The remainder R obtained by dividing x^{100} by $x^2 - 3x + 2$ is a polynomial of degree less than 2. Then R may be written as:

- (A) $2^{100} - 1$ (B) $2^{100}(x - 1) - (x - 2)$ (C) $2^{200}(x - 3)$
 (D) $x(2^{100} - 1) + 2(2^{99} - 1)$ (E) $2^{100}(x + 1) - (x + 2)$

Answer: B

Solution

Let the polynomial $Q(x)$ be the quotient when x^{100} is divided by $x^2 - 3x + 2$, and let the remainder $R = ax + b$, for some real a and b . Then we can write: $x^{100} = (x^2 - 3x + 2)Q(x) + ax + b$. Since it is hard to deal with $Q(x)$ (it is of degree 98!), we factor $x^2 - 3x + 2$ as $(x - 2)(x - 1)$ so we can eliminate $Q(x)$ by plugging in x values of 2 and 1.

$$x^{100} = (x - 2)(x - 1)Q(x) + ax + b,$$

$$2^{100} = (0)Q(2) + 2a + b,$$

$$2^{100} = 2a + b.$$

$$\text{Similarly, } 1^{100} = a + b.$$

Solving this system of equations gives $a = 2^{100} - 1$ and $b = 2 - 2^{100}$. Thus, $R = ax + b = (2^{100} - 1)x + (2 - 2^{100})$. Expanding and combining terms, we see that the answer is B.

35. (AHSME 1969) Let $L(m)$ be the x coordinate of the left end point of the intersection of the graphs of $y = x^2 - 6$ and $y = m$, where $-6 < m < 6$. Let $r = [L(-m) - L(m)]/m$. Then, as m is made arbitrarily close to zero, the value of r is:

- (A) arbitrarily close to 0
 (B) arbitrarily close to $\frac{1}{\sqrt{6}}$ (C) arbitrarily close to $\frac{2}{\sqrt{6}}$ (D) arbitrarily large (E) undetermined

Answer: B

Solution 1

Since $L(a)$ is the x coordinate of the left end point of the intersection of the graphs of $y = x^2 - 6$ and $y = a$, we can substitute a for y and find the lowest solution x .

$$x^2 - 6 = a$$

$$x = \pm\sqrt{a+6}$$

That means $L(m) = -\sqrt{m+6}$ and $L(-m) = -\sqrt{-m+6}$. That means

$$r = \frac{-\sqrt{-m+6} + \sqrt{m+6}}{m}$$

Since plugging in 0 for m results in $0/0$, there is a removable discontinuity. Multiply the fraction by $\frac{\sqrt{-m+6} + \sqrt{m+6}}{\sqrt{-m+6} + \sqrt{m+6}}$ to get

$$r = \frac{m+6 - (-m+6)}{m(\sqrt{-m+6} + \sqrt{m+6})}$$

$$r = \frac{2m}{m(\sqrt{-m+6} + \sqrt{m+6})}$$

$$r = \frac{2}{(\sqrt{-m+6} + \sqrt{m+6})}$$

Now there wouldn't be a problem plugging in 0 for m . Doing so results in $r = \frac{2}{2\sqrt{6}} = \frac{1}{\sqrt{6}}$, so the answer is (B).

Solution 2

From Solution 1, $L(m) = -\sqrt{m+6}$ and $L(-m) = -\sqrt{-m+6}$, so

$$r = \frac{-\sqrt{-m+6} + \sqrt{m+6}}{m}$$

Since m is arbitrarily close to 0, we wish to find

$$\lim_{m \rightarrow 0} \frac{-\sqrt{-m+6} + \sqrt{m+6}}{m}$$

Using L'Hopital's Rule, the limit is equivalent to

$$\lim_{m \rightarrow 0} \frac{1}{2\sqrt{-m+6}} + \frac{1}{2\sqrt{m+6}}$$

Calculating the limit shows that r is (B) arbitrarily close to $\frac{1}{\sqrt{6}}$.

§6 AHSME 1970

1. (AHSME 1970) The fourth power of $\sqrt{1 + \sqrt{1 + \sqrt{1}}}$ is:

(A) $\sqrt{2} + \sqrt{3}$ (B) $\frac{1}{2}(7 + 3\sqrt{5})$ (C) $1 + 2\sqrt{3}$ (D) 3 (E) $3 + 2\sqrt{2}$

Answer: E

Solution

We can simplify the expression to $\sqrt{1 + \sqrt{2}}$. Then we square it so it is now $1 + \sqrt{2}$.

We still have to square it one more time, getting $\boxed{(E) \ 3 + 2\sqrt{2}}$.

2. (AHSME 1970) A square and a circle have equal perimeters. The ratio of the area of the circle to the area of the square is:

(A) $\frac{4}{\pi}$ (B) $\frac{\pi}{\sqrt{2}}$ (C) $\frac{4}{1}$ (D) $\frac{\sqrt{2}}{\pi}$ (E) $\frac{\pi}{4}$

Answer: A

Solution

Let's say the circle has a circumference (or perimeter) of 4π . Since the perimeter of the square is the same as the perimeter of the circle, the side length of the square is π . That means that the area of the square is π^2 . The area of the circle is 4π . So the area of the circle over the area of the square is $\frac{4\pi}{\pi^2} \Rightarrow \frac{4}{\pi} \Rightarrow \boxed{A}$.

3. (AHSME 1970) If $x = 1 + 2^p$ and $y = 1 + 2^{-p}$, then y in terms of x is:

(A) $\frac{x+1}{x-1}$ (B) $\frac{x+2}{x-1}$ (C) $\frac{x}{x-1}$ (D) $2-x$ (E) $\frac{x-1}{x}$

Answer: C

Solution

Since we want the y expression in terms of x , let's convert the y expression. We can convert it to $1 + \frac{1}{2^p} \Rightarrow \frac{2^p+1}{2^p} \Rightarrow \frac{x}{x-1} \Rightarrow \boxed{C}$.

4. (AHSME 1970) Let S be the set of all numbers which are the sum of the squares of three consecutive integers. Then we can say that:

(A) No member of S is divisible by 2
 (B) No member of S is divisible by 3 but some member is divisible by 11
 (C) No member of S is divisible by 3 or 5
 (D) No member of S is divisible by 3 or 7
 (E) None of these

Answer: B

Solution

Consider 3 consecutive integers a, b , and c . Exactly one of these integers must be divisible by 3; WLOG, suppose a is divisible by 3. Then $a \equiv 0 \pmod{3}$, $b \equiv 1 \pmod{3}$, and $c \equiv 2 \pmod{3}$. Squaring, we have that $a^2 \equiv 0 \pmod{3}$, $b^2 \equiv 1 \pmod{3}$, and $c^2 \equiv 1 \pmod{3}$, so $a^2 + b^2 + c^2 \equiv 2 \pmod{3}$. Therefore, no member of S is divisible by 3.

Now consider 3 more consecutive integers a, b , and c , which we will consider mod 11. We will assign k such that $a \equiv k \pmod{11}$, $b \equiv k+1 \pmod{11}$, and $c \equiv k+2 \pmod{11}$. Some experimentation shows that when $k=4$, $a \equiv 4 \pmod{11}$ so $a^2 \equiv 5 \pmod{11}$. Similarly, $b \equiv 5 \pmod{11}$ so $b^2 \equiv 3 \pmod{11}$, and $c \equiv 6 \pmod{11}$ so

$c^2 \equiv 3 \pmod{11}$. Therefore, $a^2 + b^2 + c^2 \equiv 0 \pmod{11}$, so there is at least one member of S which is divisible by 11. Thus, **B** is correct.

5. (AHSME 1970) If $f(x) = \frac{x^4 + x^2}{x + 1}$, then $f(i)$, where $i = \sqrt{-1}$, is equal to:

(A) $1 + i$ (B) 1 (C) -1 (D) 0 (E) $-1 - i$

Answer: D

Solution

$i^4 = 1$ and $i^2 = -1$, so the numerator is 0. As long as the denominator is not 0, which it isn't, the answer is $0 \Rightarrow$ **D**

6. (AHSME 1970) The smallest value of $x^2 + 8x$ for real values of x is:

(A) -16.25 (B) -16 (C) -15 (D) -8 (E) None of these

Answer: B

Solution

Let's imagine this as a quadratic equation. To find the minimum or maximum value, we always need to find the vertex of the quadratic equation. The vertex of the quadratic is $\frac{-b}{2a}$ in $ax^2 + bx + c = 0$. Then to find the output, or the y value of the quadratic, we plug the vertex " x " value back into the equation. In this quadratic, $a=1$, $b=8$, and $c=0$. So the " x " value is $\frac{-b}{2a} \Rightarrow \frac{-8}{2} = -4$. Plugging it back into $x^2 + 8x$, we get $16 - 32 = -16 \Rightarrow$ **B**

7. (AHSME 1970) Inside square $ABCD$ with side s , quarter-circle arcs with radii s and centers at A and B are drawn. These arcs intersect at point X inside the square. How far is X from side CD ?

(A) $\frac{1}{2}s(\sqrt{3} + 4)$ (B) $\frac{1}{2}s\sqrt{3}$ (C) $\frac{1}{2}s(1 + \sqrt{3})$
 (D) $\frac{1}{2}s(\sqrt{3} - 1)$ (E) $\frac{1}{2}s(2 - \sqrt{3})$

Answer: E

Solution

All answers are proportional to s , so for ease, let $s = 1$.

Let $ABCD$ be oriented so that $A(0,0)$, $B(1,0)$, $C(1,1)$, $D(0,1)$.

The circle centered at A with radius 1 is $x^2 + y^2 = 1$. The circle centered at B with radius 1 is $(x - 1)^2 + y^2 = 1$. Solving each equation for $1 - y^2$ to find the intersection leads to $(x - 1)^2 = x^2$, which leads to $x = \frac{1}{2}$.

Plugging that back in to $x^2 + y^2 = 1$ leads to $y^2 = 1 - \frac{1}{4}$, or $y = \pm \frac{\sqrt{3}}{2}$. Since we want the intersection within the square where $0 < x, y < 1$, we take the positive solution, and the intersection is at $X(\frac{1}{2}, \frac{\sqrt{3}}{2})$.

The distance from X to side CD , which lies along the line $y = 1$, is $1 - \frac{\sqrt{3}}{2}$, or $\frac{2 - \sqrt{3}}{2}$. This is answer **E** when $s = 1$.

8. (AHSME 1970) If $a = \log_8 225$ and $b = \log_2 15$, then

(A) $a = \frac{1}{2}b$ (B) $a = \frac{2b}{3}$ (C) $a = b$ (D) $b = \frac{1}{2}a$ (E) $a = \frac{3b}{2}$

Answer: B

Solution

The solutions imply that finding the ratio $\frac{a}{b}$ will solve the problem. We compute $\frac{a}{b}$, use change-of-base to a neutral base, rearrange the terms, and then use the reverse-change-of-base:

$$\begin{aligned} & \frac{\log_8 225}{\log_2 15} \\ & \frac{\frac{\ln 225}{\ln 8}}{\frac{\ln 15}{\ln 2}} \\ & \frac{\ln 225 \ln 2}{\ln 15 \ln 8} \\ & \frac{\ln 225}{\ln 15} \cdot \frac{\ln 2}{\ln 8} \\ & \ln_{15} 225 \cdot \ln_8 2 \end{aligned}$$

Since $15^2 = 225$, the first logarithm is 2. Since $8^{\frac{1}{3}} = 2$, the second logarithm is $\frac{1}{3}$.

Thus, we have $\frac{a}{b} = \frac{2}{3}$, or $a = \frac{2}{3}b$, which is option **[B]**.

9. (AHSME 1970) Points P and Q are on line segment AB , and both points are on the same side of the midpoint of AB . Point P divides AB in the ratio $2 : 3$ and Q divides AB in the ratio $3 : 4$. If $PQ = 2$, then the length of segment AB is

(A) 12 (B) 28 (C) 70 (D) 75 (E) 105

Answer: C

Solution

In order, the points from left to right are A, P, Q, B . Let the lengths between successive points be $x, 2, y$, respectively.

Since $\frac{AP}{PB} = \frac{2}{3}$, we have $\frac{x}{2+y} = \frac{2}{3}$.

Since $\frac{AQ}{QB} = \frac{3}{4}$, we have $\frac{x+2}{y} = \frac{3}{4}$.

The first equation gives $x = \frac{2}{3}(2+y)$. Plugging that in to the second equation gives:

$$\frac{\frac{2}{3}(2+y)+2}{y} = \frac{3}{4}.$$

$$\frac{2}{3}(2+y) + 2 = \frac{3}{4}y$$

$$\frac{4}{3} + \frac{2}{3}y + 2 = \frac{3}{4}y$$

$$\frac{10}{3} = \frac{1}{12}y$$

$$y = 40$$

Since $x = \frac{2}{3}(2+y)$, we have $x = 28$. The length of the entire segment is $x + 2 + y$, which is 70, or option **[C]**.

10. (AHSME 1970) Let $F = .48181 \dots$ be an infinite repeating decimal with the digits 8 and 1 repeating. When F is written as a fraction in lowest terms, the denominator exceeds the numerator by

(A) 13 (B) 14 (C) 29 (D) 57 (E) 126

Answer: D

Solution

Multiplying by 100 gives $100F = 48.181818\dots$. Subtracting the first equation from the second gives $99F = 47.7$, and all the other repeating parts cancel out. This gives $F = \frac{47.7}{99} = \frac{477}{990} = \frac{159}{330} = \frac{53}{110}$. Subtracting the numerator from the denominator gives **[D]**.

11. (AHSME 1970) If two factors of $2x^3 - hx + k$ are $x + 2$ and $x - 1$, the value of $|2h - 3k|$ is

(A) 4 (B) 3 (C) 2 (D) 1 (E) 0

Answer: E

Solution

From the Remainder Theorem, we have $2(-2)^3 - h(-2) + k = 0$ and $2(1)^3 - h(1) + k = 0$. Simplifying both of those equations gives $-16 + 2h + k = 0$ and $2 - h + k = 0$. Since $k = 16 - 2h$ and $k = h - 2$, we set those equal to get:

$$16 - 2h = h - 2$$

$$3h = 18$$

$$h = 6$$

This gives $k = 4$ when substituting back into either of the two equations, and $|2h - 3k| = 0$, which is answer E.

12. (AHSME 1970) A circle with radius r is tangent to sides AB , AD , and CD of rectangle $ABCD$ and passes through the midpoint of diagonal AC . The area of the rectangle in terms of r , is

(A) $4r^2$ (B) $6r^2$ (C) $8r^2$ (D) $12r^2$ (E) $20r^2$

Answer: C

Solution

C

13. (AHSME 1970) Given the binary operation $*$ defined by $a * b = a^b$ for all positive numbers a and b . The for all positive a, b, c, n , we have

(A) $a * b = b * a$ (B) $a * (b * c) = (a * b) * c$
 (C) $(a * b^n) = (a * n) * b$ (D) $(a * b)^n = a * (bn)$ (E) None of these

Answer: D

Solution

Let $a = 2, b = 3, c = n = 4$. If all of them are false, the answer must be E. If one does not fail, we will try to prove it.

For option A, we have $2^3 = 3^2$, which is clearly false.

For option B, we have $2^{8^1} = 8^4$, which is false.

For option C, we have $2^{8^1} = 16^3$, which is false.

For option D, we have $8^4 = 2^{12}$, which is true.

The LHS is $(a^b)^n$. By the elementary definition of exponentiation, this is a^b multiplied by itself n times. Since each a^b is actually a multiplied b times, the expression $(a^b)^n$ is a multiplied by itself bn times.

The RHS is a^{bn} . This is a multiplied by itself bn times.

Thus, the LHS is always equal to the RHS, so D is the only correct statement.

14. (AHSME 1970) Consider $x^2 + px + q = 0$ where p and q are positive numbers. If the roots of this equation differ by 1, then p equals

(A) $\sqrt{4q+1}$ (B) $q-1$ (C) $-\sqrt{4q+1}$
 (D) $q+1$ (E) $\sqrt{4q-1}$

Answer: A

Solution

From the quadratic equation, the two roots of the equation are $\frac{-p \pm \sqrt{p^2 - 4q}}{2}$. The positive difference between these roots is $\sqrt{p^2 - 4q}$. Setting $\sqrt{p^2 - 4q} = 1$ and isolating p gives $\sqrt{4q + 1}$, or choice **(A)**.

15. (AHSME 1970) Lines in the xy -plane are drawn through the point $(3, 4)$ and the trisection points of the line segment joining the points $(-4, 5)$ and $(5, -1)$. One of these lines has the equation

(A) $3x - 2y - 1 = 0$ (B) $4x - 5y + 8 = 0$ (C) $5x + 2y - 23 = 0$
 (D) $x + 7y - 31 = 0$ (E) $x - 4y + 13 = 0$

Answer: E

Solution

The trisection points of $(-4, 5)$ and $(5, -1)$ can be found by trisecting the x -coordinates and the y -coordinates separately. The difference of the x -coordinates is 9, so the trisection points happen at $-4 + \frac{9}{3}$ and $-4 + \frac{9}{3} + \frac{9}{3}$, which are -1 and 2 . Similarly, the y -coordinates have a difference of 6, so the trisections happen at 3 and 1. So, the two points are $(-1, 3)$ and $(2, 1)$.

We now check which line has both $(3, 4)$ and one of the two trisection points on it. Plugging in $(x, y) = (3, 4)$ into all five of the equations works. The point $(2, 1)$ doesn't work in any of the five lines. However, $(-1, 3)$ works in line E. Thus, the answer is **(E)**.

16. (AHSME 1970) If $F(n)$ is a function such that $F(1) = F(2) = F(3) = 1$, and such that $F(n+1) = \frac{F(n) \cdot F(n-1) + 1}{F(n-2)}$ for $n \geq 3$, then $F(6)$ is equal to

(A) 2 (B) 3 (C) 7 (D) 11 (E) 26

Answer: C

Solution

Plugging in $n = 3$ gives $F(4) = \frac{F(3) \cdot F(2) + 1}{F(1)} = \frac{1 \cdot 1 + 1}{1} = 2$.

Plugging in $n = 4$ gives $F(5) = \frac{F(4) \cdot F(3) + 1}{F(2)} = \frac{2 \cdot 1 + 1}{1} = 3$.

Plugging in $n = 5$ gives $F(6) = \frac{F(5) \cdot F(4) + 1}{F(3)} = \frac{3 \cdot 2 + 1}{1} = 7$.

Thus, the answer is **(C)**.

Sidenote All the numbers in the sequence $F(n)$ are integers. In fact, the function F satisfies $F(n) = 4F(n-2) - F(n-4)$. (Prove it!).

17. (AHSME 1970) If $r \geq 0$, then for all p and q such that $pq \neq 0$ and $pr > qr$, we have

(A) $-p > -q$ (B) $-p > q$ (C) $1 > -q/p$
 (D) $1 < q/p$ (E) None of These

Answer: E

Solution

If $r > 0$ and $pr > qr$, we can divide by the positive number r and not change the inequality direction to get $p > q$. Multiplying by -1 (and flipping the inequality

sign because we're multiplying by a negative number) leads to $-p < -q$, which directly contradicts A . Thus, A is always false.

If $p < 0$ (which is possible but not guaranteed), we can divide the true statement $p > q$ by p to get $1 > \frac{q}{p}$. This contradicts D . Thus, D is sometimes false, which is bad enough to be eliminated.

If $(p, q, r) = (3, 2, 1)$, then the condition that $pr > qr$ is satisfied. However, $-p = -3$ and $q = 2$, so $-p > q$ is false for at least this case, eliminating B .

If $(p, q, r) = (2, -3, 1)$, then $pr > qr$ is also satisfied. However, $-\frac{q}{p} = 1.5$, so $1 > -\frac{q}{p}$ is false, eliminating C .

All four options do not follow from the premises, leading to $\boxed{E}$ as the correct answer.

18. (AHSME 1970) $\sqrt{3+2\sqrt{2}} - \sqrt{3-2\sqrt{2}}$ is equal to
 (A) 2 (B) $2\sqrt{3}$ (C) $4\sqrt{2}$ (D) $\sqrt{6}$ (E) $2\sqrt{2}$

Answer: A

Solution

Square the expression:

$$\begin{aligned} (\sqrt{3+2\sqrt{2}} - \sqrt{3-2\sqrt{2}})^2 &= 3 + \sqrt{2} - 2\sqrt{(3+\sqrt{2})(3-\sqrt{2})} + 3 - 2\sqrt{2} = 6 - 2\sqrt{9-8} = \\ &= 6 - 2\sqrt{1} = 4 \\ \Rightarrow \sqrt{3+2\sqrt{2}} - \sqrt{3-2\sqrt{2}} &= \sqrt{4} = 2 \Rightarrow \boxed{A} \end{aligned}$$

19. (AHSME 1970) The sum of an infinite geometric series with common ratio r such that $|r| < 1$, is 15, and the sum of the squares of the terms of this series is 45. The first term of the series is
 (A) 12 (B) 10 (C) 5 (D) 3 (E) 2

Answer: C

Solution

We know that the formula for the sum of an infinite geometric series is $S = \frac{a}{1-r}$.

So we can apply this to the conditions given by the problem.

We have two equations:

$$\begin{aligned} 15 &= \frac{a}{1-r} \\ 45 &= \frac{a^2}{1-r^2} \end{aligned}$$

We get

$$\begin{aligned} a &= 15 - 15r \\ a^2 &= 45 - 45r^2 \end{aligned}$$

$$(15 - 15r)^2 = 45 - 45r^2$$

$$270r^2 - 450r + 180 = 0$$

$$3r^2 - 5r + 2 = 0$$

$$(3r - 2)(r - 1) = 0$$

Since $|r| < 1$, $r = \frac{2}{3}$, so plug this into the equation above and we get $a = 15 - 15r = 15 - 10 = \boxed{\text{(C)} \ 5}$

Solution by **Invoker**

20. (AHSME 1970) Lines HK and BC lie in a plane. M is the midpoint of line segment BC , and BH and CK are perpendicular to HK . Then we

- (A) always have $MH = MK$
 (B) always have $MH > BK$
 (C) sometimes have $MH = MK$ but not always
 (D) always have $MH > MB$
 (E) always have $BH < BC$

Answer: A

Solution

☐ A

21. (AHSME 1970) On an auto trip, the distance read from the instrument panel was 450 miles. With snow tires on for the return trip over the same route, the reading was 440 miles. Find, to the nearest hundredth of an inch, the increase in radius of the wheels if the original radius was 15 inches.

- (A) .33 (B) .34 (C) .35 (D) .38 (E) .66

Answer: B

Solution

☐ B

22. (AHSME 1970) If the sum of the first $3n$ positive integers is 150 more than the sum of the first n positive integers, then the sum of the first $4n$ positive integers is

- (A) 300 (B) 350 (C) 400 (D) 450 (E) 600

Answer: A

Solution 1

We can setup our first equation as

$$\frac{3n(3n+1)}{2} = \frac{n(n+1)}{2} + 150$$

Simplifying we get

$$9n^2 + 3n = n^2 + n + 300 \Rightarrow 8n^2 + 2n - 300 = 0 \Rightarrow 4n^2 + n - 150 = 0$$

So our roots using the quadratic formula are

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \Rightarrow \frac{-1 \pm \sqrt{1^2 - 4 \cdot (-150) \cdot 4}}{2 \cdot 4} \Rightarrow \frac{-1 \pm \sqrt{1 + 2400}}{8} \Rightarrow 6, -25/4$$

Since the question said positive integers, $n = 6$, so $4n = 24$

$$\frac{24 \cdot 25}{2} = 300$$

☐ A

Solution 2

Expressing as an equation:

$$\frac{3n(3n+1)}{2} = \frac{n(n+1)}{2} + 150. \quad (1)$$

The sum of the first $4n$ positive integers =

$$\frac{4n(4n+1)}{2} \quad (2)$$

We will try to rearrange Equation (1) to give equation (2)

$$\begin{aligned} \frac{3n(3n+1)}{2} - \frac{n(n+1)}{2} &= 150 \\ = \frac{n(3(3n+1)-(n+1))}{2} &= 150 = \frac{n(9n+3-n-1)}{2} \\ \frac{n(8n+2)}{2} &= \frac{2n(4n+1)}{2} = 150 \\ \frac{4n(4n+1)}{2} &= 2 * 150 = 300 \end{aligned}$$

300 is the answer

A

~Melkor

23. (*AHSME 1970*) The number $10!$ (10 is written in base 10), when written in the base 12 system, ends in exactly k zeroes. The value of k is

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Answer: D

Solution

A number in base b that ends in exactly k zeros will be divisible by b^k , but not by b^{k+1} . Thus, we want to find the highest k for which $12^k | 10!$.

There are 4 factors of 3: 3, 6, 9, and an extra factor from 9.

There are 8 factors of 2: 2, 4, 6, 8, 10, an extra factor from 4, 8, and a third extra factor from 8.

So, $2^8 \cdot 3^4 = 4^4 \cdot 3^4 = 12^4$ will divide $10!$. Thus, the answer is **D**.

24. (*AHSME 1970*) An equilateral triangle and a regular hexagon have equal perimeters. If the area of the triangle is 2, then the area of the hexagon is

(A) 2 (B) 3 (C) 4 (D) 6 (E) 12

Answer: B

Solution 1

Let $ABCDEF$ be our regular hexagon, with centre O - and join AO, BO, CO, DO, EO , and FO . Note that we form six equilateral triangles with sidelength $\frac{s}{2}$, where s is the sidelength of the triangle (since the perimeter of the two polygons are equal). If the area of the original equilateral triangle is 2, then the area of each of the six smaller triangles is $1/4 \times 2 = \frac{1}{2}$ (by similarity area ratios).

Thus, the area of the hexagon is $6 \times \frac{1}{2} = 3$, hence our answer is **B**

25. (AHSME 1970) For every real number x , let $[x]$ be the greatest integer less than or equal to x . If the postal rate for first class mail is six cents for every ounce or portion thereof, then the cost in cents of first-class postage on a letter weighing W ounces is always

(A) $6W$ (B) $6[W]$ (C) $6([W] - 1)$ (D) $6([W] + 1)$ (E) $-6[-W]$

Answer: E

Solution

This question is trying to convert the floor function, which is more commonly notated as $[x]$, into the ceiling function, which is $\lceil x \rceil$. The identity is $\lceil x \rceil = -\lfloor -x \rfloor$, which can be verified graphically, or proven using the definition of floor and ceiling functions.

However, for this problem, some test values will eliminate answers. If $W = 2.5$ ounces, the cost will be 18 cents. Plugging in $W = 2.5$ into the five options gives answers of 15, 12, 6, 18, 18. This leaves options D and E as viable. If $W = 2$ ounces, the cost is 12 cents. Option D remains 18 cents, while option E gives 12 cents, the correct answer. Thus, the answer is E.

26. (AHSME 1970) The number of distinct points in the xy -plane common to the graphs of $(x + y - 5)(2x - 3y + 5) = 0$ and $(x - y + 1)(3x + 2y - 12) = 0$ is

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Answer: B

Solution 1

The graph $(x + y - 5)(2x - 3y + 5) = 0$ is the combined graphs of $x + y - 5 = 0$ and $2x - 3y + 5 = 0$. Likewise, the graph $(x - y + 1)(3x + 2y - 12) = 0$ is the combined graphs of $x - y + 1 = 0$ and $3x + 2y - 12 = 0$. All these lines intersect at one point, $(2, 3)$. Therefore, the answer is (B) 1.

Solution 2

We need to satisfy both $(x + y - 5)(2x - 3y + 5) = 0$ and $(x - y + 1)(3x + 2y - 12) = 0$. In order to do this, let us look at the first equation. Either $x + y = 5$, or $2x - 3y = -5$. For the second equation, either $x - y = -1$, or $3x + 2y = 12$. Thus, we need to solve 4 systems of equations: $x + y = 5$ and $x - y = -1$, $x + y = 5$ and $3x + 2y = 12$, $2x - 3y = -5$ and $x - y = -1$, and finally $2x - 3y = -5$ and $3x + 2y = 12$. Solving all of these systems of equations is pretty trivial, and all of them come out to be $(2, 3)$. Thus, they only intersect at 1 point, and our answer is (B) 1.

27. (AHSME 1970) In a triangle, the area is numerically equal to the perimeter. What is the radius of the inscribed circle?

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Answer: A

Solution

One of the most common formulas involving the inradius of a triangle is $A = rs$, where A is the area of the triangle, r is the inradius, and s is the semiperimeter.

The problem states that $A = p = 2s$. This means $2s = rs$, or $r = 2$, which is option A.

28. (AHSME 1970) In triangle ABC , the median from vertex A is perpendicular to the median from vertex B . If the lengths of sides AC and BC are 6 and 7 respectively, then the length of side AB is

(A) $\sqrt{17}$ (B) 4 (C) $4\frac{1}{2}$ (D) $2\sqrt{5}$ (E) $4\frac{1}{4}$

Answer: A

Solution

[A] let the midpoint be M,N (i.e. AM,BN are the medians); connecting MN we know that $AB = 2x$ and $MN = x$ hence apply Stewart's theorem in triangle ABC with median MN first and then apply Stewart's in triangle BNC with median MN

29. (AHSME 1970) It is now between 10 : 00 and 11 : 00 o'clock, and six minutes from now, the minute hand of the watch will be exactly opposite the place where the hour hand was three minutes ago. What is the exact time now?

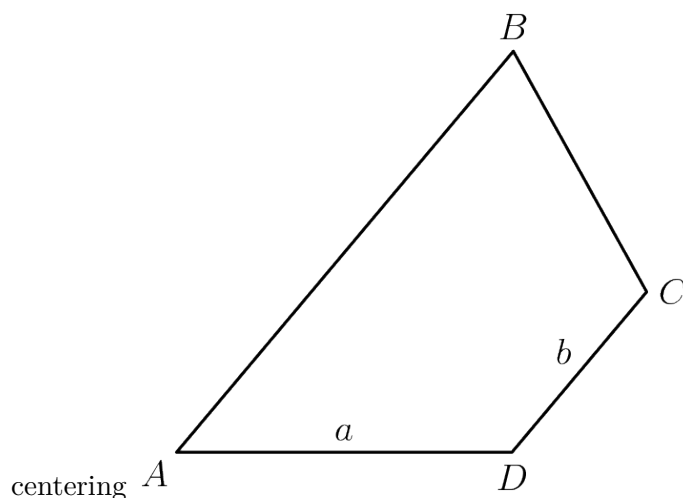
(A) $10 : 05\frac{5}{11}$ (B) $10 : 07\frac{1}{2}$ (C) 10 : 10
(D) 10 : 15 (E) $10 : 17\frac{1}{2}$

Answer: D

Solution

[D] $50 + (m-3)/12 = 30 + (m+6) = 10\text{hours} + m = 10: 15$

30. (AHSME 1970)



In the accompanying figure, segments AB and CD are parallel, the measure of $\angle D$ is twice the measure of $\angle B$, and the measures of segments AD and CD are a and b respectively. Then the measure of AB is equal to

(A) $\frac{1}{2}a + 2b$ (B) $\frac{3}{2}b + \frac{3}{4}a$ (C) $2a - b$ (D) $4b - \frac{1}{2}a$ (E) $a + b$

Answer: E

Solution

With reference to the diagram above, let E be the point on AB such that $DE \parallel BC$. Let $\angle ABC = \alpha$. We then have $\alpha = \angle AED = \angle EDC$ since $AB \parallel CD$, so $\angle ADE = \angle ADC - \angle BDC = 2\alpha - \alpha = \alpha$, which means $\triangle AED$ is isosceles.

Therefore, $AB = AE + EB = a + b$, hence our answer is [E].

31. (*AHSME 1970*) If a number is selected at random from the set of all five-digit numbers in which the sum of the digits is equal to 43, what is the probability that this number is divisible by 11?

(A) $2/5$ (B) $1/5$ (C) $1/6$ (D) $1/11$ (E) $1/15$

Answer: B

Solution

For the sums of the digits to be equal to 43, this means that the number's digits must fall under one of two categories. It either has 4 9's and a 7, or 3 9's and 2 8's. You can use casework from there to find that there are 15 5-digit numbers in which the sum of the digits is equal to 43. To find the number of ways that the number is divisible by 11, use the divisibility rule for 11, letting the number be represented by $abcde$. This gives 3 possible numbers that are divisible by 11, so the answer is $\frac{3}{15} = \frac{1}{5} = \boxed{\text{B}}$

32. (*AHSME 1970*) A and B travel around a circular track at uniform speeds in opposite directions, starting from diametrically opposite points. If they start at the same time, meet first after B has travelled 100 yards, and meet a second time 60 yards before A completes one lap, then the circumference of the track in yards is

(A) 400 (B) 440 (C) 480 (D) 560 (E) 880

Answer: C

Solution

Let x be half the circumference of the track. They first meet after B has run 100 yards, meaning that in the time B has run 100 yards, A has run $x - 100$ yards. The second time they meet is when A is 60 yards before he completes the lap. This means that in the time that A has run $2x - 60$ yards, B has run $x + 60$ yards. Because they run at uniform speeds, we can write the equation

$$\frac{100}{x - 100} = \frac{x + 60}{2x - 60}.$$

Cross multiplying,

$$200x - 6000 = x^2 - 40x - 6000$$

Adding 6000 to both sides and simplifying, we have

$$200x = x^2 - 40x$$

$$240x = x^2$$

$$x = 240.$$

Because x is only half of the circumference of the track, the answer we are looking for is $2 \cdot 240 = 480$, or $\boxed{\text{C}}$.

33. (*AHSME 1970*) Find the sum of the digits of all numerals in the sequence 1, 2, 3, 4, $\dots$, 10000.

(A) 180,001 (B) 154,756 (C) 45,001 (D) 154,755 (E) 270,001

Answer: A

Solution 1

We can find the sum using the following method. We break it down into cases. The first case is the numbers 1 to 9. The second case is the numbers 10 to 99. The

third case is the numbers 100 to 999. The fourth case is the numbers 1,000 to 9,999. And lastly, the sum of the digits in 10,000. The first case is just the sum of the numbers 1 to 9 which is, using $\frac{n(n+1)}{2}$, 45. In the second case, every number 1 to 9 is used 19 times. 10 times in the tens place, and 9 times in the ones place. So the sum is just 19(45). Similarly, in the third case, every number 1 to 9 is used 100 times in the hundreds place, 90 times in the tens place, and 90 times in the ones place, for a total sum of 280(45). By the same method, every number 1 to 9 is used 1,000 times in the thousands place, 900 times in the hundreds place, 900 times in the tens place, and 900 times in the ones place, for a total of 3700(45). Thus, our final sum is $45 + 19(45) + 280(45) + 3700(45) + 1 = 4000(45) + 1 = \boxed{\text{A)}180001}$.

Solution 2

Consider the numbers from 0000 – 9999. We have 40000 digits and each has equal an probability of being 0, 1, 2,...9. Thus, our desired sum is $4000 \left(\frac{9 \cdot 10}{2} \right) + 1 = 4000(45) + 1 = \boxed{\text{A)}180001}$.

Credit: Math1331Math

Solution 3

As in Solution 2, we consider the four digit numbers from 0000 – 9999. We see that we have $10000 \times 4 = 40000$ digits with each digit appearing equally.

Thus, the digit sum will be the average of the digits multiplied by 40000. The digit average comes out to be $\frac{0+9}{2}$, since all digits are consecutive.

So, our answer will be $\frac{9}{2} \times 40000 = 180000$. However, since we purposely did not include 10000, we add one to get our final answer as $\boxed{\text{(A)}180001}$.

Solution by davidaops.

Credit to Math1331Math

34. (AHSME 1970) The greatest integer that will divide 13511, 13903, and 14589 and leave the same remainder is
- (A) 28 (B) 49 (C) 98
 (D) an odd multiple of 7 greater than 49
 (E) an even multiple of 7 greater than 98

Answer: C

Solution

We know that 13903 minus 13511 is equivalent to 392. Additionally, 14589 minus 13903 is equivalent to 686. Since we are searching for the greatest integer that divides these three integers and leaves the same remainder, the answer resides in the greatest common factor of 686 and 392. Therefore, the answer is 98, or $\boxed{\text{C}}$

"Credit: Skuppi3"

35. (AHSME 1970) A retiring employee receives an annual pension proportional to the square root of the number of years of his service. Had he served a years more, his pension would have been p dollars greater, whereas, had he served b years more $b \neq a$, his pension would have been q dollars greater than the original annual pension. Find his annual pension in terms of a, b, p , and q .

- (A) $\frac{p^2 - q^2}{2(a - b)}$ (B) $\frac{(p - q)^2}{2\sqrt{ab}}$ (C) $\frac{ap^2 - bq^2}{2(ap - bq)}$ (D) $\frac{aq^2 - bp^2}{2(bp - aq)}$ (E) $\sqrt{(a - b)(p - q)}$

Answer: D

Solution

Note the original pension as $k\sqrt{x}$, where x is the number of years served. Then, based on the problem statement, two equations can be set up.

$$\begin{aligned}k\sqrt{x+a} &= k\sqrt{x} + p \\k\sqrt{x+b} &= k\sqrt{x} + q\end{aligned}$$

Square the first equation to get

$$k^2x + ak^2 = k^2x + 2pk\sqrt{x} + p^2.$$

Because both sides have

$$k^2x$$

, they cancel out. Similarly, the second equation will become $bk^2 = q^2 + 2qk\sqrt{x}$. Then, a can be multiplied to the second equation and b can be multiplied to the first equation so that the left side of both equations becomes abk^2 . Finally, by setting the equations equal to each other,

$$bp^2 + 2bpk\sqrt{x} = aq^2 + 2aqk\sqrt{x}.$$

Isolate $k\sqrt{x}$ to get $\boxed{\text{D}} = \frac{aq^2 - bp^2}{2(bp - aq)}$.

§7 AHSME 1971

1. (AHSME 1971) The number of digits in the number $N = 2^{12} \times 5^8$ is
(A) 9 (B) 10 (C) 11 (D) 12 (E) 20

Answer: B

Solution

$N = 2^{12} \cdot 5^8 = 2^4 \cdot 2^8 \cdot 5^8 = 2^4 \cdot (2 \cdot 5)^8 = 2^4 \cdot 10^8 = 16 \cdot 100000000$, which has $2 + 8 = 10$ digits, so the answer is $\boxed{\text{(B) } 10}$.

2. (AHSME 1971) If b men take c days to lay f bricks, then the number of days it will take c men working at the same rate to lay b bricks, is
(A) fb^2 (B) b/f^2 (C) f^2/b (D) b^2/f (E) f/b^2

Answer: D

Solution

We can use a modified version of the equation Distance = Rate $\times$ Time, which is Work Done = Rate of Work $\times$ Time Worked. In this case, the work done is the number of bricks laid, the rate of work is the number of men working, and the time worked is the number of days. With these definitions, we see from the first equation that $f = b \cdot c$. If we let X be the number of days it will take c men working at the same rate to lay b bricks, then we have the equation $b = c \cdot x$. So, $x = \frac{b}{c}$. The first equation says that $\frac{f}{c^2} = \frac{b}{c}$, which leads to $x = \frac{f}{c^2}$. This doesn't match any of our answer choices though, so we have to fiddle around a bit before we realize that $c = \frac{f}{b}$, a substitution we can make to see that $x = \frac{b^2}{f}$. Thus, the answer is

$\boxed{\text{(D) } b^2/f}$.

3. (AHSME 1971) If the point $(x, -4)$ lies on the straight line joining the points $(0, 8)$ and $(-4, 0)$ in the xy -plane, then x is equal to

(A) -2 (B) 2 (C) -8 (D) 6 (E) -6

Answer: E

Solution

Since $(x, -4)$ is on the line $(0, 8)$ and $(-4, 0)$, the slope between the latter two is the same as the slope between the former two, so $\frac{-4-8}{x-0} = \frac{8-0}{0-(-4)}$. Solving for x , we get $x = -6$, so the answer is **(E) -6** .

4. (AHSME 1971) After simple interest for two months at 5% per annum was credited, a Boy Scout Troop had a total of \$255.31 in the Council Treasury. The interest credited was a number of dollars plus the following number of cents

(A) 11 (B) 12 (C) 13 (D) 21 (E) 31

Answer: A

Solution

We can use the formula $A = P(1 + rt)$ for simple interest, where P is the principal (initial amount), A is the total, r is the interest rate per year, and t is the time in years.

We are solving for P , so we can rearrange the equation:

$$P = \frac{A}{1 + rt}$$

Plug in all the necessary numbers:

$$P = \frac{255.31}{1 + (0.05)(2/12)} = \$253.21$$

So, the interest is

$$P * (0.05)(2/12) = \$2.11$$

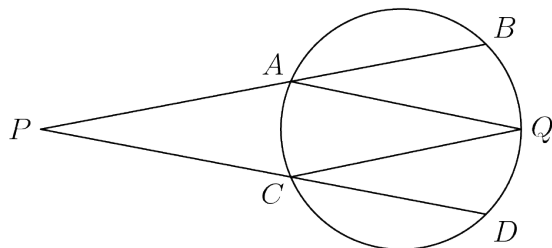
The answer is **(A) 11**

-edited by Happypuma46

If there are any mistakes, feel free to edit so that it is correct.

5. (AHSME 1971) Points A, B, Q, D , and C lie on the circle shown and the measures of arcs $\widehat{BQ}$ and $\widehat{QD}$ are 42° and 38° respectively. The sum of the measures of angles P and Q is

(A) 80° (B) 62° (C) 40° (D) 46° (E) None of these



centering

Answer: C

Solution 1

We see that the measure of P equals $(\widehat{BD} - \widehat{AC})/2$, and that the measure of Q equals $\widehat{AC}/2$. Since $\widehat{BD} = \widehat{BQ} + \widehat{QD} = 42^\circ + 38^\circ = 80^\circ$, the sum of the measures of P and Q is $\widehat{BD}/2 = 80^\circ/2 = 40^\circ \implies \boxed{\text{(C)} 40^\circ}$.

Solution 2

Arcs are measured by the angle measures of their corresponding **central angles**. Thus, the **inscribed angle** $\angle BAQ = \frac{42^\circ}{2} = 21^\circ$, and, likewise, $\angle QCD = \frac{38^\circ}{2} = 19^\circ$. Thus, by **supplementary angles**, $\angle PAQ = 180^\circ - \angle BAQ = 159^\circ$, and $\angle PCQ = 180^\circ - \angle QCD = 161^\circ$. Because the sum of the **interior angle** measures of a quadrilateral add to 360° , we see that $\angle P + \angle Q = 360^\circ - \angle PAQ - \angle PCQ = 360^\circ - 159^\circ - 161^\circ = 40^\circ$. Thus, our answer is $\boxed{\text{(C)} 40^\circ}$.

6. (AHSME 1971) Let $*$ be the symbol denoting the binary operation on the set S of all non-zero real numbers as follows: For any two numbers a and b of S , $a * b = 2ab$. Then the one of the following statements which is not true, is

- (A) $*$ is commutative over S (B) $*$ is associative over S
 (C) $\frac{1}{2}$ is an identity element for $*$ in S (D) Every element of S has an inverse for $*$
 (E) $\frac{1}{2a}$ is an inverse for $*$ of the element a of S

Answer: E

Solution

- (A) $*$ is commutative over S

$$a * b = b * a = 2ab$$

Statement A is true.

- (B) $*$ is associative over S

$$a * (b * c) = a * 2bc = 2a2bc = 4abc$$

$$(a * b) * c = 2ab * c = 2(2ab)(c) = 4abc$$

Statement B is true.

- (C) $\frac{1}{2}$ is an identity element for $*$ in S

$$a * \frac{1}{2} = a$$

Statement C is true.

- (D) From the previous answer choice, we know $\frac{1}{2}$ is the identity element for $*$ in S . For inverses . Thus, $a * b = \frac{1}{2}$, which leads to $b = \frac{1}{4a}$

Statement D is true.

- (E) $\frac{1}{2a}$ is an inverse for $*$ of the element a of S

$$\frac{1}{2a} * a = 1$$

Since the identity is $\frac{1}{2}$, not 1, we can see that statement E is false.

Hence, the answer is $\boxed{\text{(E)}}$.

-edited by coolmath34-fixed by DoctorSeventeen

7. (AHSME 1971) $2^{-(2k+1)} - 2^{-(2k-1)} + 2^{-2k}$ is equal to

(A) 2^{-2k} (B) $2^{-(2k-1)}$ (C) $-2^{-(2k+1)}$ (D) 0 (E) 2

Answer: C

Solution

By using the properties of **exponents**, we can simplify the given expression as follows to obtain our answer:

$$\begin{aligned}
 2^{-(2k+1)} - 2^{-(2k-1)} + 2^{-2k} &= 2^{-2k-1} - 2^{-2k+1} + 2^{-2k} \\
 &= \frac{2^{-2k}}{2} - 2 \cdot 2^{-2k} + 2^{-2k} \\
 &= 2^{-2k} \left(\frac{1}{2} - 2 + 1 \right) \\
 &= 2^{-2k} \left(-\frac{1}{2} \right) \\
 &= -\frac{2^{-2k}}{2} \\
 &= -2^{-2k-1} \\
 &= \boxed{\text{(C)} - 2^{-(2k+1)}}.
 \end{aligned}$$

8. (AHSME 1971) The solution set of $6x^2 + 5x < 4$ is the set of all values of x such that

(A) $-2 < x < 1$ (B) $-\frac{4}{3} < x < \frac{1}{2}$ (C) $-\frac{1}{2} < x < \frac{4}{3}$
 (D) $x < \frac{1}{2}$ or $x > -\frac{4}{3}$ (E) $x < -\frac{4}{3}$ or $x > \frac{1}{2}$

Answer: B

Solution

We are solving the inequality $6x^2 + 5x - 4 < 0$. This can be factored as

$$(2x - 1)(3x + 4) < 0$$

The graph of this inequality is a parabola facing upwards, so the **interval** between the roots satisfies the equation. This interval, $(-\frac{4}{3}, \frac{1}{2})$, is answer **(B)**.

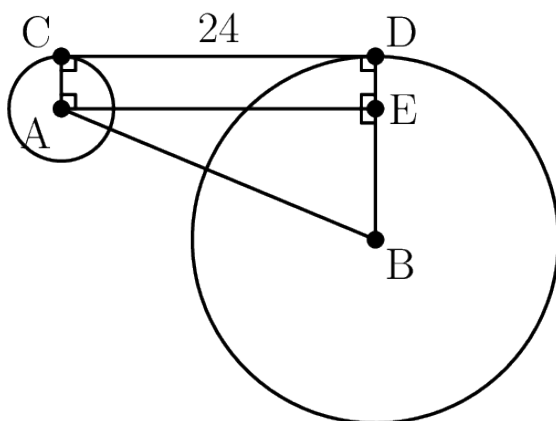
-edited by coolmath34

9. (AHSME 1971) An uncrossed belt is fitted without slack around two circular pulleys with radii of 14 inches and 4 inches. If the distance between the points of contact of the belt with the pulleys is 24 inches, then the distance between the centers of the pulleys in inches is

(A) 24 (B) $2\sqrt{119}$ (C) 25 (D) 26 (E) $4\sqrt{35}$

Answer: D

Solution



centering

Let the center of the smaller circle be A and the center of the larger circle be B . Draw $\overline{AC}$ and $\overline{BD}$, as in the diagram. We know that $\overline{AC}$ and $\overline{BD}$ are perpendicular to $\overline{CD}$ because $\overline{CD}$ is **tangent** to both circles. From the problem, we know $CD = 24$.

Draw $\overline{AE} \perp \overline{BD}$ with E on $\overline{BD}$. $ABDC$ is a **rectangle**, so, knowing the radii of the circles, we see that $BE = BD - ED = BD - AC = 14 - 4 = 10$.

Points A , E , and B form a right triangle with legs 10 and 24. We are looking for AB , which is $\sqrt{10^2 + 24^2} = 26$ by the **Pythagorean Theorem**.

Thus, our answer is **(D) 26**.

-edited by coolmath34

10. (AHSME 1971) Each of a group of 50 girls is blonde or brunette and is blue eyed or brown eyed. If 14 are blue-eyed blondes, 31 are brunettes, and 18 are brown-eyed, then the number of brown-eyed brunettes is
(A) 5 **(B) 7** **(C) 9** **(D) 11** **(E) 13**

Answer: E

Solution

There are 31 brunettes, so there are $50 - 31 = 19$ blondes. We know there are 14 blue-eyed blondes, so there are $19 - 14 = 5$ brown-eyed blondes.

Next, we know that 18 people are brown-eyed, so there are $18 - 5 = 13$ brown-eyed brunettes.

The answer is **(E) 13**.

-edited by coolmath34

11. (AHSME 1971) The numeral 47 in base a represents the same number as 74 in base b . Assuming that both bases are positive integers, the least possible value of $a + b$ written as a Roman numeral, is
(A) XIII **(B) XV** **(C) XXI** **(D) XXIV** **(E) XVI**

Answer: D

Solution

$4a + 7 = 7b + 4 \implies 7b = 4a + 3 \implies b = \frac{4a+3}{7}$. $4a + 3 \equiv 0 \implies 4a \equiv 4 \implies a \equiv 1 \pmod{7}$. The smallest possible value of a is 8. Then, $b = \frac{4 \cdot 8 + 3}{7} = \frac{35}{7} = 5$. However, the digit 7 is not valid in base 5, so we have to try a larger value. $a = 8 + 7 = 15$, gives a value of $\frac{15 \cdot 4 + 3}{7} = 9$, for b , which is valid.

$15+9 = 24$, which is XXIV as a roman numeral, and thus the answer is **(D) XXIV**.

12. (AHSME 1971) For each integer $N > 1$, there is a mathematical system in which two or more positive integers are defined to be congruent if they leave the same non-negative remainder when divided by N . If 69, 90, and 125 are congruent in one such system, then in that same system, 81 is congruent to

(A) 3 (B) 4 (C) 5 (D) 7 (E) 8

Answer: B

Solution

The "mathematical system" being alluded to is **modulo** N , so we shall be using such notation for convenience. From the problem, we know that $69 \equiv 90 \equiv 125 \pmod{N}$, so the difference between each of these numbers must be a multiple of N . $90 - 69 = 21 = 3 \cdot 7$, and $125 - 90 = 35 = 5 \cdot 7$. The only common factors between these two numbers are 1 and 7, but we know from the problem that $N \neq 1$. Thus, $N = 7$. Now, $81 \equiv 77 + 4 \equiv 4 \pmod{7}$, so our answer is **(B) 4**.

13. (AHSME 1971) If $(1.0025)^{10}$ is evaluated correct to 5 decimal places, then the digit in the fifth decimal place is

(A) 0 (B) 1 (C) 2 (D) 5 (E) 8

Answer: E

Solution

We can rewrite $(1.0025)^{10}$ as $(1 + \frac{25}{10^4})^{10}$. By the **Binomial Theorem**, we know that this expression equals $1 + \binom{10}{1} \cdot \frac{25}{10^4} + \binom{10}{2} \cdot \frac{25^2}{10^8} + \binom{10}{3} \cdot \frac{25^3}{10^{12}} + \cdots + (\frac{25}{10^4})^{10}$. We are looking for the $\frac{1}{10^5}$ term (the fifth decimal digit), so we can disregard the first two terms, because they will not affect our final result. Thus, we are left with $\binom{10}{2} \cdot \frac{25^2}{10^8} + \binom{10}{3} \cdot \frac{25^3}{10^{12}} + \cdots = \frac{45 \cdot 25^2}{10^8} + \frac{120 \cdot 25^3}{10^{12}} + \cdots = \frac{28,125}{10^8} + \frac{1,875,000}{10^{12}} + \cdots$. The first term here has a $\frac{1}{10^5}$ term of 8, and the second term does not contribute to the fifth decimal place, because it does not have enough digits in the numerator. So, we can disregard the subsequent terms, because they will not have enough digits either. Thus, the fifth digit of the expansion is **(E) 8**.

14. (AHSME 1971) The number $(2^{48} - 1)$ is exactly divisible by two numbers between 60 and 70. These numbers are

(A) 61, 63 (B) 61, 65 (C) 63, 65 (D) 63, 67 (E) 67, 69

Answer: C

Solution

Factor repeatedly with **difference of squares**.

$$2^{48} - 1 = (2^{24} + 1)(2^{12} + 1)(2^6 + 1)(2^6 - 1)$$

We only care about two terms: $2^6 + 1$ and $2^6 - 1$. These simplify to 65 and 63.

Thus, our answer is **(C) 63, 65**.

-edited by coolmath34

15. (AHSME 1971) An aquarium on a level table has rectangular faces and is 10 inches wide and 8 inches high. When it was tilted, the water in it covered an 8×10 end

but only $\frac{3}{4}$ of the rectangular bottom. The depth of the water when the bottom was again made level, was

- (A) $2\frac{1}{2}$ " (B) 3" (C) $3\frac{1}{4}$ " (D) $3\frac{1}{2}$ " (E) 4"

Answer: B

Solution

Let the third dimension of the aquarium be x , and let the height of the water when the aquarium is level be h . When the aquarium is tilted, the water forms a triangular prism. The triangular faces have height 8 and, from the problem, base $\frac{3}{4}x$. Thus, they have area $\frac{1}{2} \cdot 8 \cdot \frac{3}{4}x = 3x$, so, because the prism has length 10, the volume of the water is $10 \cdot 3x = 30x$. When the tank is level, the water forms a rectangular prism with volume $h \cdot 10 \cdot x = 10hx$. Because the amount of water in the tank is conserved, we can equate these two expressions for the volume of the water. Thus, $10hx = 30x$, so $h = \boxed{\text{(B) } 3"}$.

16. (AHSME 1971) After finding the average of 35 scores, a student carelessly included the average with the 35 scores and found the average of these 36 numbers. The ratio of the second average to the true average was

- (A) 1 : 1 (B) 35 : 36 (C) 36 : 35 (D) 2 : 1 (E) None of these

Answer: A

Solution

Assume the 35 scores are the first 35 natural numbers: $1, 2, 3, \dots, 35$.

The average of the scores is $\frac{(35)(36)}{2} = 18$.

If we add 18 to the first 35 numbers, the new average is $\frac{648}{36} = 18$ still.

The two averages are the same, therefore the answer is $\boxed{\text{(A) } 1 : 1}$.

-edited by coolmath34

17. (AHSME 1971) A circular disk is divided by $2n$ equally spaced radii ($n > 0$) and one secant line. The maximum number of non-overlapping areas into which the disk can be divided is

- (A) $2n + 1$ (B) $2n + 2$ (C) $3n - 1$ (D) $3n$ (E) $3n + 1$

Answer: E

Solution 1

Note that, before adding the secant line, we have $2n$ regions, and, starting from the edge of the disk, every time the secant line intersects a radius or the other end of the disk, a new region is created. Thus, we desire to maximize the number of radii that the secant line passes through.

The angle between any two adjacent radii must be $\frac{360^\circ}{2n} = \frac{180^\circ}{n}$. A given secant cannot pass through more than 180° of the radii, and passing through exactly 180° would imply that the secant must go through the center of the circle, and so it would not produce any new regions. Thus, to maximize the number of regions, the secant line must pass through n radii (spanning $\frac{180}{n}(n - 1)$ degrees) and thereby create $n + 1$ new regions (one for each radius and one for exiting the disk). Thus, we have a maximum of $2n + n + 1 = \boxed{\text{(E) } 3n + 1}$ regions.

Solution 2

We can draw the cases for small values of n .

$$n = 0 \rightarrow \text{areas} = 1$$

$$n = 1 \rightarrow \text{areas} = 4$$

$$n = 2 \rightarrow \text{areas} = 7$$

$$n = 3 \rightarrow \text{areas} = 10$$

It seems that for $2n$ radii, there are $3n + 1$ distinct areas. The secant line must pass through n radii for this to occur.

The answer is **(E)** $3n + 1$.

-edited by coolmath34

18. (*AHSME 1971*) The current in a river is flowing steadily at 3 miles per hour. A motor boat which travels at a constant rate in still water goes downstream 4 miles and then returns to its starting point. The trip takes one hour, excluding the time spent in turning the boat around. The ratio of the downstream to the upstream rate is

(A) 4 : 3 (B) 3 : 2 (C) 5 : 3 (D) 2 : 1 (E) 5 : 2

Answer: D

Solution

Let the boat travel over still water at rate r . Then, it travels downstream at rate $r + 3$ and upstream at rate $r - 3$. The distance travelled downstream can be expressed as $(r + 3)t$, where t is the time taken to travel downstream. Because the total time taken is 1 hour, the distance travelled upstream can be represented as $(r - 3)(1 - t)$. Because both distances are the same (4 miles), we can equate the two expressions to solve for t in terms of r :

$$\begin{aligned}(r + 3)t &= (r - 3)(1 - t) \\ rt + 3t &= r - rt - 3 + 3t \\ 2rt &= r - 3 \\ t &= \frac{r - 3}{2r}\end{aligned}$$

Recalling that the distance travelled downstream, $(r + 3)t$, is 4 miles, we can substitute the above expression for t to solve for r :

$$\begin{aligned}(r + 3)t &= 4 \\ (r + 3)\frac{r - 3}{2r} &= 4 \\ r^2 - 9 &= 8r \\ r^2 - 8r - 9 &= 0 \\ (r - 9)(r + 1) &= 0\end{aligned}$$

Because $r > 0$, $r = 9$. Thus, the ratio of the downstream rate $r + 3$ to the upstream rate $r - 3$ is $\frac{9+3}{9-3} = \frac{12}{6} = \span style="border: 1px solid black; padding: 2px;">**(D)** 2 : 1.$

19. (AHSME 1971) If the line $y = mx + 1$ intersects the ellipse $x^2 + 4y^2 = 1$ exactly once, then the value of m^2 is

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{4}{5}$ (E) $\frac{5}{6}$

Answer: C

Solution

Plug in $y = mx + 1$ into the ellipse's equation to find the intersection points:

$$x^2 + 4(mx + 1)^2 = 1$$

After simplifying, we have a quadratic in x :

$$(4m^2 + 1)x^2 + 8mx + 3 = 0$$

Because there is only one intersection point, then the quadratic has only one solution. This can only happen when the **discriminant** is 0.

$$\Delta = b^2 - 4ac = (8m)^2 - (4)(1 + 4m^2)(3) = 0$$

Solving, we find $m^2 = \boxed{\text{(C)} \frac{3}{4}}$.

-edited by coolmath34

20. (AHSME 1971) The sum of the squares of the roots of the equation $x^2 + 2hx = 3$ is 10. The absolute value of h is equal to

(A) -1 (B) $\frac{1}{2}$ (C) $\frac{3}{2}$ (D) 2 (E) None of these

Answer: E

Solution 1

We can rewrite the equation as $x^2 + 2hx - 3 = 0$. By **Vieta's Formulas**, the sum of the roots is $-2h$ and the product of the roots is -3 .

Let the two roots be r and s . Note that

$$r^2 + s^2 = (r + s)^2 - 2rs = (-2h)^2 - 2(-3)$$

Therefore, $4h^2 + 6 = 10$ and $h = \pm 1$. This doesn't match any of the answer choices, so the answer is $\boxed{\text{(E)} \text{ None of these}}$.

-edited by coolmath34

Solution 2

The given equation can be rewritten as $x^2 + 2hx - 3 = 0$. By **Vieta's Formulas**, we know that the sum of the roots is $-2h$. Thus, by **Newton's sums**, we have the following equation:

$$\begin{aligned} 10 + (2h)(-2h) + 2(-3) &= 0 \\ 10 - 4h^2 - 6 &= 0 \\ -4h^2 &= -4 \\ h^2 &= 1 \\ |h| &= 1 \end{aligned}$$

Thus, our answer is $\boxed{\text{(E)} \text{ None of these}}$.

21. (*AHSME 1971*) If $\log_2(\log_3(\log_4 x)) = \log_3(\log_4(\log_2 y)) = \log_4(\log_2(\log_3 z)) = 0$, then the sum $x + y + z$ is equal to

(A) 50 (B) 58 (C) 89 (D) 111 (E) 1296

Answer: C

Solution

If $\log_x n = 0$, then $n = 1$. So, we can rewrite this equation:

$$(\log_3(\log_4 x)) = (\log_4(\log_2 y)) = (\log_2(\log_3 z)) = 1$$

Solve individually for each variable.

$$x = 4^3 = 64$$

$$y = 2^4 = 16$$

$$z = 3^2 = 9$$

Therefore, $x + y + z = 89$.

The answer is (C) 89.

-edited by coolmath34

22. (*AHSME 1971*) If w is one of the imaginary roots of the equation $x^3 = 1$, then the product $(1 - w + w^2)(1 + w - w^2)$ is equal to

(A) 4 (B) w (C) 2 (D) w^2 (E) 1

Answer: A

Solution 1

Expanding the given expression yields $1 + w - w^2 - w - w^2 + w^3 + w^2 + w^3 - w^4 = 1 - w^2 + 2w^3 - w^4$. Recalling that $w^3 = 1$, we see that this expression equals $1 - w^2 + 2 - w = 4 - (1 + w + w^2)$. By the properties of **roots of unity** $\neq 1$, $w^2 + w + 1 = 0$, so the given expression equals (A) 4.

Solution 2 (not recommended)

Suppose $w = e^{i\frac{2\pi}{3}} = \cos(\frac{2\pi}{3}) + i\sin(\frac{2\pi}{3}) = \frac{-1+i\sqrt{3}}{2}$. Substituting this into the given expression, we can calculate the result:

$$\begin{aligned} (1 - w + w^2)(1 + w - w^2) &= (1 - \frac{-1+i\sqrt{3}}{2} + (\frac{-1+i\sqrt{3}}{2})^2)(1 + \frac{-1+i\sqrt{3}}{2} - (\frac{-1+i\sqrt{3}}{2})^2) \\ &= (1 - \frac{-1+i\sqrt{3}}{2} + \frac{1-3-2i\sqrt{3}}{4})(1 + \frac{-1+i\sqrt{3}}{2} - \frac{1-3-2i\sqrt{3}}{4}) \\ &= (1 - \frac{-1+i\sqrt{3}}{2} + \frac{-1-i\sqrt{3}}{2})(1 + \frac{-1+i\sqrt{3}}{2} - \frac{-1-i\sqrt{3}}{2}) \\ &= (1 - 2(\frac{i\sqrt{3}}{2}))(1 + 2(\frac{i\sqrt{3}}{2})) \\ &= 1^2 - (i\sqrt{3})^2 \\ &= 1 + 3 \\ &= 4 \end{aligned}$$

Thus, our answer is (A) 4.

23. (AHSME 1971) Teams A and B are playing a series of games. If the odds for either to win any game are even and Team A must win two or Team B three games to win the series, then the odds favoring Team A to win the series are

(A) 11 to 5 (B) 5 to 2 (C) 8 to 3 (D) 3 to 2 (E) 13 to 6

Answer: A

Solution

We have two cases: one where A wins the first game and the other where A loses.

Case 1: In the $\frac{1}{2}$ chance that A wins the first game, A simply needs to win at least 1 of the next 3 games. We see that the probability of A losing the next 3 games is $(\frac{1}{2})^3 = \frac{1}{8}$, so, by **complementary counting**, the probability that A wins at least 1 of the next 3 games is $1 - \frac{1}{8} = \frac{7}{8}$.

Case 2: In the $\frac{1}{2}$ chance that A loses the first game, both teams need to win 2 games, so A 's advantage completely disappears. Thus, the probability that A wins the series from here is $\frac{1}{2}$.

Combining the information from the two cases, we see that A 's probability of winning the series is $\frac{1}{2} \cdot \frac{7}{8} + \frac{1}{2} \cdot \frac{1}{2} = \frac{7}{16} + \frac{4}{16} = \frac{11}{16}$. Thus, our answer is **(A) 11 to 5**.

24. (AHSME 1971)

$$\begin{array}{ccccccccc}
 & & & & & & 1 & & & & & & & & & & \\
 & & & & & & & & & & & 1 & & & & & \\
 & & & & & & 1 & & & & 1 & & & & & & \\
 & & & & & 1 & & & 2 & & & 1 & & & & & \\
 & & & 1 & & & 3 & & & 3 & & & 1 & & & & \\
 & & 1 & & & & & & & & & & & & & & \\
 & 1 & & & 4 & & 6 & & 4 & & & & 1 & & & & \\
 & & & & & & & & & & & & & & & &
 \end{array}$$

centering

etc.

Pascal's triangle is an array of positive integers(See figure), in which the first row is 1, the second row is two 1's, each row begins and ends with 1, and the k^{th} number in any row when it is not 1, is the sum of the k^{th} and $(k-1)^{\text{th}}$ numbers in the immediately preceding row. The quotient of the number of numbers in the first n rows which are not 1's and the number of 1's is

(A) $\frac{n^2 - n}{2n - 1}$ (B) $\frac{n^2 - n}{4n - 2}$ (C) $\frac{n^2 - 2n}{2n - 1}$ (D) $\frac{n^2 - 3n + 2}{4n - 2}$ (E) None of these

Answer: D

Solution

The first row has 1 one, and all the other rows have 2 ones. Thus, the total number of ones in the first n rows is $1 + 2(n-1) = 2n - 1$. The total number of elements in the first n rows is $1 + 2 + 3 + \dots + n$, which is the n^{th} **triangular number**, $\frac{n(n+1)}{2}$. Thus, the number of elements excluding ones is $\frac{n(n+1)}{2} - (2n - 1)$. Thus, our desired

quotient is $\frac{\frac{n(n+1)}{2} - (2n-1)}{2n-1} = \frac{n^2 + n - 2(2n-1)}{2(2n-1)} = \frac{n^2 - 3n + 2}{4n - 2}$. **(D)**

25. (AHSME 1971) A teen age boy wrote his own age after his father's. From this new four place number, he subtracted the absolute value of the difference of their ages to get 4,289. The sum of their ages was

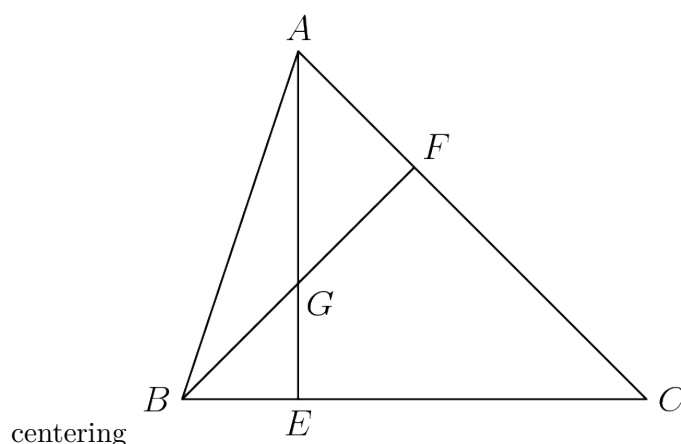
(A) 48 (B) 52 (C) 56 (D) 59 (E) 64

Answer: D

Solution

Because the father's age is a two digit number, we know that the father's age must be 43 so that the second two digits of the original number can be a number in the teens. Let the son's age be x . We know that the original number is $4300 + x$, and the positive difference between their ages is $43 - x$. Thus, we have the equation $4300 + x - (43 - x) = 4289$, which yields $x = 16$. Thus, the sum of the two ages is $43 + 16 = \boxed{\text{(D) } 59}$.

26. (AHSME 1971)



In $\triangle ABC$, point F divides side AC in the ratio $1 : 2$. Let E be the point of intersection of side BC and AG where G is the midpoint of BF . The point E divides side BC in the ratio

(A) $1 : 4$ (B) $1 : 3$ (C) $2 : 5$ (D) $4 : 11$ (E) $3 : 8$

Answer: B

Solution 1

We will use **mass points** to solve this problem. AC is in the ratio $1 : 2$, so we will assign a mass of 2 to point A , a mass of 1 to point C , and a mass of 3 to point F . We also know that G is the midpoint of BF , so $BG : GF = 1 : 1$. F has a mass of 3, so B also has a mass of 3.

In line BC , B has a mass of 3 and C has a mass of 1. Therefore, $BE : EC = 1 : 3$.

The answer is $\boxed{\text{(B) } 1 : 3}$.

-edited by coolmath34

Solution 2

By **Menelaus' Theorem** on $\triangle AGF$ and line $\overleftrightarrow{BC}$, we know that $\frac{AE}{GE} \cdot \frac{GB}{FB} \cdot \frac{FC}{AC} = 1$. Because G is the midpoint of FB , we know that $\frac{GB}{FB} = \frac{1}{2}$. Furthermore, from the problem, we know $\frac{FC}{AC} = \frac{2}{3}$, so, by substitution, our first equation becomes $\frac{AE}{GE} \cdot \frac{1}{2} \cdot \frac{2}{3} = 1$, so $\frac{AE}{GE} = 3$, and therefore $\frac{AE}{AG} = \frac{3}{2}$.

Using Menelaus again on $\triangle BEG$ and line $\overleftrightarrow{AC}$ reveals that $\frac{BC}{EC} \cdot \frac{EA}{GA} \cdot \frac{GF}{BF} = 1$. We know that $\frac{EA}{GA} = \frac{AE}{AG} = \frac{3}{2}$ and $\frac{GF}{BF} = \frac{1}{2}$. Thus, our equation becomes $\frac{BC}{EC} \cdot \frac{3}{2} \cdot \frac{1}{2} = 1$, and so $\frac{BC}{EC} = \frac{4}{3}$. Because $\frac{BC}{EC} = \frac{BE+EC}{EC} = 1 + \frac{BE}{EC}$, we see that $\frac{BE}{EC} = \frac{1}{3}$, so our answer is **(B)** $1 : 3$.

27. (AHSME 1971) A box contains chips, each of which is red, white, or blue. The number of blue chips is at least half the number of white chips, and at most one third the number of red chips. The number which are white or blue is at least 55. The minimum number of red chips is

(A) 24 (B) 33 (C) 45 (D) 54 (E) 57

Answer: E

Solution

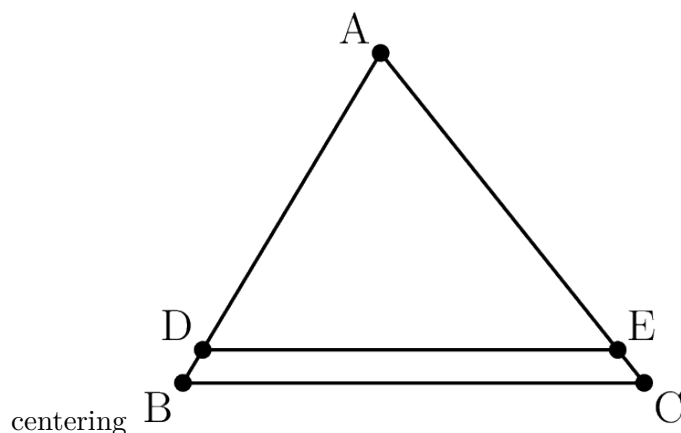
Let the number of white be $2x$. The number of blue is then $x - y$ for some constant y . So we want $2x + x - y = 55 \rightarrow 3x - y = 55$. We take mod 3 to find y . $55 = 1 \pmod{3}$, so blue is 1 more than half white. The number of whites is then 36, and the number of blues is 19. So $19 * 3 = \mathbf{(E) 57}$

28. (AHSME 1971) Nine lines parallel to the base of a triangle divide the other sides each into 10 equal segments and the area into 10 distinct parts. If the area of the largest of these parts is 38, then the area of the original triangle is

(A) 180 (B) 190 (C) 200 (D) 210 (E) 240

Answer: C

Solution



Let the triangle be $\triangle ABC$ with base $\overline{BC}$ and longest parallel segment $\overline{DE}$ with D on $\overline{AB}$ and E on $\overline{AC}$, as in the diagram.

By the properties of **transversals**, we have $\angle ABC = \angle ADE$. Thus, by **AA Similarity**, we have $\triangle ADE \sim \triangle ABC$ (because they share $\angle BAC$). From the problem, we know that $\frac{AE}{AC} = \frac{9}{10}$, so, by similarity, $\frac{DE}{BC} = \frac{9}{10}$, and so $DE = \frac{9}{10}BC$.

Now, let $[\triangle ABC] = A$. Because $\frac{DE}{BC} = \frac{9}{10}$, we know that $\frac{[\triangle ADE]}{[\triangle ABC]} = \left(\frac{9}{10}\right)^2 = \frac{81}{100}$. From the problem, $[BCED] = 38$, so $A = [\triangle ADE] + 38 = \frac{9}{10}A + 38$. Solving for A yields $A = \mathbf{(C) 200}$.

29. (AHSME 1971) Given the progression $10^{\frac{1}{11}}, 10^{\frac{2}{11}}, 10^{\frac{3}{11}}, 10^{\frac{4}{11}}, \dots, 10^{\frac{n}{11}}$. The least positive integer n such that the product of the first n terms of the progression exceeds 100,000 is

(A) 7 (B) 8 (C) 9 (D) 10 (E) 11

Answer: E

Solution

By the rules of **exponents**, the product of the first n terms of the sequence equals $10^{\frac{1}{11} + \frac{2}{11} + \dots + \frac{n}{11}}$. From here, we can set up the equation $10^{\frac{1}{11} + \frac{2}{11} + \dots + \frac{n}{11}} = 100,000 = 10^5$, which simplifies to $\frac{1}{11} + \frac{2}{11} + \dots + \frac{n}{11} = 5$, or $1 + 2 + 3 \dots n = 55$. By the formula for **triangular numbers**, $\frac{n(n+1)}{2} = 55$. Thus, $n^2 + n - 110 = 0$, or $(n+11)(n-10) = 0$. Because $n > 0$, only $n = 10$ makes the product equal 100,000. However, we have to strictly exceed 100,000, so our answer is **(E) 11**.

30. (AHSME 1971) Given the linear fractional transformation of x into $f_1(x) = \frac{2x-1}{x+1}$. Define $f_{n+1}(x) = f_1(f_n(x))$ for $n = 1, 2, 3, \dots$. Assuming that $f_{35}(x) = f_5(x)$, it follows that $f_{28}(x)$ is equal to

(A) x (B) $\frac{1}{x}$ (C) $\frac{x-1}{x}$ (D) $\frac{1}{1-x}$ (E) None of these

Answer: D

Solution 1

Extend the definition of $f_n(x)$ to $n \leq 0$: Let $f_{n-1}(x)$ be the function such that $f_1(f_{n-1}(x)) = f_n(x)$. From the problem, $f_5(x) = f_{35}(x)$, so the functions $f_1(x), f_2(x), \dots$ must repeat in a cycle whose length is a cycle which is a divisor of $35 - 5 = 30$. Thus, if $f_a(x) = f_b(x)$ for integers a and b , we know that $a \equiv b$ **modulo** 30. Thus, because $28 \equiv -2 \pmod{30}$, we know that $f_{-2}(x) = f_{28}(x)$.

It is clear that $f_0(x) = x$, because $f_1(f_0(x)) = f_1(x)$.

Let $f_{-1}(x) = y$. Then, we know that $f_1(y) = f_0(x) = x$, so we have the following equation we can solve for y :

$$\begin{aligned}\frac{2y-1}{y+1} &= x \\ 2y-1 &= xy+x \\ y(2-x) &= x+1 \\ y &= \frac{x+1}{2-x}\end{aligned}$$

Let $f_{-2}(x) = z$. Then, we know that $f_1(z) = f_{-1}(x) = \frac{x+1}{2-x}$, so we have the following equation we can solve for z :

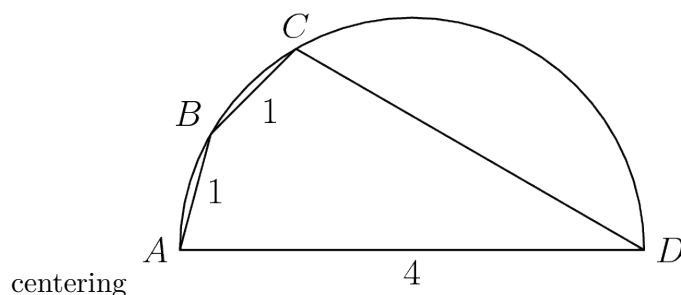
$$\begin{aligned}\frac{2z-1}{z+1} &= \frac{x+1}{2-x} \\ 4z-2-2xz+x &= xz+x+z+1 \\ 3z-3xz &= 3 \\ z(1-x) &= 1 \\ z &= \frac{1}{1-x}\end{aligned}$$

We derived earlier the fact that $z = f_{-2}(x) = f_{28}(x)$, so $f_{28}(x) = \boxed{\text{(D)} \frac{1}{1-x}}$.

Solution 2

Keep solving for the next function in the sequence of $f_n(x)$ while being sure not to make silly algebra mistakes. This process reveals that $f_6(x) = x$, so $f_7(x) = f_1(x)$, and the cycle of functions repeats **modulo** 6. Because $28 \equiv 4 \pmod{6}$, we know that $f_{28}(x) = f_4(x) = \boxed{\text{(D)} \frac{1}{1-x}}$, which we calculated on the way to deducing that $f_6(x) = x$.

31. (AHSME 1971)



centering

Quadrilateral $ABCD$ is inscribed in a circle with side AD , a diameter of length 4. If sides AB and BC each have length 1, then side CD has length

- (A) $\frac{7}{2}$ (B) $\frac{5\sqrt{2}}{2}$ (C) $\sqrt{11}$ (D) $\sqrt{13}$ (E) $2\sqrt{3}$

Answer: A

Solution 1

Note that the length 4 forms a semicircle. We can then use the Law of Cosines. Take the center and form a line segment with the other two points.

Let's find the cosine of angle AOB . The cosine of that angle is $7/8$. We use the double cosine angle to find angle AOC :

$$\cos(2x) = 2\cos^2(x) - 1$$

Therefore, $\cos(2x) = 17/32$

Applying the Law of Cosines to the triangle DOC , we get

$$CD^2 = 2^2 + 2^2 - 2(4)(-17/32) \text{ (because } \cos(x) = -\cos(180 - x)\text{)}$$

Therefore, $CD^2 = 8 + 17/4$

and we find that $CD = \boxed{\text{(A)} 7/2}$!

Solution 2

By the properties of a circle, we see that angle DCA and angle DBA are both right angles. Thus, if we formulate the diagonal lengths of the cyclic quadrilateral using the Pythagorean theorem, we can finish the problem with Ptolemy's Theorem.

With that train of thought, we see that

$$BD^2 = AD^2 - AB^2$$

$$BD^2 = 4^2 - 1^2$$

$$BD^2 = 15 \Rightarrow BD = \sqrt{15}$$

To let us formulate that rest of our lengths, let $CD = x$. Then, similar to the above,

$$AC^2 = AD^2 - CD^2$$

$$AC^2 = 4^2 - x^2 = 16 - x^2$$

$$AC = \sqrt{16 - x^2}$$

Now that we have all of the side lengths and diagonal lengths in one variable, we use Ptolemy's Theorem to finish from here:

$$4 \cdot 1 + x \cdot 1 = \sqrt{15}\sqrt{16 - x^2}$$

$$x + 4 = \sqrt{240 - 15x^2}$$

$$x^2 + 8x + 16 = 240 - 15x^2$$

$$16x^2 + 8x - 224 = 0$$

$$2x^2 + x - 28 = 0$$

$$x = \frac{-1 \pm \sqrt{1 - (4)(2)(-28)}}{2(2)} = \frac{-1 \pm \sqrt{225}}{4} = \frac{-1 \pm 15}{4}$$

Since the negative solution is extraneous, we see that our solution is $x = \frac{7}{2} \Rightarrow \boxed{\text{(A)}}$

32. (AHSME 1971) If $s = (1 + 2^{-\frac{1}{32}})(1 + 2^{-\frac{1}{16}})(1 + 2^{-\frac{1}{8}})(1 + 2^{-\frac{1}{4}})(1 + 2^{-\frac{1}{2}})$, then s is equal to

- (A) $\frac{1}{2}(1 - 2^{-\frac{1}{32}})^{-1}$ (B) $(1 - 2^{-\frac{1}{32}})^{-1}$ (C) $1 - 2^{-\frac{1}{32}}$
 (D) $\frac{1}{2}(1 - 2^{-\frac{1}{32}})$ (E) $\frac{1}{2}$

Answer: A

Solution 1

Multiply both sides of the given equation by $(1 - 2^{-\frac{1}{32}})$. Using **difference of squares** reveals that $(1 - 2^{-\frac{1}{32}})(1 + 2^{-\frac{1}{32}}) = (1 - 2^{-\frac{1}{16}})$. Thus, we can use difference of squares again on the next term, $(1 + 2^{-\frac{1}{16}})$, and the terms after that until we get

$(1 + 2^{-\frac{1}{32}})s = (1 - 2^{-\frac{1}{2}})(1 + 2^{-\frac{1}{2}}) = 1 - 2^{-1} = \frac{1}{2}$. Dividing both sides by $(1 + 2^{-\frac{1}{32}})$ reveals that $s = \boxed{\text{(A)} \frac{1}{2}(1 + 2^{-\frac{1}{32}})^{-1}}$.

Solution 2 (50-50 guess)

For any two reals a and x with $a > 0$, $a^x > 0$. Thus, we know that all of the terms $2^{-\frac{1}{x}} > 0$, so s is the product of 5 real terms greater than 1. Thus, $s > 1$, so we can eliminate options (C), (D), and (E). Now, we have a 50% chance of guessing the right answer, $\boxed{\text{(A)} \frac{1}{2}(1 + 2^{-\frac{1}{32}})^{-1}}$.

33. (AHSME 1971) If P is the product of n quantities in Geometric Progression, S their sum, and S' the sum of their reciprocals, then P in terms of S, S' , and n is
 (A) $(SS')^{\frac{1}{2}n}$ (B) $(S/S')^{\frac{1}{2}n}$ (C) $(SS')^{n-2}$ (D) $(S/S')^n$ (E) $(S/S')^{\frac{1}{2}(n-1)}$

Answer: B

Solution 1

Let the **geometric sequence** have first term a and common ratio R . Then, the first n terms of the sequence are $a, aR, aR^2, \dots, aR^{n-1}$. The product of these terms P is $a^n R^{1+2+\dots+n-1} = a^n R^{\frac{(n-1)n}{2}}$ by the formula for **triangular numbers**. Using the **sum formula** reveals that $S = a \cdot \frac{1-R^n}{1-R}$.

We know that $S' = \frac{1}{a} + \frac{1}{aR} + \dots + \frac{1}{aR^{n-1}}$. Combining fractions reveals that $S' = \frac{aR^{n-1} + aR^{n-2} + \dots + aR + a}{a^2 R^{n-1}} = \frac{S}{a^2 R^{n-1}}$. Note that this denominator looks suspiciously similar to our formula for P . In fact, $(a^2 + R^{n-1})^{\frac{n}{2}} = a^n + R^{\frac{(n-1)n}{2}} = P$. Because $(S/S')^{\frac{n}{2}} = (a^2 + R^{n-1})^{\frac{n}{2}} = P$, our answer is $\boxed{\text{(B)} (S/S')^{\frac{1}{2}n}}$.

Solution 2 (Answer Choices)

We can just look at a very specific case: 1, 2, 4, 8. Here, $n = 4$, $P = 64$, $S = 15$, and $S' = \frac{30}{16} = \frac{15}{8}$.

Then, plug in values of S, S' , and n into each of the answer choices and see if it matches the product.

Answer choice $\boxed{\text{(B)} (S/S')^{\frac{1}{2}n}}$ works: $(\frac{15}{8})^2 = 64$.

-edited by coolmath34

Solution 3 (Answer Choices)

We can use **dimensional analysis** to cut down our answer choices. Suppose that each of the terms in the geometric progression is in units of meters. Then, S should have units of meters, S' units of $\frac{1}{\text{meters}}$ and P units of meters ^{n} . Therefore, SS' is unitless, so we can eliminate options (A) and (C). S/S' has units meters², so, to equal P (which has units meters ^{n}), the exponent needs to be $\frac{n}{2}$. The only remaining answer choice which satisfies this constraint is $\boxed{\text{(B)} (S/S')^{\frac{1}{2}n}}$.

34. (AHSME 1971) An ordinary clock in a factory is running slow so that the minute hand passes the hour hand at the usual dial position(12 o'clock, etc.) but only every 69 minutes. At time and one-half for overtime, the extra pay to which a \$4.00 per hour worker should be entitled after working a normal 8 hour day by that slow running clock, is
 (A) \$2.30 (B) \$2.60 (C) \$2.80 (D) \$3.00 (E) \$3.30

Answer: B

Solution

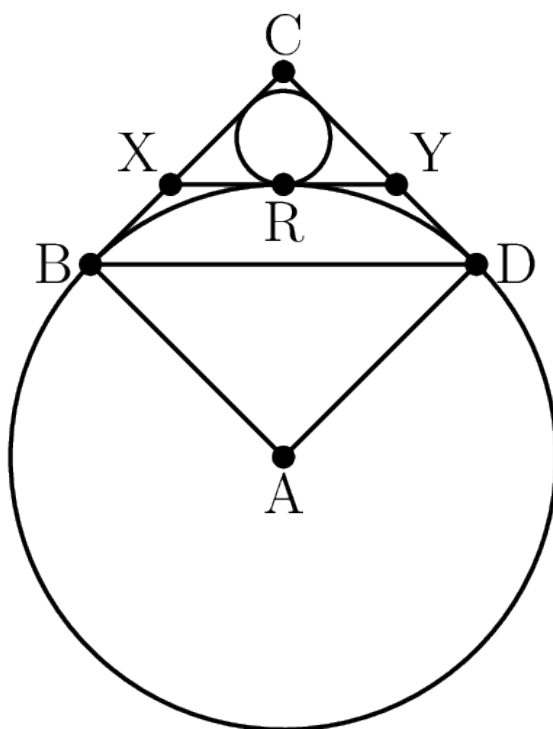
If the clock were functioning properly, the minute hand would have an angular velocity of $1 \frac{\text{rev}}{\text{hr}}$, and the hour hand would have an angular velocity of $\frac{1}{12} \frac{\text{rev}}{\text{hr}}$. Thus, the minute hand would move at a velocity of $1 - \frac{1}{12} = \frac{11}{12} \frac{\text{rev}}{\text{hr}}$ relative to the hour hand. Thus, the minute hand would pass the hour hand every $\frac{12}{11}$ hours, or $\frac{720}{11}$ minutes (after the minute hand has made a full revolution relative to the hour hand). Because the slow clock takes 69 minutes to do this, one hour to this clock is actually $\frac{69}{\frac{720}{11}} = \frac{253}{240}$ hours. Thus, 8 hours on the slow clock is actually $8 \cdot \frac{253}{240} = \frac{253}{30}$ hours. Thus, the workers worked for an extra $\frac{253}{30} - 8 = \frac{13}{30}$ hours. Because, from the problem, the workers are paid 1.5 times their usual wage when working overtime, they earn $\$4 \cdot 1.5 = \6 per hour in overtime. Therefore, they should receive $\frac{13}{30} \cdot 6 = \frac{13}{5} = 2.6$ extra dollars, so our answer is **(B) \$2.60**.

35. (AHSME 1971) Each circle in an infinite sequence with decreasing radii is tangent externally to the one following it and to both sides of a given right angle. The ratio of the area of the first circle to the sum of areas of all other circles in the sequence, is

(A) $(4 + 3\sqrt{2}) : 4$ (B) $9\sqrt{2} : 2$ (C) $(16 + 12\sqrt{2}) : 1$
 (D) $(2 + 2\sqrt{2}) : 1$ (E) $(3 + 2\sqrt{2}) : 1$

Answer: C

Solution 1



centering

WLOG, let the radius of the largest circle be 1 (we can do this because scaling the diagram preserves the ratio of areas). Let the largest circle be ω_1 and the second largest circle be ω_2 . As in the diagram above, let ω_1 have center A with points of tangency at B and D . Let the tangents from these two points meet at C . From

the problem, we know that $\angle BCD = 90^\circ$. Let the common tangent of ω_1 and ω_2 intersect $\overline{BC}$ and $\overline{DC}$ at X and Y , respectively. Let ω_2 intersect $\overline{XY}$ at R .

By the **Two Tangent Theorem**, $CB = CD$. Thus, $\triangle BCD$ is an isosceles right triangle, so $\angle CBD = \angle CDB = 45^\circ$. $\angle CBD$ is a **tangent-chord angle** in ω_1 , so $\angle BAD = 2 \cdot \angle CDB = 90^\circ$. Thus, because $ABCD$ has 4 right angles and 2 adjacent sides congruent, it is a square. Thus, $BC = CD = 1$.

ω_2 is the **incircle** of $\triangle CXY$, because it is tangent to all its sides. Let $CX = x$ and s be the **semiperimeter** of $\triangle CXY$. Because $\overline{XY} \parallel \overline{BD}$, we see that $\angle CXY = \angle CBD = 45^\circ$ and $\angle CYX = \angle CDB = 45^\circ$. Thus, $\triangle CXY$ is an isosceles right triangle, so $XY = x\sqrt{2}$. By the properties of an incircle's touchpoints, $XR = s - CY = s - x$ and $YR = s - CX = s - x$. Thus, R is the midpoint of $\overline{XY}$, so $RX = XY = \frac{\sqrt{2}}{2}x$.

Because $\overline{XY}$ is the common tangent of ω_1 and ω_2 , R is on ω_1 . Thus, by the Two Tangent Theorem, $XB = XR = \frac{\sqrt{2}}{2}x$.

Now, we have the equation $1 = BC = BX + XC = \frac{\sqrt{2}}{2}x + x$, which yields $x = \frac{1}{1 + \frac{\sqrt{2}}{2}} = \frac{2(2 - \sqrt{2})}{(2 + \sqrt{2})(2 - \sqrt{2})} = 2 - \sqrt{2}$. We know that $s = \frac{x + x + x\sqrt{2}}{2} = x + \frac{\sqrt{2}}{2}x$. Let the **inradius** of $\triangle CXY$ be r . Now the formula $A = rs$ in $\triangle CXY$ gives us the equation $\frac{x^2}{2} = r(x + \frac{\sqrt{2}}{2}x)$, which yields $r = \frac{x}{(2 + \sqrt{2})} = \frac{(2 - \sqrt{2})^2}{(2 + \sqrt{2})(2 - \sqrt{2})} = \frac{4 - 4\sqrt{2} + 2}{4 - 2} = 3 - 2\sqrt{2}$.

Because the ratio of the radii of ω_1 and ω_2 is $\frac{3 - 2\sqrt{2}}{1} = 3 - 2\sqrt{2}$, the ratio of their areas is $(3 - 2\sqrt{2})^2 = 9 - 12\sqrt{2} + 8 = 17 - 12\sqrt{2}$. This ratio is the same as that of the area of ω_2 to the third largest circle, the third to the fourth, etc., because removing the largest circle gives a scaled-down version of the same problem. Thus, the total area of the circles is given by the **geometric sequence** starting at π (the area of ω_1) with common ratio $17 - 12\sqrt{2} \approx 17 - 12 \cdot 1.4 = 0.2 < 1$, which, by the infinite **geometric series** formula, has an infinite sum of $\pi \cdot \frac{1}{1 - (17 - 12\sqrt{2})} = \pi \cdot \frac{(12\sqrt{2} + 16)}{(12\sqrt{2} - 16)(12\sqrt{2} + 16)} = \pi \cdot \frac{3\sqrt{2} + 4}{8}$. However, we are looking for the ratio of the area of ω_1 to that of all of the circles *excluding* ω_1 , so we have to subtract π . This reveals that the area of the other circles is $\pi \cdot \frac{3\sqrt{2} - 4}{8}$. Now, our desired answer is the ratio $\frac{\pi}{\pi \cdot \frac{3\sqrt{2} - 4}{8}} = \frac{8(3\sqrt{2} + 4)}{(3\sqrt{2} - 4)(3\sqrt{2} + 4)} = \frac{8(3\sqrt{2} + 4)}{18 - 16} = 4(3\sqrt{2} + 4) = 12\sqrt{2} + 16$. Thus, we choose **(C)** $(16 + 12\sqrt{2}) : 1$.

Solution 2 (Informed guess)

Draw a good diagram, and especially make sure that you are drawing a right angle. Then, you will realize that the big circle completely dwarfs all of the other circles. Answer choice (C), $16 + 12\sqrt{2}$, is about $16 + 12 \cdot 1.4 = 32.8$, whereas the greatest of the other answer choices is about 6.3, which is unreasonably small based on our diagram. Thus, we choose answer **(C)** $(16 + 12\sqrt{2}) : 1$.

§8 AHSME 1972

1. (AHSME 1972) The lengths in inches of the three sides of each of four triangles I , II , III , and IV are as follows:

I	3, 4, and 5	III	7, 24, and 25
II	4, $7\frac{1}{2}$, and $8\frac{1}{2}$	IV	$3\frac{1}{2}$, $4\frac{1}{2}$, and $5\frac{1}{2}$.

Of these four given triangles, the only right triangles are

- (A) I and II (B) I and III (C) I and IV
 (D) I, II, and III (E) I, II, and IV

Answer: D

Solution

Let's test out the four triangles:

$$\text{I: } 3^2 + 4^2 = 5^2 \text{ (true)}$$

$$\text{II: } 4^2 + 7.5^2 = 8.5^2 \text{ (true)}$$

$$\text{III: } 7^2 + 24^2 = 25^2$$

The only answer choice with I, II, and III in the answer is D $\rightarrow$ I, II, and III.

2. (*AHSME 1972*) If a dealer could get his goods for 8% less while keeping his selling price fixed, his profit, based on cost, would be increased to $(x + 10)\%$ from his present profit of $x\%$, which is

- (A) 12% (B) 15% (C) 30% (D) 50% (E) 75%

Answer: B

Solution

If c is the cost, then $0.92c$ is the reduced cost. Selling price is equal to cost plus profit:

$$c(1 + 0.01x) = 0.92c(1 + 0.01(x + 10))$$

Solving for x yields $x = 15$. The answer is (B).

-edited by coolmath34

3. (*AHSME 1972*) If $x = \frac{1 - i\sqrt{3}}{2}$ where $i = \sqrt{-1}$, then $\frac{1}{x^2 - x}$ is equal to

- (A) -2 (B) -1 (C) $1 + i\sqrt{3}$ (D) 1 (E) 2

Answer: B

Solution

Using DeMoivre's theorem, we can calculate $x^2 = \frac{1+i\sqrt{3}}{2}$. The denominator is therefore -1 which makes the answer

$$\boxed{\text{(B)} \quad -1}.$$

lopkiloinm

4. (*AHSME 1972*) The number of solutions to $\{1, 2\} \subseteq X \subseteq \{1, 2, 3, 4, 5\}$, where X is a subset of $\{1, 2, 3, 4, 5\}$ is

- (A) 2 (B) 4 (C) 6 (D) 8 (E) None of these

Answer: D

Solution

X has to contain $\{1, 2\}$, so only $\{3, 4, 5\}$ matters. There are two choices for the elements; the element is either in X or outside of X . With this combinatorics in mind, the answer is simply $2^3 = \boxed{\text{(D)} \quad 8}$. lopkiloinm

5. (AHSME 1972) From among $2^{1/2}, 3^{1/3}, 8^{1/8}, 9^{1/9}$ those which have the greatest and the next to the greatest values, in that order, are
 (A) $3^{1/3}, 2^{1/2}$ (B) $3^{1/3}, 8^{1/8}$ (C) $3^{1/3}, 9^{1/9}$ (D) $8^{1/8}, 9^{1/9}$
 (E) None of these

Answer: A

Solution

$8^{1/8}$ and $9^{1/9}$ are obviously too small. We must then compare $3^{1/3}$ with $2^{1/2}$. Raising both to the power of 6 gives 9 and 8 respectively. Our answer is therefore

(A) $3^{1/3}, 2^{1/2}$. lopkiloinm

6. (AHSME 1972) If $3^{2x} + 9 = 10(3^x)$, then the value of $(x^2 + 1)$ is
 (A) 1 only (B) 5 only (C) 1 or 5 (D) 2 (E) 10

Answer: C

Solution

We can rearrange to get $3^{2x} - 10(3^x) + 9 = 0$. Factor or use the quadratic formula to find that $3^x = 9$ or 1, which makes $x = 2$ or 0 and our answer

(C) 1 or 5.

7. (AHSME 1972) If $yz : zx : xy = 1 : 2 : 3$, then $\frac{x}{yz} : \frac{y}{zx}$ is equal to
 (A) 3 : 2 (B) 1 : 2 (C) 1 : 4 (D) 2 : 1 (E) 4 : 1

Answer: E

Solution

We want to find

$$\frac{x}{yz} : \frac{y}{zx} = \frac{x}{yz} \cdot \frac{zx}{y} = \frac{x^2}{y^2}.$$

We are given $yz : zx = 1 : 2$, and dividing both sides by z gives $y : x = 1 : 2$. Therefore,

$$\frac{x^2}{y^2} = \frac{1}{\frac{1}{2^2}} = 4.$$

The answer is (E).

-edited by coolmath34

8. (AHSME 1972) If $|x - \log y| = x + \log y$ where x and $\log y$ are real, then
 (A) $x = 0$ (B) $y = 1$ (C) $x = 0$ and $y = 1$
 (D) $x(y - 1) = 0$ (E) None of these

Answer: D

Solution

We have two cases: x is positive and x is negative.

$$\text{Positive} \quad x - \log y = x + \log y \quad \rightarrow \quad 2 \log y = 0 \rightarrow y = 1$$

$$\text{Negative} \quad -x + \log y = x + \log y \quad \rightarrow \quad 2x = 0 \rightarrow x = 0$$

We can write $x(y - 1) = 0$ to show that $x = 0, y = 1$, or both.

The answer is **(D)**.

-edited by coolmath34

9. (*AHSME 1972*) Ann and Sue bought identical boxes of stationery. Ann used hers to write 1-sheet letters and Sue used hers to write 3-sheet letters. Ann used all the envelopes and had 50 sheets of paper left, while Sue used all of the sheets of paper and had 50 envelopes left. The number of sheets of paper in each box was

(A) 150 **(B)** 125 **(C)** 120 **(D)** 100 **(E)** 80

Answer: A

Solution

Let S be the number of sheets of paper and E be the number of envelopes. We can write two equations:

$$S - E = 50$$

$$E - S/3 = 50$$

Solving yields $(E, S) = (100, 150)$.

The answer is **(A)**.

-edited by coolmath34

10. (*AHSME 1972*) For x real, the inequality $1 \leq |x - 2| \leq 7$ is equivalent to

(A) $x \leq 1$ or $x \geq 3$ **(B)** $1 \leq x \leq 3$ **(C)** $-5 \leq x \leq 9$
(D) $-5 \leq x \leq 1$ or $3 \leq x \leq 9$ **(E)** $-6 \leq x \leq 1$ or $3 \leq x \leq 10$

Answer: D

Solution

We can split the inequality into two smaller inequalities and solve them individually.

$$|x - 2| \geq 1 \quad \rightarrow \quad x \geq 3 \quad \text{and} \quad x \leq 1$$

$$|x - 2| \leq 7 \quad \rightarrow \quad x \leq 9 \quad \text{and} \quad x \geq -5$$

Combining these inequalities, we get $x \in [-5, 1] \cup [3, 9]$.

The answer is **(D)**.

-edited by coolmath34

11. (*AHSME 1972*) The value(s) of y for which the following pair of equations $x^2 + y^2 - 16 = 0$ and $x^2 - 3y + 12 = 0$ may have a real common solution, are

(A) 4 only **(B)** $-7, 4$ **(C)** 0, 4 **(D)** no y **(E)** all y

Answer: A

Solution (Quick)

Checking the answer choices, we see that only 4 is the viable choice. Therefore, the answer is A

12. (AHSME 1972) The number of cubic feet in the volume of a cube is the same as the number of square inches in its surface area. The length of the edge expressed as a number of feet is

(A) 6 (B) 864 (C) 1728 (D) 6×1728 (E) 2304

Answer: B

Solution

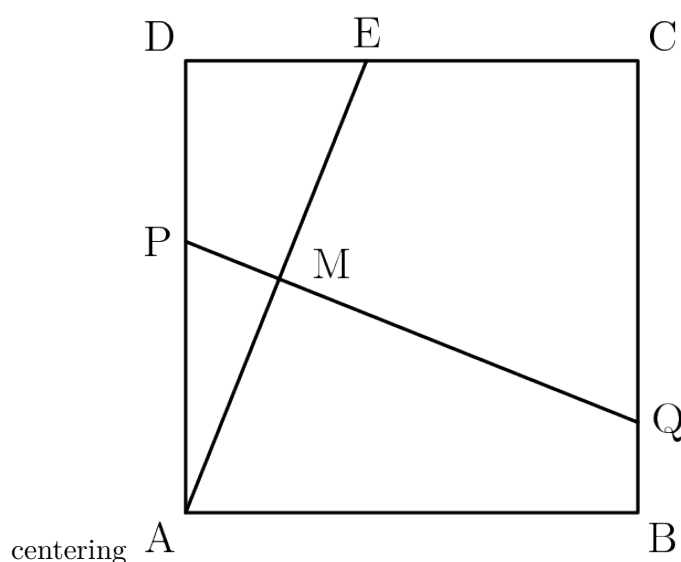
If the side length of a cube is s feet the volume is s^3 cubic feet and the surface area is $6(12s^2)$ square inches. The volume and surface are numerically equal, so we can write

$$s^3 = 6(12s^2)$$

Solving yields $s = 864$.

The answer is (B).

13. (AHSME 1972)



Inside square $ABCD$ (See figure) with sides of length 12 inches, segment AE is drawn where E is the point on DC which is 5 inches from D . The perpendicular bisector of AE is drawn and intersects AE , AD , and BC at points M , P , and Q respectively. The ratio of segment PM to MQ is

(A) 5 : 12 (B) 5 : 13 (C) 5 : 19 (D) 1 : 4 (E) 5 : 21

Answer: C

Solution

Let the line passing through M parallel to AB intersect AD and BC and S and T respectively. Since M is the midpoint of AE , $SM = \frac{5}{2}$ and $TM = 12 - \frac{5}{2} = \frac{19}{2}$. Since $\triangle PSM \sim \triangle QTM$, $PM : MQ = SM : MT = 5 : 19$, hence our answer is C.

14. (AHSME 1972) A triangle has angles of 30° and 45° . If the side opposite the 45° angle has length 8, then the side opposite the 30° angle has length

(A) 4 (B) $4\sqrt{2}$ (C) $4\sqrt{3}$ (D) $4\sqrt{6}$ (E) 6

Answer: B

Solution

This triangle can be split into smaller 30-60-90 and 45-45-90 triangles. The side opposite the 45° angle has length 8, so the 30-60-90 triangle has sides 4, $4\sqrt{3}$, and 8.

One of the legs of the 45-45-90 triangles is 4, so the hypotenuse is $4\sqrt{2}$. This is also the side opposite the 30° angle, so the answer is **(B)**.

-edited by coolmath34

Solution 2

Use **Law of Sines** to get $\frac{8}{\frac{\sqrt{2}}{2}} = \frac{x}{\frac{1}{2}}$. The answer is **(B)** $4\sqrt{2}$.

-aopspandy

15. (*AHSME 1972*) A contractor estimated that one of his two bricklayers would take 9 hours to build a certain wall and the other 10 hours. However, he knew from experience that when they worked together, their combined output fell by 10 bricks per hour. Being in a hurry, he put both men on the job and found that it took exactly 5 hours to build the wall. The number of bricks in the wall was

(A) 500 (B) 550 (C) 900 (D) 950 (E) 960

Answer: C

Solution

Every hour, the contractors lay $x/9$ and $x/10$ bricks, assuming a total of x bricks in the wall. Together, $x/5$ bricks are laid per hour:

$$x = 5(x/9 + x/10 - 10)$$

There are a total of $x = 900$ bricks, and the answer is **(C)** 900.

-edited by coolmath34

16. (*AHSME 1972*) There are two positive numbers that may be inserted between 3 and 9 such that the first three are in geometric progression while the last three are in arithmetic progression. The sum of those two positive numbers is

(A) $13\frac{1}{2}$ (B) $11\frac{1}{4}$ (C) $10\frac{1}{2}$ (D) 10 (E) $9\frac{1}{2}$

Answer: B

Solution 1

Let a be first and b be second. We can then get equations based on our knowledge: $b - a = 9 - b$ and $b/a = a/3$. We then get $b = (9 + a)/2$ from our first equation and we substitute that into the second to get $\frac{9+a}{2a} = a/3$ which simplifies to $2a^2 - 3a - 27 = 0$ which be $(2a - 9)(a + 3) = 0$ and $a = 9/2$. Then $b = 27/4$. Their

sum is **(B)** $11\frac{1}{4}$. lopkiloinm

Solution 2

Like in Solution 1, let a be the first number and b be the second. From the problem, we have that

$$a = 3r \quad (1)$$

$$b = 3r^2 \quad (2)$$

$$9 = a + 2d \quad (3)$$

for some common difference d and common ratio r . We can use these to obtain the following:

$$3r + d = 3r^2 \quad (\text{from (1), (2), and the definition of an arithmetic sequence}) \quad (4)$$

$$9 = 3r + 2d \quad (\text{from (1) and (3)}) \quad (5)$$

Solving (5) for d and substituting into (4), we find that $3r + \frac{9-3r}{2} = 3r^2$ or $6r + (9 - 3r) = 6r^2$. Moving every term to the right side and factoring, we get that $3(2r - 3)(r + 1) = 0$. Because a and b must be positive, the common ratio cannot be -1 , so we ascertain that the common ratio is $3/2$. From here, we find

$$\text{that } a = 3 \cdot \frac{3}{2} = \frac{9}{2}, \text{ and } b = \frac{9}{2} \cdot \frac{3}{2} = 27/4. \text{ Then their sum is } \boxed{\text{(B) } 11\frac{1}{4}}.$$

17. (AHSME 1972) A piece of string is cut in two at a point selected at random. The probability that the longer piece is at least x times as large as the shorter piece is

$$\text{(A) } \frac{1}{2} \quad \text{(B) } \frac{2}{x} \quad \text{(C) } \frac{1}{x+1} \quad \text{(D) } \frac{1}{x} \quad \text{(E) } \frac{2}{x+1}$$

Answer: E

Solution

The string could be represented by a line plot segment on the interval $(0, x + 1)$. The interval $(0, 1)$ is a solution as well as $(x, x + 1)$. The probability is therefore

$$\boxed{\text{(E) } \frac{2}{x+1}}. \quad \text{lopkiloinm}$$

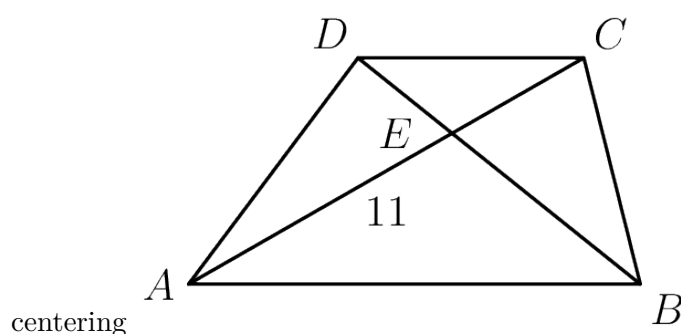
18. (AHSME 1972) Let $ABCD$ be a trapezoid with the measure of base AB twice that of base DC , and let E be the point of intersection of the diagonals. If the measure of diagonal AC is 11, then that of segment EC is equal to

$$\text{(A) } 3\frac{2}{3} \quad \text{(B) } 3\frac{3}{4} \quad \text{(C) } 4 \quad \text{(D) } 3\frac{1}{2} \quad \text{(E) } 3$$

Answer: A

Solution

We begin with a diagram:



The bases of a trapezoid are parallel by definition, so $\angle EDC$ and $\angle EBA$ are alternate interior angles, and therefore equal. We have the same setup with $\angle ECD$ and $\angle EAB$, meaning that $\triangle ABE \sim \triangle CDE$ by AA Similarity. We could've also used the fact that $\angle BEA$ and $\angle DEC$ are vertical angles.

With this information, we can setup a ratio of corresponding sides:

$$\frac{AB}{CD} = \frac{AE}{CE} \implies \frac{2CD}{CD} = \frac{11 - CE}{CE}.$$

And simplify from there:

$$\begin{aligned}\frac{2CD}{CD} &= \frac{11 - CE}{CE} \\ 2 &= \frac{11 - CE}{CE} \\ 2CE &= 11 - CE \\ 3CE &= 11 \\ CE &= \frac{11}{3} = 3\frac{2}{3}.\end{aligned}$$

Therefore, our answer is (A) $3\frac{2}{3}$.

19. (AHSME 1972) The sum of the first n terms of the sequence 1, $(1 + 2)$, $(1 + 2 + 2^2)$, $\dots$ $(1 + 2 + 2^2 + \dots + 2^{n-1})$ in terms of n is

(A) 2^n (B) $2^n - n$ (C) $2^{n+1} - n$ (D) $2^{n+1} - n - 2$ (E) $n \cdot 2^n$

Answer: D

Solution

We begin by evaluating the sum of the first n terms of the sequence with 1, 2, and 3. We find that the sum of the first term of the sequence is 1, the sum of the first two terms is 4, and the sum of the first three terms 11. From here, we can simply guess and check the answer choices and will find that only D works.

20. (AHSME 1972) If $\tan x = \frac{2ab}{a^2 - b^2}$ where $a > b > 0$ and $0^\circ < x < 90^\circ$, then $\sin x$ is equal to

(A) $\frac{a}{b}$ (B) $\frac{b}{a}$ (C) $\frac{\sqrt{a^2 - b^2}}{2a}$ (D) $\frac{\sqrt{a^2 - b^2}}{2ab}$ (E) $\frac{2ab}{a^2 + b^2}$

Answer: E

Solution

We start by letting $\tan x = \frac{\sin x}{\cos x}$ so that our equation is now:

$$\frac{\sin x}{\cos x} = \frac{2ab}{a^2 - b^2}$$

Multiplying through and rearranging gives us the equation:

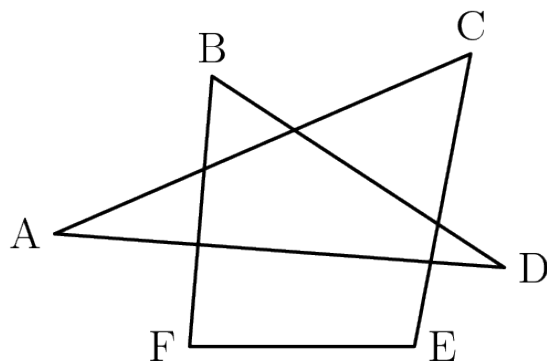
$$\cos x = \frac{a^2 - b^2}{2ab} * \sin x$$

We now apply the Pythagorean identity $\sin^2 x + \cos^2 x = 1$, using our substitution:

$$\left(\frac{a^2 - b^2}{2ab} * \sin x \right)^2 + \sin^2 x = 1$$

We can isolate $\sin x$ without worrying about division by 0 since $a \neq b \neq 0$ our final answer is (E) $\frac{2ab}{a^2 + b^2}$

21. (AHSME 1972)



centering

If the sum of the measures in degrees of angles A , B , C , D , E and F in the figure above is $90n$, then n is equal to

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Answer: C

Solution

Let P be the intersection of AC and BD .

Since $\angle APD = \angle BPC$ are vertical angles, the angle sums of $\triangle PAD$ and $\triangle PBC$ give:

$$\angle A + \angle D = 180^\circ - \angle APD$$

$$\angle B + \angle C = 180^\circ - \angle BPC$$

Adding these equations yields:

$$\angle A + \angle B + \angle C + \angle D = 360^\circ - 2\angle BPC$$

Consider quadrilateral $PFEC$. Its interior angle sum gives $\angle E + \angle F + \angle BPC + \text{adj angles} = 360^\circ$, which simplifies the total sum of all six angles to:

$$\angle A + \angle B + \angle C + \angle D + \angle E + \angle F = 360^\circ$$

We are given that the sum equals $90n^\circ$:

$$90n = 360 \implies n = 4$$

Thus, n is equal to 4, which corresponds to (C).

22. (AHSME 1972) If $a \pm bi$ ($b \neq 0$) are imaginary roots of the equation $x^3 + qx + r = 0$ where a , b , q , and r are real numbers, then q in terms of a and b is

- (A) $a^2 + b^2$ (B) $2a^2 - b^2$ (C) $b^2 - a^2$ (D) $b^2 - 2a^2$ (E) $b^2 - 3a^2$

Answer: E

Solution

We have that $x^3 + qx + r$ is a cubic, indicating that it has 3 roots. Two of these roots is given to you, namely $a \pm bi$, where $i = \sqrt{-1}$ so call the third root ω .

We then have the polynomial $x^3 + qx + r$ to be equivalent in the form

$$(x - (a + bi))(x - (a - bi))(x - \omega)$$

Which can be simplified in the associative property into

$$((x - a) + bi)((x - a) - bi)(x - \omega)$$

In which if we set α as $x - a$ and β as bi we can use difference of squares to get

$$\begin{aligned} &\implies (\alpha + \beta)(\alpha - \beta)(x - \omega) \\ &\implies (\alpha^2 - \beta^2)(x - \omega) \\ &\text{substitution of } \alpha \text{ and } \beta \\ &\implies ((x - a)^2 - (bi)^2)(x - \omega) \\ &\implies (x^2 - 2ax + a^2 + b^2)(x - \omega) \end{aligned}$$

Expanding gives us

$$\implies x^2(x - \omega) - 2ax(x - \omega) + a^2(x - \omega) + b^2(x - \omega)$$

in which simplification and rearranging gives us

$$x^3 - x^2\omega - 2ax^2 + 2ax\omega + a^2x + b^2x - a^2\omega - b^2\omega$$

Notice how there exists no x^2 term, indicating that the coefficients of the x^2 term must equal 0. Consequently, we get

$$(-\omega - 2a)x^2 = 0x^2$$

$$-\omega - 2a = 0$$

$$-2a = \omega$$

Additionally, notice how q is the coefficient of the x term, so we have

$$2ax\omega + a^2x + b^2x = qx$$

$$(2a\omega + a^2 + b^2)x = qx$$

$$2a\omega + a^2 + b^2 = q$$

using definition of ω

$$2a(-2a) + a^2 + b^2 = q$$

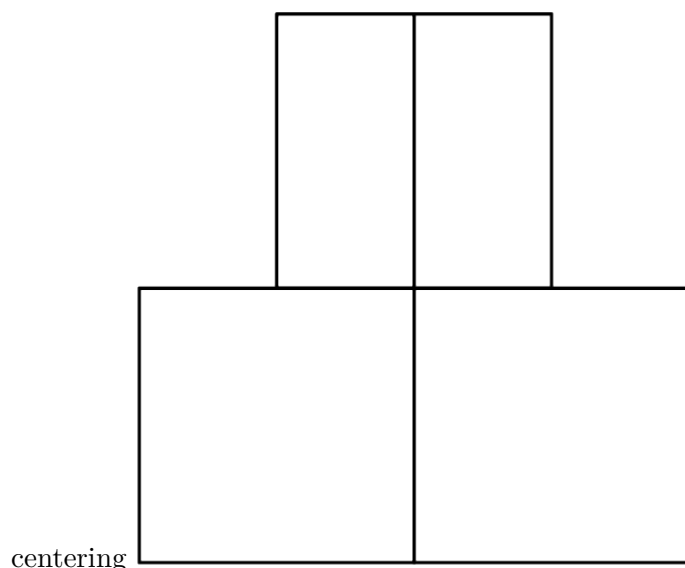
$$-4a^2 + a^2 + b^2 = q$$

$$-3a^2 + b^2 = q$$

$$b^2 - 3a^2 = q$$

Our answer is clearly **(E)** $b^2 - 3a^2$

23. (AHSME 1972)

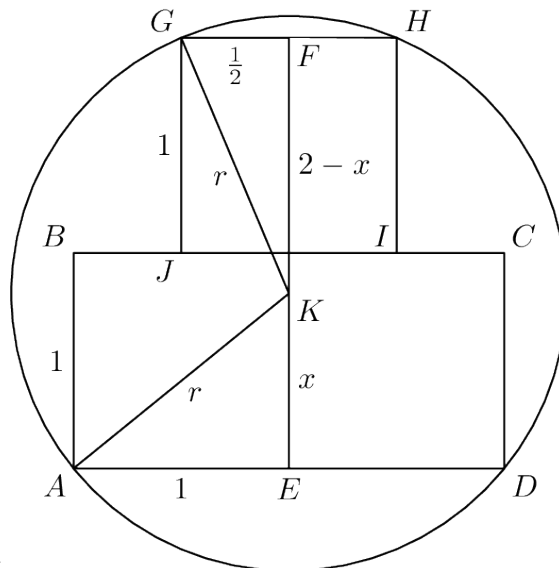


The radius of the smallest circle containing the symmetric figure composed of the 3 unit squares shown above is

- (A) $\sqrt{2}$ (B) $\sqrt{1.25}$ (C) 1.25 (D) $\frac{5\sqrt{17}}{16}$ (E) None of these

Answer: D

Solution 1



Draw lines from points A and G to the center of the circle K . As shown from the diagram, r is the radius of the circumscribed circle. Set $\overline{EK}$ to x , so as $\overline{EF}$ is $1 + 1 = 2$, $\overline{FK}$ is $2 - x$. Then, $\triangle FGK$ and $\triangle AEK$ are both right, so an equation can be formed with the Pythagorean Theorem. The radii (hypotenuses) are equal, so the following equation can be made and solved:

$$\left(\frac{1}{2}\right)^2 + (2 - x)^2 = 1^2 + x^2$$

$$\begin{aligned}\frac{1}{4} + (-x + 2)^2 &= 1 + x^2 \\ \frac{1}{4} + x^2 - 4x + 4 &= 1 + x^2 \\ \frac{1}{4} - 4x + 4 &= 1 \\ -4x &= -3 - \frac{1}{4} \\ 4x &= \frac{13}{4} \\ x &= \frac{13}{16}\end{aligned}$$

x is solved, so plugging it in to the original formula yields:

$$\begin{aligned}&\sqrt{1^2 + \frac{13^2}{16}} \\ &= \sqrt{1 + \frac{169}{256}} \\ &= \sqrt{\frac{425}{256}} \\ &= \frac{\sqrt{425}}{\sqrt{256}} \\ &= \frac{5\sqrt{17}}{16} \\ &= \boxed{\text{(D)}}.\end{aligned}$$

airbus-a321

24. (*AHSME 1972*) A man walked a certain distance at a constant rate. If he had gone $\frac{1}{2}$ mile per hour faster, he would have walked the distance in four-fifths of the time; if he had gone $\frac{1}{2}$ mile per hour slower, he would have been $2\frac{1}{2}$ hours longer on the road. The distance in miles he walked was
- (A) $13\frac{1}{2}$ (B) 15 (C) $17\frac{1}{2}$ (D) 20 (E) 25

Answer: B

Solution

We can make three equations out of the information, and since the distances are the same, we can equate these equations.

$$\frac{4t}{5}\left(x + \frac{1}{2}\right) = xt = \left(t + \frac{5}{2}\right)\left(x - \frac{1}{2}\right)$$

where x is the man's rate and t is the time it takes him.

Looking at the first two parts of the equations,

$$\frac{4t}{5}\left(x + \frac{1}{2}\right) = xt$$

we note that we can solve for x . Solving for x , we get $x = 2$.

Now we look at the last two parts of the equation:

$$xt = (t + \frac{5}{2})(x - \frac{1}{2})$$

we note that we can solve for t and we get $t = \frac{15}{2}$. We want the find the distance, which is $xt = \boxed{15}$.

-edited for readability

25. (AHSME 1972) Inscribed in a circle is a quadrilateral having sides of lengths 25, 39, 52, and 60 taken consecutively. The diameter of this circle has length
(A) 62 (B) 63 (C) 65 (D) 66 (E) 69

Answer: C

Solution

We note that $25^2 + 60^2 = 65^2$ and $39^2 + 52^2 = 65^2$ so our answer is $\boxed{C}$.

-Pleaseletmewin

Alternate Solution:

Let's call $\overline{AB} = 25$, $\overline{BC} = 39$, $\overline{CD} = 52$, $\overline{DA} = 60$. Let's call $\overline{BD} = x$ and $\angle DAB = y$. By LoC we get the relations

$$\begin{aligned} x^2 &= 25^2 + 60^2 - 3000 \cos(y) \\ x^2 &= 39^2 + 52^2 - 4056 \cos(180 - y) \end{aligned}$$

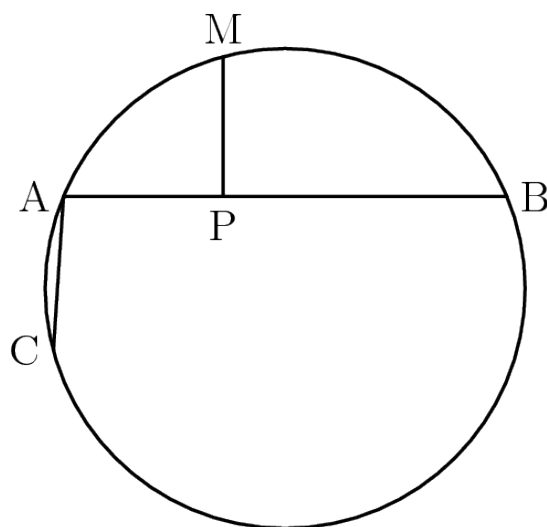
If we do a bit of computation we get $x^2 = 4225 - 3000 \cos(y)$, and $x^2 = 4225 - 4056 \cos(y)$. This means that $4056 \cos(y) = 3000 \cos(180 - y)$.

We know that $\cos(180 - y) = -\cos(y)$ so substituting back in we get $4056 \cos(y) = -3000 \cos(y)$. We can clearly see that the only solution of this is $\cos(y) = 0$ or $y = 90$. This then means that $\angle BAD = 90$ and $\angle BCD = 90$. If a triangle is a right triangle and is inscribed in a circle then the diameter is the hypotenuse.

This means that the diameter is $\sqrt{4225} = 65$ so our answer is $\boxed{C}$.

-Bole (edited for easier readability)

26. (AHSME 1972)



centering

In the circle above, M is the midpoint of arc CAB and segment MP is perpendicular to chord AB at P . If the measure of chord AC is x and that of segment AP is $(x + 1)$, then segment PB has measure equal to

- (A) $3x + 2$ (B) $3x + 1$ (C) $2x + 3$ (D) $2x + 2$ (E) $2x + 1$

Answer: E

Solution

E

27. (AHSME 1972) If the area of $\triangle ABC$ is 64 square units and the geometric mean (mean proportional) between sides AB and AC is 12 inches, then $\sin A$ is equal to

- (A) $\frac{\sqrt{3}}{2}$ (B) $\frac{3}{5}$ (C) $\frac{4}{5}$ (D) $\frac{8}{9}$ (E) $\frac{15}{17}$

Answer: D

Solution

Draw Diagram later

We can let $AB = s$ and $AC = r$. We can also say that the area of a triangle is $\frac{1}{2}rs \sin A$, which we know, is 64. We also know that the geometric mean of r and s is 12, so $\sqrt{rs} = 12$.

Squaring both sides gives us that $rs = 144$. We can substitute this value into the equation we had earlier. This gives us

$$\frac{1}{2} \times 144 \times \sin A = 64$$

$$72 \sin A = 64$$

$$\sin A = \frac{64}{72} = \frac{8}{9} \Rightarrow \boxed{\text{D}}$$

-edited for readability

28. (AHSME 1972) A circular disc with diameter D is placed on an 8×8 checkerboard with width D so that the centers coincide. The number of checkerboard squares which are completely covered by the disc is

- (A) 48 (B) 44 (C) 40 (D) 36 (E) 32

Answer: E

Solution

Consider the upper right half of the grid, which consists of a 4×4 section of the checkerboard and a quarter-circle of radius 4. We can draw this as a coordinate grid and shade in the complete squares. There are 8 squares in the upper right corner, so there are $8 \cdot 4 = \boxed{32}$ whole squares in total.

The answer is (E).

-edited by coolmath34

29. (AHSME 1972) If $f(x) = \log \left(\frac{1+x}{1-x} \right)$ for $-1 < x < 1$, then $f \left(\frac{3x+x^3}{1+3x^2} \right)$ in terms of $f(x)$ is

- (A) $-f(x)$ (B) $2f(x)$ (C) $3f(x)$ (D) $[f(x)]^2$
 (E) $[f(x)]^3 - f(x)$

Answer: C

Solution

For simplicity, we can write $f(x) = \log\left(\frac{1+x}{1-x}\right)$ as $f(x) = \log(1+x) - \log(1-x)$ via the divisibility log property. From here we can plug in $\frac{3x+x^3}{1+3x^2}$ for x which becomes:

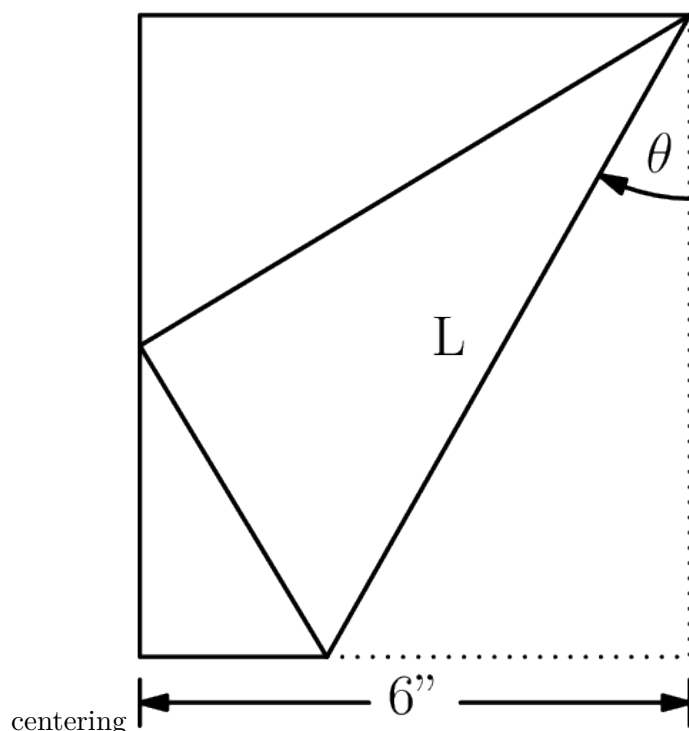
$$\begin{aligned} \log\left(1 + \frac{3x+x^3}{1+3x^2}\right) - \log\left(1 - \frac{3x+x^3}{1+3x^2}\right) &\implies \\ \log\left(\frac{1+3x^2}{1+3x^2} + \frac{3x+x^3}{1+3x^2}\right) - \log\left(\frac{1+3x^2}{1+3x^2} - \frac{3x+x^3}{1+3x^2}\right) &\implies \\ \log\left(\frac{x^3+3x^2+3x+1}{1+3x^2}\right) - \log\left(\frac{-x^3+3x^2-3x+1}{1+3x^2}\right) &\implies \\ \log\left(\frac{(x+1)^3}{1+3x^2}\right) - \log\left(\frac{(-x+1)^3}{1+3x^2}\right) \end{aligned}$$

Now, if we bring the two "log" terms back together ($f(x) = \log\left(\frac{\frac{(x+1)^3}{3x^2+1}}{\frac{(-x+1)^3}{3x^2+1}}\right)$), it becomes apparent that the two denominators cancel out, making for:

$$f(x) = \log\left(\frac{(x+1)^3}{(-x+1)^3}\right) \implies$$

$f(x) = 3\log\left(\frac{1+x}{1-x}\right)$ via the power law. We can then identify that $\frac{1+x}{1-x}$ is $f(x)$, and by substitution, the answer is $3f(x)$ or **(C)**

30. (AHSME 1972)

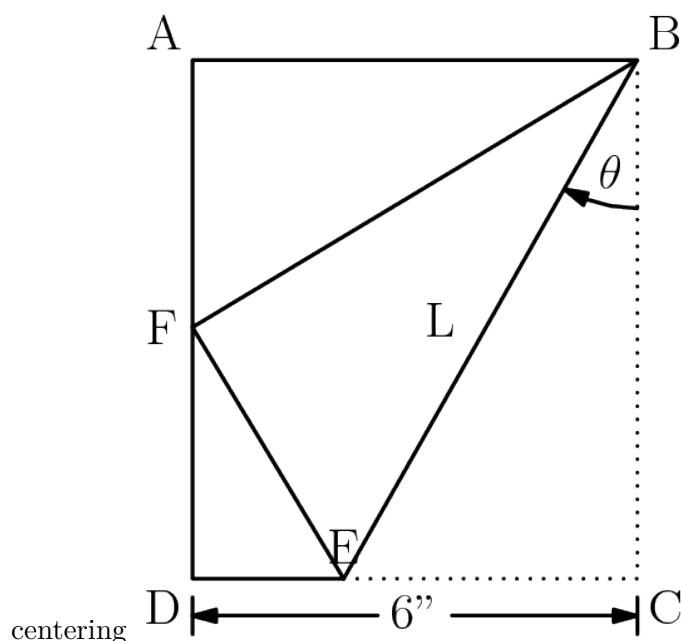


A rectangular piece of paper 6 inches wide is folded as in the diagram so that one corner touches the opposite side. The length in inches of the crease L in terms of angle θ is

- (A) $3 \sec^2 \theta \csc \theta$ (B) $6 \sin \theta \sec \theta$ (C) $3 \sec \theta \csc \theta$ (D) $6 \sec \theta \csc^2 \theta$ (E) None of these

Answer: A

Solution



Let the rectangle $ABCD$ have crease BE with E on CD , and let F be on AD such that F is a reflection of C over BE . Notice that triangles ABF and DEF are similar, so by setting $CE = EF = x$ with $DE = 6 - x$, giving $DF = 2\sqrt{3x - 9}$ we have that $AF = \frac{18-3x}{\sqrt{3x-9}}$. Noticing that $BC = BF = x \cot \theta$ gives $\frac{(18-3x)^2}{3x-9} + 36 = x^2 \cot^2 \theta \Rightarrow \frac{3(x-6)^2}{x-3} + 36 = \frac{3x^2}{x-3} = x^2 \cot^2 \theta$. $x = \frac{3 \cot^2 \theta + 3}{\cot^2 \theta} = \frac{3 \csc^2 \theta}{\cot^2 \theta} = 3 \sec^2 \theta$. Noticing that $BE = x \sqrt{\cot^2 \theta + 1} = x \csc \theta$ gives the answer to be $3 \sec^2 \theta \csc \theta$.

Thanks to xiej for correcting my mistake!

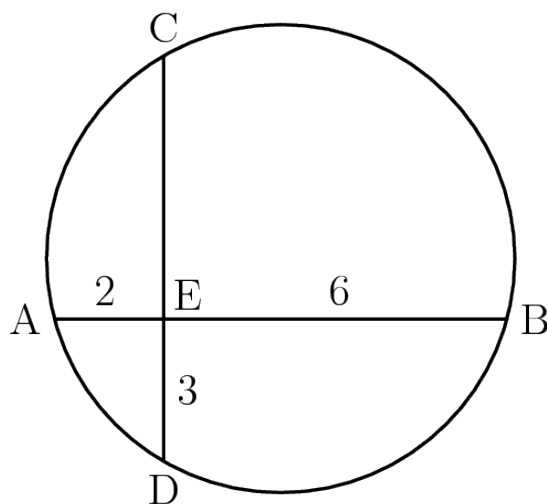
31. (AHSME 1972) When the number 2^{1000} is divided by 13, the remainder in the division is
 (A) 1 (B) 2 (C) 3 (D) 7 (E) 11

Answer: C

Solution

By **Fermat's Little Theorem**, we know that $2^{1000} \equiv 2^{1000 \pmod{12}} \pmod{13}$. However, we find that $1000 \equiv 4 \pmod{12}$, so $2^{1000} \equiv 2^4 = 16 \equiv 3 \pmod{13}$, so the answer is (C).

32. (AHSME 1972)



centering

Chords AB and CD in the circle above intersect at E and are perpendicular to each other. If segments AE , EB , and ED have measures 2, 3, and 6 respectively, then the length of the diameter of the circle is

- (A) $4\sqrt{5}$ (B) $\sqrt{65}$ (C) $2\sqrt{17}$ (D) $3\sqrt{7}$ (E) $6\sqrt{2}$

Answer: B

Solution

Using the chord theorem we can immediately solve for $\overline{CE}$ which will help us. The chord theorem states that if two chords, $\overline{AB}$ and $\overline{CD}$, intersect at lets say E , then $\overline{AE} \cdot \overline{EB} = \overline{CE} \cdot \overline{ED}$. Now that we know this we can label length $\overline{CE}$ as η . We now have the equation $3\eta = 2 \cdot 6 \implies 3\eta = 12 \implies \eta = 4$. We now know the chord lengths 7 and 8. We now label the center of the circle as O . If the circle was on the graph, with O being right on the origin, then make a $x = 0$ line cutting through the center. This bisects chord AB at point M . This means $\overline{AM} = 4$ and $\overline{EM} = 2$. If we now draw a " $y = 0$ " line which bisects chord CD at point N making $\overline{CN} = \frac{7}{2}$. If we now move up $\overline{EM}$ to $\overline{ON}$ then we can find the radius($\overline{CO}$) using Pythagorean Theorem.

$$\sqrt{\overline{EM}^2 + \overline{ON}^2} = r$$

$$\sqrt{2^2 + \left(\frac{7}{2}\right)^2}$$

$$\sqrt{4 + \frac{49}{4}}$$

$$\sqrt{\frac{65}{4}}$$

$$\frac{\sqrt{65}}{2} = r$$

Now that we have r , the diameter is just two times of that, so (B) $\sqrt{65}$

33. (AHSME 1972) The minimum value of the quotient of a (base ten) number of three different non-zero digits divided by the sum of its digits is

- (A) 9.7 (B) 10.1 (C) 10.5 (D) 10.9 (E) 20.5

Answer: C

Solution

The answer we are looking for can be expressed as $\frac{100a + 10b + c}{a + b + c}$. This is equivalent to $1 + \frac{99a + 9b}{a + b + c}$. Because we are trying to minimize our solution, we set $c = 9$, so we have $1 + \frac{99a + 9b}{a + b + 9}$. This is equal to $1 + \frac{9a + 9b + 81}{a + b + 9} + \frac{90a - 81}{a + b + 9}$, which simplifies to $10 + \frac{90a - 81}{a + b + 9}$. Since each digit is unique, we set b to 8, leaving us with $10 + \frac{90a - 81}{a + 17}$. Clearly, a should be minimized, so $a = 1$ and our answer is

(C) 10.5.

34. (AHSME 1972) Three times Dick's age plus Tom's age equals twice Harry's age. Double the cube of Harry's age is equal to three times the cube of Dick's age added to the cube of Tom's age. Their respective ages are relatively prime to each other. The sum of the squares of their ages is

(A) 42 (B) 46 (C) 122 (D) 290 (E) 326

Answer: A

Solution

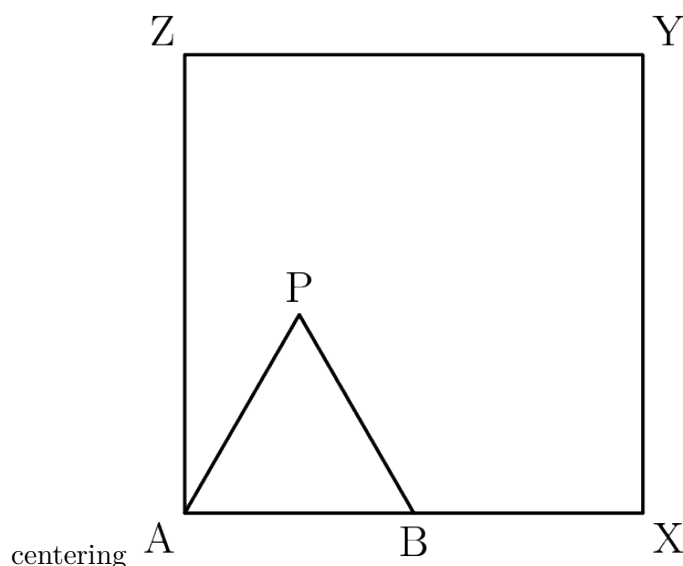
$$t = 2h - 3d$$

$$3d^3 + t^3 = 2h^3$$

First, substitute in t into the second equation and get $3d^3 + 8h^3 - 36h^2d + 54hd^2 - 27d^3 = 2h^3$. That turns into $h^3 - 6h^2d + 9hd^2 - 4d^3 = 0$ which is factored into $(h - 4d)(h - d)^2 = 0$. WLOG, $d = 1$ and consequently $h = 4$. Then $t = 8 - 3 = 5$. Everything appears to be relatively prime already. The answer is thus $1 + 16 + 25 =$

(A) 42. lopkiloim

35. (AHSME 1972)



Equilateral triangle ABP (see figure) with side AB of length 2 inches is placed inside square $AXYZ$ with side of length 4 inches so that B is on side AX . The

triangle is rotated clockwise about B , then P , and so on along the sides of the square until P returns to its original position. The length of the path in inches traversed by vertex P is equal to

- (A) $20\pi/3$ (B) $32\pi/3$ (C) 12π (D) $40\pi/3$ (E) 15π

Answer: D

Solution

D

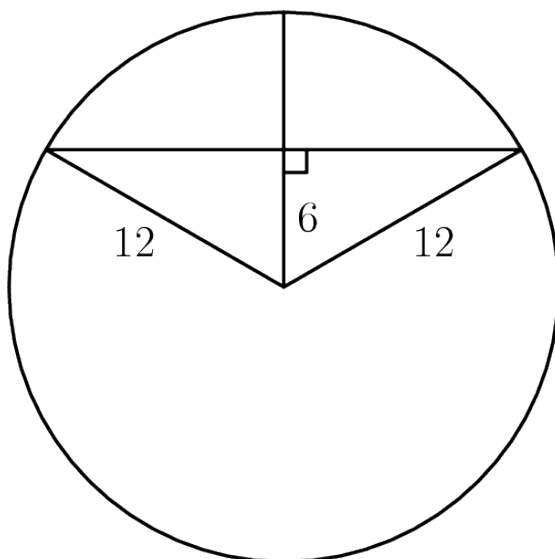
§9 AHSME 1973

1. (AHSME 1973) A chord which is the perpendicular bisector of a radius of length 12 in a circle, has length

- (A) $3\sqrt{3}$ (B) 27 (C) $6\sqrt{3}$ (D) $12\sqrt{3}$ (E) none of these

Answer: D

Solution



centering

Draw a diagram as shown. Using the **Pythagorean Theorem** (or by using 30-60-90 triangles), half of the chord length is $6\sqrt{3}$, so the chord's length is **(D) $12\sqrt{3}$** . One can also find the length directly by using the **Law of Cosines**.

2. (AHSME 1973) One thousand unit cubes are fastened together to form a large cube with edge length 10 units; this is painted and then separated into the original cubes. The number of these unit cubes which have at least one face painted is

- (A) 600 (B) 520 (C) 488 (D) 480 (E) 400

Answer: C

Solution 1

The total number of cubes is 10^3 or 1000. Because each surface of the large cube is one cube deep, the number of the unpainted cubes is $8^3 = 512$, since we subtract two from the side lengths of the cube itself, and cube it to find the volume of

that cube. So there are $1000 - 512 = \boxed{\text{(C)} 488}$ cubes that have at least one face painted.

Solution 2

Each face has 100 cubes, so multiply by six to get 600. However, we overcounted each small cube on the edge but not on corner of the big cube once and each small cube on the corner of the big cube twice. Thus, there are $600 - (12 \cdot 8 + 2 \cdot 8) = \boxed{\text{(C)} 488}$ cubes that have at least one face painted.

3. (*AHSME 1973*) The stronger Goldbach conjecture states that any even integer greater than 7 can be written as the sum of two different prime numbers. For such representations of the even number 126, the largest possible difference between the two primes is
- (A) 112 (B) 100 (C) 92 (D) 88 (E) 80

Answer: B

Solution (Trial and Error)

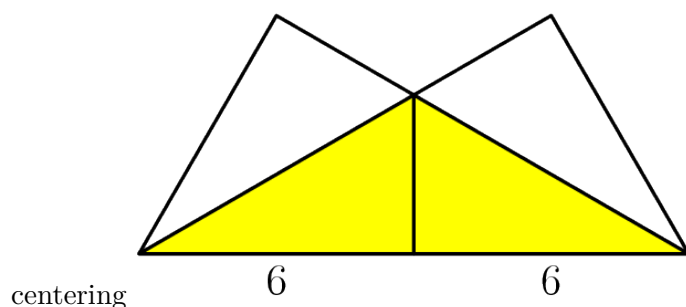
We can guess and check small primes, subtract it from 126, and see if the result is a prime because the further away the two numbers are, the greater the difference will be. Since $126 = 2 \cdot 3^2 \cdot 7$, we can eliminate 2, 3, and 7 as an option because subtracting these would result in a composite number.

If we subtract 5, then the resulting number is 121, which is not prime. If we subtract 11, then the resulting number is 115, which is also not prime. But when we subtract 13, the resulting number is 113, a prime number. The largest possible difference is $113 - 13 = \boxed{\text{(B)} 100}$.

4. (*AHSME 1973*) Two congruent 30-60-90 are placed so that they overlap partly and their hypotenuses coincide. If the hypotenuse of each triangle is 12, the area common to both triangles is
- (A) $6\sqrt{3}$ (B) $8\sqrt{3}$ (C) $9\sqrt{3}$ (D) $12\sqrt{3}$ (E) 24

Answer: D

Solution



Note that the altitude of the shared region bisects the hypotenuse of the original two right triangles (this can be confirmed by using AAS on the two right triangles with that altitude of the shared region). By using 30-60-90 triangles, the altitude's length is $2\sqrt{3}$. The area of the shared region is $\frac{1}{2} \cdot 12 \cdot 2\sqrt{3} = \boxed{\text{(D)} 12\sqrt{3}}$.

5. (*AHSME 1973*) Of the following five statements, I to V, about the binary operation of averaging (arithmetic mean),

- I. Averaging is associative
 II. Averaging is commutative
 III. Averaging distributes over addition
 IV. Addition distributes over averaging
 V. Averaging has an identity element

those which are always true are

- (A) All (B) I and II only (C) II and III only
 (D) II and IV only (E) II and V only

Answer: D

Solution

We can consider each statement independently and see which ones are true. The average of two numbers x and y is $avg(x, y) = \frac{x+y}{2}$.

Statement I

$$avg(avg(x, y), z) = avg(x, avg(y, z))$$

$$\frac{\frac{\frac{x+y}{2} + z}{2}}{2} = \frac{x + \frac{\frac{y+z}{2}}{2}}{2}$$

$$\frac{x + y + 2z}{4} = \frac{2x + y + z}{4}$$

The above statement is not true for all (x, y, z) , so Statement I is false.

Statement II

$$avg(x, y) = avg(y, x)$$

$$\frac{x + y}{2} = \frac{y + x}{2}$$

Since addition is commutative, the above is true for all (x, y) , so Statement II is true.

Statement III

$$avg(x, y + z) = avg(x, y) + avg(x, z)$$

$$\frac{x + y + z}{2} = \frac{x + y}{2} + \frac{x + z}{2}$$

$$\frac{x + y + z}{2} = \frac{2x + y + z}{2}$$

The above statement is not true for all (x, y, z) , so Statement III is false.

Statement IV

$$x + avg(y, z) = avg(x + y, x + z)$$

$$x + \frac{y + z}{2} = \frac{2x + y + z}{2}$$

$$\frac{2x + y + z}{2} = \frac{2x + y + z}{2}$$

The left side and the right side are equivalent expressions, so Statement IV is true.

Statement V Let i be the identity element. If such i exists, then the following is true:

$$avg(x, i) = x$$

$$\begin{aligned}\frac{x+i}{2} &= x \\ x+i &= 2x \\ i &= x\end{aligned}$$

However, since x is a variable, but i must be a constant, fixed number, there is no identity element for the binary operation of averaging. Therefore, Statement V is false.

Statements II and IV are true, so the answer is **(D)II and IV only**.

6. (*AHSME 1973*) If 554 is the base b representation of the square of the number whose base b representation is 24, then b , when written in base 10, equals
(A) 6 **(B) 8** **(C) 12** **(D) 14** **(E) 16**

Answer: C

Solution

Write out the numbers using the definition of **base numbers**.

$$554_b = 5b^2 + 5b + 4$$

$$24_b = 2b + 4$$

Since $554_b = (24_b)^2$, we can write an equation.

$$5b^2 + 5b + 4 = (2b + 4)^2$$

$$5b^2 + 5b + 4 = 4b^2 + 16b + 16$$

$$b^2 - 11b - 12 = 0$$

$$(b - 12)(b + 1) = 0$$

Since base numbers must be positive, b in base 10 equals **(C) 12**.

7. (*AHSME 1973*) The sum of all integers between 50 and 350 which end in 1 is
(A) 5880 **(B) 5539** **(C) 5208** **(D) 4877** **(E) 4566**

Answer: A

Solution

The numbers that we are adding are 51, 61, 71, ..., 341. The numbers are part of an arithmetic series with first term 51, last term 341, common difference 10, and 30 terms. Using the arithmetic series formula, the sum of the terms is $\frac{30 \cdot 392}{2} = \mathbf{(A) 5880}$.

8. (*AHSME 1973*) If 1 pint of paint is needed to paint a statue 6 ft. high, then the number of pints it will take to paint (to the same thickness) 540 statues similar to the original but only 1 ft. high is
(A) 90 **(B) 72** **(C) 45** **(D) 30** **(E) 15**

Answer: E

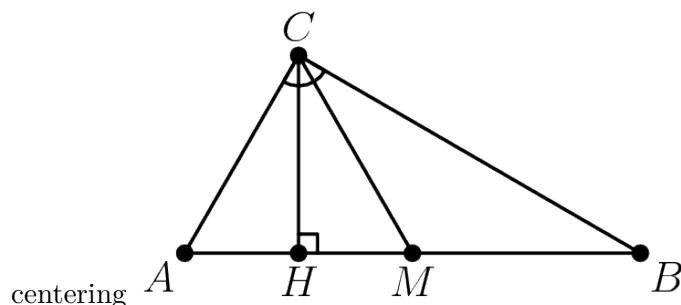
Solution

The statues are similar, and since the height of the smaller statue is $\frac{1}{6}$ of the original statue, the surface area is $\frac{1}{36}$ of the original statue. Thus, $\frac{1}{36}$ pints of paint is needed for one 1 ft. statue, so painting 540 of these statues requires $\frac{540}{36} = \mathbf{(E) 15}$ pints of paint.

9. (AHSME 1973) In $\triangle ABC$ with right angle at C , altitude CH and median CM trisect the right angle. If the area of $\triangle CHM$ is K , then the area of $\triangle ABC$ is
 (A) $6K$ (B) $4\sqrt{3}K$ (C) $3\sqrt{3}K$ (D) $3K$ (E) $4K$

Answer: E

Solution



Draw diagram as shown (note that A and B can be interchanged, but it doesn't change the solution).

Note that because CM is a median, $AM = BM$. Also, by ASA Congruency, $\triangle CHA \cong \triangle CHM$, so $AH = HM$. That means $HM = \frac{1}{4} \cdot AB$, and since $\triangle CHM$ and $\triangle ABC$ share an altitude, $[ABC] = \boxed{\text{(E)} 4K}$.

10. (AHSME 1973) If n is a real number, then the simultaneous system

$$nx + y = 1$$

$$ny + z = 1$$

$$x + nz = 1$$

has no solution if and only if n is equal to

- (A) -1 (B) 0 (C) 1 (D) 0 or 1 (E) $\frac{1}{2}$

Answer: A

Solution 1

Add up all three equations to get

$$(n+1)(x+y+z) = 3$$

If $n = -1$, then equation results in $0 = 3$. That has no solutions, so the answer is $\boxed{\text{(A)}}$.

Solution 2

From the first equation,

$$y = 1 - nx$$

Substitute that in the second equation to get

$$n(1 - nx) + z = 1$$

$$z = 1 - n + n^2x$$

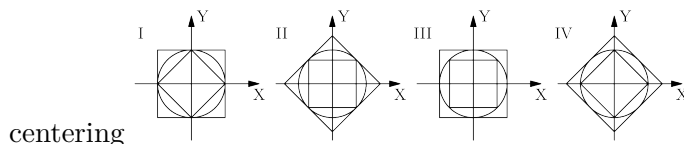
Substitute that in the third equation to get

$$x + n(1 - n + n^2x) = 1$$

$$(1 + n^3)x + n - n^2 = 1$$

If $n = -1$, then equation results in $-2 = 1$. That has no solutions, so the answer is **(A)**.

11. (AHSME 1973) A circle with a circumscribed and an inscribed square centered at the origin of a rectangular coordinate system with positive x and y axes and is shown in each figure to below.



centering

The inequalities

$$|x| + |y| \leq \sqrt{2(x^2 + y^2)} \leq 2\text{Max}(|x|, |y|)$$

are represented geometrically* by the figure numbered

- (A)** I **(B)** II **(C)** III **(D)** IV **(E)** none of these

* An inequality of the form $f(x, y) \leq g(x, y)$, for all x and y is represented geometrically by a figure showing the containment

{The set of points (x, y) such that $g(x, y) \leq a$ } $\subset$
 {The set of points (x, y) such that $f(x, y) \leq a$ }

for a typical real number a .

Answer: B

Solution

First, note that the following inequality

$$|x| + |y| \leq \sqrt{2(x^2 + y^2)} \leq 2\text{Max}(|x|, |y|)$$

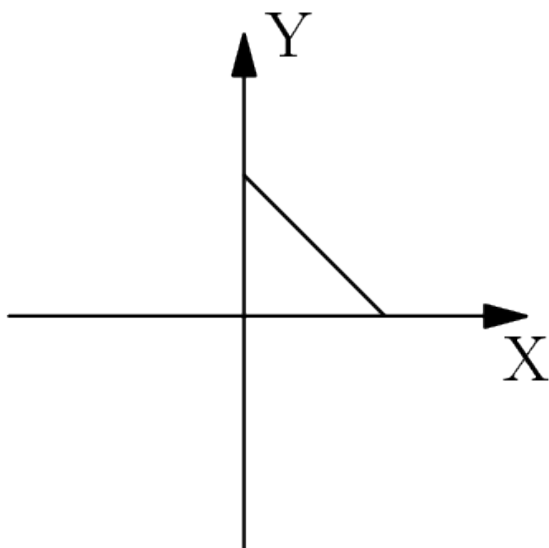
represents the graphs $|x| + |y| = a$, $\sqrt{2(x^2 + y^2)} = a$, and $2\text{Max}(|x|, |y|) = a$. We don't actually have to worry about the inequality, we just want to find the picture that graphs those 3 equations for some constant a .

WLOG, we can set $a = 1$. To make our life easier, we can also focus entirely on the first quadrant, since all 4 graphs have different shapes in the first quadrant. Therefore, $x \geq 0$ and $y \geq 0$.

First, we have $|x| + |y| = 1$. Since $x \geq 0$ and $y \geq 0$, this becomes $x + y = 1$. After a bit of rearranging, we get

$$y = -x + 1$$

.



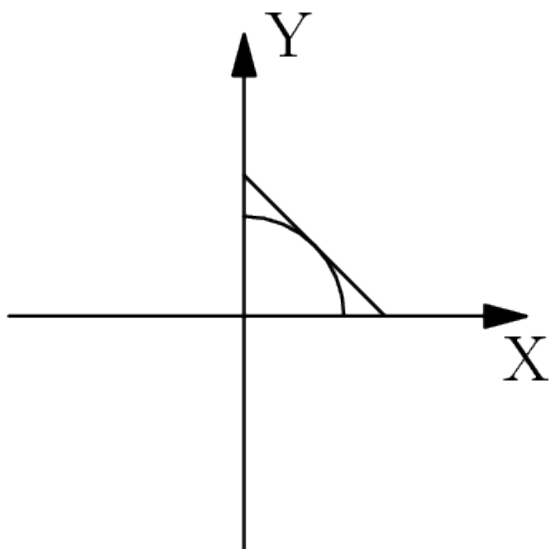
centering

This eliminates *III*, which does not have a diagonal line in the first quadrant.

Next, we have $\sqrt{2(x^2 + y^2)} = 1$. Squaring both sides and dividing by 2, we get

$$x^2 + y^2 = \frac{\sqrt{2}^2}{2}$$

Note that this describes a circle with the center $(0,0)$ and radius $\frac{\sqrt{2}}{2}$. We can find that $(\frac{1}{2}, \frac{1}{2})$ is on the circle, which is also on the line $y = -x + 1$. Therefore, the circle intersects the line at that point. We can graph this as follows:



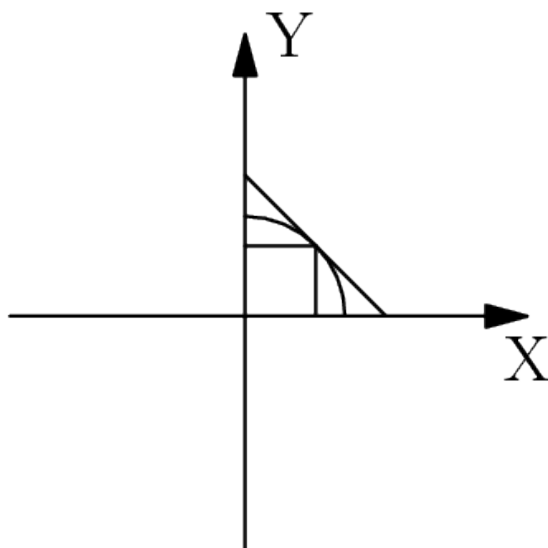
centering

This eliminates *I*, which does not include the intersection point listed above.

Finally, we have $2\text{Max}(|x|, |y|) = 1$. Rearranging and substituting x for $|x|$ and likewise for y , we get

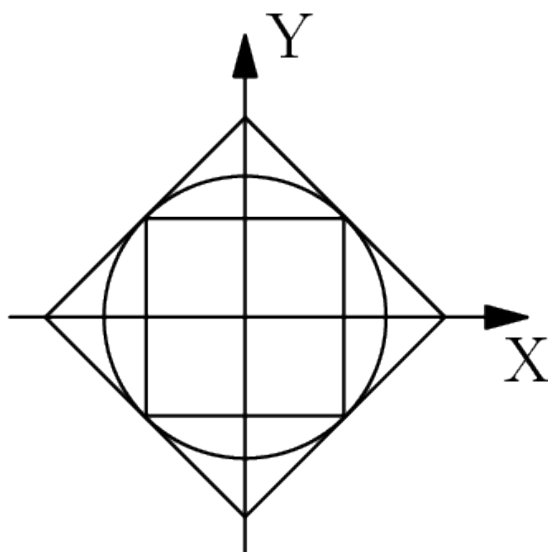
$$\text{Max}(x, y) = \frac{1}{2}$$

Note that this equation means that either $x \leq y = \frac{1}{2}$ or $y \leq x = \frac{1}{2}$. Therefore, from $x = 0$ to $x = \frac{1}{2}$, the graph looks like the line $y = \frac{1}{2}$. Then, from $y = 0$ to $y = \frac{1}{2}$, the graph looks like the line $x = \frac{1}{2}$. Graphing this, we get:



centering

Thus, we can eliminate *IV*. However, at this point, we don't know if the answer is *II* or none of the graphs, since this entire time we were only graphing in the first quadrant to simplify things. However, note that no matter what sign x or y is, the equations all surround the variables with absolute value signs or exponents and make the values positive. Therefore, the graphs are symmetric about the x and y axes. As a result, the final graph looks like this:



centering

Therefore, we can choose **B** as our answer.

12. (AHSME 1973) The average (arithmetic mean) age of a group consisting of doctors and lawyers is 40. If the doctors average 35 and the lawyers 50 years old, then the ratio of the numbers of doctors to the number of lawyers is
- (A) 3 : 2 (B) 3 : 1 (C) 2 : 3 (D) 2 : 1 (E) 1 : 2

Answer: D

Solution

Let d be the number of doctors and l be the number of lawyers. The average age of the doctors is 35, so the sum of the doctors' ages is $35d$. The average age of the lawyers is 50, so the sum of the lawyers' ages is $50l$. With the two information, we can write an equation.

$$\begin{aligned}\frac{35d + 50l}{d + l} &= 40 \\ 35d + 50l &= 40d + 40l \\ 10l &= 5d \\ \frac{d}{l} &= 2\end{aligned}$$

The ratio of the number of doctors to the number of lawyers is (D) 2 : 1.

13. (AHSME 1973) The fraction $\frac{2(\sqrt{2}+\sqrt{6})}{3\sqrt{2+\sqrt{3}}}$ is equal to

(A) $\frac{2\sqrt{2}}{3}$ (B) 1 (C) $\frac{2\sqrt{3}}{3}$ (D) $\frac{4}{3}$ (E) $\frac{16}{9}$

Answer: D

Solution

Squaring the expression and taking the positive square root (since numerator and denominator are positive) of the result yields

$$\begin{aligned}&\sqrt{\left(\frac{2(\sqrt{2} + \sqrt{6})}{3\sqrt{2 + \sqrt{3}}}\right)^2} \\&\sqrt{\frac{4(2 + 4\sqrt{3} + 6)}{9(2 + \sqrt{3})}} \\&\sqrt{\frac{4(8 + 4\sqrt{3})}{9(2 + \sqrt{3})}} \\&\sqrt{\frac{16}{9}} \\&\frac{4}{3}\end{aligned}$$

The answer is (D).

14. (AHSME 1973) Each valve A , B , and C , when open, releases water into a tank at its own constant rate. With all three valves open, the tank fills in 1 hour, with only valves A and C open it takes 1.5 hours, and with only valves B and C open it takes 2 hours. The number of hours required with only valves A and B open is
- (A) 1.1 (B) 1.15 (C) 1.2 (D) 1.25 (E) 1.75

Answer: C

Solution

Let the rate of water flowing through valve A be a , the rate of water flowing through valve B be b , and the rate of water flowing through valve C be c . **WLOG**, let the

volume of the tank be 1 liter, and let the units for the rates be liters per hour. With this information, we can write three equations.

$$\frac{1}{a+b+c} = 1$$

$$\frac{1}{a+c} = \frac{3}{2}$$

$$\frac{1}{b+c} = 2$$

Manipulate each equation to get

$$1 = a + b + c$$

$$\frac{2}{3} = a + c$$

$$\frac{1}{2} = b + c$$

Solving for a yields $a = \frac{1}{2}$, and solving for b yields $b = \frac{1}{3}$. The number of hours to fill the tub with only valves A and B on is $\frac{1}{\frac{1}{2} + \frac{1}{3}} = \frac{6}{5} = \boxed{\text{(C) } 1.2}$.

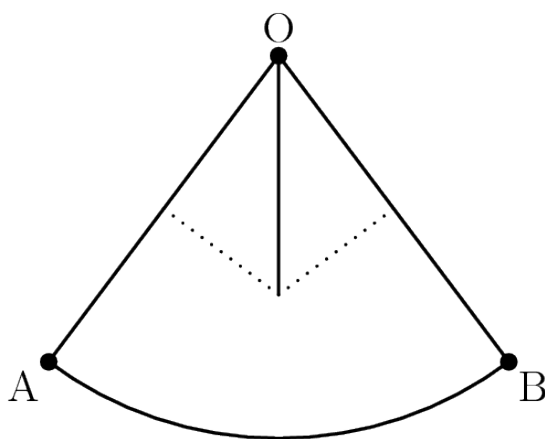
15. (AHSME 1973) A sector with acute central angle θ is cut from a circle of radius 6. The radius of the circle circumscribed about the sector is

(A) $3 \cos \theta$ (B) $3 \sec \theta$ (C) $3 \cos \frac{1}{2}\theta$ (D) $3 \sec \frac{1}{2}\theta$ (E) 3

Answer: D

Solution

Let O be the center of the circle and A, B be two points on the circle such that $\angle AOB = \theta$. If the circle circumscribes the sector, then the circle must circumscribe $\triangle AOB$.



centering

Draw the perpendicular bisectors of OA and OB and mark the intersection as point C , and draw a line from C to O . By HL Congruency and CPCTC, $\angle AOC = \angle BOC = \theta/2$.

Let R be the circumradius of the triangle. Using the definition of cosine for right triangles,

$$\cos(\theta/2) = \frac{3}{R}$$

$$R = \frac{3}{\cos(\theta/2)}$$

$$R = 3 \sec(\theta/2)$$

Answer choices A, C, and E are smaller, so they are eliminated. However, as θ approaches 90° , the value $3 \sec \theta$ would approach infinity while $3 \sec \frac{1}{2}\theta$ would approach $\frac{3\sqrt{2}}{2}$. A super large circle would definitely not be a circumcircle if θ is close to 90° , so we can confirm that the answer is **(D)** $3 \sec \frac{1}{2}\theta$.

16. (AHSME 1973) If the sum of all the angles except one of a convex polygon is 2190° , then the number of sides of the polygon must be

(A) 13 (B) 15 (C) 17 (D) 19 (E) 21

Answer: B

Solution

Let n be the number of sides in the polygon. The number of interior angles in the polygon is $180(n - 2)$. We know that the sum of all but one of them is 2190° , so the sum of all the angles is more than that.

$$180(n - 2) > 2190$$

$$n - 2 > 12\frac{1}{6}$$

$$n > 14\frac{1}{6}$$

The sum of the angles in a 15-sided polygon is 2340° , making the remaining angle 150° . The angles of a convex polygon are all less than 180° , and since adding one more side means adding 180° to the measure of the remaining angle, we can confirm that there are **(B)** 15 sides in the polygon.

17. (AHSME 1973) If θ is an acute angle and $\sin \frac{1}{2}\theta = \sqrt{\frac{x-1}{2x}}$, then $\tan \theta$ equals

(A) x (B) $\frac{1}{x}$ (C) $\frac{\sqrt{x-1}}{x+1}$ (D) $\frac{\sqrt{x^2-1}}{x}$ (E) $\sqrt{x^2-1}$

Answer: E

Solution

Since $\frac{\theta}{2}$ is acute and $\cos \frac{\theta}{2} = \sqrt{1 - \sin^2(\frac{\theta}{2})}$,

$$\cos \frac{\theta}{2} = \sqrt{1 - \left(\frac{x-1}{2x}\right)}$$

$$\cos \frac{\theta}{2} = \sqrt{\frac{x+1}{2x}}$$

Using the definition of tangent,

$$\tan \frac{\theta}{2} = \sqrt{\frac{x-1}{2x}} \div \sqrt{\frac{x+1}{2x}}$$

$$\tan \frac{\theta}{2} = \sqrt{\frac{x-1}{x+1}}$$

Finally, by using the Double Angle Identity for Tangent,

$$\tan \theta = \frac{2 \cdot \sqrt{\frac{x-1}{x+1}}}{1 - \frac{x-1}{x+1}}$$

$$\tan \theta = \frac{\frac{2\sqrt{x^2-1}}{x+1}}{\frac{2}{x+1}}$$

$$\tan \theta = \sqrt{x^2 - 1}$$

The answer is **(E)**.

18. (AHSME 1973) If $p \geq 5$ is a prime number, then 24 divides $p^2 - 1$ without remainder

(A) never (B) sometimes only (C) always
(D) only if $p = 5$ (E) none of these

Answer: C

Solution

Starting with some experimentation, substituting $p = 5$ results in 24, substituting $p = 7$ results in 48, and substituting $p = 11$ results in 120. For these primes, the resulting numbers are multiples of 24.

To show that all primes we devise the following proof:

Proof: $p \geq 5$ result in $p^2 - 1$ being a multiple of 24, we can use modular arithmetic. Note that $p^2 - 1 = (p + 1)(p - 1)$. Since $p^2 \equiv 0, 1, 2, 3 \pmod{4}$, $p^2 - 1$ is a multiple of 8. Also, since $p^2 \equiv 0, 1, 2 \pmod{3}$, $p^2 - 1$ is a multiple of 3. Thus, $p^2 - 1$ is a multiple of 24, so the answer is **(C)**.

19. (AHSME 1973) Define $n_a!$ for n and a positive to be

$$n_a! = n(n - a)(n - 2a)(n - 3a) \dots (n - ka)$$

where k is the greatest integer for which $n > ka$. Then the quotient $72_8! / 18_2!$ is equal to

(A) 4^5 (B) 4^6 (C) 4^8 (D) 4^9 (E) 4^{12}

Answer: D

Solution

Using the definition of $n_a!$, the quotient can be rewritten as

$$\frac{72 \cdot 64 \cdot 56 \cdots 8}{18 \cdot 16 \cdot 14 \cdots 2}$$

Note that for a given integer x , $\frac{72-8x}{18-2x} = 4$. Since $0 \leq x \leq 8$, the quotient simplifies to **(D)** 4^9 .

20. (AHSME 1973) A cowboy is 4 miles south of a stream which flows due east. He is also 8 miles west and 7 miles north of his cabin. He wishes to water his horse at the stream and return home. The shortest distance (in miles) he can travel and accomplish this is

(A) $4 + \sqrt{185}$ (B) 16 (C) 17 (D) 18 (E) $\sqrt{32} + \sqrt{137}$

Answer: C

Solution

First, you draw a reflection of the cowboy across the river. Then, you draw the straight line from the "cowboy" to his cabin. This will be a 8, 15, 17 Pythagorean triple, so the answer is 17, which is **(C)**.

21. (*AHSME 1973*) The number of sets of two or more consecutive positive integers whose sum is 100 is

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Answer: B

Solution

If the first number of a group of n consecutive numbers is a , the n^{th} number is $a + n - 1$. We know that the sum of the group of numbers is 100, so

$$\frac{n(2a + n - 1)}{2} = 100$$

$$2a + n - 1 = \frac{200}{n}$$

$$2a = 1 - n + \frac{200}{n}$$

We know that n and a are positive integers, so we check values of n that are a factor of 200. Of these values, the only ones that result in a positive integer a is when $n = 5$ or when $n = 8$, so there are **(B) 2** sets of two or more consecutive positive integers whose sum is 100.

22. (*AHSME 1973*) The set of all real solutions of the inequality

$$|x - 1| + |x + 2| < 5$$

is

(A) $x \in (-3, 2)$ (B) $x \in (-1, 2)$ (C) $x \in (-2, 1)$
 (D) $x \in \left(-\frac{3}{2}, \frac{7}{2}\right)$ (E) $\emptyset$ (empty)

[Note: I updated the notation on this problem.]

Answer: A

Solution

We can do casework upon the value of x . First, consider the case where both absolute values are positive, which is when $x \geq 1$. In this case, the equation becomes $2x + 1 < 5$. This turns into $x < 2$. Combining this with our original assumption, we get the solutions $1 \leq x < 2$.

Next, consider the case when both absolute values are negative, which is when $x < -2$. This yields $-1 - 2x < 5$, or $x > -3$. Combining this with our original assumption, we get $-3 < x < -2$.

The next case is when the first absolute value is positive and the second is negative. This occurs when $x \geq 1$ and $x < -2$. Obviously, this has no solutions, since the inequalities do not overlap.

The final case is when the first absolute value is negative and the second is positive. This occurs when $x < 1$ and $x \geq -2$. This yields $3 < 5$, which is always true. Therefore, we also get the solutions $-2 \leq x < 1$.

Therefore, after combining all of our solutions, we get the range $-3 < x < 2$, which is **(A)**.

23. (*AHSME 1973*) There are two cards; one is red on both sides and the other is red on one side and blue on the other. The cards have the same probability ($1/2$) of being chosen, and one is chosen and placed on the table. If the upper side of the card on the table is red, then the probability that the under-side is also red is

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$ (E) $\frac{3}{4}$

Answer: D

Solution

There are three red faces, and two are on the card that is completely red, so our answer is $\frac{2}{3}$, which is **(D)**.

24. (*AHSME 1973*) The check for a luncheon of 3 sandwiches, 7 cups of coffee and one piece of pie came to \$3.15. The check for a luncheon consisting of 4 sandwiches, 10 cups of coffee and one piece of pie came to \$4.20 at the same place. The cost of a luncheon consisting of one sandwich, one cup of coffee, and one piece of pie at the same place will come to

(A) \$1.70 (B) \$1.65 (C) \$1.20 (D) \$1.05 (E) \$0.95

Answer: D

Solution

Let s be the cost of one sandwich, c be the cost of one cup of coffee, and p be the price of one piece of pie. With the information,

$$3s + 7c + p = 3.15$$

$$4s + 10c + p = 4.20$$

Subtract the first equation from the second to get

$$s + 3c = 1.05$$

That means $s = 1.05 - 3c$. Substituting it back in the second equation results in.

$$4.20 - 12c + 10c + p = 4.20$$

Solving for p yields $p = 2c$. With the substitutions, the cost of one sandwich, one cup of coffee, and one slice of pie is $(1.05 - 3c) + c + (2c) = \mathbf{(D) \$1.05}$.

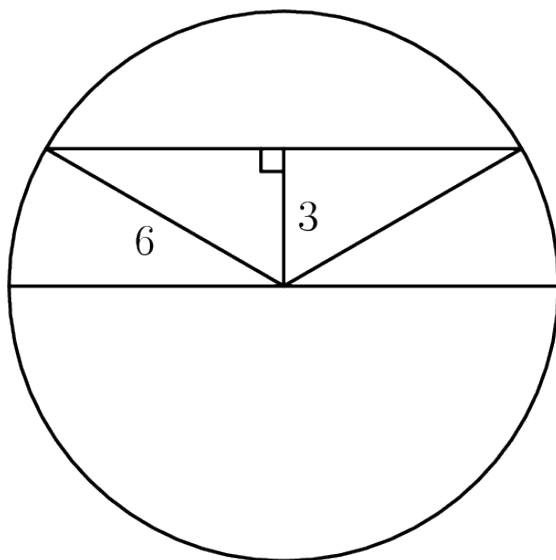
25. (*AHSME 1973*) A circular grass plot 12 feet in diameter is cut by a straight gravel path 3 feet wide, one edge of which passes through the center of the plot. The number of square feet in the remaining grass area is

(A) $36\pi - 34$ (B) $30\pi - 15$ (C) $36\pi - 33$

(D) $35\pi - 9\sqrt{3}$ (E) $30\pi - 9\sqrt{3}$

Answer: E

Solution



centering

The wanted area can be divided into a semicircle and a circular segment. The area of the semicircle is $\frac{1}{2} \cdot \pi \cdot 6^2 = 18\pi$ square feet. To find the area of the circular segment, note that by using 30-60-90 triangles (or **Law of Cosines**), the chord's length is $6\sqrt{3}$ ft, and the angle of the sector is 120° . That means the area of the circular segment is $(\frac{1}{3} \cdot \pi \cdot 6^2) - (\frac{1}{2} \cdot 6\sqrt{3} \cdot 3) = 12\pi - 9\sqrt{3}$ square feet. Thus, the area of the remaining grassy area is **(E)** $30\pi - 9\sqrt{3}$ square feet.

26. (*AHSME 1973*) The number of terms in an A.P. (Arithmetic Progression) is even. The sum of the odd and even-numbered terms are 24 and 30, respectively. If the last term exceeds the first by 10.5, the number of terms in the A.P. is
- (A) 20 (B) 18 (C) 12 (D) 10 (E) 8

Answer: E

Solution

Let a be the first term, n be the number of terms, and d be the common difference. That means the last term is $a + r(n - 1)$.

We can write an equation on the difference between the last and first term based on the conditions.

$$a + r(n - 1) - a = 10.5$$

$$rn - r = 10.5$$

Also, half of the terms add up to 24 while the other half of the terms add up to 30, so

$$24 + r\frac{n}{2} = 30$$

$$nr = 12$$

Substituting the value back to a previous equation,

$$12 - r = 10.5$$

$$r = 1.5$$

Substituting to a previous equation again,

$$1.5n - 1.5 = 10.5$$

$$n = 8$$

Thus, there are (E) 8 terms in the arithmetic sequence.

27. (*AHSME 1973*) Cars A and B travel the same distance. Car A travels half that distance at u miles per hour and half at v miles per hour. Car B travels half the time at u miles per hour and half at v miles per hour. The average speed of Car A is x miles per hour and that of Car B is y miles per hour. Then we always have

(A) $x \leq y$ (B) $x \geq y$ (C) $x = y$ (D) $x < y$ (E) $x > y$

Answer: A

Solution

Let t be the total number of time in hours that Car B took to drive the distance. This means that for half the time Car B traveled $\frac{ut}{2}$ miles and for half the time Car B traveled $\frac{vt}{2}$ miles. That means the total distance traveled is $\frac{ut+vt}{2}$ miles, so

$$y = \frac{\frac{ut+vt}{2}}{t}$$

$$y = \frac{u+v}{2}$$

Because both cars traveled the same distance, for half the distance Car A took $\frac{ut+vt}{4u}$ hours and for half the distance Car A took $\frac{ut+vt}{4v}$ hours. That means

$$x = \frac{\frac{ut+vt}{2}}{\frac{ut+vt}{4u} + \frac{ut+vt}{4v}}$$

$$x = \frac{2}{\frac{1}{u} + \frac{1}{v}}$$

In order to compare x and y , we need to compare the values that x and y are equal to.

$$\frac{2}{\frac{1}{u} + \frac{1}{v}} \bigcirc \frac{u+v}{2}$$

Since the denominators of both numbers are positive, cross-multiplying won't change the comparison sign.

$$4 \bigcirc 2 + \frac{u}{v} + \frac{v}{u}$$

$$2 \bigcirc \frac{u}{v} + \frac{v}{u}$$

Because uv is positive, the comparison sign does not need to be changed either.

$$2uv \bigcirc u^2 + v^2$$

$$0 \bigcirc (u-v)^2$$

By the **Trivial Inequality**, $0 \leq (u-v)^2$. All of the steps are reversible, so (A) $x \leq y$. This can be confirmed by testing values of u and v .

28. (AHSME 1973) If a , b , and c are in geometric progression (G.P.) with $1 < a < b < c$ and $n > 1$ is an integer, then $\log_a n$, $\log_b n$, $\log_c n$ form a sequence
- (A) which is a G.P
- (B) which is an arithmetic progression (A.P)
- (C) in which the reciprocals of the terms form an A.P
- (D) in which the second and third terms are the n th powers of the first and second respectively
- (E) none of these

Answer: C

Solution

Using the change of base formula, the three logarithmic terms can be written as

$$\frac{\log n}{\log a}, \frac{\log n}{\log b}, \frac{\log n}{\log c}$$

Since a , b , and c are members of a geometric sequence, $b = ar$ and $c = ar^2$. That means the three logarithmic terms can be rewritten as

$$\frac{\log n}{\log a}, \frac{\log n}{\log ar}, \frac{\log n}{\log ar^2}$$

$$\frac{\log n}{\log a}, \frac{\log n}{\log a + \log r}, \frac{\log n}{\log a + 2\log r}$$

Note that if we take the reciprocals of each term, the next term can be derived from the previous term by adding $\frac{\log r}{\log n}$, so the answer is **(C)**.

29. (AHSME 1973) Two boys start moving from the same point A on a circular track but in opposite directions. Their speeds are 5 ft. per second and 9 ft. per second. If they start at the same time and finish when they first meet at the point A again, then the number of times they meet, excluding the start and finish, is
- (A) 13 (B) 25 (C) 44 (D) infinity (E) none of these

Answer: A

Solution

Let d be the length of the track in feet and x be the number of laps that one of the boys did, so time one of the boys traveled before the two finish is $\frac{dx}{5}$. Since the time elapsed for both boys is equal, one boy ran $5(\frac{dx}{5})$ feet while the other boy ran $9(\frac{dx}{5})$ feet. Because both finished at the starting point, both ran an integral number of laps, so $5(\frac{dx}{5})$ and $9(\frac{dx}{5})$ are multiples of d . Because both stopped when both met at the start for the first time, $x = 5$.

Note that between the time a runner finishes a lap and a runner (can be same) finishes a lap, both runners must meet each other. When $0 < x \leq 5$ and either $5(\frac{dx}{5})$ or $9(\frac{dx}{5})$ is a multiple of d , one of the runners completed a lap. This is achieved when $x = \frac{5}{9}, 1, \frac{10}{9}, \frac{15}{9}, 2, \frac{20}{9}, \frac{25}{9}, 3, \frac{30}{9}, \frac{35}{9}, 4, \frac{40}{9}, 5$, so the two meet each other (excluding start and finish) a total of **(A) 13** times.

30. (AHSME 1973) Let $[t]$ denote the greatest integer $\leq t$ where $t \geq 0$ and $S = \{(x, y) : (x - T)^2 + y^2 \leq T^2 \text{ where } T = t - [t]\}$. Then we have
- (A) the point $(0, 0)$ does not belong to S for any t

- (B) $0 \leq \text{Area } S \leq \pi$ for all t
 (C) S is contained in the first quadrant for all $t \geq 5$
 (D) the center of S for any t is on the line $y = x$
 (E) none of the other statements is true

Answer: B

Solution

The region S is a circle radius T and center $(T, 0)$. Since $T = t - [t]$, $0 \leq T < 1$. That means the area of the circle is less than π , and since the region can also be just a dot (achieved when t is integer), the answer is **(B)**.

31. (AHSME 1973) In the following equation, each of the letters represents uniquely a different digit in base ten:

$$(YE) \cdot (ME) = TTT$$

The sum $E + M + T + Y$ equals

- (A) 19 (B) 20 (C) 21 (D) 22 (E) 24

Answer: C

Solution

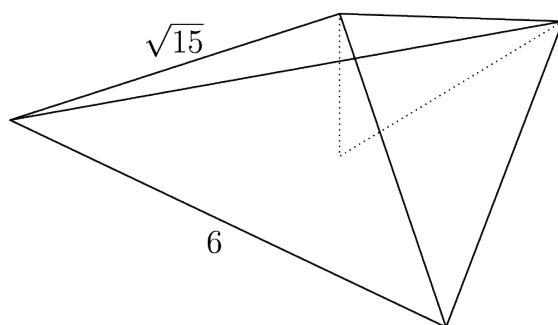
The right side of the equation can be rewritten as $111T = 37 \cdot 3T$. With trial and error and prime factorization as a guide, we can test different digits of T to see if we can find two two-digit numbers that have the same units digit and multiply to $111T$.

The only possibility that works is $37 \cdot 27 = 999$. That means $E + M + T + Y =$ **(C) 21**.

32. (AHSME 1973) The volume of a pyramid whose base is an equilateral triangle of side length 6 and whose other edges are each of length $\sqrt{15}$ is
- (A) 9 (B) $9/2$ (C) $27/2$ (D) $\frac{9\sqrt{3}}{2}$ (E) none of these

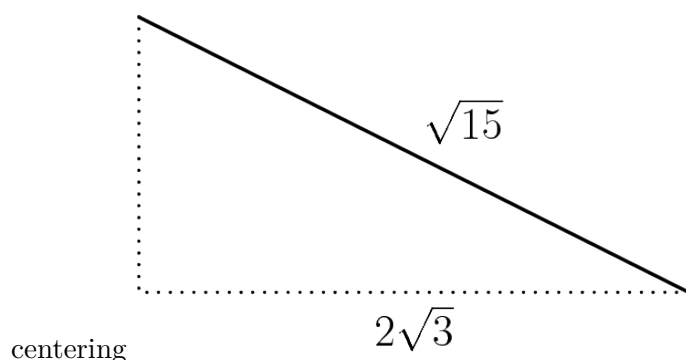
Answer: A

Solution



centering

Draw an altitude towards the equilateral triangle base. By symmetry (this can also be proved by HL), the base of the altitude is equidistant from the three points of the equilateral triangle. This means that the distance from the base of the altitude to one of the points of the equilateral triangle is $2\sqrt{3}$.



Using the **Pythagorean Theorem**, the length of the altitude is $\sqrt{3}$, so the volume of the triangular pyramid is $\frac{1}{3} \cdot \frac{6^2 \cdot \sqrt{3}}{4} \cdot \sqrt{3} = \boxed{\text{(A) } 9}$.

33. (*AHSME 1973*) When one ounce of water is added to a mixture of acid and water, the new mixture is 20% acid. When one ounce of acid is added to the new mixture, the result is $33\frac{1}{3}\%$ acid. The percentage of acid in the original mixture is
- (A) 22% (B) 24% (C) 25% (D) 30% (E) $33\frac{1}{3}\%$

Answer: C

Solution

Let a be the original number of ounces of acid and w be the original number of ounces of water. We can write two equations since we know the percentage of acid after some water and acid.

$$\frac{a}{a + w + 1} = \frac{1}{5}$$

$$\frac{a + 1}{a + w + 2} = \frac{1}{3}$$

Cross-multiply to get rid of the fractions.

$$5a = a + w + 1$$

$$3a + 3 = a + w + 1$$

Solve the system to get $a = 1$ and $w = 3$. The percentage of acid in the original mixture is $\frac{1}{1+3} = \boxed{\text{(C) } 25\%}$.

34. (*AHSME 1973*) A plane flew straight against a wind between two towns in 84 minutes and returned with that wind in 9 minutes less than it would take in still air. The number of minutes (2 answers) for the return trip was
- (A) 54 or 18 (B) 60 or 15 (C) 63 or 12 (D) 72 or 36 (E) 75 or 20

Answer: C

Solution

Let d be the distance between the two towns, p be the speed of the plane, and w be the speed of the wind. Since the time it took to fly against the wind towards the other town is 84 minutes,

$$\frac{d}{p - w} = 84$$

$$d = 84p - 84w$$

Since flying with the wind on the return trip takes 9 minutes less compared to the return trip without wind,

$$\frac{d}{p+w} + 9 = \frac{d}{p}$$

Substitute d and substitute p and w

$$\frac{84p - 84w}{p+w} + 9 = \frac{84p - 84w}{p}$$

$$\frac{93p - 75w}{p+w} = \frac{84p - 84w}{p}$$

$$93p^2 - 75pw = 84p^2 + 84pw - 84pw - 84w^2$$

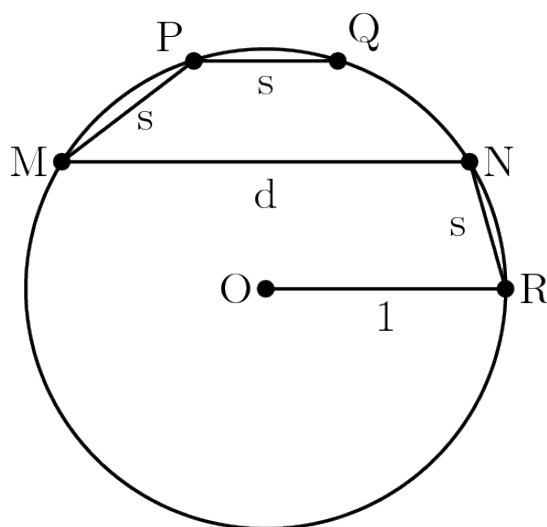
$$9p^2 - 75pw + 84w^2 = 0$$

$$3p^2 - 25pw + 28w^2 = 0$$

$$(3p - 4w)(p - 7w) = 0$$

That means $p = \frac{4}{3}w$ or $p = 7w$. If $p = \frac{4}{3}w$, then $\frac{112w - 84w}{\frac{7}{3}w} = 12$. If $p = 7w$, then $\frac{84(7w) - 84w}{8w} = 63$. The number of minutes for the return trip can be **(C) 63 or 12**.

35. (AHSME 1973) In the unit circle shown in the figure, chords PQ and MN are parallel to the unit radius OR of the circle with center at O . Chords MP , PQ , and NR are each s units long and chord MN is d units long.



centering

Of the three equations

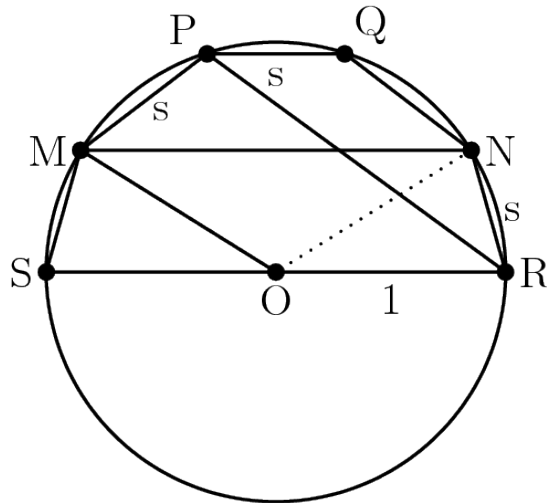
$$\text{I. } d - s = 1, \quad \text{II. } ds = 1, \quad \text{III. } d^2 - s^2 = \sqrt{5}$$

those which are necessarily true are

- (A) I only (B) II only (C) III only (D) I and II only (E) I, II and III

Answer: E

Solution 1



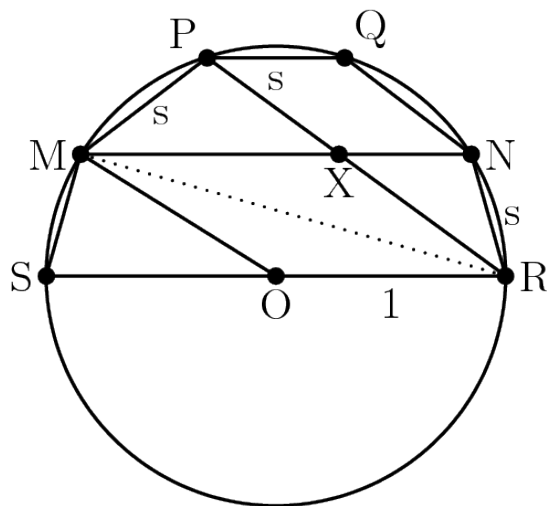
centering

First, let S be on circle O so RS is a diameter. In order to prove that the three statements are true (or false), we first show that $SM = QN = s$, and then we examine each statement one by one.

Lemma 1: $SM = QN = s$ Since $OM = ON$, by the Base Angle Theorem, $\angle OMN = \angle ONM$. By the Alternate Interior Angles Theorem, $\angle MOS = \angle NMO = \angle MNO = \angle NOR$, making $\triangle MOS \cong \triangle NOR$ by SAS Congruency. That means $SM = NR = s$ by CPCTC.

Because $PQNM$ is a **cyclic quadrilateral**, $\angle MPQ + \angle QNM = 180^\circ$, but we are given that PQ is parallel to MN , so $\angle MPQ + \angle PMN = 180^\circ$. Therefore, $\angle PMN = \angle QNM$. That makes $PQNM$ an **isosceles trapezoid**, so $MP = QN = s$.

Lemma 2: Showing Statement I is (or isn't) true



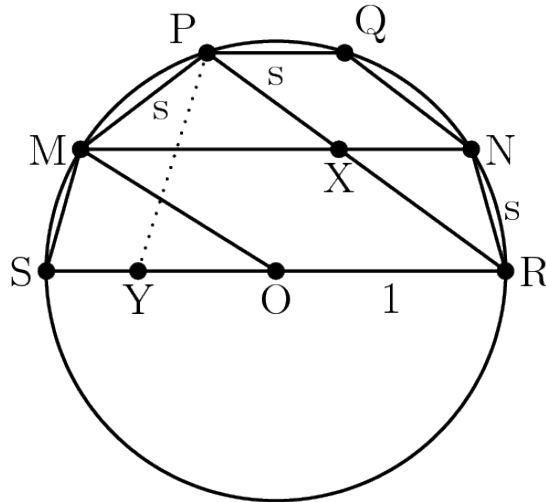
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Let X be the intersection of MN and PR . By SSS Congruency, $\triangle POQ \cong \triangle QON \cong \triangle NOR$, so $\angle PQN = \angle QNR$. We know that $PQNR$ is a cyclic quadrilateral, so $\angle PQN + \angle PRN = \angle QNR + \angle PRN = 180^\circ$, so $QN \parallel PR$. That makes $PQNX$ a **parallelogram**, so $XN = s$. Thus, $MX = d - s$.

In addition, $PX = s$, and by the Base Angle Theorem and **Vertical Angle Theorem**, $\angle PMX = \angle MXP = \angle NXR = \angle NRX$. That means by AAS Congruency, $\triangle MXP \cong \triangle RXN$, so $MX = XR = d - s$.

By the Base Angle Theorem and the Alternating Interior Angle Theorem, $\angle XRM = \angle XMR = \angle MRO = \angle RMO$, so by ASA Congruency, $\angle MXR = \angle MOR$. Thus, $MX = OR = d - s = 1$. Statement I is true.

Lemma 3: Showing Statement II is (or isn't) true



centering

From Lemma 2, we have $PX + XR = s + (d - s) = d$. Draw point Y on SR such that $YR = d$, making $YO = s$.

Since $PQ \parallel YO$, we have $\angle QPO = \angle POY$. Additionally, $PQ = YO$ and $PO = PO$, so by SAS Congruency, $\triangle PQY \cong \triangle OYQ$. That means $PY = OQ = 1$.

Since $\angle PRS$ is an inscribed angle, $\angle POS = 2 \cdot \angle PRS$. Additionally, $\triangle SOM \cong \triangle MOP$, so MO bisects $\angle SOP$. Thus, $\angle SOM = \angle PRS$, making $\triangle PRY \sim \triangle MOS$ by SAS Similarity.

By using the similarity, we find that

$$\frac{1}{s} = \frac{d}{1}$$

$$ds = 1$$

Thus, Statement II is true.

Lemma 4: Showing Statement III is (or isn't) true From Lemmas 2 and 3, we have $d - s = 1$ and $ds = 1$. Squaring the first equation results in

$$d^2 - 2ds + s^2 = 1.$$

Adding $4ds = 4$ to both sides results in

$$d^2 + 2ds + s^2 = 5$$

$$(d + s)^2 = 5$$

Since $d + s$ is positive, we find that $d + s = \sqrt{5}$, which confirms that Statement III is true.

In summary, all three statements are true, so the answer is (E).

Solution 2

It is a well-known fact that any cyclic trapezoid has its legs equal. Therefore, $QN = s$. Now, extend OR to meet the circle again at T . By similar reasoning, $MT = s$. Furthermore, since $\angle MOT, \angle POM, \angle POQ, \angle QON$, and $\angle NOR$ sum to 180 and are equal, they have measure 36 degrees. It trivially follows that $2 \sin 18 = s \iff s = 2 \left(\frac{-1+\sqrt{5}}{4} \right)$. Dropping the altitude from O to MN , we see that $d = 2 \sin 54 \iff d = 2 \left(\frac{1+\sqrt{5}}{4} \right)$. Therefore, $d - s = 1$, $ds = 1$, and $d^2 - s^2 = \sqrt{5}$.

§10 AHSME 1974

1. (AHSME 1974) If $x \neq 0$ or 4 and $y \neq 0$ or 6, then $\frac{2}{x} + \frac{3}{y} = \frac{1}{2}$ is equivalent to

- (A) $4x + 3y = xy$ (B) $y = \frac{4x}{6-y}$ (C) $\frac{x}{2} + \frac{y}{3} = 2$
 (D) $\frac{4y}{y-6} = x$ (E) none of these

Answer: D

Solution

To clear out the fractions, we multiply both sides of the equation by xy to get $2y + 3x = \frac{xy}{2}$, or $4y + 6x = xy$. From this, we have $4y = xy - 6x$. We can factor an x out of the RHS, and so $4y = x(y - 6) \implies \frac{4y}{y-6} = x$, **[D]**.

2. (AHSME 1974) Let x_1 and x_2 be such that $x_1 \neq x_2$ and $3x_i^2 - hx_i = b$, $i = 1, 2$. Then $x_1 + x_2$ equals

- (A) $-\frac{h}{3}$ (B) $\frac{h}{3}$ (C) $\frac{b}{3}$ (D) $2b$ (E) $-\frac{b}{3}$

Answer: B

Solution

Notice that x_1 and x_2 are the distinct solutions to the quadratic $3x^2 - hx - b = 0$. By Vieta, the sum of the roots of this quadratic is the negation of the coefficient of the linear term divided by the coefficient of the quadratic term, so in this case $-\frac{-h}{3} = \frac{h}{3}$, **[B]**.

3. (AHSME 1974) The coefficient of x^7 in the polynomial expansion of

$$(1 + 2x - x^2)^4$$

is

- (A) -8 (B) 12 (C) 6 (D) -12 (E) none of these

Answer: A

Solution 1

Let's write out the multiplication, so that it becomes easier to see.

$$(1 + 2x - x^2)(1 + 2x - x^2)(1 + 2x - x^2)(1 + 2x - x^2)$$

We can now see that the only way to get an x^7 is by taking three $-x^2$ and one $2x$. There are $\binom{4}{1} = 4$ ways to pick which term the $2x$ comes from, and the coefficient of each one is $(-1)^3(2) = -2$. Therefore, the coefficient of x^7 is $(4)(-2) = -8$, **[A]**.

Solution 2

$$(x^2 - 2x - 1)^4 = (x^2 - 2x + 1 - 2)^4 = ((x-1)^2 - (\sqrt{2})^2)^4 = (x-1-\sqrt{2})^4(x-1+\sqrt{2})^4$$

Each of the two polynomials can have terms of order 4,3,2,1,0. The only way to obtain a term of order 7 is either a term of order 4 from the first and 3 from the second, or the other way around. Notice that the terms of order 4 must necessarily have coefficient 1. Also keeping in mind that the coefficient of x^3 in $(x+a)^4$ is $4a$, we have:

- Coefficient of x^3 (by extension x^7) in the first polynomial $= 4 \times (-1 - \sqrt{2})$;
- Coefficient of x^3 (by extension x^7) in the second polynomial is $4 \times (-1 + \sqrt{2})$.

So the coefficient of x^7 in the expansion is $(-4 - 4\sqrt{2}) + (-4 + 4\sqrt{2}) = -4 - 4 = -8$, A.

4. (AHSME 1974) What is the remainder when $x^{51} + 51$ is divided by $x + 1$?
 (A) 0 (B) 1 (C) 49 (D) 50 (E) 51

Answer: D

Solution 1

From the **Remainder Theorem**, the remainder when $x^{51} + 51$ is divided by $x + 1$ is $(-1)^{51} + 51 = -1 + 51 = 50$, D.

Solution 2

Notice that $x^{51} + 51$ is equal to $(x^{51} + 1) + 50$. Since $x^{51} + 1 = (x + 1)(x^{50} - x^{49} + \dots + x^2 - x + 1)$, this part of the polynomial is divisible by $x + 1$. Thus, the remainder is (D) 50.

5. (AHSME 1974) Given a quadrilateral $ABCD$ inscribed in a circle with side AB extended beyond B to point E , if $\angle BAD = 92^\circ$ and $\angle ADC = 68^\circ$, find $\angle EBC$.
 (A) 66° (B) 68° (C) 70° (D) 88° (E) 92°

Answer: B

Solution

Since $ABCD$ is cyclic, opposite angles must sum to 180° . Therefore, $\angle ADC + \angle ABC = 180^\circ$, and $\angle ABC = 180^\circ - \angle ADC = 180^\circ - 68^\circ = 112^\circ$. Notice also that $\angle ABC$ and $\angle CBE$ form a linear pair, and so they sum to 180° . Therefore, $\angle EBC = 180^\circ - \angle ABC = 180^\circ - 112^\circ = 68^\circ$, B. Notice that the answer didn't even depend on $\angle BAD$.

6. (AHSME 1974) For positive real numbers x and y define $x * y = \frac{x \cdot y}{x + y}$, then
 (A) "*" is commutative but not associative
 (B) "*" is associative but not commutative
 (C) "*" is neither commutative nor associative
 (D) "*" is commutative and associative
 (E) none of these

Answer: D

Solution

First, let's check for commutivity. We have

$$y * x = \frac{y \cdot x}{y + x} = \frac{x \cdot y}{x + y} = x * y$$

, so $*$ is commutative.

Now we check for associativity. We have

$$(x * y) * z = \left(\frac{x \cdot y}{x + y} \right) * z = \frac{\frac{x \cdot y}{x + y} \cdot z}{\frac{x \cdot y}{x + y} + z} = \frac{x \cdot y \cdot z}{x \cdot y + z(x + y)} = \frac{x \cdot y \cdot z}{x \cdot y + y \cdot z + x \cdot z}.$$

Also,

$$x * (y * z) = x * \left(\frac{y \cdot z}{y + z} \right) = \frac{x \cdot \frac{y \cdot z}{y + z}}{x + \frac{y \cdot z}{y + z}} = \frac{x \cdot y \cdot z}{x(y + z) + y \cdot z} = \frac{x \cdot y \cdot z}{x \cdot y + y \cdot z + x \cdot z} = (x * y) * z,$$

and so $*$ is also associative. $\boxed{D}$

7. (AHSME 1974) A town's population increased by 1,200 people, and then this new population decreased by 11%. The town now had 32 less people than it did before the 1,200 increase. What is the original population?

(A) 1,200 (B) 11,200 (C) 9,968 (D) 10,000 (E) none of these

Answer: D

Solution

Let n be the original population. When 1,200 people came, the new population was $n + 1200$. Then, since the new population decreased by 11%, we must multiply this new population by .89 to get the final population. This is equal to $(0.89)(n + 1200) = 0.89n + 1068$. We're given that this is 32 less than the original population, n . Therefore, $0.89n + 1068 = n - 32 \implies 0.11n = 1100 \implies n = \frac{1100}{0.11} = 10000$, $\boxed{D}$.

8. (AHSME 1974) What is the smallest prime number dividing the sum $3^{11} + 5^{13}$?

(A) 2 (B) 3 (C) 5 (D) $3^{11} + 5^{13}$ (E) none of these

Answer: A

Solution

Since we want to find the smallest prime dividing the sum, we start with the smallest prime and move up, so first we try 2. Notice that 3^{11} and 5^{13} are both odd, so their sum must be even. This means that 2 must divide $3^{11} + 5^{13}$, and so since 2 is the smallest prime, our answer must be 2, $\boxed{A}$.

9. (AHSME 1974) The integers greater than one are arranged in five columns as follows:

	2	3	4	5
9	8	7	6	
	10	11	12	13
17	16	15	14	
	.	.	.	.

(Four consecutive integers appear in each row; in the first, third and other odd numbered rows, the integers appear in the last four columns and increase from left

to right; in the second, fourth and other even numbered rows, the integers appear in the first four columns and increase from right to left.)

In which column will the number 1,000 fall?

- (A) first (B) second (C) third (D) fourth (E) fifth

Answer: B

Solution

We try pairing numbers with the column number they're in.

Number	Column Number
2	2
3	3
4	4
5	5
6	4
7	3
8	2
9	1
10	2
11	3
12	4

Now we can see a pattern. The column number starts at 2, goes up to 5, then goes back down to 1, and repeats. Each of these blocks has a length of 7, so the column number has a period of 8. Since $1000 \equiv 8 \pmod{8}$, 1000 is in the same column as 8, which is the second column. B

10. (AHSME 1974) What is the smallest integral value of k such that

$$2x(kx - 4) - x^2 + 6 = 0$$

has no real roots?

- (A) -1 (B) 2 (C) 3 (D) 4 (E) 5

Answer: B

Solution

Expanding, we have $2kx^2 - 8x - x^2 + 6 = 0$, or $(2k - 1)x^2 - 8x + 6 = 0$. For this quadratic not to have real roots, it must have a negative discriminant. Therefore, $(-8)^2 - 4(2k - 1)(6) < 0 \implies 64 - 48k + 24 < 0 \implies k > \frac{11}{6}$. From here, we can easily see that the smallest integral value of k is 2, B.

11. (AHSME 1974) If (a, b) and (c, d) are two points on the line whose equation is $y = mx + k$, then the distance between (a, b) and (c, d) , in terms of a, c , and m is

- (A) $|a - c|\sqrt{1 + m^2}$ (B) $|a + c|\sqrt{1 + m^2}$ (C) $\frac{|a - c|}{\sqrt{1 + m^2}}$
 (D) $|a - c|(1 + m^2)$ (E) $|a - c||m|$

Answer: A

Solution

Notice that since (a, b) is on $y = mx + k$, we have $b = am + k$. Similarly, $d = cm + k$. Using the distance formula, the distance between the points (a, b) and (c, d) is

$$\begin{aligned}\sqrt{(a-c)^2 + (b-d)^2} &= \sqrt{(a-c)^2 + (am+k-cm-k)^2} \\ &= \sqrt{(a-c)^2 + m^2(a-c)^2} \\ &= |a-c|\sqrt{1+m^2}.\end{aligned}$$

And so the answer is A.

12. (AHSME 1974) If $g(x) = 1 - x^2$ and $f(g(x)) = \frac{1-x^2}{x^2}$ when $x \neq 0$, then $f(1/2)$ equals

(A) $3/4$ (B) 1 (C) 3 (D) $\sqrt{2}/2$ (E) $\sqrt{2}$

Answer: B

Solution

Notice that if we can find a y such that $g(y) = \frac{1}{2}$, then we can plug that into the second equation to get $f\left(\frac{1}{2}\right) = f(g(y)) = \frac{1-y^2}{y^2}$. Thus, we let $1-y^2 = \frac{1}{2} \implies y^2 = \frac{1}{2}$. Notice that, since our final expression only involves y^2 , we don't need to take the square root. Thus, $f\left(\frac{1}{2}\right) = \frac{1-y^2}{y^2} = \frac{1-\frac{1}{2}}{\frac{1}{2}} = 1$, B.

13. (AHSME 1974) Which of the following is equivalent to "If P is true, then Q is false."?

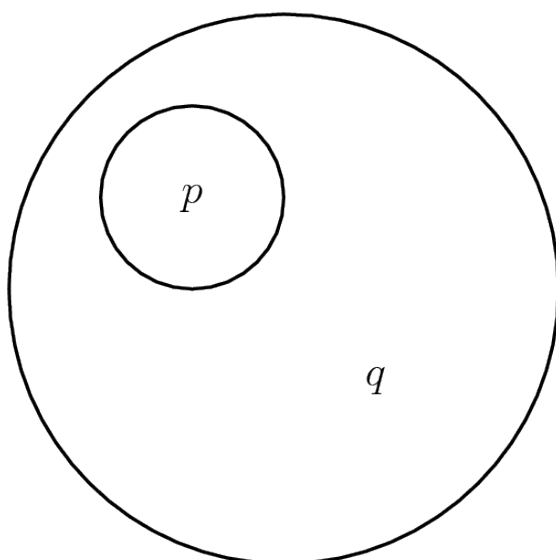
(A) "P is true or Q is false."
 (B) "If Q is false then P is true."
 (C) "If P is false then Q is true."
 (D) "If Q is true then P is false."
 (E) "If Q is true then P is true."

Answer: D

Solution

Remember that a statement is logically equivalent to its contrapositive, which is formed by first negating the hypothesis and conclusion and then switching them. In this case, the contrapositive of "If P is true, then Q is false." is "If Q is true, then P is false." D

The fact that a statement's contrapositive is logically equivalent to it can easily be seen from a venn diagram argument.



centering

From this venn diagram, clearly "If p , then q ." is true. However, since p is fully contained in q , the statement "If not q , then not p ." is also true, and so a statement and its contrapositive are equivalent.

Solution

Let's consider the following case of dogs. If $p = \text{true}$ then the dog is 100% blue. If $q = \text{true}$ then the dog is 100% red.

This is a valid solution to if P is true, then Q is false, or if the dog is 100% blue then the dog is not 100% red.

Now let's look at the options: (A) "P is true or Q is false." $\rightarrow$ If the dog is 100% blue or the dog is not 100% red. This statement makes no sense.

(B) "If Q is false then P is true." $\rightarrow$ If the dog is not 100% red, then the dog is 100% blue. This does not make sense, as the dog could be other colors, like white, or black, or yellow. My dog is brown!

(C) "If P is false then Q is true." $\rightarrow$ If the dog is not 100% blue, then the dog is 100% red. Again, this fails for other colors of dogs. This is dogsiscrimination!

(D) "If Q is true then P is false." $\rightarrow$ If the dog is 100% red, then it is not 100% blue. This statement makes logical sense, and is correct.

(E) "If Q is true then P is true." $\rightarrow$ If the dog is 100% red, then the dog is 100% blue. This does not make sense, because we know the color of the dog, a dog cannot be two colors at once. Actually in quantum mechanics maybe, but not for this AHSME problem. (This example is not the best way to test this answer choice if there were a different question so mind that by TFEA)

The statement that makes the most sense is D

14. (AHSME 1974) Which statement is correct?

(A) If $x < 0$, then $x^2 > x$. (B) If $x^2 > 0$, then $x > 0$.

(C) If $x^2 > x$, then $x > 0$. (D) If $x^2 > x$, then $x < 0$.

(E) If $x < 1$, then $x^2 < x$.

Answer: A

Solution

We begin by going statement by statement.

For (A), $x^2 > 0$ for all real x , and so $x^2 > 0 > x$. Therefore, **A** is true.

We can give counterexamples for all other statements:

For (B), $x = -1$

For (C), $x = -1$

For (D), $x = 2$

For (E), $x = -1$

15. (AHSME 1974) If $x < -2$, then $|1 - |1 + x||$ equals

(A) $2 + x$ (B) $-2 - x$ (C) x (D) $-x$ (E) -2

Answer: B

Solution

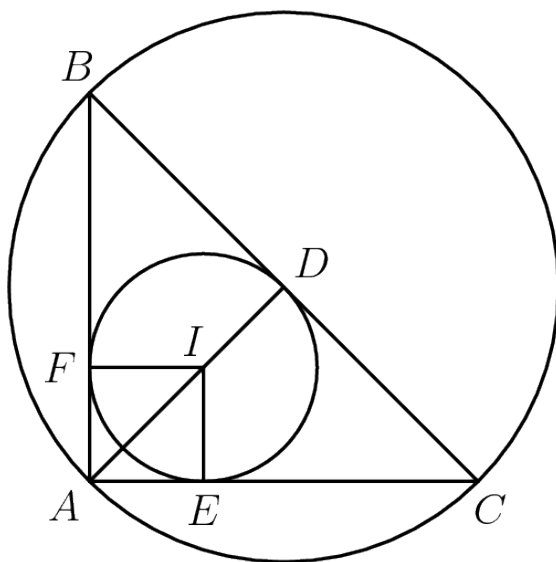
Notice that, for $a < 0$, $|a| = -a$, and for $a > 0$, $|a| = a$.

Since $x < -2$, $1 + x < 1 - 2 < 0$, so $|1 + x| = -x - 1$. Therefore, $1 - |1 + x| = 1 + x + 1 = x + 2 < -2 + 2 = 0$, and so $|1 - |1 + x|| = |x + 2| = -2 - x$, **B**.

16. (AHSME 1974) A circle of radius r is inscribed in a right isosceles triangle, and a circle of radius R is circumscribed about the triangle. Then R/r equals

(A) $1 + \sqrt{2}$ (B) $\frac{2+\sqrt{2}}{2}$ (C) $\frac{\sqrt{2}-1}{2}$ (D) $\frac{1+\sqrt{2}}{2}$ (E) $2(2 - \sqrt{2})$

Answer: A

Solution

centering

Label the points as in the figure above. Let the side length $AB = AC = s$. Therefore, $BC = s\sqrt{2}$. Since the circumradius of a right triangle is equal to half of the length of the hypotenuse, we have $R = \frac{s\sqrt{2}}{2}$.

Now to find the inradius. Notice that $IFAE$ is a square with side length r , and also $AD = R$. Therefore, $s = AD = AI + ID = r\sqrt{2} + r$, and so $r = \frac{R}{\sqrt{2}+1}$.

Finally, $\frac{R}{r} = 1 + \sqrt{2}$, **A**.

17. (AHSME 1974) If $i^2 = -1$, then $(1+i)^{20} - (1-i)^{20}$ equals
 (A) -1024 (B) $-1024i$ (C) 0 (D) 1024 (E) $1024i$

Answer: C

Solution

Notice that $(1+i)^2 = 2i$ and $(1-i)^2 = -2i$. Therefore,

$$(1+i)^{20} - (1-i)^{20} = (2i)^{10} - (-2i)^{10} = (2i)^{10} - (2i)^{10} = 0, \boxed{\text{C}}.$$

18. (AHSME 1974) If $\log_8 3 = p$ and $\log_3 5 = q$, then, in terms of p and q , $\log_{10} 5$ equals
 (A) pq (B) $\frac{3p+q}{5}$ (C) $\frac{1+3pq}{p+q}$ (D) $\frac{3pq}{1+3pq}$ (E) $p^2 + q^2$

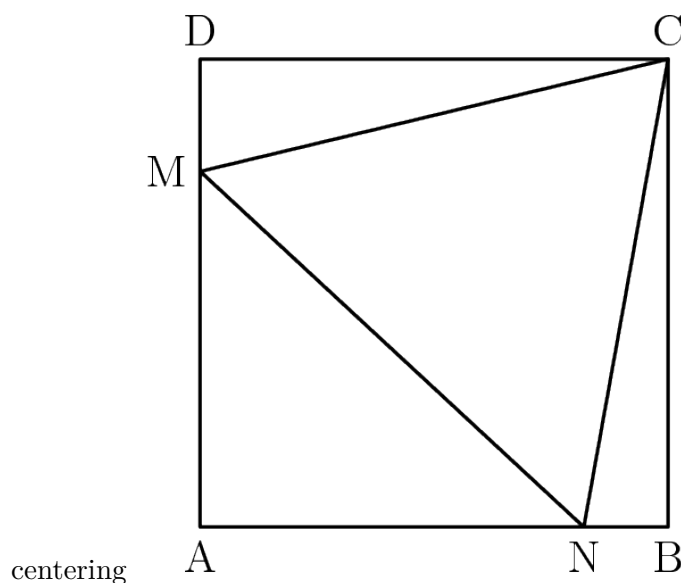
Answer: D

Solution

Notice that $\frac{1}{p} = \log_3 8$, so it would probably be easier to work in base 3. From change of base, $\log_{10} 5 = \frac{\log_3 5}{\log_3 10}$. We're given that $\log_3 5 = q$, so now we just need to find $\log_3 10$.

We have $\frac{1}{p} = \log_3 8 = 3 \log_3 2$, so $\log_3 2 = \frac{1}{3p}$. Also, $\log_3 10 = \log_3 5 + \log_3 2 = q + \frac{1}{3p}$. Therefore, $\log_{10} 5 = \frac{q}{q + \frac{1}{3p}} = \frac{3pq}{3pq+1}, \boxed{\text{D}}.$

19. (AHSME 1974) In the adjoining figure $ABCD$ is a square and CMN is an equilateral triangle. If the area of $ABCD$ is one square inch, then the area of CMN in square inches is



- (A) $2\sqrt{3} - 3$ (B) $1 - \frac{\sqrt{3}}{3}$ (C) $\frac{\sqrt{3}}{4}$ (D) $\frac{\sqrt{2}}{3}$ (E) $4 - 2\sqrt{3}$

Answer: A

Solution

Let $BN = x$ so that $AN = 1 - x$. From the Pythagorean Theorem on $\triangle NBC$, we get $CN = \sqrt{x^2 + 1}$, and from the Pythagorean Theorem on $\triangle AMN$, we get $MN = (1-x)\sqrt{2}$. Since $\triangle CMN$ is equilateral, we must have $\sqrt{x^2 + 1} =$

$(1-x)\sqrt{2} \implies x^2 + 1 = 2x^2 - 4x + 2 \implies x^2 - 4x + 1 = 0$. From the Pythagorean Theorem, we get $x = 2 - \sqrt{3}$, since we want the root that's less than 1.

Therefore, $CN^2 = x^2 + 1 = (2 - \sqrt{3})^2 + 1 = 8 - 4\sqrt{3}$. The area of an equilateral triangle with side length x is equal to $\frac{x^2\sqrt{3}}{4}$, so the area of $\triangle CMN$ is $\frac{(8-4\sqrt{3})(\sqrt{3})}{4} = 2\sqrt{3} - 3$, **[A]**.

Solution 2 (Visualization + Using Ratios)

We know there is only one way to fit an equilateral triangle into a square: one of its vertices is a corner of the square and the other two vertices fall on opposite sides (try to imagine it in your head). It must be symmetrical along a diagonal of the square.

Thus $\triangle CNB \sim \triangle CDM$ where both are $15^\circ - 75^\circ - 90^\circ$ triangles since $\angle MCN = 60^\circ$

Using the $15^\circ - 75^\circ - 90^\circ$ ratios $(2 - \sqrt{3} : 1 : \sqrt{6} - \sqrt{2})$, we have $CN = \sqrt{6} - \sqrt{2}$.

Thus $[\triangle CMN] = \frac{\sqrt{3}}{4}(\sqrt{6} - \sqrt{2})^2 = \frac{\sqrt{3}}{4}(8 - 4\sqrt{3}) = 2\sqrt{3} - 3 \Rightarrow$ **[A]**.

$\sim$ Jonysun

20. (AHSME 1974) Let

$$T = \frac{1}{3-\sqrt{8}} - \frac{1}{\sqrt{8}-\sqrt{7}} + \frac{1}{\sqrt{7}-\sqrt{6}} - \frac{1}{\sqrt{6}-\sqrt{5}} + \frac{1}{\sqrt{5}-2}.$$

Then

- (A) $T < 1$ (B) $T = 1$ (C) $1 < T < 2$ (D) $T > 2$

(E) $T = \frac{1}{(3-\sqrt{8})(\sqrt{8}-\sqrt{7})(\sqrt{7}-\sqrt{6})(\sqrt{6}-\sqrt{5})(\sqrt{5}-2)}$

Answer: D

Solution

Let's try to rationalize $\frac{1}{\sqrt{n+1}-\sqrt{n}}$. Multiplying the numerator and denominator by $\sqrt{n+1} + \sqrt{n}$ gives us $\frac{1}{\sqrt{n+1}-\sqrt{n}} = \sqrt{n+1} + \sqrt{n}$.

Therefore,

$$T = (3 + \sqrt{8}) - (\sqrt{8} + \sqrt{7}) + (\sqrt{7} + \sqrt{6}) - (\sqrt{6} + \sqrt{5}) + (\sqrt{5} + 2) = 5.$$

Hence the answer is **[D]**.

21. (AHSME 1974) In a geometric series of positive terms the difference between the fifth and fourth terms is 576, and the difference between the second and first terms is 9. What is the sum of the first five terms of this series?

- (A) 1061 (B) 1023 (C) 1024 (D) 768 (E) none of these

Answer: B

Solution

Let the first term be a and the common ratio be r . Therefore, the second term is ar , the fourth term is ar^3 , and the fifth term is ar^4 . We're given that $ar^4 - ar^3 = 576 \implies ar^3(r - 1) = 576$ and $ar - a = 9 \implies a(r - 1) = 9$. Dividing this first equation by this second one, we get $r^3 = \frac{576}{9} = 64 \implies r = 4$. Therefore, $a(4 - 1) = 9$, so $a = 3$.

Therefore, the first five terms of this series are 3, 12, 48, 192, 768, and their sum is 1023, **[B]**.

22. (AHSME 1974) The minimum of $\sin \frac{A}{2} - \sqrt{3} \cos \frac{A}{2}$ is attained when A is
 (A) -180° (B) 60° (C) 120° (D) 0° (E) none of these

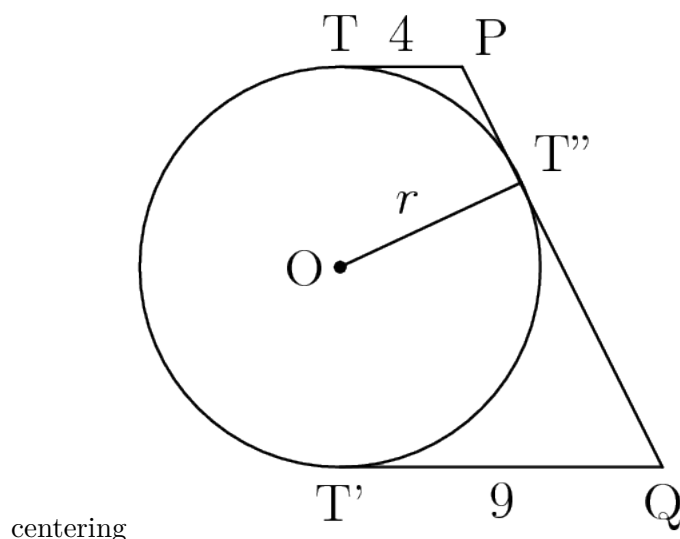
Answer: E

Solution

Define a new number B such that $\cos(B) = \frac{1}{2}$ and $\sin(B) = -\frac{\sqrt{3}}{2}$. Notice that $B = \frac{5\pi}{3}$. Therefore, we have $\sin \frac{A}{2} - \sqrt{3} \cos \frac{A}{2} = 2 \cos(B) \sin \left(\frac{A}{2}\right) + 2 \sin(B) \cos \left(\frac{A}{2}\right)$.

Recognizing the angle sum formula, we can rewrite this as $\sin \frac{A}{2} - \sqrt{3} \cos \frac{A}{2} = 2 \sin \left(\frac{A}{2} + B\right)$. We can now clearly see that the minimum is -2 , and this is achieved when $\frac{A}{2} + B = \frac{3\pi}{2}$. Plugging in $B = \frac{5\pi}{3}$, we get $\frac{A}{2} = -\frac{\pi}{6} \implies A = -\frac{\pi}{3} = -60^\circ$. E

23. (AHSME 1974) In the adjoining figure TP and $T'Q$ are parallel tangents to a circle of radius r , with T and T' the points of tangency. $PT''Q$ is a third tangent with T'' as a point of tangency. If $TP = 4$ and $T'Q = 9$ then r is

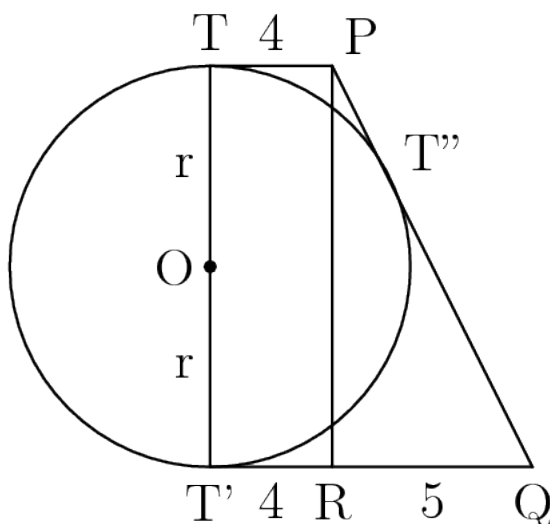


centering

- (A) $25/6$ (B) 6 (C) $25/4$
 (D) a number other than $25/6, 6, 25/4$
 (E) not determinable from the given information

Answer: B

Solution



centering

Drop the perpendicular from P to $T'Q$ and let the foot be R . Note that $PTT'R$ is a rectangle. Also, from Two Tangents, $PT'' = 4$ and $QT'' = 9$, so $PQ = 13$. Therefore, from the Pythagorean Theorem on $\triangle PRQ$, $PR = \sqrt{13^2 - 5^2} = 12$. We now see that $r = \frac{PR}{2} = 6$, **B**.

24. (AHSME 1974) A fair die is rolled six times. The probability of rolling at least a five at least five times is

(A) $\frac{13}{729}$ (B) $\frac{12}{729}$ (C) $\frac{2}{729}$ (D) $\frac{3}{729}$ (E) none of these

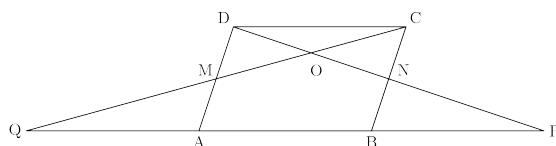
Answer: A

Solution

The probability of rolling at least a five on any one roll is $\frac{2}{6} = \frac{1}{3}$. If there are exactly 5 fives or sixes rolled, there are $\binom{6}{5} = 6$ ways to pick which of the rolls are the fives and sixes, and so the probability in this case is $6 \left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right) = \frac{12}{729}$. If there are exactly 6 fives or sixes rolled, then there is only one way to pick which of the rolls are fives and sixes, so the probability in this case is $\left(\frac{1}{3}\right)^6 = \frac{1}{729}$.

Therefore, the total probability is $\frac{12}{729} + \frac{1}{729} = \frac{13}{729}$, **A**.

25. (AHSME 1974) In parallelogram $ABCD$ of the accompanying diagram, line DP is drawn bisecting BC at N and meeting AB (extended) at P . From vertex C , line CQ is drawn bisecting side AD at M and meeting AB (extended) at Q . Lines DP and CQ meet at O . If the area of parallelogram $ABCD$ is k , then the area of the triangle QPO is equal to



centering

(A) k (B) $\frac{6k}{5}$ (C) $\frac{9k}{8}$ (D) $\frac{5k}{4}$ (E) $2k$

Answer: C

Solution

Note that

$$[QPO] = [QAM] + [PBN] + [AMONB] = [AMONB] + [MDC] + [NCD]$$

$$= [AMONB] + [MDC] + [NOC] + [DOC] = [ABCD] + [DOC] = k + [DOC].$$

Also, note that $DCNM$ is a parallelogram, and so $[DOC] = \frac{1}{4}[DCNM] = \frac{1}{8}[ABCD] = \frac{k}{8}$.

Therefore, $[QPO] = k + \frac{k}{8} = \frac{9k}{8}$, C.

26. (AHSME 1974) The number of distinct positive integral divisors of $(30)^4$ excluding 1 and $(30)^4$ is

(A) 100 (B) 125 (C) 123 (D) 30 (E) none of these

Answer: C

Solution

The prime factorization of 30 is $2 \cdot 3 \cdot 5$, so the prime factorization of 30^4 is $2^4 \cdot 3^4 \cdot 5^4$. Therefore, the number of positive divisors of 30^4 is $(4+1)(4+1)(4+1) = 125$. However, we have to subtract 2 to account for 1 and 30^4 , so our final answer is $125 - 2 = 123$, C.

soln by RNVAA

27. (AHSME 1974) If $f(x) = 3x + 2$ for all real x , then the statement: " $|f(x) + 4| < a$ whenever $|x + 2| < b$ and $a > 0$ and $b > 0$ " is true when

(A) $b \leq a/3$ (B) $b > a/3$ (C) $a \leq b/3$ (D) $a > b/3$

(E) The statement is never true.

Answer: A

Solution

Plugging in $f(x) = 3x + 2$ into $|f(x) + 4| < a$, we get $|3x + 6| = |3(x + 2)| = 3|x + 2| < a$. Therefore, $|x + 2| < \frac{a}{3}$. Therefore, we want $|x + 2| < b \implies |x + 2| < \frac{a}{3}$. This is clearly only true when $b \leq \frac{a}{3}$, A.

28. (AHSME 1974) Which of the following is satisfied by all numbers x of the form

$$x = \frac{a_1}{3} + \frac{a_2}{3^2} + \cdots + \frac{a_{25}}{3^{25}}$$

where a_1 is 0 or 2, a_2 is 0 or 2, ..., a_{25} is 0 or 2?

(A) $0 \leq x < 1/3$ (B) $1/3 \leq x < 2/3$ (C) $2/3 \leq x < 1$

(D) $0 \leq x < 1/3$ or $2/3 \leq x < 1$ (E) $1/2 \leq x \leq 3/4$

Answer: D

Solution

We can separate this into two cases.

Case 1: $a_1 = 0$ Here, clearly the minimum sum is when all of the a_i are 0, so $0 \leq x$. The maximum sum is when the rest of the a_i are all 2. In this case,

$$\sum_{n=2}^{25} \frac{2}{3^n} < 2 \sum_{n=2}^{\infty} \frac{1}{3^n} = 2 \left(\frac{\frac{1}{9}}{1 - \frac{1}{3}} \right) = \frac{1}{3}.$$

Therefore, in this case, $0 \leq x < 1/3$.

Case 2: $a_1 = 2$ Here, again, the minimum sum is when the rest of the a_i are all 0, in which case the minimum sum is $\frac{2}{3}$. The maximum sum is attained when all of the a_i are 2. In that case,

$$\sum_{n=1}^{25} \frac{2}{3^n} < 2 \sum_{n=1}^{\infty} \frac{1}{3^n} = 2 \left(\frac{\frac{1}{3}}{1 - \frac{1}{3}} \right) = 1.$$

Therefore, in this case, $2/3 \leq x < 1$.

Thus, we must have $0 \leq x < 1/3$ or $2/3 \leq x < 1$, D.

29. (AHSME 1974) For $p = 1, 2, \dots, 10$ let S_p be the sum of the first 40 terms of the arithmetic progression whose first term is p and whose common difference is $2p - 1$; then $S_1 + S_2 + \dots + S_{10}$ is
- (A) 80000 (B) 80200 (C) 80400 (D) 80600 (E) 80800

Answer: B

Solution

The 40th term of an arithmetic progression with a first term p and a common difference $2p - 1$ is $p + 39(2p - 1) = 79p - 39$. Therefore, the sum of the first 40 terms of such a progression is $\frac{40}{2}(79p - 39 + p) = 1600p - 780$.

We now want to evaluate $\sum_{p=1}^{10} (1600p - 780)$.

$$\begin{aligned} \sum_{p=1}^{10} (1600p - 780) &= 1600 \sum_{p=1}^{10} (p) - \sum_{p=1}^{10} (780) \\ &= (1600) \left(\frac{10 \cdot 11}{2} \right) - (780)(10) = 88000 - 7800 = 80200, \quad \boxed{\text{B}}. \end{aligned}$$

30. (AHSME 1974) A line segment is divided so that the lesser part is to the greater part as the greater part is to the whole. If R is the ratio of the lesser part to the greater part, then the value of

$$R^{[R(R^2+R^{-1})+R^{-1}]} + R^{-1}$$

is

- (A) 2 (B) $2R$ (C) R^{-1} (D) $2 + R^{-1}$ (E) $2 + R$

Answer: A

Solution

Let w be the length of the shorter segment and l be the length of the longer segment. We're given that $\frac{w}{l} = \frac{l}{w+l}$. Cross-multiplying, we find that $w^2 + wl = l^2 \implies w^2 + wl - l^2 = 0$. Now we divide both sides by l^2 to get $\left(\frac{w}{l}\right)^2 + \left(\frac{w}{l}\right) - 1 = 0$. Therefore, $R^2 + R - 1 = 0$.

From this, we have $R^2 = -R + 1$. Dividing both sides by R , we get $R = -1 + \frac{1}{R} \implies R^{-1} = R + 1$. Therefore, $R^2 + R^{-1} = -R + 1 + R + 1 = 2$. Finally, we have

$$R^{[R(R^2+R^{-1})+R^{-1}]} + R^{-1} = R^{[R^2+R^{-1}]} + R^{-1} = R^2 + R^{-1} = \boxed{\text{(A)}2}.$$

§11 AHSME 1975

1. (AHSME 1975) The value of $\frac{1}{2 - \frac{1}{2 - \frac{1}{2 - \frac{1}{2}}}}$ is

(A) $3/4$ (B) $4/5$ (C) $5/6$ (D) $6/7$ (E) $6/5$

Answer: B

Solution

Solution by e_power_pi_times_i

Calculating, we find that $\frac{1}{2 - \frac{1}{2 - \frac{1}{2 - \frac{1}{2}}}} = \frac{1}{2 - \frac{1}{2 - \frac{2}{3}}} = \frac{1}{2 - \frac{3}{4}} = \frac{1}{\frac{5}{4}} = \boxed{\text{(B)} \frac{4}{5}}.$

2. (AHSME 1975) For which real values of m are the simultaneous equations

$$y = mx + 3$$

$$y = (2m - 1)x + 4$$

satisfied by at least one pair of real numbers (x, y) ?

(A) all m (B) all $m \neq 0$ (C) all $m \neq 1/2$ (D) all $m \neq 1$
 (E) no values of m

Answer: D

Solution 1

Setting the two expressions for y equal to each other yields $mx + 3 = (2m - 1)x + 4$, which simplifies to $(m - 1)x + 1 = 0$. Therefore $x = \frac{1}{1 - m}$, so x is a real number if and only if $\boxed{\text{(D)} m \neq 1}$.

-e_power_pi_times_i

Solution 2

Any two lines with different slopes intersect. So if $m \neq 2m - 1$, that is, $m \neq 1$, the lines intersect and thus the system has one solution.

If $m = 1$, however, the lines will have equations $y = x + 3$ and $y = x + 4$. These lines have the same slope but different y -intercepts, so they are parallel and thus the system has no solutions.

Therefore, the system has at least one solution if and only if $\boxed{\text{(D)} m \neq 1}$.

-j314andrews

3. (AHSME 1975) Which of the following inequalities are satisfied for all real numbers a, b, c, x, y, z which satisfy the conditions $x < a, y < b$, and $z < c$?

I. $xy + yz + zx < ab + bc + ca$

II. $x^2 + y^2 + z^2 < a^2 + b^2 + c^2$

III. $xyz < abc$

(A) None are satisfied. (B) I only (C) II only (D) III only (E) All are satisfied.

Answer: A

Solution

Solution by e_power_pi_times_i

Notice if a , b , and c are 0, then we can find x , y , and z to disprove I and II. For example, if $(a, b, c, x, y, z) = (0, 0, 0, -1, -1, -1)$, then I and II are disproved. If $(a, b, c, x, y, z) = (0, 1, 2, -1, -1, 1)$, then III is disproved. Therefore the answer is **(A)** None are satisfied.

4. (AHSME 1975) If the side of one square is the diagonal of a second square, what is the ratio of the area of the first square to the area of the second?

(A) 2 (B) $\sqrt{2}$ (C) $1/2$ (D) $2\sqrt{2}$ (E) 4

Answer: A

Solution

Solution by e_power_pi_times_i

Denote the side of one square as s . Then the diagonal of the second square is s , so the side of the second square is $\frac{s\sqrt{2}}{2}$. The area of the second square is $\frac{1}{2}s^2$, so the ratio of the areas is $\frac{s^2}{\frac{1}{2}s^2} = \mathbf{(A) 2}$.

5. (AHSME 1975) The polynomial $(x + y)^9$ is expanded in decreasing powers of x . The second and third terms have equal values when evaluated at $x = p$ and $y = q$, where p and q are positive numbers whose sum is one. What is the value of p ?

(A) $1/5$ (B) $4/5$ (C) $1/4$ (D) $3/4$ (E) $8/9$

Answer: B

Solution

Solution by e_power_pi_times_i

The second and third term of $(x + y)^9$ is $9x^8y$ and $36x^7y^2$, respectively. For them to be equal when $x = p$, $\frac{p}{4} = y$. For them to be equal when $y = q$, $x = 4q$. Then $p + \frac{p}{4} = q + 4q$, so $\frac{5p}{4} = 5q$, which simplifies to $p = 4q$. Since $p + q = 1$, $p = \mathbf{(B) 4/5}$.

6. (AHSME 1975) The sum of the first eighty positive odd integers subtracted from the sum of the first eighty positive even integers is

(A) 0 (B) 20 (C) 40 (D) 60 (E) 80

Answer: E

Solution

Solution by e_power_pi_times_i

When the n th odd positive integer is subtracted from the n th even positive integer, the result is 1. Therefore the sum of the first eighty positive odd integers subtracted from the sum of the first eighty positive even integers is $80 \cdot 1 = \mathbf{(E) 80}$.

7. (AHSME 1975) For which non-zero real numbers x is $\frac{|x-|x||}{x}$ a positive integer?

(A) for negative x only
(B) for positive x only

- (C) only for x an even integer
 (D) for all non-zero real numbers x
 (E) for no non-zero real numbers x

Answer: E

Solution

Solution by e_power_pi_times_i

Notice that if x is negative, then the whole thing would amount to a negative number. Also notice that if x is positive, then $|x - |x||$ would be 0, hence the whole thing would amount to 0. Therefore, $\frac{|x - |x||}{x}$ is positive **(E)** for no non-zero real numbers x .

8. (AHSME 1975) If the statement "All shirts in this store are on sale." is false, then which of the following statements must be true?
- I. All shirts in this store are at non-sale prices.
 II. There is some shirt in this store not on sale.
 III. No shirt in this store is on sale.
 IV. Not all shirts in this store are on sale.

(A) II only (B) IV only (C) I and III only (D) II and IV only (E) I, II and IV only

Answer: D

Solution

Solution by e_power_pi_times_i

If not all the shirts in the store are on sale, then the answer is **(D)** II and IV only.

9. (AHSME 1975) Let $a_1, a_2, \dots$ and $b_1, b_2, \dots$ be arithmetic progressions such that $a_1 = 25, b_1 = 75$, and $a_{100} + b_{100} = 100$. Find the sum of the first hundred terms of the progression $a_1 + b_1, a_2 + b_2, \dots$
- (A) 0 (B) 100 (C) 10,000 (D) 505,000
 (E) not enough information given to solve the problem

Answer: C

Solution

Notice that a_{100} and b_{100} are $25 + 99k_1$ and $75 + 99k_2$, respectively. Therefore $k_2 = -k_1$. Now notice that $a_n + b_n = 25 + k_1(n - 1) + 75 + k_2(n - 1) = 100 + k_1(n - 1) - k_1(n - 1) = 100$. The sum of the first 100 terms is $100 \cdot 100 =$ **(C)** 10,000.

10. (AHSME 1975) The sum of the digits in base ten of $(10^{4n^2+8} + 1)^2$, where n is a positive integer, is
- (A) 4 (B) $4n$ (C) $2 + 2n$ (D) $4n^2$ (E) $n^2 + n + 2$

Answer: A

Solution

We see that the result of this expression will always be in the form (100 some number of zeros 001)². Multiplying these together yields:

$$110 \text{ some number of zeros } 011.$$

This works because of the way they are multiplied. Therefore, the answer is **(A)** 4.

11. (*AHSME 1975*) Let P be an interior point of circle K other than the center of K . Form all chords of K which pass through P , and determine their midpoints. The locus of these midpoints is

(A) a circle with one point deleted
 (B) a circle if the distance from P to the center of K is less than one half the radius of K ; otherwise a circular arc of less than 360°
 (C) a semicircle with one point deleted
 (D) a semicircle (E) a circle

Answer: E

Solution 1

It is (E) a circle.

The midpoints of the chord seems to be on the circle. For example, draw a chord passing through P . It is on the circle PK , where PK is the diameter. To prove this, we can use the fact that The perpendicular bisector of any chord of a circle passes through the center of the circle.

"Hence, the perpendicular through A passes through K . Since angle $PAK=90$, angle PAK is inscribed in the semicircular arc PK "

(Volume 2 , Chapter 20)

12. (*AHSME 1975*) If $a \neq b$, $a^3 - b^3 = 19x^3$, and $a - b = x$, which of the following conclusions is correct?

(A) $a = 3x$ (B) $a = 3x$ or $a = -2x$ (C) $a = -3x$ or $a = 2x$
 (D) $a = 3x$ or $a = 2x$ (E) $a = 2x$

Answer: B

Solution

We can factor $a^3 - b^3 = 19x^3$ into:

$$(a - b)(a^2 + ab + b^2) = 19x^3.$$

Substituting yields:

$$x(a^2 + ab + b^2) = 19x^3$$

$$a^2 + ab + b^2 = 19x^2.$$

This is equal to:

$$(a - b)^2 + 3ab = 19x^2$$

$$x^2 + 3ab = 19x^2$$

$$ab = 6x^2.$$

Checking with the possible answers, along with $a - b = x$ yields the only answer to be (B) : $a = 3x$ or $a = -2x$.

13. (*AHSME 1975*) The equation $x^6 - 3x^5 - 6x^3 - x + 8$ has
 (A) no real roots (B) exactly two distinct negative roots
 (C) exactly one negative root (D) no negative roots, but at least one positive root
 (E) none of these

Answer: D

Solution 1 (Really Smart)

Let $P(x) = x^6 - 3x^5 - 6x^3 - x + 8$. When $x < 0$, $P(x) > 0$. Therefore, there are no negative roots.

Notice that $P(1) = -1$ and $P(0) = 8$. There must be at least one positive root between 0 and 1, therefore the answer is **(D)**.

Sidenote : If anyone is wondering what this method is called,

It is called the **\textbf{\textit{squeeze theorem}}**

14. (AHSME 1975) If the *whatsis* is *so* when the *whosis* is *is* and the *so* and *so* is *is* · *so*, what is the *whosis* · *whatsis* when the *whosis* is *so*, the *so* and *so* is *so* · *so*, and the *is* is two (*whatsis*, *whosis*, *is*, and *so* are variables taking positive values)?
(A) *whosis* · *is* · *so* **(B)** *whosis* **(C)** *is* **(D)** *so* **(E)** *so* and *so*

Answer: E

Solution

From the problem, we are given:

$$\textit{whatsis} = \textit{so}$$

$$\textit{whosis} = \textit{is}$$

$$\textit{so} + \textit{so} = \textit{is} \cdot \textit{so}$$

We want to find what *whatsis* · *whosis* is when *whosis* = *so*, *so* + *so* = *so* · *so*, and *is* = 2.

Since *is* = 2 and *so* + *so* = *is* · *so* = *so* · *so* (from the given), that means *so* = 2. Now we know that *whatsis* = 2 and *whosis* = 2 as well because they are equal to *so* and *is*, respectively.

Plugging in the values, we get *whatsis* · *whosis* = *so* · *is* = 4. This also means that *whatsis* · *whosis* = *so* + *so*.

Taking a look at the answer choices, we see that our answer is **(E) so and so**.
 jiang147369

15. (AHSME 1975) In the sequence of numbers 1, 3, 2, ... each term after the first two is equal to the term preceding it minus the term preceding that. The sum of the first one hundred terms of the sequence is
(A) 5 **(B)** 4 **(C)** 2 **(D)** 1 **(E)** -1

Answer: A

Solution

First, write a few terms of the sequence: 1, 3, 2, -1, -3, -2, 1, 3, 2, ... Notice how the pattern repeats every six terms and every six terms have a sum of 0. Then, find that the $16 \cdot 6 = 96$ th term is -2 and the sum of the all those previous terms is 0. Then, write the 97th to the 100th terms down: 1, 3, 2, -1 and add them up to get the sum of **(A) 5**

16. (AHSME 1975) If the first term of an infinite geometric series is a positive integer, the common ratio is the reciprocal of a positive integer, and the sum of the series is 3, then the sum of the first two terms of the series is
(A) $1/3$ **(B)** $2/3$ **(C)** $8/3$ **(D)** 2 **(E)** $9/2$

Answer: C

Solution

Let a and r be the first term and common ratio of the series, respectively. Then $r = \frac{1}{b}$ for some positive integer b .

The sum of the series is $\frac{a}{1-r} = \frac{a}{1-\frac{1}{b}} = \frac{ab}{b-1} = 3$.

So $ab = 3(b-1) = 3b-3$, and thus $ab-3b = b(a-3) = -3$.

Since b and $a-3$ are both integers, b is a factor of 3. So either $b = 1$ or $b = 3$. But if $b = 1$, then $a = 0$ which is not positive. So $b = 3$ and thus $a = 2$.

Therefore, $a = 2$ and $r = \frac{1}{3}$. So the sum of the first 2 terms of the series is

$$2 + \frac{2}{3} = \boxed{\text{(C)} \frac{8}{3}}.$$

17. (AHSME 1975) A man can commute either by train or by bus. If he goes to work on the train in the morning, he comes home on the bus in the afternoon; and if he comes home in the afternoon on the train, he took the bus in the morning. During a total of x working days, the man took the bus to work in the morning 8 times, came home by bus in the afternoon 15 times, and commuted by train (either morning or afternoon) 9 times. Find x .

(A) 19 (B) 18 (C) 17 (D) 16
(E) not enough information given to solve the problem

Answer: D

Solution 1

Each day the man takes 2 trips: a trip to work and a trip back home. Since the man took $8 + 15 + 9 = 32$ trips, he worked $32 \div 2 = \boxed{\text{(D)} 16}$ days.

Solution 2

The man has three possible combinations of transportation:

Morning train, Afternoon bus (m.t., a.b.)

Morning bus, Afternoon train (m.b., a.t.)

Morning bus, Afternoon bus (m.b, a.b.)

.

Let y be the number of times the man takes the a.t.. Then, $9 - y$ is the number of times he takes the m.t.. Keep in mind that m.b. = y and a.b. = $9 - y$.

Let z be the number of times the man takes the m.b. and a.b.. Now, we get the two equations

$$y + z = 8$$

and

$$9 - y + z = 15.$$

Solving the system of equations, we get $y = 1$ and $z = 7$.

So during the x working days, the man took the (m.t., a.b.) on $9 - 1 = 8$ days, the (m.b., a.t.) on 1 day, and the (m.b., a.b.) on 7 days.

Therefore, $x = 8 + 1 + 7 = \boxed{\text{(D)} 16}$. jiang147369

18. (AHSME 1975) A positive integer N with three digits in its base ten representation is chosen at random, with each three digit number having an equal chance of being chosen. The probability that $\log_2 N$ is an integer is

(A) 0 (B) $3/899$ (C) $1/225$ (D) $1/300$ (E) $1/450$

Answer: D

Solution

In order for $\log_2 N$ to be an integer, N has to be an integer power of 2. Since N is a three-digit number in decimal form, N can only be $2^7 = 128$, $2^8 = 256$, or $2^9 = 512$. From 100 to 999, there are 900 three-digit numbers in total.

The probability of choosing 128, 256, or 512 from 900 numbers is $\frac{3}{900} = \frac{1}{300}$, which gives us the answer (D) $1/300$. jiang147369

19. (AHSME 1975) Which positive numbers x satisfy the equation $(\log_3 x)(\log_x 5) = \log_3 5$?

(A) 3 and 5 only (B) 3, 5, and 15 only
 (C) only numbers of the form $5^n \cdot 3^m$, where n and m are positive integers
 (D) all positive $x \neq 1$ (E) none of these

Answer: D

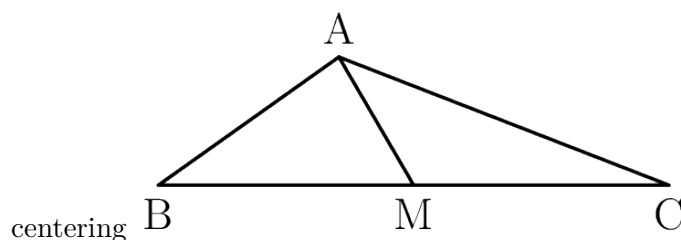
Solution

Note that $\log_3 x$ is defined if and only if x is positive, while $\log_x 5$ is defined if and only if x is positive but not equal to 1.

If x is positive and not equal to 1, by the change of base formula, $(\log_3 x)(\log_x 5) = \frac{\log x}{\log 3} \cdot \frac{\log 5}{\log x} = \frac{\log 5}{\log 3} = \log_3 5$.

Thus the set of all positive numbers x satisfying the equation is (D) all positive $x \neq 1$.

20. (AHSME 1975) In the adjoining figure $\triangle ABC$ is such that $AB = 4$ and $AC = 8$. If M is the midpoint of BC and $AM = 3$, what is the length of BC ?



(A) $2\sqrt{26}$ (B) $2\sqrt{31}$ (C) 9 (D) $4 + 2\sqrt{13}$
 (E) not enough information given to solve the problem

Answer: B

Solution

Let $BM = CM = x$. Then, by Stewart's Theorem, we have

$$\begin{aligned} 2x^3 + 18x &= 16x + 64x \\ \implies x^2 + 9 &= 40 \\ \implies x &= \sqrt{31} \implies BC = \boxed{2\sqrt{31}}. \end{aligned}$$

The answer is B.-brainiacmaniac31

21. (AHSME 1975) Suppose $f(x)$ is defined for all real numbers x ; $f(x) > 0$ for all x , and $f(a)f(b) = f(a+b)$ for all a and b . Which of the following statements is true?

I. $f(0) = 1$

II. $f(-a) = 1/f(a)$ for all a

III. $f(a) = \sqrt[3]{f(3a)}$ for all a

IV. $f(b) > f(a)$ if $b > a$

(A) III and IV only (B) I, III, and IV only (C) I, II, and IV only (D) I, II, and III only

Answer: D

Solution

I : $f(0) = 1$

Let $b = 0$. Our equation becomes $f(a)f(0) = f(a)$, so $f(0) = 1$. Therefore *I* is always true.

II : $f(-a) = \frac{1}{f(a)}$ for all a

Let $b = -a$. Our equation becomes $f(a)f(-a) = f(0) = 1 \rightarrow f(-a) = \frac{1}{f(a)}$. Therefore *II* is always true.

III : $f(a) = \sqrt[3]{f(3a)}$ for all a

First let $b = a$. We get $f(a)f(a) = f(2a)$. Now let $b = 2a$, giving us $f(3a) = f(a)f(2a) = f(a)^3 \rightarrow f(a) = \sqrt[3]{f(3a)}$. Therefore *III* is always true.

IV : $f(b) > f(a)$ if $b > a$ This is false. Let $f(x) = 2^{-x}$, for example. It satisfies the conditions but makes *IV* false. Therefore *IV* is not always true.

Since *I*, *II*, and *III* are true, the answer is (D).

- makol7

Note: For *IV*, one may also disprove it with simply $f(x) = 1$.

22. (AHSME 1975) If p and q are primes and $x^2 - px + q = 0$ has distinct positive integral roots, then which of the following statements are true?

I. The difference of the roots is odd.

II. At least one root is prime.

III. $p^2 - q$ is prime.

IV. $p + q$ is prime

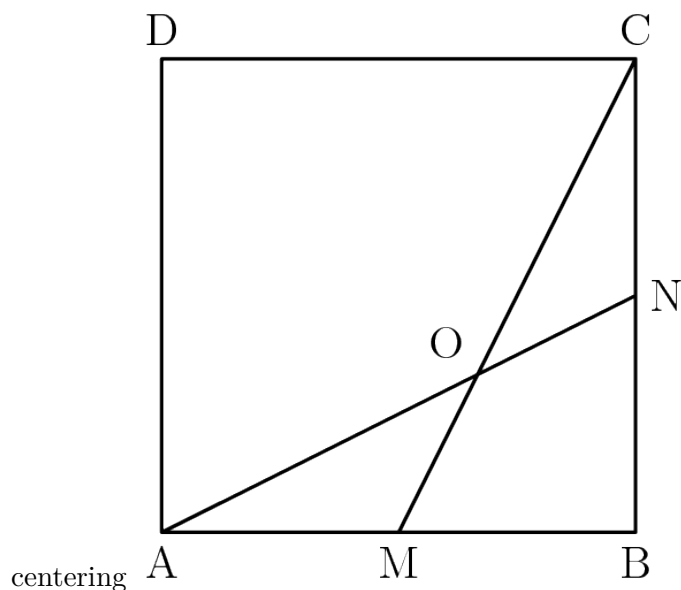
(A) I only (B) II only (C) II and III only (D) I, II and IV only (E) All are true.

Answer: E

Solution

Since the roots are both positive integers, we can say that $x^2 - px + q = (x-1)(x-q)$ since q only has 2 divisors. Thus, the roots are 1 and q and $p = q + 1$. The only two primes which differ by 1 are 2, 3 so $p = 3$ and $q = 2$. *I* is true because $3 - 2 = 1$. *II* is true because one of the roots is 2 which is prime. *III* is true because $3^2 - 2 = 7$ is prime. *IV* is true because $2 + 3 = 5$ is prime. Thus, the answer is (E).-brainiacmaniac31

23. (AHSME 1975) In the adjoining figure AB and BC are adjacent sides of square $ABCD$; M is the midpoint of AB ; N is the midpoint of BC ; and AN and CM intersect at O . The ratio of the area of $A OCD$ to the area of $ABCD$ is



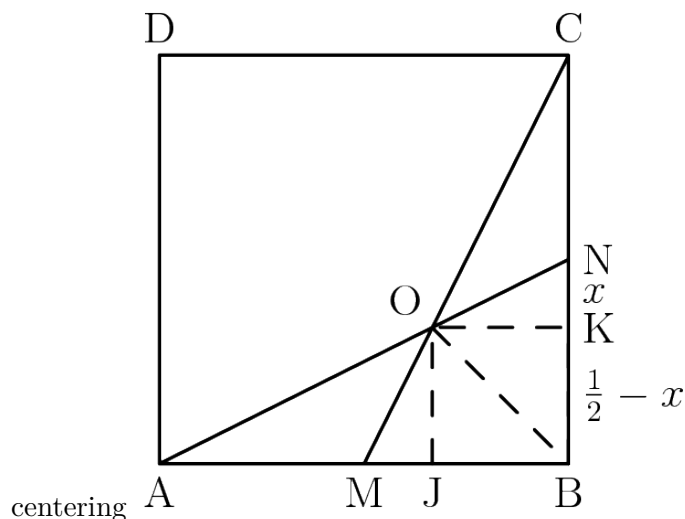
- (A) $\frac{5}{6}$ (B) $\frac{3}{4}$ (C) $\frac{2}{3}$ (D) $\frac{\sqrt{3}}{2}$ (E) $\frac{(\sqrt{3}-1)}{2}$

Answer: C

Solution

First, let's draw a few auxiliary lines. Drop altitudes from O to AB and from O to BC . We can label the points as J and K , respectively. This forms square $OKBJ$. Connect OB .

Without loss of generality, set the side length of the square $ABCD$ equal to 1. Let $NK = x$, and since N is the midpoint of CB , KB would be $\frac{1}{2} - x$. With the same reasoning, $MJ = x$ and $JB = \frac{1}{2} - x$.



We can also see that $\triangle COK$ is similar to $\triangle CMB$. That means $\frac{CK}{OK} = \frac{CB}{MB}$.

Plugging in the values, we get: $\frac{\frac{1}{2}+x}{\frac{1}{2}-x} = \frac{1}{\frac{1}{2}}$. Solving, we find that $x = \frac{1}{6}$. Then, $OK = \frac{1}{2} - \frac{1}{6} = \frac{1}{3}$. The area of $\triangle COB$ and $\triangle AOB$ together would be $2(\frac{1}{2} \cdot 1 \cdot \frac{1}{3}) = \frac{1}{3}$. Subtract this area from the total area of $ABCD$ to get the area of $AOCD$.

So, $A OCD = 1 - \frac{1}{3} = \frac{2}{3}$. The question asks for the ratio of the area of $A OCD$ to the area of $ABCD$, which is $\frac{\frac{2}{3}}{1} = \frac{2}{3}$.

The answer is (C) $\frac{2}{3}$. jiang147369

24. (AHSME 1975) In $\triangle ABC$, $\angle C = \theta$ and $\angle B = 2\theta$, where $0^\circ < \theta < 60^\circ$. The circle with center A and radius AB intersects AC at D and intersects BC , extended if necessary, at B and at E (E may coincide with B). Then $EC = AD$

(A) for no values of θ (B) only if $\theta = 45^\circ$ (C) only if $0^\circ < \theta \leq 45^\circ$
 (D) only if $45^\circ \leq \theta < 60^\circ$ (E) for all θ such that $0^\circ < \theta < 60^\circ$

Answer: E

Solution

Since $AD = AE$, we know $EC = AD$ if and only if triangle ACE is isosceles and $\angle ACE = \angle CAE$. Letting $\angle ACE = \theta$, we want to find when $\angle CAE = \theta$. We know $\angle ABC = \angle AEB = 2\theta$, so $\angle EAB = 180 - 4\theta$. We also know $\angle CAB = 180 - 3\theta$, and since $\angle CAE = \angle CAB - \angle EAB$, $\angle CAE = \theta$. Since we now know that $\angle CAE = \angle ACE$ regardless of θ , we have $0^\circ < \theta < 60^\circ$, or E.

25. (AHSME 1975) A woman, her brother, her son and her daughter are chess players (all relations by birth). The worst player's twin (who is one of the four players) and the best player are of opposite sex. The worst player and the best player are the same age. Who is the worst player?

(A) the woman (B) her son (C) her brother (D) her daughter
 (E) No solution is consistent with the given information

Answer: B

Solution 1

Since the worst player, the worst player's twin, and the best player are all different people of the same age, 3 of the 4 players are the same age. Since the woman is older than her 2 children, she is not one of these 3 players. Therefore, her brother and her 2 children are the same age.

So the worst player and worst player's twin must be the woman's 2 children, while the best player must be the woman's brother. Therefore, the worst player's twin is the woman's daughter while the worst player is (B) her son.

-j314andrews

Solution 2

We know that the worst player's twin and the best player are of opposite sex, and the worst and best players are the same age.

Case 1: Suppose the daughter was the worst player. The son would have to be her twin because nobody else can be her twin. Then, that means the son and the best player are of opposite sex. But we initially made the daughter the worst player, so she cannot also be the best player. That only leaves the woman to be the best player. However, the daughter and the woman are not the same age. Therefore, the worst player cannot be the daughter.

Case 2: Suppose the woman was the worst player. Suppose again that her and her brother are twins. Her brother and the best player are of opposite sex. Using

the same logic as in Case 1, the best player would be the daughter. The woman and the daughter are not the same age, so the worst player cannot be the woman.

Case 3: Suppose the brother was the worst player and that him and the woman are twins. The woman and the best player are of opposite sex, so the best player would be the son. If the brother is the woman's twin (our initial assumption), then the son is not the same age as the brother, which means that the son is not the best player, and the brother is not the worst player.

However, the woman and the brother do not necessarily have to be twins. For this case, we had to assume they were twins or else there was no way to continue the problem, since the brother would have no twin.

Case 4: Suppose the son was the worst player. The daughter and the best player are of opposite sex, so the best player is the brother. The woman and her brother do not have to be twins. So it is, in fact, possible that the brother and the son are the same age.

From this, we conclude our answer as (B) her son. jiang147369

26. (AHSME 1975) In acute $\triangle ABC$ the bisector of $\angle A$ meets side BC at D . The circle with center B and radius BD intersects side AB at M ; and the circle with center C and radius CD intersects side AC at N . Then it is always true that

(A) $\angle CND + \angle BMD - \angle DAC = 120^\circ$ (B) $AMDN$ is a trapezoid (C) BC is parallel to MN
 (D) $AM - AN = \frac{3(DB-DC)}{2}$ (E) $AB - AC = \frac{3(DB-DC)}{2}$

Answer: C

Solution(Very cheesy and sketch)

First by the angle bisector theorem $BM = BD = \frac{AB \cdot BC}{AC + BC} = AB \cdot \frac{BC}{AB + AC}$. Similarly, $CN = AC \cdot \frac{BC}{AB + AC}$. We thus have that $AM = AB - BM = AB - AB \cdot \frac{BC}{AB + AC} = AB \cdot \frac{AB + AC - BC}{AB + AC}$ and $AN = AC - CN = AC - AC \cdot \frac{BC}{AB + AC} = AC \cdot \frac{AB + AC - BC}{AB + AC}$. Finally, since A, M, B and A, N, C are all collinear, there exists a homothety between triangles BAC and MAN with scale factor $\frac{AB + AC - BC}{AB + AC}$. Thus, $BC \parallel MN$, or (C).

–AndrewZhong2012–

27. (AHSME 1975) If p, q and r are distinct roots of $x^3 - x^2 + x - 2 = 0$, then $p^3 + q^3 + r^3$ equals
 (A) -1 (B) 1 (C) 3 (D) 5 (E) none of these

Answer: E

Solution 1

If p is a root of $x^3 - x^2 + x - 2 = 0$, then $p^3 - p^2 + p - 2 = 0$, or

$$p^3 = p^2 - p + 2.$$

Similarly, $q^3 = q^2 - q + 2$, and $r^3 = r^2 - r + 2$, so

$$p^3 + q^3 + r^3 = (p^2 + q^2 + r^2) - (p + q + r) + 6.$$

By Vieta's formulas, $p + q + r = 1$, $pq + pr + qr = 1$, and $pqr = 2$. Squaring the equation $p + q + r = 1$, we get

$$p^2 + q^2 + r^2 + 2pq + 2pr + 2qr = 1.$$

Subtracting $2pq + 2pr + 2qr = 2$, we get

$$p^2 + q^2 + r^2 = -1.$$

Therefore, $p^3 + q^3 + r^3 = (p^2 + q^2 + r^2) - (p + q + r) + 6 = (-1) - 1 + 6 = \boxed{4}$. The answer is (E).

Solution 2(Faster)

We know that $p^3 + q^3 + r^3 = (p + q + r)(p^2 + q^2 + r^2 - pq - qr - pr) + 3pqr$. By Vieta's formulas, $p + q + r = 1$, $pqr = 2$, and $pq + qr + pr = 1$. So if we can find $p^2 + q^2 + r^2$, we are done. Notice that $(p + q + r)^2 = p^2 + q^2 + r^2 + 2pq + 2qr + 2pr$, so $p^2 + q^2 + r^2 = (p + q + r)^2 - 2(pq + qr + pr) = 1^2 - 2 \cdot 1 = -1$, which means that $p^3 + q^3 + r^3 = 1 \cdot -2 + 3 \cdot 2 = \boxed{(E) 4}$

Solution 3 (Beginner's Solution)

Use Vieta's formulas to get $p + q + r = 1$, $pq + qr + pr = 1$, and $pqr = 2$.

Square $p + q + r = 1$, and get $p^2 + q^2 + r^2 + 2pq + 2pr + 2qr = 1$

Substitute $pq + qr + pr = 1$ and simplify to get $-1 = p^2 + q^2 + r^2$

After that, multiply both sides by $1 = p + q + r$, to get $-1 = p^3 + q^3 + r^3 + p^2q + q^2r + p^2r + q^2r + r^2p + r^2q$

Then, factor out pq , qr , and pr : $-1 = p^3 + q^3 + r^3 + pq(p + q) + qr(q + r) + pr(p + r)$

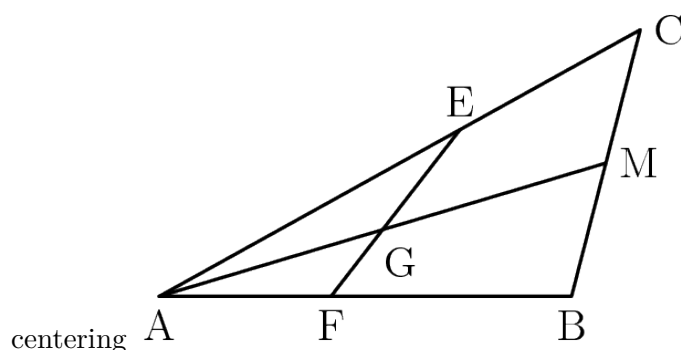
Then, substitute the first equation into $p + q$, $q + r$, and $p + r$. $-1 = p^3 + q^3 + r^3 + pq(1 - r) + qr(1 - p) + pr(1 - q)$

Then, multiply it out: $-1 = p^3 + q^3 + r^3 + pq + qr + pr - 3pqr$

After that, substitute the equations $pq + qr + pr = 1$ and $pqr = 2$: $-1 = p^3 + q^3 + r^3 + 1 - 6$

Solving that, you get $p^3 + q^3 + r^3 = \boxed{(E) 4}$ EZ PZ Ms.Lemon SQUEEZY

28. (AHSME 1975) In $\triangle ABC$ shown in the adjoining figure, M is the midpoint of side BC , $AB = 12$ and $AC = 16$. Points E and F are taken on AC and AB , respectively, and lines EF and AM intersect at G . If $AE = 2AF$ then $\frac{EG}{GF}$ equals



- (A) $\frac{3}{2}$ (B) $\frac{4}{3}$ (C) $\frac{5}{4}$ (D) $\frac{6}{5}$
 (E) not enough information to solve the problem

Answer: A

Solution

Here, we use Mass Points. Let $AF = x$. We then have $AE = 2x$, $EC = 16 - 2x$, and $FB = 12 - x$. Let B have a mass of 2. Since M is the midpoint, C also has a mass of 2. Looking at segment AB , we have

$$2 \cdot (12 - x) = mA_{AB} \cdot x$$

So

$$mA_{AB} = \frac{24 - 2x}{x}$$

Looking at segment AC , we have

$$2 \cdot (16 - 2x) = mA_{AC} \cdot 2x$$

So

$$mA_{AC} = \frac{16 - 2x}{x}$$

From this, we get

$$mE = \frac{16 - 2x}{x} + 2 \Rightarrow mE = \frac{16}{x}$$

and

$$mF = \frac{24 - 2x}{x} + 2 \Rightarrow mF = \frac{24}{x}$$

We want the value of $\frac{EG}{GF}$. This can be written as

$$\frac{EG}{GF} = \frac{mF}{mE}$$

Thus

$$\frac{mF}{mE} = \frac{\frac{24}{x}}{\frac{16}{x}} = \frac{3}{2}$$

A

Solution 2

Since we only care about a ratio EG/GF , and since we are given M being the midpoint of $\overline{BC}$, we realize we can conveniently also choose E to be the midpoint of $\overline{AC}$. (we're free to choose any point E on $\overline{AC}$ as long as AE is twice AF , the constraint given in the problem). This means $AE = 16/2 = 8$, and $AF = 4$. We then connect ME which creates similar triangles $\triangle EMC$ and $\triangle ABC$ by SAS, and thus generates parallel lines $\overline{EM}$ and $\overline{AB}$. This also immediately gives us similar triangles $\triangle AFG \sim \triangle MEG$, $\Rightarrow EG/FG = ME/AF = 6/4 = \boxed{3/2}$ (note that $ME = 6$ because ME/AB is in 1 : 2 ratio).

Solution 3

In order to find $\frac{EG}{FG}$, we can apply the law of sine to this model. Let:

$$\angle MAC = \alpha, \angle MAB = \beta, \angle AGE = \gamma, \angle AMC = \delta$$

Then, in the $\triangle AMC$ and $\triangle AMB$:

$$\frac{MC}{\sin \alpha} = \frac{AC}{\sin \delta}, \frac{MB}{\sin \beta} = \frac{AB}{\sin \delta} = \frac{MC}{\sin \alpha} \cdot \frac{3}{4}, \frac{\sin \alpha}{\sin \beta} = \frac{3}{4}$$

In the $\triangle AGE$ and $\triangle AGF$:

$$\frac{FG}{\sin \beta} = \frac{AF}{\sin \gamma}, \frac{EG}{\sin \alpha} = \frac{AE}{\sin \gamma}, \frac{2FG}{\sin \beta} = \frac{EG}{\sin \alpha}, \frac{EG}{FG} = \frac{2 \sin \alpha}{\sin \beta} = \frac{3}{2}$$

Hence, our answer is A.

-VSN

29. (AHSME 1975) What is the smallest integer larger than $(\sqrt{3} + \sqrt{2})^6$?

(A) 972 (B) 971 (C) 970 (D) 969 (E) 968

Answer: C

Solution(Brute Force)

$(\sqrt{3} + \sqrt{2})^6 = (5 + 2\sqrt{6})^3 = (5 + 2\sqrt{6})(49 + 20\sqrt{6}) = (485 + 198\sqrt{6})$ Then, find that $\sqrt{6}$ is about 2.449. Finally, multiply and add to find that the smallest integer higher is (C) 970

Solution(The Real Way - Binomial Theorem)

Let's evaluate $(\sqrt{3} + \sqrt{2})^6 + (\sqrt{3} - \sqrt{2})^6$. We see that all the irrational terms cancel. Then, using binomial theorem, we evaluate all the rational terms in the first expression to get 485. Then, the sum of the rational parts of the 2nd term will be 485 as well. Then, we get a total of 970 and since $(\sqrt{3} - \sqrt{2})^6 < 1$, the greatest integer greater than our original expression is (C) 970

30. (AHSME 1975) Let $x = \cos 36^\circ - \cos 72^\circ$. Then x equals

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) $3 - \sqrt{6}$ (D) $2\sqrt{3} - 3$ (E) none of these

Answer: B

Solution 1

Using the difference to product identity, we find that $x = \cos 36^\circ - \cos 72^\circ$ is equivalent to

$$\begin{aligned} x &= -2 \sin \frac{(36^\circ + 72^\circ)}{2} \sin \frac{(36^\circ - 72^\circ)}{2} \implies \\ x &= -2 \sin 54^\circ \sin(-18^\circ). \end{aligned}$$

Since sine is an odd function, we find that $\sin(-18^\circ) = -\sin 18^\circ$, and thus $-2 \sin 54^\circ \sin(-18^\circ) = 2 \sin 54^\circ \sin 18^\circ$. Using the property $\sin(90^\circ - a) = \cos a$, we find

$$\begin{aligned} x &= 2 \cos(90^\circ - 54^\circ) \cos(90^\circ - 18^\circ) \implies \\ x &= 2 \cos 36^\circ \cos 72^\circ. \end{aligned}$$

We multiply the entire expression by $\sin 36^\circ$ and use the double angle identity of sine twice to find

$$\begin{aligned} x \sin 36^\circ &= 2 \sin 36^\circ \cos 36^\circ \cos 72^\circ \implies \\ x \sin 36^\circ &= \sin 72^\circ \cos 72^\circ \implies \\ x \sin 36^\circ &= \frac{1}{2} \sin 144^\circ. \end{aligned}$$

Using the property $\sin(180^\circ - a) = \sin a$, we find $\sin 144^\circ = \sin 36^\circ$. Substituting this back into the equation, we have

$$x \sin 36^\circ = \frac{1}{2} \sin 36^\circ.$$

Dividing both sides by $\sin 36^\circ$, we have

$$x = \boxed{\text{(B)} \frac{1}{2}}$$

Solution 2

We observe that 72° is twice of 36° , so evaluating $\cos 36^\circ$ and using the double angle identify is enough to find $\cos 36^\circ - \cos 72^\circ$

We first let $n = \cos 36^\circ$. Notice that $36^\circ/4 = 9^\circ$ and $36^\circ = 45^\circ - 9^\circ$. Rewriting $\cos 36^\circ$ as $\cos(45^\circ - 9^\circ)$, we then use the difference of angles identify to find

$$n = \cos 36^\circ = \cos(45^\circ - 9^\circ) = \cos 45^\circ \cos 9^\circ - \sin 45^\circ \sin 9^\circ = \frac{\sqrt{2}}{2}(\cos 9^\circ - \sin 9^\circ)$$

Rearranged to get

$$\sqrt{2}n = \cos 9^\circ - \sin 9^\circ$$

Using the half-angle identify twice, and noticing that $\cos 9^\circ$ is positive, we get

$$\cos 9^\circ = \cos \frac{18^\circ}{2} = \sqrt{\frac{1 + \cos 18^\circ}{2}} = \sqrt{\frac{1 + \sqrt{\frac{1 + \cos 36^\circ}{2}}}{2}}$$

Also,

$$\sin 9^\circ = \sqrt{1 - \cos^2 9^\circ} = \sqrt{1 - \frac{1 + \sqrt{\frac{1 + \cos 36^\circ}{2}}}{2}} = \sqrt{\frac{1 - \sqrt{\frac{1 + \cos 36^\circ}{2}}}{2}}$$

Thus we can get

$$\sqrt{2}n = \sqrt{\frac{1 + \sqrt{\frac{1 + \cos 36^\circ}{2}}}{2}} - \sqrt{\frac{1 - \sqrt{\frac{1 + \cos 36^\circ}{2}}}{2}}$$

Squaring both sides and simplifying, this results in

$$2n^2 = 1 - \sqrt{\frac{1 - n}{2}}$$

From which we get

$$2n^2 - 1 = -\sqrt{\frac{1 - n}{2}}$$

Squaring both sides again,

$$4n^4 - 4n^2 + 1 = \frac{1 - n}{2}$$

Multiplying both sides by 2 and rearranging,

$$8n^4 - 8n^2 + n + 1 = 0$$

Factoring out $8n^2$ from the first two terms and applying difference of squares,

$$8n^2(n + 1)(n - 1) + n + 1 = 0$$

We know that $n = \cos 36^\circ$ which is not equal to -1 , so we divide both sides by $n + 1$

$$8n^3 - 8n^2 + 1 = 0$$

Using the rational root test, we find that $n = \frac{1}{2}$ is a solution of this equation. However, we also know that $\cos 36^\circ$ is not equal to $\frac{1}{2}$, so we divide both sides by $n - \frac{1}{2}$ to get

$$8n^2 - 4n - 2 = 0$$

Using the quadratic formula, we find that

$$n = \frac{1 \pm \sqrt{5}}{4}$$

We know n must be positive, which gives us

$$n = \frac{1 + \sqrt{5}}{4}$$

Thus, using the double angle identity, we can find

$$x = \cos 36^\circ - \cos 72^\circ = \cos 36^\circ - 2\cos^2 36^\circ + 1 = n - 2n^2 + 1 = \boxed{\text{(B)} \frac{1}{2}}$$

-SaraLiu24

Solution 3 (Roots of Unity)

Let $w = \cos 72^\circ - i \sin 72^\circ$. Then w is a fifth root of unity, so $w^5 = 1$. So $w^5 - 1 = (w - 1)(w^4 + w^3 + w^2 + w + 1) = 0$. Since $w - 1 \neq 0$, $w^4 + w^3 + w^2 + w + 1 = 0$.

Both the real and imaginary parts of $w^4 + w^3 + w^2 + w + 1$ must be equal to 0, and the real part of $w^4 + w^3 + w^2 + w + 1$ can also be expressed as $\cos 288^\circ + \cos 216^\circ + \cos 144^\circ + \cos 72^\circ + 1$. Also, $\cos 288^\circ = \cos 72^\circ$ and $\cos 216^\circ = \cos 144^\circ = -\cos 36^\circ$.

So $\cos 288^\circ + \cos 216^\circ + \cos 144^\circ + \cos 72^\circ + 1 = \cos 72^\circ - \cos 36^\circ - \cos 36^\circ + \cos 72^\circ + 1 = 2(\cos 72^\circ - \cos 36^\circ) + 1 = 0$. Therefore, $\cos 72^\circ - \cos 36^\circ = -\frac{1}{2}$ and

$$\cos 36^\circ - \cos 72^\circ = \boxed{\text{(B)} \frac{1}{2}}.$$

-j314andrews

§12 AHSME 1976

1. (AHSME 1976) If one minus the reciprocal of $(1 - x)$ equals the reciprocal of $(1 - x)$, then x equals

(A) -2 (B) -1 (C) $1/2$ (D) 2 (E) 3

Answer: B

Solution

The reciprocal of $(1 - x)$ is $\frac{1}{1-x}$, so our equation is

$$1 - \frac{1}{1-x} = \frac{1}{1-x},$$

which is equivalent to $\frac{1}{1-x} = \frac{1}{2}$. So, $1 - x = 2$ and $x = -1 \Rightarrow \text{(B)}$. MathJams

2. (AHSME 1976) For how many real numbers x is $\sqrt{-(x+1)^2}$ a real number?

(A) none (B) one (C) two
(D) a finite number greater than two (E) ∞

Answer: B

Solution

$\sqrt{-(x+1)^2}$ is a real number, if and only if $-(x+1)^2$ is nonnegative. Since $(x+1)^2$ is always nonnegative, $-(x+1)^2$ is nonnegative only when $-(x+1)^2 = 0$, or when $x = -1 \Rightarrow \text{(B)}$. MathJams

3. (AHSME 1976) The sum of the distances from one vertex of a square with sides of length 2 to the midpoints of each of the sides of the square is

(A) $2\sqrt{5}$ (B) $2 + \sqrt{3}$ (C) $2 + 2\sqrt{3}$ (D) $2 + \sqrt{5}$ (E) $2 + 2\sqrt{5}$

Answer: E

Solution

The lengths to the side are $1, \sqrt{2^2 + 1^2}, \sqrt{2^2 + 1^2}, 1$, respectively. Therefore, the sum is $\boxed{\text{(E)} 2 + 2\sqrt{5}}$. MathJams

4. (AHSME 1976) Let a geometric progression with n terms have first term one, common ratio r and sum s , where r and s are not zero. The sum of the geometric progression formed by replacing each term of the original progression by its reciprocal is

(A) $\frac{1}{s}$ (B) $\frac{1}{r^n s}$ (C) $\frac{s}{r^{n-1}}$ (D) $\frac{r^n}{s}$ (E) $\frac{r^{n-1}}{s}$

Answer: C

Solution

The sum of a geometric series with n terms, initial term a , and ratio r is $\frac{a(1-r^n)}{1-r}$. So, $s = \frac{(1-r^n)}{1-r}$. Our initial sequence is $1, r, r^2, \dots, r^n$, and replacing each term with its reciprocal gives us the sequence $1, \frac{1}{r}, \frac{1}{r^2}, \dots, \frac{1}{r^n}$. The sum is now $\frac{1 - (\frac{1}{r^n})}{1 - \frac{1}{r}} = \frac{\frac{(1-r^n)}{r^n}}{\frac{r-1}{r}} = \frac{(1-r^n)}{r^n} \cdot \frac{r}{r-1} = \frac{s}{r^{n-1}} \Rightarrow \text{(C)}$. MathJams

5. (AHSME 1976) How many integers greater than 10 and less than 100, written in base-10 notation, are increased by 9 when their digits are reversed?

(A) 0 (B) 1 (C) 8 (D) 9 (E) 10

Answer: C

Solution

Let our two digit number be $\overline{ab}$, where a is the tens digit, and b is the ones digit. So, $\overline{ab} = 10a + b$. When we reverse our digits, it becomes $10b + a$. So, $10a + b + 9 = 10b + a \Rightarrow a - b = 1$. So, our numbers are 12, 23, 34, 45, 56, 67, 78, 89 $\Rightarrow \text{(C)}$. MathJams

6. (AHSME 1976) If c is a real number and the negative of one of the solutions of $x^2 - 3x + c = 0$ is a solution of $x^2 + 3x - c = 0$, then the solutions of $x^2 - 3x + c = 0$ are

(A) 1, 2 (B) $-1, -2$ (C) 0, 3 (D) 0, -3 (E) $\frac{3}{2}, \frac{3}{2}$

Answer: C

Solution

We let the roots of the first equation be r, s and the roots of the second equation be $s, -t$. By Vieta's Formulas, $r + s = 3$ and $s - t = -3$, $rs = c$ and $-st = c$. So, $r = t$. Thus, $t + s = 3$, $s - t = 3$, so $t = 0$, and $s = 3 \Rightarrow \text{(C)}$. MathJams

7. (AHSME 1976) If x is a real number, then the quantity $(1 - |x|)(1 + x)$ is positive if and only if

(A) $|x| < 1$ (B) $|x| > 1$ (C) $x < -1$ or $-1 < x < 1$
(D) $x < 1$ (E) $x < -1$

Answer: C

Solution

We divide our solution into three cases: that of $x < -1$, that of $-1 < x < 1$, and that of $x > 1$. (When $x = -1$ or $x = 1$, the expression is zero, therefore not positive.)

If $x < -1$, then the first factor is negative, and the second factor is also negative.

If $-1 < x < 1$, then the first factor is positive, and the second factor is also positive.

If $x > 1$, the first factor is negative, but the second factor is positive.

Combining this with the rules for signs and multiplication, we find that the expression is positive when $x < -1$ or when $-1 < x < 1$, so our answer is (C) and we are done.

8. (AHSME 1976) A point in the plane, both of whose rectangular coordinates are integers with absolute values less than or equal to four, is chosen at random, with all such points having an equal probability of being chosen. What is the probability that the distance from the point to the origin is at most two units?
- (A) $\frac{13}{81}$ (B) $\frac{15}{81}$ (C) $\frac{13}{64}$ (D) $\frac{\pi}{16}$ (E) the square of a rational number

Answer: A

Solution 1

The rectangular region will be an 8x8 grid which is centered at the origin. Since the distance from the point chosen to the origin is at most two units, the favorable region will NOT be the 2×2 grid centered at the origin. It will be a circle of radius two units centered at the origin. The area of that circle is pi, and the area of the total region is 64. Therefore, the answer is (D) $\frac{\pi}{16}$

9. (AHSME 1976) In triangle ABC , D is the midpoint of AB ; E is the midpoint of DB ; and F is the midpoint of BC . If the area of $\triangle ABC$ is 96, then the area of $\triangle AEF$ is
- (A) 16 (B) 24 (C) 32 (D) 36 (E) 48

Answer: D

Solution

Let $[ABC]$ denote the area of ABC . Since F is the midpoint of BC , we have $[ABF] = [ACF] = \frac{96}{2} = 48$. D is the midpoint of AB , so $[ADF] = [BDF] = \frac{48}{2} = 24$. Now we just need to find $[DEF]$, since $[AEF] = [ADF] + [DEF]$. In $\triangle DBF$, E is the midpoint of DB , so $[DEF] = [EBF] = \frac{24}{2} = 12$. Thus,

$$[AEF] = [ADF] + [DEF] = 24 + 12 = 36$$

D

10. (AHSME 1976) If m , n , p , and q are real numbers and $f(x) = mx + n$ and $g(x) = px + q$, then the equation $f(g(x)) = g(f(x))$ has a solution
- (A) for all choices of m , n , p , and q
 (B) if and only if $m = p$ and $n = q$
 (C) if and only if $mq - np = 0$

- (D) if and only if $n(1 - p) - q(1 - m) = 0$
 (E) if and only if $(1 - n)(1 - p) - (1 - q)(1 - m) = 0$

Answer: D

Solution

$$f(g(x) = g(f(x)) \implies m(px + q) + n = p(mx + n) + q$$

This simplifies to $mq + n = pn + q$ or $n(1 - p) - q(1 - m) = 0$.

The answer is (D).

11. (*AHSME 1976*) Which of the following statements is (are) equivalent to the statement "If the pink elephant on planet alpha has purple eyes, then the wild pig on planet beta does not have a long nose"?

- I. "If the wild pig on planet beta has a long nose, then the pink elephant on planet alpha has purple eyes."
 II. "If the pink elephant on planet alpha does not have purple eyes, then the wild pig on planet beta does not have a long nose."
 III. "If the wild pig on planet beta has a long nose, then the pink elephant on planet alpha does not have purple eyes."
 IV. "The pink elephant on planet alpha does not have purple eyes, or the wild pig on planet beta does not have a long nose."

- (A) I. and II. only (B) III. and IV. only (C) II. and IV. only (D) II. and III. only

Answer: B

Solution

The following statements are all logically equivalent.

- I. P implies Q (pink elephant on planet alpha has purple eyes $\implies$ wild pig on planet beta does not have a long nose)
 III. Not Q implies not P (contrapositive)
 IV. Not P nor Q (inverse)

I is the given statement, so III and IV work. The answer is (B).

12. (*AHSME 1976*) A supermarket has 128 crates of apples. Each crate contains at least 120 apples and at most 144 apples. What is the largest integer n such that there must be at least n crates containing the same number of apples?

- (A) 4 (B) 5 (C) 6 (D) 24 (E) 25

Answer: C

Solution

To find the largest number of "repeated" crates necessary, we must account for all the possibilities of the number of apples in each crate. Since each crate contains a minimum of 120 apples and a maximum of 144 apples, there are $144 - 120 + 1 = 25$ different amounts possible for the number of apples per crate.

Now, we have to count for the worst case scenario: the 25 amounts are repeated as many times as possible.

25 can go into 128 exactly 5 times because $5 \cdot 25 = 125$, which is less than 128. This leaves a remainder of 3 crates.

The worst case scenario would be that these 3 crates have a different number of apples each. It doesn't actually matter how many apples are in these 3 crates because any of the 25 values would be repeated again anyway. So, the answer is

$5 + 1 =$ (C) 6 jiang147369

13. (AHSME 1976) If x cows give $x + 1$ cans of milk in $x + 2$ days, how many days will it take $x + 3$ cows to give $x + 5$ cans of milk?

- (A) $\frac{x(x+2)(x+5)}{(x+1)(x+3)}$ (B) $\frac{x(x+1)(x+5)}{(x+2)(x+3)}$
 (C) $\frac{(x+1)(x+3)(x+5)}{x(x+2)}$ (D) $\frac{(x+1)(x+3)}{x(x+2)(x+5)}$
 (E) none of these

Answer: A

Solution

First, the problem states:

x cows $x + 1$ cans $x + 2$ days

Multiply the current number of cows by $\frac{x+3}{x}$ to get the number of cows you want, and divide the number of days by the same amount because an increase in cows will cause a decrease in time.

$$x \cdot \frac{x+3}{x} \quad x + 1 \quad (x + 2) \cdot \frac{x}{x+3}$$

Finally, multiply the current amount of cans by $\frac{x+5}{x+1}$ to get the number of cans you want, and multiply the number of days by the same amount because an increase in cans will cause an increase in time.

$$x \cdot \frac{x+3}{x} \quad (x + 1) \cdot \frac{x+5}{x+1} \quad (x + 2) \cdot \frac{x}{x+3} \cdot \frac{x+5}{x+1}$$

Therefore, our answer is (A) $\frac{x(x+2)(x+5)}{(x+1)(x+3)}$ jiang147369

14. (AHSME 1976) The measures of the interior angles of a convex polygon are in arithmetic progression. If the smallest angle is 100° , and the largest is 140° , then the number of sides the polygon has is

- (A) 6 (B) 8 (C) 10 (D) 11 (E) 12

Answer: A

Solution

Let n equal the number of sides the polygon has. The sum of all the interior angles of a polygon is: $180(n - 2)$.

The formula for an arithmetic series is $\frac{n(a_1 + a_n)}{2}$. Set this equal to $180(n - 2)$ and solve. In this case, $a_1 = 100$ and $a_n = 140$.

Our equation becomes $\frac{n(100+140)}{2} = 180(n - 2) \Rightarrow 240n = 360(n - 2) \Rightarrow 120n = 720$.

Simplifying, we get $n =$ (A) 6 jiang147369

15. (AHSME 1976) If r is the remainder when each of the numbers 1059, 1417, and 2312 is divided by d , where d is an integer greater than 1, then $d - r$ equals

- (A) 1 (B) 15 (C) 179 (D) $d - 15$ (E) $d - 1$

Answer: B

Solution

We are given these congruences:

$$1059 \equiv r \pmod{d} \quad (\text{i})$$

$$1417 \equiv r \pmod{d} \quad (\text{ii})$$

$$2312 \equiv r \pmod{d} \quad (\text{iii})$$

Let's make a new congruence by subtracting (i) from (ii), which results in

$$358 \equiv 0 \pmod{d}.$$

Subtract (ii) from (iii) to get

$$895 \equiv 0 \pmod{d}.$$

Now we know that 358 and 895 are both multiples of d . Their prime factorizations are $358 = 2 \cdot 179$ and $895 = 5 \cdot 179$, so their common factor is 179, which means $d = 179$.

Plug $d = 179$ back into any of the original congruences to get $r = 164$. Then, $d - r = 179 - 164 = \boxed{\text{(B) } 15}$. [jiang147369](#)

16. (AHSME 1976) In triangles ABC and DEF , lengths AC , BC , DF , and EF are all equal. Length AB is twice the length of the altitude of $\triangle DEF$ from F to DE . Which of the following statements is (are) true?

- I. $\angle ACB$ and $\angle DFE$ must be complementary.
- II. $\angle ACB$ and $\angle DFE$ must be supplementary.
- III. The area of $\triangle ABC$ must equal the area of $\triangle DEF$.
- IV. The area of $\triangle ABC$ must equal twice the area of $\triangle DEF$.

(A) II. only (B) III. only (C) IV. only (D) I. and III. only (E) II. and III. only

Answer: E

Solution

Let $AC = BC = DF = EF = x$ and let the altitude from F to $\overline{DE}$ have length y . This gives that $AB = 2y$. Since $\triangle ABC$ and $\triangle DEF$ are isosceles, we have that $\angle ACB = 2\sin^{-1}\left(\frac{y}{x}\right)$ and $\angle DFE = 2\cos^{-1}\left(\frac{y}{x}\right)$. Since the sum of the inverse sine and cosine of any possible sine/cosine fraction is 90° , we have that $m\angle ACB + m\angle DFE = 180^\circ$. Thus, $\angle ACB$ and $\angle DFE$ are supplementary, so **II.** is true while **I.** is not.

Applying the Pythagorean Theorem gives that the length of the altitude from C to $\overline{AB}$ is $\sqrt{x^2 - y^2}$ and $DE = 2\sqrt{x^2 - y^2}$. This means that $[ABC] = [DEF] = y\sqrt{x^2 - y^2}$. So we also have that **III.** is true while **IV.** is not.

Thus, our answer is **(E) II. and III. only.**

17. (AHSME 1976) If θ is an acute angle, and $\sin 2\theta = a$, then $\sin \theta + \cos \theta$ equals

- (A) $\sqrt{a+1}$ (B) $(\sqrt{2}-1)a+1$ (C) $\sqrt{a+1}-\sqrt{a^2-a}$
 (D) $\sqrt{a+1}+\sqrt{a^2-a}$ (E) $\sqrt{a+1}+a^2-a$

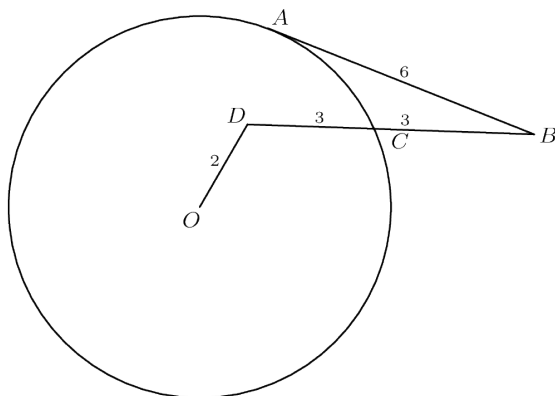
Answer: A

Solution

Let $x = \sin \theta + \cos \theta$, so we want to find x . First, square the expression to get $x^2 = \sin^2 \theta + 2\sin \theta \cos \theta + \cos^2 \theta$. Recall that $\sin^2 \theta + \cos^2 \theta = 1$ and $\sin 2\theta = 2\sin \theta \cos \theta$. Plugging these in, the equation simplifies to $x^2 = 1 + \sin 2\theta$. Given that $\sin 2\theta = a$, the equation becomes $x^2 = 1 + a$. Take the square root of both sides to get $x = \sqrt{1+a}$.

Hence, the answer is $\boxed{\text{(A) } \sqrt{a+1}}$ [jiang147369](#)

18. (AHSME 1976)



centering

In the adjoining figure, AB is tangent at A to the circle with center O ; point D is interior to the circle; and DB intersects the circle at C . If $BC = DC = 3$, $OD = 2$, and $AB = 6$, then the radius of the circle is

- (A) $3 + \sqrt{3}$ (B) $15/\pi$ (C) $9/2$ (D) $2\sqrt{6}$ (E) $\sqrt{22}$

Answer: E

Solution

Extend $\overline{BD}$ until it touches the opposite side of the circle, say at point E . By power of a point, we have $AB^2 = (BC)(BE)$, so $BE = \frac{6^2}{3} = 12$. Therefore, $DE = 12 - 3 - 3 = 6$.

Now extend $\overline{OD}$ in both directions so that it intersects the circle in two points (in other words, draw the diameter of the circle that contains OD). Let the two endpoints of this diameter be P and Q , where Q is closer to C . Again use power of a point. We have $(PD)(DQ) = (CD)(DE) = 3 \cdot 6 = 18$. But if the radius of the circle is r , we see that $PD = r + 2$ and $DQ = r - 2$, so we have the equation $(r + 2)(r - 2) = 18$. Solving gives $r = \sqrt{22}$.

19. (AHSME 1976) A polynomial $p(x)$ has remainder three when divided by $x - 1$ and remainder five when divided by $x - 3$. The remainder when $p(x)$ is divided by $(x - 1)(x - 3)$ is

- (A) $x - 2$ (B) $x + 2$ (C) 2 (D) 8 (E) 15

Answer: B

Solution

We know that $p(1) = 3$ and $p(3) = 5$. We write $p(x)$ as $p(x) = q(x)(x - 1)(x - 3) + r(x)$, where $r(x) = ax + b$. Plugging in $x = 1$, we get $a + b = 3$. Plugging in $x = 3$, we know $3a + b = 5$. We have a systems of equations, where we can solve that $a = 1$ and $b = 2$. So, our answer is $1(x) + (2) = x + 2 \Rightarrow$ (B). MathJams

1976 AHSME Problems

20. (AHSME 1976) Let a , b , and x be positive real numbers distinct from one. Then $4(\log_a x)^2 + 3(\log_b x)^2 = 8(\log_a x)(\log_b x)$

- (A) for all values of a , b , and x
 (B) if and only if $a = b^2$
 (C) if and only if $b = a^2$

(D) if and only if $x = ab$

(E) for none of these

Answer: E

Solution

Because $\log_m n = \frac{\log n}{\log m}$, $4(\log_a x)^2 + 3(\log_b x)^2 = \frac{4(\log x)^2}{(\log a)^2} + \frac{3(\log x)^2}{(\log b)^2} = \frac{(\log x)^2(4(\log a)^2 + 3(\log b)^2)}{(\log a \log b)^2}$

Therefore,

$$\frac{(\log x)^2(4(\log a)^2 + 3(\log b)^2)}{(\log a \log b)^2} = \frac{8(\log x)^2}{\log a \log b}$$

$$\frac{4(\log a)^2 + 3(\log b)^2}{\log a \log b} = 8$$

$$\frac{4(\log a)^2}{\log a \log b} + \frac{3(\log b)^2}{\log a \log b} = 8$$

$$4 \log_b a + 3 \log_a b = 8$$

Now let $k = \log_b a$. Our equation becomes

$$4k + \frac{3}{k} = 8$$

$$4k^2 - 8k + 3 = 0$$

$$k = \frac{8 \pm \sqrt{64 - 4(4)(3)}}{8}$$

$$k = \frac{1}{2}, \frac{3}{2}$$

Therefore, either $b = a^2$ or $b = \sqrt[3]{a^2}$. Since no option matches both of these solutions, the answer is **(E)**.

- mako17

21. (AHSME 1976) What is the smallest positive odd integer n such that the product $2^{1/7} 2^{3/7} \dots 2^{(2n+1)/7}$ is greater than 1000? (In the product the denominators of the exponents are all sevens, and the numerators are the successive odd integers from 1 to $2n + 1$.)

(A) 7 (B) 9 (C) 11 (D) 17 (E) 19

Answer: B

Solution

Combine the terms in the product to get $2^{\frac{1+3+5+\dots+(2n-1)+(2n+1)}{7}}$.

The exponent can be simplified to

$$\frac{1 + 3 + 5 + \dots + (2n - 1) + (2n + 1)}{7} \Rightarrow \frac{\frac{n(1+(2n+1))}{2}}{7} \Rightarrow \frac{n^2}{7}.$$

We want this inequality to be true with the smallest positive odd integer value of n :

$$2^{\frac{n^2}{7}} > 1000.$$

Now, let's test the answer choices. For $n = 7$, we have $2^{49/7} = 2^7 < 1000$. For $n = 9$, we have $2^{81/7} > 2^{10} > 1000$.

So our answer is (B) 9. [jiang147369](#)

22. (AHSME 1976) Given an equilateral triangle with side of length s , consider the locus of all points P in the plane of the triangle such that the sum of the squares of the distances from P to the vertices of the triangle is a fixed number a . This locus
- (A) is a circle if $a > s^2$
 - (B) contains only three points if $a = 2s^2$ and is a circle if $a > 2s^2$
 - (C) is a circle with positive radius only if $s^2 < a < 2s^2$
 - (D) contains only a finite number of points for any value of a
 - (E) is none of these

Answer: A

Solution

If we define point $P = (x, y)$, and the three points of the triangle as $P_1 = (x_1, y_1)$, $P_2 = (x_2, y_2)$, and $P_3 = (x_3, y_3)$, then we have two equations to work with.

First, P must satisfy the equation:

$$|P - P_1|^2 + |P - P_2|^2 + |P - P_3|^2 = [(x - x_1)^2 + (y - y_1)^2] + [(x - x_2)^2 + (y - y_2)^2] + [(x - x_3)^2 + (y - y_3)^2] = a$$

Second, the fact that the three sides of the triangle all have length s gives us: $|P_2 - P_1|^2 + |P_3 - P_2|^2 + |P_1 - P_3|^2 = [(x_2 - x_1)^2 + (y_2 - y_1)^2] + [(x_3 - x_2)^2 + (y_3 - y_2)^2] + [(x_1 - x_3)^2 + (y_1 - y_3)^2] = 3s^2$

Define the center of the triangle as $P_C = (x_C, y_C)$.

Note that $x_C = \frac{x_1 + x_2 + x_3}{3}$ and $y_C = \frac{y_1 + y_2 + y_3}{3}$.

Also note that the distance between P_C and each of the triangle vertices is $\frac{\sqrt{3}s}{3}$, so:

$$|P_C - P_1|^2 + |P_C - P_2|^2 + |P_C - P_3|^2 = [(x_C - x_1)^2 + (y_C - y_1)^2] + [(x_C - x_2)^2 + (y_C - y_2)^2] + [(x_C - x_3)^2 + (y_C - y_3)^2] = s^2$$

Rewrite $(x - x_1) = (x - x_C) + (x_C - x_1)$, and similarly for $(y - y_1)$, $(x - x_2)$, etc.

Now expanding the first sum and using the second sum, we get:

$$a = 3[(x - x_C)^2 + (y - y_C)^2] + s^2 + 2(x - x_C)[(x_C - x_1) + (x_C - x_2) + (x_C - x_3)] + 2(y - y_C)[(y_C - y_1) + (y_C - y_2) + (y_C - y_3)]$$

Because P_C is centroid of the triangle, its coordinates are the averages of the coordinates of the three points. Thus, $3x_C = x_1 + x_2 + x_3$, and the latter parts of the above equations are zero.

$$\text{Thus, } a - s^2 = 3[(x - x_C)^2 + (y - y_C)^2]$$

This means that when $a > s^2$, the locus of points P will be a circle centered at P_C .

Therefore, the correct answer is (A).

This problem could be also solved with coordinate geometry, where we place P_1 at the origin and P_2 along the positive x-axis, or even let P_C be the origin and place P_1 along the negative x-axis.

23. (AHSME 1976) For integers k and n such that $1 \leq k < n$, let $C_k^n = \frac{n!}{k!(n-k)!}$. Then $\left(\frac{n-2k-1}{k+1}\right) C_k^n$ is an integer

- (A) for all k and n
 (B) for all even values of k and n , but not for all k and n
 (C) for all odd values of k and n , but not for all k and n
 (D) if $k = 1$ or $n = 1$, but not for all odd values k and n
 (E) if n is divisible by k , but not for all even values k and n

Answer: A

Solution

We know $C_k^n = \binom{n}{k}$, so let's rewrite the expression as $\binom{\frac{n-2k-1}{k+1}}{\frac{n-2k-1}{k+1}}$. Notice that

$$n - 2k - 1 = n - 2(k + 1) + 1 = (n + 1) - 2(k + 1).$$

This allows us to rewrite the expression as

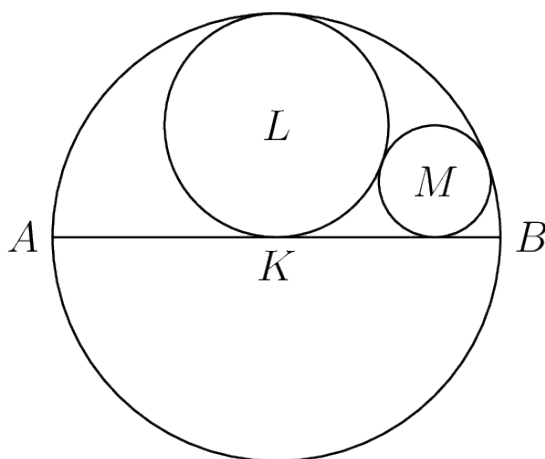
$$\binom{(n+1)-2(k+1)}{k+1} \binom{n}{k}.$$

From here, we just have to do some algebra to get

$$\begin{aligned} \binom{(n+1)-2(k+1)}{k+1} \binom{n}{k} &= \left(\frac{n+1}{k+1} - 2 \right) \frac{n!}{k!(n-k)!} \\ &= \frac{(n+1)!}{(k+1)!(n-k)!} - 2 \cdot \frac{n!}{k!(n-k)!} \\ &= \binom{n+1}{k+1} - 2 \binom{n}{k}, \end{aligned}$$

so $\binom{\frac{n-2k-1}{k+1}}{\frac{n-2k-1}{k+1}} C_k^n$ is an integer (A) for all k and n . jiang147369

24. (AHSME 1976) In the adjoining figure, circle K has diameter AB ; circle L is tangent to circle K and to AB at the center of circle K ; and circle M tangent to circle K , to circle L and AB . The ratio of the area of circle K to the area of circle M is



centering

- (A) 12 (B) 14 (C) 16 (D) 18 (E) not an integer

Answer: C

Let R and r be the radius of $\odot K$ and the radius of $\odot M$, respectively. It follows that the radius of $\odot L$ is $\frac{R}{2}$.

[illegible]

In right $\triangle KPM$, we have $KP = r$ and $KM = R - r$. By the Pythagorean Theorem, we get $PM^2 = (R - r)^2 - r^2$.

We equate the expressions for PM^2 , then simplify:

Therefore, the ratio of the area of $\odot K$ to the area of $\odot M$ is $\frac{\pi R^2}{\pi r^2} = \left(\frac{R}{r}\right)^2 = \boxed{(\mathbf{C}) \ 16}$.

- (A)** if $k = 1$
(B) if $k = 2$, but not if $k = 1$
(C) if $k = 3$, but not if $k = 2$
(D) if $k = 4$, but not if $k = 3$
(E) for no value of k

Solution

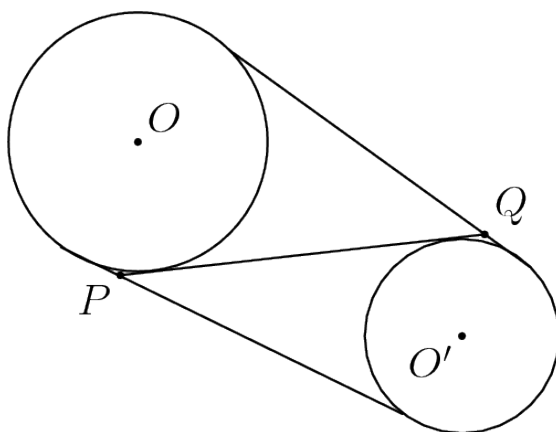
Note that

$$\begin{aligned}
 \Delta^1(u_n) &= [(n+1)^3 + (n+1)] - [n^3 + n] \\
 &= [n^3 + 3n^2 + 4n + 2] - [n^3 + n] \\
 &= 3n^2 + 3n + 2, \\
 \Delta^2(u_n) &= [3(n+1)^2 + 3(n+1) + 2] - [3n^2 + 3n + 2] \\
 &= [3n^2 + 9n + 8] - [3n^2 + 3n + 2] \\
 &= 6n + 6, \\
 \Delta^3(u_n) &= [6(n+1) + 6] - [6n + 6] \\
 &= [6n + 12] - [6n + 6] \\
 &= 6, \\
 \Delta^4(u_n) &= 6 - 6 \\
 &= 0.
 \end{aligned}$$

Therefore, the answer is (D).

More generally, we have $\Delta^k(u_n) = 0$ for all n if $k \geq 4$.

26. (AHSME 1976)



centering

In the adjoining figure, every point of circle O' is exterior to circle O . Let P and Q be the points of intersection of an internal common tangent with the two external common tangents. Then the length of PQ is

- (A) the average of the lengths of the internal and external common tangents
- (B) equal to the length of an external common tangent if and only if circles O and O' have equal radii
- (C) always equal to the length of an external common tangent
- (D) greater than the length of an external common tangent
- (E) the geometric mean of the lengths of the internal and external common tangents

Answer: C

Solution

By the **Two Tangent Theorem**, the segments drawn from point P tangent to circle O are congruent. Let them have length p . Similarly, the two segments drawn from point Q tangent to circle O' are congruent. Let them have length q .

Let the length of the two external common tangents be x . We know that the two segments drawn from point P tangent to circle O' are congruent. One of these tangents (the one on the external common tangent) has length $x - p$, so the other (the one on the internal common tangent) also has length $x - p$. Doing the same thing for point Q and circle O shows that the segments drawn from Q tangent to circle O has length $x - q$. Thus, $PQ = \frac{x-p+x-q+p+q}{2} = \frac{2x}{2} = x$, because the numerator of this fraction counts the length of $\overline{PQ}$ twice. Because x is the length of the external common tangent, our answer is (C) always equal to the length of an external common tangent.

27. (AHSME 1976) If

$$N = \frac{\sqrt{\sqrt{5}+2} + \sqrt{\sqrt{5}-2}}{\sqrt{\sqrt{5}+1}} - \sqrt{3-2\sqrt{2}},$$

then N equals

- (A) 1 (B) $2\sqrt{2} - 1$ (C) $\frac{\sqrt{5}}{2}$ (D) $\sqrt{\frac{5}{2}}$ (E) none of these

Answer: A

Solution 1

Let $x = \frac{\sqrt{\sqrt{5}+2} + \sqrt{\sqrt{5}-2}}{\sqrt{\sqrt{5}+1}}$ and $y = \sqrt{3-2\sqrt{2}}$.

Note that

$$\begin{aligned} x^2 &= \frac{\left(\sqrt{\sqrt{5}+2} + \sqrt{\sqrt{5}-2}\right)^2}{\left(\sqrt{\sqrt{5}+1}\right)^2} \\ &= \frac{(\sqrt{5}+2) + 2 \cdot \left(\sqrt{\sqrt{5}+2} \cdot \sqrt{\sqrt{5}-2}\right) + (\sqrt{5}-2)}{\sqrt{5}+1} \\ &= \frac{(\sqrt{5}+2) + 2 + (\sqrt{5}-2)}{\sqrt{5}+1} \\ &= \frac{2\sqrt{5}+2}{\sqrt{5}+1} \\ &= 2. \end{aligned}$$

Since $x > 0$, we have $x = \sqrt{2}$.

On the other hand, note that

$$\begin{aligned} y^2 &= 3 - 2\sqrt{2} \\ &= 2 - 2\sqrt{2} + 1 \\ &= (\sqrt{2} - 1)^2. \end{aligned}$$

Since $y > 0$, we have $y = \sqrt{2} - 1$.

Finally, the answer is

$$N = x - y = \boxed{\text{(A) } 1}.$$

Solution 2

Let $x = \frac{\sqrt{\sqrt{5}+2} + \sqrt{\sqrt{5}-2}}{\sqrt{\sqrt{5}+1}}$ and $y = \sqrt{3-2\sqrt{2}}$.

Note that

$$x = \frac{\sqrt{\sqrt{5}+2} + \sqrt{\sqrt{5}-2}}{\sqrt{\sqrt{5}+1}} \cdot \frac{\sqrt{\sqrt{5}-1}}{\sqrt{\sqrt{5}-1}} = \frac{\sqrt{3+\sqrt{5}} + \sqrt{7-3\sqrt{5}}}{2}. \quad (\star)$$

We rewrite each term in the numerator separately:

Let $\sqrt{a} + \sqrt{b} = \sqrt{3 + \sqrt{5}}$ for some nonnegative rational numbers a and b . We square both sides of this equation, then rearrange:

$$a + b + 2\sqrt{ab} = 3 + \sqrt{5}$$

$$a + b + \sqrt{4ab} = 3 + \sqrt{5}.$$

It follows that

$$a + b = 3,$$

$$ab = \frac{5}{4}.$$

By inspection, we get $\{a, b\} = \left\{\frac{1}{2}, \frac{5}{2}\right\}$. Alternatively, we conclude that a and b are the solutions to the quadratic equation $t^2 - 3t + \frac{5}{4} = 0$ by Vieta's Formulas, in which $t = \frac{1}{2}, \frac{5}{2}$. Therefore, we obtain

$$\sqrt{3 + \sqrt{5}} = \sqrt{\frac{1}{2}} + \sqrt{\frac{5}{2}} = \frac{\sqrt{2}}{2} + \frac{\sqrt{10}}{2}.$$

28. Similarly, we obtain

$$\sqrt{7 - 3\sqrt{5}} = \frac{3\sqrt{2}}{2} - \frac{\sqrt{10}}{2}.$$

Substituting these results into $(\star)$, we have

$$x = \frac{\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{10}}{2}\right) + \left(\frac{3\sqrt{2}}{2} - \frac{\sqrt{10}}{2}\right)}{2} = \sqrt{2}.$$

On the other hand, we have

$$y = \sqrt{2} - 1$$

by the argument of either Solution 1 or Solution 2.

Finally, the answer is

$$N = x - y = \boxed{\text{(A) } 1}.$$

29. (AHSME 1976) Lines $L_1, L_2, \dots, L_{100}$ are distinct. All lines L_{4n}, n a positive integer, are parallel to each other. All lines L_{4n-3}, n a positive integer, pass through a given point A . The maximum number of points of intersection of pairs of lines from the complete set $\{L_1, L_2, \dots, L_{100}\}$ is

(A) 4350 (B) 4351 (C) 4900 (D) 4901 (E) 9851

Answer: B

Solution

We partition $\{L_1, L_2, \dots, L_{100}\}$ into three sets. Let

$$\begin{aligned} X &= \{L_n \mid n \equiv 0 \pmod{4}\}, \\ Y &= \{L_n \mid n \equiv 1 \pmod{4}\}, \\ Z &= \{L_n \mid n \equiv 2, 3 \pmod{4}\}, \end{aligned}$$

from which $|X| = |Y| = 25$ and $|Z| = 50$.

Any two distinct lines can intersect at most once. To maximize the number of points of intersection, note that each point must be passed by exactly two lines. If three or more lines pass through the same point, then we can create more points of intersection by translating the lines.

We construct the sets one by one:

- a) We construct all lines in set X . Since the lines in set X are parallel to each other, they have 0 points of intersection.
- b) We construct all lines in set Y . The lines in set Y have 1 point of intersection with each other, namely A .

Moreover, each line in set Y can intersect each line in set X once. So, there are $25^2 = 625$ points of intersection.

Now, we have $1 + 625 = 626$ additional points of intersection.

- c) We construct all lines in set Z . The lines in set Z can have $\binom{50}{2} = 1225$ points of intersection with each other.

Moreover, each line in set Z can intersect each line in sets X and Y once. So, there are $50^2 = 2500$ points of intersection.

Now, we have $1225 + 2500 = 3725$ additional points of intersection.

Together, the answer is $626 + 3725 = \boxed{\text{(B)} 4351}$.

30. (AHSME 1976) Ann and Barbara were comparing their ages and found that Barbara is as old as Ann was when Barbara was as old as Ann had been when Barbara was half as old as Ann is. If the sum of their present ages is 44 years, then Ann's age is
 (A) 22 (B) 24 (C) 25 (D) 26 (E) 28

Answer: B

Solution

This problem is very wordy. Nonetheless, let a and b be Ann and Barbara's current ages, respectively. We are given that $a + b = 44$. Let y equal the difference between their ages, so $y = a - b$. Know that y is constant because the difference between their ages will always be the same.

Now, let's tackle the equation: $b = \text{Ann's age when Barbara was Ann's age when Barbara was } \frac{a}{2}$. When Barbara was $\frac{a}{2}$ years old, Ann was $\frac{a}{2} + y$ years old. So the equation becomes $b = \text{Ann's age when Barbara was } \frac{a}{2} + y$. Adding on their age difference again, we get $b = \frac{a}{2} + y + y \Rightarrow b = \frac{a}{2} + 2y$. Substitute $a - b$ back in for y to get $b = \frac{a}{2} + 2(a - b)$. Simplify: $2b = a + 4(a - b) \Rightarrow 6b = 5a$. Solving b in terms of a , we have $b = \frac{5a}{6}$. Substitute that back into the first equation of $a + b = 44$ to get $\frac{11a}{6} = 44$. Solve for a , and the answer is $\boxed{\text{(B)} 24}$. [jiang147369](#)

31. (*AHSME 1976*) How many distinct ordered triples (x, y, z) satisfy the following equations?

$$\begin{aligned}x + 2y + 4z &= 12 \\xy + 4yz + 2xz &= 22 \\xyz &= 6\end{aligned}$$

(A) none (B) 1 (C) 2 (D) 4 (E) 6

Answer: E

Solution

The first equation suggests the substitution $(a, b, c) = (x, 2y, 4z)$, from which $(x, y, z) = \left(a, \frac{b}{2}, \frac{c}{4}\right)$.

We rewrite the given equations in terms of a, b , and c :

$$\begin{aligned}a + b + c &= 12, \\ \frac{ab}{2} + \frac{bc}{2} + \frac{ac}{2} &= 22, \\ \frac{abc}{8} &= 6.\end{aligned}$$

We clear fractions in these equations:

$$\begin{aligned}a + b + c &= 12, \\ ab + ac + bc &= 44, \\ abc &= 48.\end{aligned}$$

By Vieta's formulas, note that a, b , and c are the roots of the equation

$$r^3 - 12r^2 + 44r - 48 = 0,$$

which factors as

$$(r - 2)(r - 4)(r - 6) = 0.$$

It follows that $\{a, b, c\} = \{2, 4, 6\}$. For each permutation of (a, b, c) , we find the corresponding ordered triple $(x, y, z) = \left(a, \frac{b}{2}, \frac{c}{4}\right)$, as shown below:

a	b	c	x	y	z
2	4	6	2	2	3/2
2	6	4	2	3	1
4	2	6	4	1	3/2
4	6	2	4	3	1/2
6	2	4	6	1	1
6	4	2	6	2	1/2

So, there are (E) 6 such ordered triples (x, y, z) .

§13 AHSME 1977

1. (AHSME 1977) If $y = 2x$ and $z = 2y$, then $x + y + z$ equals

(A) x (B) $3x$ (C) $5x$ (D) $7x$ (E) $9x$

Answer: D

Solution

Solution by e_power_pi_times_i

$$x + y + z = x + (2x) + (4x) = \boxed{\text{(D)} 7x}$$

2. (AHSME 1977) Which one of the following statements is false? All equilateral triangles are

(A) equiangular (B) isosceles (C) regular polygons
(D) congruent to each other (E) similar to each other

Answer: D

Solution

Solution by e_power_pi_times_i

Equilateral triangles can be in different sizes, therefore they are not (D) congruent to each other

3. (AHSME 1977) A man has 2.73 in pennies, nickels, dimes, quarters and half dollars. If he has an equal number of each, how many coins does he have?

(A) 3 (B) 5 (C) 9 (D) 10 (E) 15

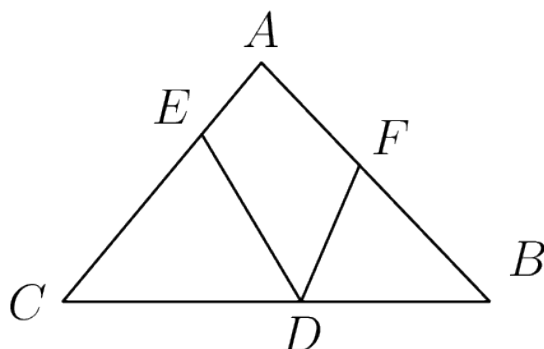
Answer: E

Solution

Solution by e_power_pi_times_i

Denote the number of pennies, nickels, dimes, quarters, and half-dollars by x . Then $x + 5x + 10x + 25x + 50x = 273$, and $x = 3$. The total amount of coins is $5x$, which is **(E)** 15.

4. (AHSME 1977)



centering

In triangle ABC , $AB = AC$ and $\angle A = 80^\circ$. If points D , E , and F lie on sides BC , AC and AB , respectively, and $CE = CD$ and $BF = BD$, then $\angle EDF$ equals

(A) 30° (B) 40° (C) 50° (D) 65° (E) none of these

Answer: C

Solution

Solution by e_power_pi_times_i

Because $\angle A = 80^\circ$, $\angle B = \angle C = 50^\circ$. Then, because triangles CDE and BDF are isosceles, $\angle CDE = \angle BDF = 65^\circ$. $\angle EDF = 180^\circ - 2(65^\circ) = \text{(C)} 50^\circ$

5. (AHSME 1977) The set of all points P such that the sum of the (undirected) distances from P to two fixed points A and B equals the distance between A and B is

(A) the line segment from A to B (B) the line passing through A and B
 (C) the perpendicular bisector of the line segment from A to B
 (D) an ellipse having positive area (E) a parabola

Answer: A

Solution

Solution by e_power_pi_times_i

The answer is (A) because P has to be on the line segment AB in order to satisfy $PA + PB = AB$.

6. (AHSME 1977) If x, y and $2x + \frac{y}{2}$ are not zero, then $(2x + \frac{y}{2})^{-1} \left[(2x)^{-1} + (\frac{y}{2})^{-1} \right]$ equals

(A) 1 (B) xy^{-1} (C) $x^{-1}y$ (D) $(xy)^{-1}$ (E) none of these

Answer: D

Solution

We can write $(2x + \frac{y}{2})^{-1} = \left(\frac{4x+y}{2} \right)^{-1} = \frac{2}{4x+y}$.

Then, the expression simplifies to

$$\left(2x + \frac{y}{2} \right)^{-1} \left[(2x)^{-1} + \left(\frac{y}{2} \right)^{-1} \right] \Rightarrow \frac{2}{4x+y} \left(\frac{1}{2x} + \frac{2}{y} \right) \Rightarrow \frac{2}{4x+y} \left(\frac{y+4x}{2xy} \right) \Rightarrow \frac{1}{xy}.$$

Thus, our answer is (D) $(xy)^{-1}$. [jiang147369](#)

7. (AHSME 1977) If $t = \frac{1}{1-\sqrt[4]{2}}$, then t equals

(A) $(1 - \sqrt[4]{2})(2 - \sqrt{2})$ (B) $(1 - \sqrt[4]{2})(1 + \sqrt{2})$ (C) $(1 + \sqrt[4]{2})(1 - \sqrt{2})$
 (D) $(1 + \sqrt[4]{2})(1 + \sqrt{2})$ (E) $-(1 + \sqrt[4]{2})(1 + \sqrt{2})$

Answer: E

Solution

Solution by e_power_pi_times_i

$$t = \frac{1}{1 - \sqrt[4]{2}} = \left(\frac{1}{1 - \sqrt[4]{2}} \right) \left(\frac{1 + \sqrt[4]{2}}{1 + \sqrt[4]{2}} \right) = \left(\frac{1 + \sqrt[4]{2}}{1 - \sqrt{2}} \right) \left(\frac{1 + \sqrt{2}}{1 + \sqrt{2}} \right) = \frac{(1 + \sqrt[4]{2})(1 + \sqrt{2})}{-1} = \text{(E)} - (1 + \sqrt[4]{2})(1 + \sqrt{2}).$$

8. (AHSME 1977) For every triple (a, b, c) of non-zero real numbers, form the number $\frac{a}{|a|} + \frac{b}{|b|} + \frac{c}{|c|} + \frac{abc}{|abc|}$. The set of all numbers formed is

(A) 0 (B) $\{-4, 0, 4\}$ (C) $\{-4, -2, 0, 2, 4\}$ (D) $\{-4, -2, 2, 4\}$ (E) none of these

Answer: B

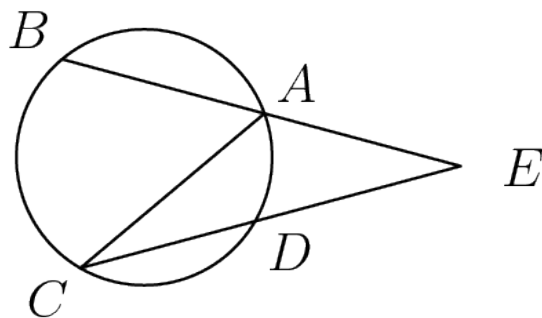
Solution

Solution by e_power_pi_times_i

$\frac{x}{|x|} = 1$ or -1 depending whether x is positive or negative. If a , b , and c are positive, then the entire thing amounts to 4. If one of the three is negative and the other two positive, the answer is 0. If two of the three is negative and one is positive, the answer is 0. If all three are negative, the answer is -4 . Therefore the set is **(B)** $\{-4, 0, 4\}$.

This is very similar to https://artofproblemsolving.com/wiki/index.php/2017_AMC_8_Problems/Problem_9

9. (AHSME 1977)



centering

In the adjoining figure $\angle E = 40^\circ$ and arc AB , arc BC , and arc CD all have equal length. Find the measure of $\angle ACD$.

- (A) 10° (B) 15° (C) 20° (D) $\left(\frac{45}{2}\right)^\circ$ (E) 30°

Answer: B

Solution

Solution by e_power_pi_times_i

If arcs AB , BC , and CD are congruent, then $\angle ACB = \angle BDC = \angle CBD = \theta$. Because $ABCD$ is cyclic, $\angle CAD = \angle CBD = \theta$, and $\angle ADB = \angle ACB = \theta$. Then, $\angle EAD = \angle EDA = \frac{180^\circ - 40^\circ}{2} = 70^\circ$. $\theta = 55^\circ$. $\angle ACD = 180^\circ - 55^\circ - 110^\circ = \mathbf{(B) 15^\circ}$.

10. (AHSME 1977) If $(3x - 1)^7 = a_7x^7 + a_6x^6 + \cdots + a_0$, then $a_7 + a_6 + \cdots + a_0$ equals

- (A) 0 (B) 1 (C) 64 (D) -64 (E) 128

Answer: E

Solution

Solution by e_power_pi_times_i

Notice that if $x = 1$, then $a_7x^7 + a_6x^6 + \cdots + a_0 = a_7 + a_6 + \cdots + a_0$. Therefore the answer is $(3(1) - 1)^7 = \mathbf{(E) 128}$.

11. (AHSME 1977) For each real number x , let $[x]$ be the largest integer not exceeding x (i.e., the integer n such that $n \leq x < n + 1$). Which of the following statements is (are) true?

I. $[x + 1] = [x] + 1$ for all x

II. $[x + y] = [x] + [y]$ for all x and y

III. $[xy] = [x][y]$ for all x and y

- (A) none (B) I only (C) I and II only (D) III only (E) all

Answer: B

Solution

Solution by e_power_pi_times_i

Notice that $[x]$ is just $\lfloor x \rfloor$. We see that **I** is true, as adding by one does not change the fraction part of the number. Similarly, **II** is false, because $\lfloor x + y \rfloor$ does not always equal $\lfloor x \rfloor + \lfloor y \rfloor$ (If both numbers were 1.5, $\lfloor 1.5 + 1.5 \rfloor = 3$, $\lfloor 1.5 \rfloor + \lfloor 1.5 \rfloor = 2$). Because **I** is true and **II** is false, the answer is **(B) I only**.

12. (AHSME 1977) Al's age is 16 more than the sum of Bob's age and Carl's age, and the square of Al's age is 1632 more than the square of the sum of Bob's age and Carl's age. What is the sum of the ages of Al, Bob, and Carl?

(A) 64 (B) 94 (C) 96 (D) 102 (E) 140

Answer: D

Solution

Solution by e_power_pi_times_i

Denote Al's age, Bob's age, and Carl's age by a , b , and c , respectively. Then, $a = 16 + b + c$ and $a^2 = 1632 + b^2 + c^2$. Substituting the first equation into the second, $(16 + b + c)^2 = b^2 + c^2 + 2bc + 32b + 32c + 256 = b^2 + c^2 + 1632$. Thus, $bc + 16b + 16c = 688$, and $(b + 16)(c + 16) = 944$. Since $944 = 2^4 \cdot 59$, $(b, c) = (0, 43)$ or $(43, 0)$. Then $a + b + c = 2b + 2c + 16 = \mathbf{(D) 102}$.

13. (AHSME 1977) If $a_1, a_2, a_3, \dots$ is a sequence of positive numbers such that $a_{n+2} = a_n a_{n+1}$ for all positive integers n , then the sequence $a_1, a_2, a_3, \dots$ is a geometric progression

(A) for all positive values of a_1 and a_2
 (B) if and only if $a_1 = a_2$
 (C) if and only if $a_1 = 1$
 (D) if and only if $a_2 = 1$
 (E) if and only if $a_1 = a_2 = 1$

Answer: E

Solution

Solution by e_power_pi_times_i

The first few terms are $a_1, a_2, a_1 a_2, a_1 a_2^2, a_1^2 a_2^3, \dots$. If this is a geometric progression, $\frac{a_2}{a_1} = a_1 = a_2 = a_1 a_2$. $a_1 = 0, 1$, $a_2 = 0, 1$. Since this is a sequence of positive integers, then the answer must be **(E) if and only if $a_1 = a_2 = 1$** .

14. (AHSME 1977) How many pairs (m, n) of integers satisfy the equation $m + n = mn$?

(A) 1 (B) 2 (C) 3 (D) 4 (E) more than 4

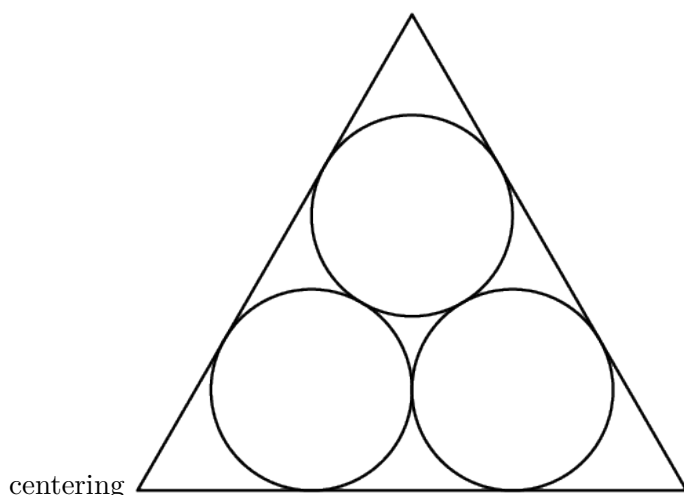
Answer: B

Solution

Solution by e_power_pi_times_i

If $m + n = mn$, $mn - m - n = (m - 1)(n - 1) - 1 = 0$. Then $(m - 1)(n - 1) = 1$, and $(m, n) = (2, 2)$ or $(0, 0)$. The answer is **(B) 2**.

15. (AHSME 1977)



centering

Each of the three circles in the adjoining figure is externally tangent to the other two, and each side of the triangle is tangent to two of the circles. If each circle has radius three, then the perimeter of the triangle is

- (A) $36 + 9\sqrt{2}$ (B) $36 + 6\sqrt{3}$ (C) $36 + 9\sqrt{3}$ (D) $18 + 18\sqrt{3}$ (E) 45

Answer: D

Solution

Solution by e_power_pi_times_i

Draw perpendicular lines from their if goe[radii of the circles to the sides of the triangle, and lines from the radii of the circles to the vertices of the triangle. Because the triangle is equilateral, the lines divide the big triangle into a small triangle, three rectangles, and six small $30-60-90$ triangles. The Longer length of the rectangles is 6 because it is the radius of one circle + the radius of another $(3+3)$. The length of one side is $2(3\sqrt{3}) + 6 = 6\sqrt{3} + 6$. The perimeter is (D) $18\sqrt{3} + 18$.

16. (AHSME 1977) If $i^2 = -1$, then the sum

$$\cos 45^\circ + i \cos 135^\circ + \cdots + i^n \cos (45 + 90n)^\circ + \cdots + i^{40} \cos 3645^\circ$$

equals

- (A) $\frac{\sqrt{2}}{2}$ (B) $-10i\sqrt{2}$ (C) $\frac{21\sqrt{2}}{2}$
 (D) $\frac{\sqrt{2}}{2}(21 - 20i)$ (E) $\frac{\sqrt{2}}{2}(21 + 20i)$

Answer: D

Solution

Solution by e_power_pi_times_i

Notice that the sequence repeats every 4 terms. These 4 terms are $\cos(45^\circ)$, $i \cos(135^\circ)$, $-\cos(225^\circ)$, and $-i \cos(315^\circ)$. This is because $\cos(\theta) = \cos(360^\circ - \theta)$. Also $\cos(\theta) = -\cos(180^\circ - \theta)$, so the 4 terms are just $\cos(45^\circ)$, $-i \cos(45^\circ)$, $\cos(45^\circ)$, and $-i \cos(45^\circ)$, which adds up to $\sqrt{2} - i\sqrt{2}$. The sequence repeats 10 times, and an extra $i^{40} \cos(3645^\circ)$.

The sum is $10(\sqrt{2} - i\sqrt{2}) + \frac{\sqrt{2}}{2} = \frac{21\sqrt{2}}{2} - 10i\sqrt{2} =$ (D) $\frac{\sqrt{2}}{2}(21 - 20i)$.

17. (AHSME 1977) Three fair dice are tossed at random (i.e., all faces have the same probability of coming up). What is the probability that the three numbers turned up can be arranged to form an arithmetic progression with common difference one?

(A) $\frac{1}{6}$ (B) $\frac{1}{9}$ (C) $\frac{1}{27}$ (D) $\frac{1}{54}$ (E) $\frac{7}{36}$

Answer: B

Solution

Solution by e_power_pi_times_i

The denominator of the probability is $6^3 = 216$. We are asked to find the probability that the three numbers are consecutive. The three numbers can be arranged in 6 ways, and there are 4 triplets of consecutive numbers from 1 – 6. The probability is

$$\frac{6 \cdot 4}{216} = \frac{24}{216} = \boxed{\text{(B)} \frac{1}{9}}.$$

18. (AHSME 1977) If $y = (\log_2 3)(\log_3 4) \cdots (\log_n [n+1]) \cdots (\log_{31} 32)$ then

(A) $4 < y < 5$ (B) $y = 5$ (C) $5 < y < 6$ (D) $y = 6$
 (E) $6 < y < 7$

Answer: B

Solution

Solution by e_power_pi_times_i

Note that $\log_a b = \frac{\log b}{\log a}$. Then $y = \left(\frac{\log 3}{\log 2}\right)\left(\frac{\log 4}{\log 3}\right) \cdots \left(\frac{\log 32}{\log 31}\right) = \frac{\log 32}{\log 2} = \log_2 32 = \boxed{\text{(B)} y = 5}$.

19. (AHSME 1977) Let E be the point of intersection of the diagonals of convex quadrilateral $ABCD$, and let P, Q, R , and S be the centers of the circles circumscribing triangles ABE, BCE, CDE , and ADE , respectively. Then

(A) $PQRS$ is a parallelogram
 (B) $PQRS$ is a parallelogram if and only if $ABCD$ is a rhombus
 (C) $PQRS$ is a parallelogram if and only if $ABCD$ is a rectangle
 (D) $PQRS$ is a parallelogram if and only if $ABCD$ is a parallelogram
 (E) none of the above are true

Answer: A

Solution 1

Let l_1, l_2, l_3 , and l_4 be the perpendicular bisectors of AE, BE, CE , and DE respectively. Then $P = l_1 \cap l_2, Q = l_2 \cap l_3, R = l_3 \cap l_4$, and $S = l_4 \cap l_1$. Thus $l_1 = \overleftrightarrow{PS}, l_2 = \overleftrightarrow{PQ}, l_3 = \overleftrightarrow{QR}$, and $l_4 = \overleftrightarrow{RS}$.

Since $\overleftrightarrow{PS}$ and $\overleftrightarrow{QR}$ are both perpendicular to $\overleftrightarrow{AC}$, $\overleftrightarrow{PS} \parallel \overleftrightarrow{QR}$. Also, since $\overleftrightarrow{PQ}$ and $\overleftrightarrow{RS}$ are both perpendicular to $\overleftrightarrow{BD}$, $\overleftrightarrow{PQ} \parallel \overleftrightarrow{RS}$.

Because $PQRS$ has two pairs of parallel opposite sides, $\boxed{\text{(A)} PQRS \text{ is a parallelogram}}$.

-j314andrews

Solution 2

Let L, M, N , and O be the intersections of AC and PS, BD and PQ, CA and QR , and DB and RS respectively. Since the **circumcenter** of a triangle is determined by the intersection of its perpendicular bisectors, each of the pairs mentioned earlier

22. (AHSME 1977) If $f(x)$ is a real valued function of the real variable x , and $f(x)$ is not identically zero, and for all a and b $f(a+b) + f(a-b) = 2f(a) + 2f(b)$, then for all x and y

- (A) $f(0) = 1$ (B) $f(-x) = -f(x)$ (C) $f(-x) = f(x)$
 (D) $f(x+y) = f(x) + f(y)$
 (E) there is a positive real number T such that $f(x+T) = f(x)$

Answer: C

Solution

Substituting $(a, b) = (0, 0)$ yields $2f(0) = 4f(0)$, so $f(0) = 0$.

Substituting $(a, b) = (0, x)$ yields $f(x) + f(-x) = 2f(-x)$ so $\boxed{\text{(C)} f(x) = f(-x)}$ for all x .

Note: $f(x) = x^2$ satisfies the original equation and may be used to show that the other answer choices are incorrect.

23. (AHSME 1977) If the solutions of the equation $x^2 + px + q = 0$ are the cubes of the solutions of the equation $x^2 + mx + n = 0$, then

- (A) $p = m^3 + 3mn$ (B) $p = m^3 - 3mn$ (C) $p + q = m^3$
 (D) $\left(\frac{m}{n}\right)^2 = \frac{p}{q}$ (E) none of these

Answer: B

Solution

Let r_1 and r_2 be the roots of the equation $x^2 + mx + n = 0$. Then, r_1^3 and r_2^3 are the roots of $x^2 + px + q = 0$. Applying **Vieta's Formulas** to the first equation, we have $r_1 + r_2 = -m$ and $r_1 r_2 = n$. Similarly, from the second equation, we have $r_1^3 + r_2^3 = -p$ and $r_1^3 r_2^3 = q$. Cubing $r_1 + r_2 = -m$, we get $r_1^3 + r_2^3 + 3r_1 r_2 (r_1 + r_2) = -m^3$. Plugging in from the other equations gives $-p + 3n(-m) = -m^3$. Rearranging we have $\boxed{\text{(B)} p = m^3 - 3mn}$.

24. (AHSME 1977) Find the sum $\frac{1}{1(3)} + \frac{1}{3(5)} + \cdots + \frac{1}{(2n-1)(2n+1)} + \cdots + \frac{1}{255(257)}$.

- (A) $\frac{127}{255}$ (B) $\frac{128}{255}$ (C) $\frac{1}{2}$ (D) $\frac{128}{257}$ (E) $\frac{129}{257}$

Answer: D

Solution

Note that $\frac{1}{(2n-1)(2n+1)} = \frac{1/(2n-1) - 1/(2n+1)}{2}$. Indeed, we find the series telescopes and is equal to $\frac{1 - \frac{1}{257}}{2}$, which is evidently $\boxed{\frac{128}{257}}$

25. (AHSME 1977) Determine the largest positive integer n such that $1005!$ is divisible by 10^n .

- (A) 102 (B) 112 (C) 249 (D) 502 (E) none of these

Answer: E

Solution

We first observe that since there will be more 2s than 5s in $1005!$, we are looking for the largest n such that 5^n divides $1005!$. We will use the fact that:

$$n = \left\lfloor \frac{1005}{5^1} \right\rfloor + \left\lfloor \frac{1005}{5^2} \right\rfloor + \left\lfloor \frac{1005}{5^3} \right\rfloor + \cdots$$

(This is an application of Legendre's formula).

From $k = 5$ and onwards, $\left\lfloor \frac{1005}{5^k} \right\rfloor = 0$. Thus, our calculation becomes

$$n = \left\lfloor \frac{1005}{5^1} \right\rfloor + \left\lfloor \frac{1005}{5^2} \right\rfloor + \left\lfloor \frac{1005}{5^3} \right\rfloor + \left\lfloor \frac{1005}{5^4} \right\rfloor$$

$$n = \left\lfloor \frac{1005}{5} \right\rfloor + \left\lfloor \frac{1005}{25} \right\rfloor + \left\lfloor \frac{1005}{125} \right\rfloor + \left\lfloor \frac{1005}{625} \right\rfloor$$

$$n = 201 + 40 + 8 + 1 = \boxed{250}$$

Since none of the answer choices equal 250, the answer is **(E)**.

- makol7

26. (AHSME 1977) Let a, b, c , and d be the lengths of sides MN, NP, PQ , and QM , respectively, of quadrilateral $MNPQ$. If A is the area of $MNPQ$, then

- (A) $A = \left(\frac{a+c}{2}\right)\left(\frac{b+d}{2}\right)$ if and only if $MNPQ$ is convex
 (B) $A = \left(\frac{a+c}{2}\right)\left(\frac{b+d}{2}\right)$ if and only if $MNPQ$ is a rectangle
 (C) $A \leq \left(\frac{a+c}{2}\right)\left(\frac{b+d}{2}\right)$ if and only if $MNPQ$ is a rectangle
 (D) $A \leq \left(\frac{a+c}{2}\right)\left(\frac{b+d}{2}\right)$ if and only if $MNPQ$ is a parallelogram
 (E) $A \geq \left(\frac{a+c}{2}\right)\left(\frac{b+d}{2}\right)$ if and only if $MNPQ$ is a parallelogram

Answer: B

27. (AHSME 1977) There are two spherical balls of different sizes lying in two corners of a rectangular room, each touching two walls and the floor. If there is a point on each ball which is 5 inches from each wall which that ball touches and 10 inches from the floor, then the sum of the diameters of the balls is

- (A) 20 inches (B) 30 inches (C) 40 inches (D) 60 inches
 (E) not determined by the given information

Answer: C

Solution

Let r be the radius of one of the balls, in inches, O be the center of the ball, and A be the point which is 5 inches away from each wall and 10 inches above the floor. Define a three-dimensional Cartesian coordinate system such that a corner of the room is the origin, the floor is in the positive xy -plane, one wall is in the positive yz -plane, and another wall is in the positive xz -plane.

Then O is at (r, r, r) and A is at $(5, 5, 10)$. By the distance formula, $OA^2 = (r-5)^2 + (r-5)^2 + (r-10)^2 = r^2$. This can be simplified to $2(r-15)(r-5) = 0$, which has solutions $r = 15$ and $r = 5$. So one ball has diameter 30 inches while the other has diameter 10 inches, thus the sum is **(C) 40** inches.

-j314andrews

28. (AHSME 1977) Let $g(x) = x^5 + x^4 + x^3 + x^2 + x + 1$. What is the remainder when the polynomial $g(x^{12})$ is divided by the polynomial $g(x)$?

- (A) 6 (B) $5 - x$ (C) $4 - x + x^2$ (D) $3 - x + x^2 - x^3$
 (E) $2 - x + x^2 - x^3 + x^4$

Answer: A

Solution 1

Let $r(x)$ be the remainder when $g(x^{12})$ is divided by $g(x)$. Then $r(x)$ is the unique polynomial such that

$$g(x^{12}) - r(x) = x^{60} + x^{48} + x^{36} + x^{24} + x^{12} + 1 - r(x)$$

is divisible by $g(x) = x^5 + x^4 + x^3 + x^2 + x + 1$, and $\deg r(x) < 5$.

Note that $(x-1)(x^5 + x^4 + x^3 + x^2 + x + 1) = x^6 - 1$ is a multiple of $g(x)$. Also,

$$g(x^{12}) - 6 = x^{60} + x^{48} + x^{36} + x^{24} + x^{12} - 5 = (x^{60} - 1) + (x^{48} - 1) + (x^{36} - 1) + (x^{24} - 1) + (x^{12} - 1).$$

Each term is a multiple of $x^6 - 1$. For example,

$$x^{60} - 1 = (x^6 - 1)(x^{54} + x^{48} + \cdots + x^6 + 1).$$

Hence, $g(x^{12}) - 6$ is a multiple of $x^6 - 1$, which means that $g(x^{12}) - 6$ is a multiple of $g(x)$. Therefore, the remainder is $\boxed{6}$. The answer is (A).

Solution 2

We express the quotient and remainder as follows.

$$g(x^{12}) = Q(x)g(x) + R(x)$$

Note that the solutions to $g(x)$ correspond to the 6th roots of unity, excluding 1. Hence, we have $x^6 = 1$, allowing us to set:

$$g(x^{12}) = 6$$

$$g(x) = 0$$

We have 5 values of x that return $R(x) = 6$. However, $g(x)$ is quintic, implying the remainder is of degree at most 4. Since there are 5 solutions, the only possibility is that the remainder is a constant $\boxed{6}$.

Solution 3

We can use the Chinese remainder theorem over $\mathbb{Q}[x]$. Since $g(x) = (x+1)(x^2 - x + 1)(x^2 + x + 1)$, $\mathbb{Q}[x]/g \cong \mathbb{Q}[x]/(x+1) \times \mathbb{Q}[x]/(x^2 - x + 1) \times \mathbb{Q}[x]/(x^2 + x + 1)$. This means that if we can find the remainder of $g(x^{12})$ modulo $x+1$, $x^2 - x + 1$, $x^2 + x + 1$, we can reconstruct the remainder modulo g . We can further use that each factor is irreducible and that if $p(x)$ is an irreducible polynomial over $\mathbb{Q}$ with root α , $\mathbb{Q}[x]/p \cong \mathbb{Q}(\alpha)$ so to evaluate the remainders of $g(x^{12})$, we just need to evaluate it on one of the roots of the irreducible factors. The first factor has root -1 , the second has roots the primitive sixth roots of unity, and the third as roots the primitive cube roots of unity (this is easily seen as $g(x)(x-1) = x^6 - 1$). Evaluating $g(x^{12})$ on each of these values yields $g(1) = 6$ so the remainder is 6 on each factor on the right of the isomorphism. Hence, by the Chinese remainder theorem, the remainder modulo g must be $\boxed{6}$ as well.

Solution 4 (Remainder Theorem)

First, we factor $g(x)$ as $(x^3 + 1)(x^2 + x + 1)$. Substituting to find $g(x^{12})$, we have

$$g(x^{12}) = (x^{36} + 1)(x^{24} + x^{12} + 1).$$

We want the remainder of

$$\frac{g(x^{12})}{g(x)} = \frac{(x^{36} + 1)(x^{24} + x^{12} + 1)}{(x^3 + 1)(x^2 + x + 1)}.$$

Splitting it into two expressions, we can find the remainder of

$$\frac{x^{36} + 1}{x^3 + 1} \quad \text{and} \quad \frac{x^{24} + x^{12} + 1}{x^2 + x + 1},$$

then multiply the results.

- First expression:

The roots of $x^3 + 1$ are -1 , $\text{cis}(60^\circ)$, and $\text{cis}(-60^\circ)$. Plugging each of these roots into $x^{36} + 1$ yields 2. Since the remainder must have degree at most 2 and takes the constant value 2 at three distinct points, the remainder is $r(x) = 2$.

- Second expression:

The roots of $x^2 + x + 1$ are $\text{cis}(120^\circ)$ and $\text{cis}(-120^\circ)$. Plugging each of these roots into $x^{24} + x^{12} + 1$ yields 3. Since the remainder must have degree at most 1 and takes the constant value 3 at two distinct points, the remainder is $r(x) = 3$.

Multiplying the two remainders, we get $2 \cdot 3 = 6$, or **(A)**. Voidling

29. (AHSME 1977) Find the smallest integer n such that $(x^2 + y^2 + z^2)^2 \leq n(x^4 + y^4 + z^4)$ for all real numbers x, y , and z .

- (A) 2 (B) 3 (C) 4 (D) 6 (E) There is no such integer n

Answer: B

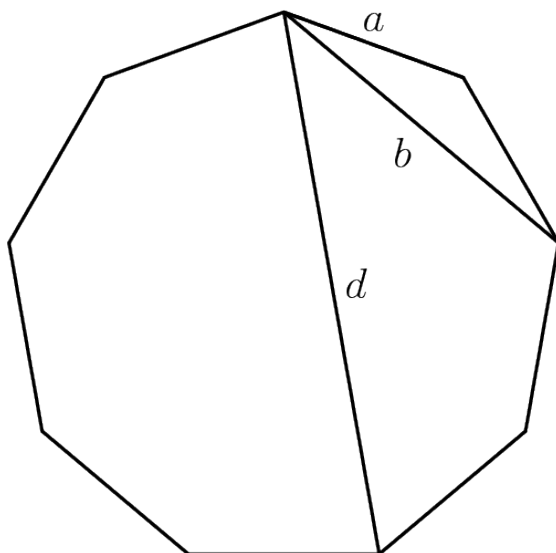
Solution (Official MAA)

Let $a = x^2$, $b = y^2$, $c = z^2$. Then

$$\begin{aligned} 0 &\leq (a - b)^2 + (b - c)^2 + (c - a)^2 \\ &\quad \frac{ab + bc + ca}{a^2 + b^2 + c^2} \leq 1; \\ &\quad \frac{a^2 + b^2 + c^2 + 2(ab + bc + ca)}{a^2 + b^2 + c^2} \leq 3; \\ &\quad (a + b + c)^2 \leq 3(a^2 + b^2 + c^2). \end{aligned}$$

Therefore $n \leq 3$. Choosing $a = b = c > 0$ shows n is not less than three.

30. (AHSME 1977)



centering

If a , b , and d are the lengths of a side, a shortest diagonal and a longest diagonal, respectively, of a regular nonagon (see adjoining figure), then

- (A) $d = a + b$ (B) $d^2 = a^2 + b^2$ (C) $d^2 = a^2 + ab + b^2$
 (D) $b = \frac{a+d}{2}$ (E) $b^2 = ad$

Answer: A

Solution 1

By the law of cosines we can get the following expressions for d^2 and b^2 :

$$d^2 = 2b^2(1 - \cos(100^\circ))$$

$$b^2 = 2a^2(1 - \cos(140^\circ))$$

We can substitute what we got for b^2 into the expression for d^2 :

$$d^2 = 4a^2(1 - \cos(140^\circ))(1 - \cos(100^\circ)) = 4a^2(1 - (\cos(100^\circ) + \cos(140^\circ)) + \cos(100^\circ)\cos(140^\circ))$$

Now apply sum-to-product and product-to-sum identities:

$$d^2 = 4a^2(1 - (2\cos(120^\circ)\cos(20^\circ)) + \frac{1}{2}(\cos(240^\circ) + \cos(40^\circ)))$$

Simplifying further gives us:

$$d^2 = 4a^2(1 + \cos(20^\circ) - \frac{1}{4} + \frac{1}{2}\cos(40^\circ))$$

After using the fact that $\cos(40^\circ) = 2\cos^2(20^\circ) - 1$, it's not hard to see that the expression in the parentheses is equal to $(\cos(20^\circ) + \frac{1}{2})^2$. So we can square-root both sides to find the expression for d :

$$d = 2a(\cos(20^\circ) + \frac{1}{2})$$

Now let's look at the expression for b^2 . We can apply the reverse of the double angle identity to show that $1 - \cos(140^\circ)$ equals $2 \sin^2(70^\circ)$. So if we square root the entire expression we get that

$$b = 2a \sin(70^\circ) = 2a \cos(20^\circ)$$

We now have everything in terms of a . Luckily when we consider choice **A** we can verify without much work that this must be the answer.

Solution by harita19

- NOTE: a much easier solution exists by drawing some lines and recognizing that the nonagon is cyclic, but for those of use who use algebra in every geometry problem, this is the best solution.

Solution 2

Using Van Schooten's Theorem, we draw an equilateral triangle in the nonagon to see that $a + b = d$.

Solution 3

Reflect b such that it connects the left vertex of the a to the 2nd vertex counterclockwise. This length is also b . Notice that when connecting the two endpoints of the b segments, we achieve a length of d . We now have an isosceles trapezoid with bases a and d .

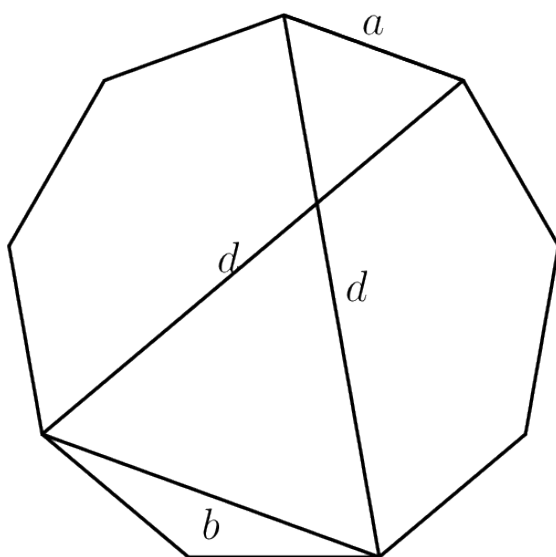
Next, we can find the angle between the a and the b segments as follows. The interior angle of a regular nonagon measures $180 - \frac{360}{9} = 140^\circ$. Thus, the isosceles triangle formed from two of the side lengths of the nonagon and b has angles 140 , $\frac{180-140}{2}$, and $\frac{180-140}{2} = 20$. Then, the angle between a and b is $140 - 20 = 120$. We can therefore deduce that the other angles are 120 , 60 , and 60 degrees.

We can drop a perpendicular from both endpoints of segment a to segment d to get a rectangle and two 30-60-90 triangles. Each of the triangles has a shorter leg length of $\frac{d-a}{2}$, and hence, we can deduce that $\frac{d-a}{2} \cdot 2 = b$. Hence, the correct answer is **(A)** $d = a + b$

- Solution by scjh999999

Solution 4

Draw chords of length a , b , and d to create two triangles as follows.



centering

Because each edge of a nonagon has a central angle of 40° , and inscribed angles are half the corresponding central angle, the two triangles thus created are equilateral. This means that $d = a + b$.

- Solution by Dr.B.

§14 AHSME 1978

1. (AHSME 1978) If $1 - \frac{4}{x} + \frac{4}{x^2} = 0$, then $\frac{2}{x}$ equals

(A) -1 (B) 1 (C) 2 (D) -1 or 2 (E) -1 or -2

Answer: B

Solution 1

By guessing and checking, 2 works. $\frac{2}{x} = \boxed{\text{(B)} 1}$ awin

Solution 2

Multiplying each side by x^2 , we get $x^2 - 4x + 4 = 0$. Factoring, we get $(x-2)(x-2) = 0$. Therefore, $x = 2$. $\frac{2}{x} = \boxed{\text{(B)} 1}$ awin

Solution 3

Directly factoring, we get $(1 - \frac{2}{x})^2 = 0$. Thus $\frac{2}{x}$ must equal $\boxed{\text{(B)} 1}$

2. (AHSME 1978) If four times the reciprocal of the circumference of a circle equals the diameter of the circle, then the area of the circle is

(A) $\frac{1}{\pi^2}$ (B) $\frac{1}{\pi}$ (C) 1 (D) π (E) π^2

Answer: C

Solution 1

Creating equations, we get $4 \cdot \frac{1}{2\pi r} = 2r$. Simplifying, we get $\frac{1}{\pi r} = r$. Multiplying each side by r , we get $\frac{1}{\pi} = r^2$. Because the formula of the area of a circle is πr^2 , we multiply each side by π to get $1 = \pi r^2$. Therefore, our answer is $\boxed{\text{(C)} 1}$

3. (AHSME 1978) For all non-zero numbers x and y such that $x = 1/y$, $(x - \frac{1}{x})(y + \frac{1}{y})$ equals

(A) $2x^2$ (B) $2y^2$ (C) $x^2 + y^2$ (D) $x^2 - y^2$ (E) $y^2 - x^2$

Answer: D

Solution 1

Using substitution, we can substitute y into the equation in the first parentheses. Therefore, we'll get

$$\left(x - \frac{1}{\frac{1}{y}}\right) \left(y + \frac{1}{y}\right)$$

Because $x = \frac{1}{y}$, we can also see that $y = \frac{1}{x}$. Using substitution again, we can substitute x into the second equation getting

$$\left(x - \frac{1}{\frac{1}{y}}\right) \left(y + \frac{1}{\frac{1}{x}}\right)$$

Simplifying, we get $(x - y)(y + x)$. Multiplying, we get $\boxed{\text{(D)} x^2 - y^2}$

4. (AHSME 1978) If $a = 1$, $b = 10$, $c = 100$, and $d = 1000$, then $(a + b + c - d) + (a + b - c + d) + (a - b + c + d) + (-a + b + c + d)$ is equal to
 (A) 1111 (B) 2222 (C) 3333 (D) 1212 (E) 4242

Answer: B

Solution 1

Adding all four of the terms up, we can see that it equals

$$2(a + b + c + d)$$

This is equal to $2(1111) = \boxed{\text{(B) } 2222}$ awin

5. (AHSME 1978) Four boys bought a boat for \$60. The first boy paid one half of the sum of the amounts paid by the other boys; the second boy paid one third of the sum of the amounts paid by the other boys; and the third boy paid one fourth of the sum of the amounts paid by the other boys. How much did the fourth boy pay?
 (A) \$10 (B) \$12 (C) \$13 (D) \$14 (E) \$15

Answer: C

Solution 1

If the first boy paid one half of the sum of the amounts paid by the other boys, that means that he paid $\frac{1}{3}$ of the total. If the second boy paid one third of the sum of the amounts paid by the other boys, that means that he paid $\frac{1}{4}$ of the total. If the third boy paid one fourth of the sum of the amounts paid by the other boys, that means that he paid $\frac{1}{5}$ of the total.

Summing it up, we get $\$20 + \$15 + \$12 = \47 . Therefore, our answer is $\$60 - \$47 = \boxed{\text{(C) } \$13}$ awin

6. (AHSME 1978) The number of distinct pairs (x, y) of real numbers satisfying both of the following equations:

$$x = x^2 + y^2 \quad y = 2xy$$

is

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Answer: E

Solution

If $x = x^2 + y^2$ and $y = 2xy$, then we can break this into two cases.

Case 1: $y = 0$

If $y = 0$, then $x = x^2$ and $0 = 0$

Therefore, $x = 0$ or $x = 1$

This yields 2 solutions

Case 2: $x = \frac{1}{2}$

If $x = \frac{1}{2}$, this means that $y = y$, and $\frac{1}{2} = \frac{1}{4} + y^2$.

Because y can be negative or positive, this yields $y = \frac{1}{2}$ or $y = -\frac{1}{2}$

This yields another 2 solutions.

$$2 + 2 = \boxed{\text{(E) } 4}$$

7. (AHSME 1978) Opposite sides of a regular hexagon are 12 inches apart. The length of each side, in inches, is

(A) 7.5 (B) $6\sqrt{2}$ (C) $5\sqrt{2}$ (D) $\frac{9}{2}\sqrt{3}$ (E) $4\sqrt{3}$

Answer: E

Solution

Draw a perpendicular through the midpoint of the line of length 12 such that it passes through a vertex. We now have created 2 $30-60-90$ triangles. Using the ratios, we get that the hypotenuse is $6 \times \frac{2}{\sqrt{3}} = \frac{12}{\sqrt{3}} = 4\sqrt{3} \rightarrow \boxed{\text{(E)}}$

8. (AHSME 1978) If $x \neq y$ and the sequences x, a_1, a_2, y and x, b_1, b_2, b_3, y each are in arithmetic progression, then $(a_2 - a_1)/(b_2 - b_1)$ equals

(A) $\frac{2}{3}$ (B) $\frac{3}{4}$ (C) 1 (D) $\frac{4}{3}$ (E) $\frac{3}{2}$

Answer: D

Solution

WLOG, let $x = 2$ and $y = 5$. From the first sequence, we get

$$2, 3, 4, 5$$

so $a_2 - a_1 = 1$ From the second sequence, we get

$$2, 2 + r, 2 + 2r, 2 + 3r, 5$$

so $2 + 4r = 5$ and $r = \frac{3}{4}$ Thus, we have

$$2, 2\frac{3}{4}, 3\frac{1}{2}, \dots$$

and $b_2 - b_1 = \frac{3}{4}$ So $\frac{\frac{1}{3}}{\frac{3}{4}} = \frac{4}{3}$

$\boxed{D}$

9. (AHSME 1978) If $x < 0$, then $|x - \sqrt{(x-1)^2}|$ equals

(A) 1 (B) $1 - 2x$ (C) $-2x - 1$ (D) $1 + 2x$ (E) $2x - 1$

Answer: B

Solution

We have $\sqrt{x^2} = |x|$, so we rewrite the expression as follows.

$$|x - \sqrt{(x-1)^2}| = |x - |x-1||$$

We know that $x < 0$, so $x - 1 < 0$. Thus, we can rewrite $|x - 1|$ as $1 - x$. So

$$|x - |x-1|| = |x - (1-x)| = |2x-1|$$

.Since $x < 0$, $2x - 1 < 0$. Thus, we can write this as

$$|2x-1| = 1-2x$$

$\boxed{B}$

10. (AHSME 1978) If B is a point on circle C with center P , then the set of all points A in the plane of circle C such that the distance between A and B is less than or equal to the distance between A and any other point on circle C is
- (A) the line segment from P to B
 (B) the ray beginning at P and passing through B
 (C) a ray beginning at B
 (D) a circle whose center is P
 (E) a circle whose center is B

Answer: B

Solution

Begin by drawing circle P and point B . To satisfy the conditions of the problem, A needs to be in a position where it is closer to B . This can only happen if A and B are on the same line, so we choose from answer choices A and B.

We can pick some arbitrary point A outside circle P that is collinear with B and see that the conditions still hold, so the answer is **(B)**.

11. (AHSME 1978) If r is positive and the line whose equation is $x + y = r$ is tangent to the circle whose equation is $x^2 + y^2 = r$, then r equals
- (A) $\frac{1}{2}$ (B) 1 (C) 2 (D) $\sqrt{2}$ (E) $2\sqrt{2}$

Answer: C

Solution

The circle $x^2 + y^2 = r$ has center $(0,0)$ and radius $\sqrt{r}$. Therefore, if the line $x + y = r$ is tangent to the circle $x^2 + y^2 = r$, then the distance between $(0,0)$ and the line $x + y = r$ is $\sqrt{r}$.

The distance between $(0,0)$ and the line $x + y = r$ is

$$\frac{|0 + 0 - r|}{\sqrt{1^2 + 1^2}} = \frac{r}{\sqrt{2}}.$$

Hence,

$$\frac{r}{\sqrt{2}} = \sqrt{r}.$$

Then $r = \sqrt{r} \cdot \sqrt{2}$, so $\sqrt{r} = \sqrt{2}$, which means $r = \boxed{2}$ or (C), 2.

12. (AHSME 1978) In $\triangle ADE$, $\angle ADE = 140^\circ$, points B and C lie on sides AD and AE , respectively, and points A, B, C, D, E are distinct.* If lengths AB, BC, CD , and DE are all equal, then the measure of $\angle EAD$ is

- The specification that points A, B, C, D, E be distinct was not included in the original statement of the problem.

If $B = D$, then $C = E$ and $\angle EAD = 20^\circ$.

- (A) 5° (B) 6° (C) 7.5° (D) 8° (E) 10°

Answer: E

13. (AHSME 1978) If a, b, c , and d are non-zero numbers such that c and d are the solutions of $x^2 + ax + b = 0$ and a and b are the solutions of $x^2 + cx + d = 0$, then $a + b + c + d$ equals

- (A) 0 (B) -2 (C) 2 (D) 4 (E) $(-1 + \sqrt{5})/2$

Answer: B

Solution

By Vieta's formulas, $c + d = -a$, $cd = b$, $a + b = -c$, and $ab = d$. From the equation $c + d = -a$, $d = -a - c$, and from the equation $a + b = -c$, $b = -a - c$, so $b = d$.

Then from the equation $cd = b$, $cb = b$. Since b is nonzero, we can divide both sides of the equation by b to get $c = 1$. Similarly, from the equation $ab = d$, $ab = b$, so $a = 1$. Then $b = d = -a - c = -2$. Therefore, $a + b + c + d = 1 + (-2) + 1 + (-2) = \boxed{-2}$. The answer is (B).

14. (AHSME 1978) If an integer $n > 8$ is a solution of the equation $x^2 - ax + b = 0$ and the representation of a in the base- n number system is 18, then the base- n representation of b is

(A) 18 (B) 20 (C) 80 (D) 81 (E) 280

Answer: C

Solution

Assuming the solutions to the equation are n and m , by Vieta's formulas, $n_n + m_n = 18_n$.

$n_n = 10_n$, so $10_n + m_n = 18_n$.

$$m_n = 8_n$$

.

Also by Vieta's formulas, $n_n \cdot m_n = b_n$.

$$10_n \cdot 8_n = \boxed{80_n}$$

.

The answer is (C) 80

15. (AHSME 1978) If $\sin x + \cos x = 1/5$ and $0 \leq x < \pi$, then $\tan x$ is

(A) $-\frac{4}{3}$ (B) $-\frac{3}{4}$ (C) $\frac{3}{4}$ (D) $\frac{4}{3}$
 (E) not completely determined by the given information

Answer: A

Solution

Squaring the equation, we get

$$\sin^2 x + \cos^2 x + 2 \sin x \cos x = \frac{1}{25} \Rightarrow 2 \sin x \cos x = -\frac{24}{25} \Rightarrow \sin x \cos x = -\frac{12}{25}$$

Recall that $(a - b)^2 = (a + b)^2 - 4ab$, so

$$(\sin x - \cos x)^2 = (\sin x + \cos x)^2 - 4 \sin x \cos x$$

$$(\sin x - \cos x)^2 = \frac{1}{25} - 4\left(-\frac{12}{25}\right)$$

$$(\sin x - \cos x) = \frac{7}{5}$$

We can now solve for the values of $\sin x$ and $\cos x$

$$\sin x + \cos x = \frac{1}{5}$$

$$\sin x - \cos x = \frac{7}{5}$$

$$2 \sin x = \frac{8}{5} \Rightarrow \sin x = \frac{4}{5}, \cos x = -\frac{3}{5}$$

Since $\tan x = \frac{\sin x}{\cos x}$, we have

$$\tan x = \frac{\frac{4}{5}}{-\frac{3}{5}} = -\frac{4}{3}$$

A

16. (AHSME 1978) In a room containing N people, $N > 3$, at least one person has not shaken hands with everyone else in the room. What is the maximum number of people in the room that could have shaken hands with everyone else?

(A) 0 (B) 1 (C) $N - 1$ (D) N (E) none of these

Answer: E

Solution

We assume that one person hasn't shaken hands with the other N people, meaning that they only had shaken hands with $N-1$ people. However, this doesn't make sense! There will be 1 person (from the $N-1$) people who hasn't shaken hands with the first person, meaning that they also only had shaken hands with $N-1$ people. Therefore, there is a minimum of 2 people in the room that haven't shaken hands with one person. Therefore, the maximum is $N-2$ people, so the answer is E

17. (AHSME 1978) If k is a positive number and f is a function such that, for every positive number x , $[f(x^2 + 1)]^{\sqrt{x}} = k$; then, for every positive number y , $\left[f\left(\frac{9+y^2}{y^2}\right)\right]^{\sqrt{\frac{12}{y}}}$ is equal to

(A) $\sqrt{k}$ (B) $2k$ (C) $k\sqrt{k}$ (D) k^2 (E) $y\sqrt{k}$

Answer: D

Solution

We are given that

$$[f(x^2 + 1)]^{\sqrt{x}} = k$$

We can rewrite $\frac{9+y^2}{y^2}$ as $\frac{9}{y^2} + 1$. Thus, our function is now

$$\left[f\left(\frac{9}{y^2} + 1\right)\right]^{\sqrt{\frac{12}{y}}} = k$$

$$\Rightarrow \left[f\left(\frac{9}{y^2} + 1\right)\right]^{\sqrt{\frac{3}{y}} \cdot 4} = k$$

$$\Rightarrow ([f(\frac{9}{y^2} + 1)]^{\sqrt{\frac{3}{y}}})^4 = (k)^4$$

$$\Rightarrow ([f(\frac{9}{y^2} + 1)]^{\sqrt{\frac{3}{y}}})^2 = (k)^2 = k^2$$

Thus, the answer is $k^2 \Rightarrow \boxed{D}$.

18. (AHSME 1978) What is the smallest positive integer n such that $\sqrt{n} - \sqrt{n-1} < .01$?
 (A) 2499 (B) 2500 (C) 2501 (D) 10,000 (E) There is no such integer

Answer: C

Solution

Adding $\sqrt{n-1}$ to both sides, we get

$$\sqrt{n} < \sqrt{n-1} + 0.01.$$

Squaring both sides, we get

$$n < n-1 + 0.02\sqrt{n-1} + 0.0001,$$

which simplifies to

$$0.9999 < 0.02\sqrt{n-1},$$

or

$$\sqrt{n-1} > 49.995.$$

Squaring both sides again, we get

$$n-1 > 2499.500025,$$

so $n > 2500.500025$. The smallest positive integer n that satisfies this inequality is $\boxed{2501}$.

alternative Taking reciprocals and flipping the inequality we get

$$\sqrt{n} + \sqrt{n-1} > 100$$

Which is easy to see the answer is

$\boxed{2501}$.

-bjump

19. (AHSME 1978) A positive integer n not exceeding 100 is chosen in such a way that if $n \leq 50$, then the probability of choosing n is p , and if $n > 50$, then the probability of choosing n is $3p$. The probability that a perfect square is chosen is
 (A) .05 (B) .065 (C) .08 (D) .09 (E) .1

Answer: C

Solution

Let's say that we will have 3 slips for every number not exceeding 100 but bigger than 50. This is to account for the $3p$ probability part. Let's now say that we will only have one slip for each number below or equal to 50. The probability(or p) will then be $\frac{1}{200}$. Now let's have all the squares under 50, which are 1, 4, 9, 16, 25, 36, 49. The probability for these are $\frac{7}{200}$. The numbers above 50 that are squares are 64, 81, 100. We then need to multiply the probability by 3 so the probability of these are $\frac{9}{200}$. The answer is $\frac{7}{200} + \frac{9}{200} = 0.08 \Rightarrow \boxed{(C)}$.

20. (AHSME 1978) If a, b, c are non-zero real numbers such that

$$\frac{a+b-c}{c} = \frac{a-b+c}{b} = \frac{-a+b+c}{a},$$

and

$$x = \frac{(a+b)(b+c)(c+a)}{abc},$$

and $x < 0$, then x equals

- (A) -1 (B) -2 (C) -4 (D) -6 (E) -8

Answer: A

Solution

From the equation

$$\frac{a+b-c}{c} = \frac{a-b+c}{b} = \frac{-a+b+c}{a},$$

we add 2 to each fraction to get

$$\frac{a+b+c}{c} = \frac{a+b+c}{b} = \frac{a+b+c}{a}.$$

We perform casework on $a+b+c$:

- If $a+b+c \neq 0$, then $a=b=c$, from which $x = \frac{(2a)(2a)(2a)}{a^3} = 8$. However, this contradicts the precondition $x < 0$.
- If $a+b+c = 0$, then $x = \frac{(-c)(-a)(-b)}{abc} = \boxed{\text{(A)} -1}$.

21. (AHSME 1978) For all positive numbers x distinct from 1,

$$\frac{1}{\log_3(x)} + \frac{1}{\log_4(x)} + \frac{1}{\log_5(x)}$$

equals

- (A) $\frac{1}{\log_{60}(x)}$
 (B) $\frac{1}{\log_x(60)}$
 (C) $\frac{1}{(\log_3(x))(\log_4(x))(\log_5(x))}$
 (D) $\frac{12}{\log_3(x)+\log_4(x)+\log_5(x)}$
 (E) $\frac{\log_2(x)}{\log_3(x)\log_5(x)} + \frac{\log_3(x)}{\log_2(x)\log_5(x)} + \frac{\log_5(x)}{\log_2(x)\log_3(x)}$

Answer: A

Solution

$$\frac{1}{\log_3(x)} + \frac{1}{\log_4(x)} + \frac{1}{\log_5(x)} \implies \frac{1}{\frac{\log(x)}{\log(3)}} + \frac{1}{\frac{\log(x)}{\log(4)}} + \frac{1}{\frac{\log(x)}{\log(5)}} \implies \frac{\log(3) + \log(4) + \log(5)}{\log(x)} = \frac{\log(60)}{\log(x)} \implies 1$$

Thus, the answer is (A)

22. (AHSME 1978) The following four statements, and only these are found on a card:

On this card exactly one statement is false.
 On this card exactly two statements are false.
 On this card exactly three statements are false.
 On this card exactly four statements are false.

centering

(Assume each statement is either true or false.) Among them the number of false statements is exactly

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Answer: D

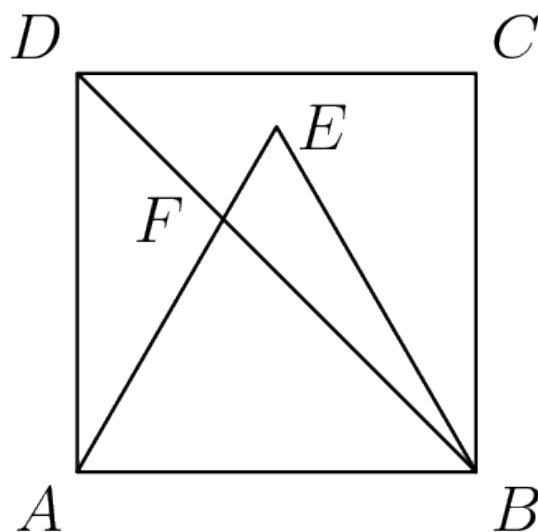
Solution 1

Any 2 of these statements contradict each other, so at most 1 statement is true. Thus at least 3 statements are false. However, if all 4 statements were false, then the last statement would be true, which would be a contradiction. Therefore, there are (D) 3 false statements. This is possible if only the third statement is true.

Solution 2

If all of them are false, that would mean that the 4th one is false too. Therefore, E is not the correct answer. If exactly 3 of them are false, that would mean that only 1 statement is true. This is correct since if only 1 statement is true, the card that is true is the one that has 3 of these statements are false. If we have 1 or 2 false statements, that would mean that there is more than 1 true statement. Therefore, our answer is (D).

23. (AHSME 1978)



centering

Vertex E of equilateral $\triangle ABE$ is in the interior of square $ABCD$, and F is the point of intersection of diagonal BD and line segment AE . If length AB is $\sqrt{1 + \sqrt{3}}$ then the area of $\triangle ABF$ is

- (A) 1 (B) $\frac{\sqrt{2}}{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) $4 - 2\sqrt{3}$ (E) $\frac{1}{2} + \frac{\sqrt{3}}{4}$

Answer: C

Solution

Place square $ABCD$ on the coordinate plane with A at the origin.

In polar form, line BD is $r \sin(\theta) = \sqrt{1 + \sqrt{3}} - r \cos(\theta)$ and line AF is $\theta = \frac{\pi}{3}$.

This means that the length from the origin to the intersection (r) is $r \frac{\sqrt{3}}{2} = \sqrt{1 + \sqrt{3}} - \frac{r}{2}$

Solving for r , you get: $r = \frac{2}{\sqrt{1 + \sqrt{3}}}$

Using the formula for area of a triangle ($A = \frac{ab \sin C}{2}$), you get $A = \frac{(\frac{2}{\sqrt{1 + \sqrt{3}}})(\sqrt{1 + \sqrt{3}}) \sin(\frac{\pi}{3})}{2} = \frac{\sqrt{3}}{2}$

Getting **C** as the answer

24. (AHSME 1978) If the distinct non-zero numbers $x(y - z)$, $y(z - x)$, $z(x - y)$ form a geometric progression with common ratio r , then r satisfies the equation

- (A) $r^2 + r + 1 = 0$ (B) $r^2 - r + 1 = 0$ (C) $r^4 + r^2 - 1 = 0$
 (D) $(r + 1)^4 + r = 0$ (E) $(r - 1)^4 + r = 0$

Answer: A

Solution

Let the geometric progression be a , ar , ar^2 , so $a = x(y - z)$, $ar = y(z - x)$, and $ar^2 = z(x - y)$. Adding these equations, we get

$$a + ar + ar^2 = xy - xz + yz - xy + xz - yz = 0.$$

Since a is nonzero, we can divide by a , to get $\boxed{r^2 + r + 1 = 0}$. The answer is (A).

25. (AHSME 1978) Let a be a positive number. Consider the set S of all points whose rectangular coordinates (x, y) satisfy all of the following conditions:

(i) $\frac{a}{2} \leq x \leq 2a$ (ii) $\frac{a}{2} \leq y \leq 2a$ (iii) $x + y \geq a$

(iv) $x + a \geq y$ (v) $y + a \geq x$

The boundary of set S is a polygon with

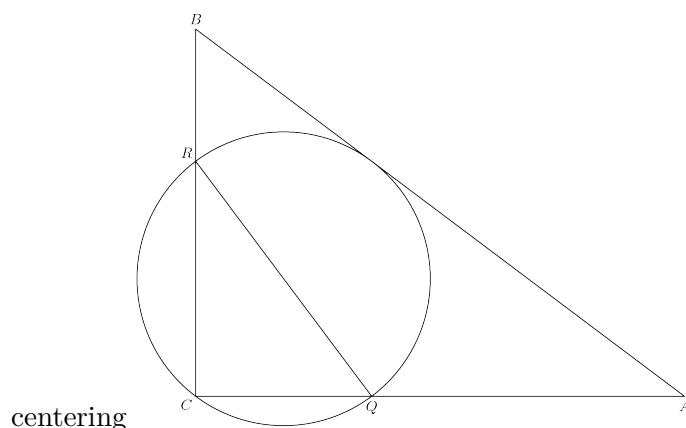
- (A) 3 sides (B) 4 sides (C) 5 sides (D) 6 sides (E) 7 sides

Answer: D

Solution

Draw a picture. The answer is $\boxed{\text{D}}$.

26. (AHSME 1978)



In $\triangle ABC$, $AB = 10$, $AC = 8$ and $BC = 6$. Circle P is the circle with smallest radius which passes through C and is tangent to AB . Let Q and R be the points of intersection, distinct from C , of circle P with sides AC and BC , respectively. The length of segment QR is

- (A) 4.75 (B) 4.8 (C) 5 (D) $4\sqrt{2}$ (E) $3\sqrt{3}$

Answer: B

Solution 1

Let D be the foot of the altitude from C to $\overline{AB}$, and let $h = CD$. Since the area of $\triangle ABC$ is $\frac{1}{2} \cdot 8 \cdot 6 = 24$, it follows that $\frac{1}{2} \cdot 10 \cdot h = 24$, so

27. (AHSME 1978) There is more than one integer greater than 1 which, when divided by any integer k such that $2 \leq k \leq 11$, has a remainder of 1. What is the difference between the two smallest such integers?

- (A) 2310 (B) 2311 (C) 27,720 (D) 27,721 (E) none of these

Answer: C

Solution

Let this integer be n . We have $n \equiv 1 \pmod{2}$, $n \equiv 1 \pmod{3}$, $n \equiv 1 \pmod{4} \dots n \equiv 1 \pmod{11}$. Recall that if

$$a \equiv b \pmod{c}$$

and

$$a \equiv b \pmod{d}$$

then

$$a \equiv b \pmod{\text{lcm}(c, d)}$$

We see that since $n \equiv 1 \pmod{2}$, $n \equiv 1 \pmod{3}$, $n \equiv 1 \pmod{4} \dots n \equiv 1 \pmod{11}$. We have

$$n \equiv 1 \pmod{\text{lcm}(2, 3, \dots, 10, 11)}$$

From 2 to 11, 8 contains the largest power of 2, 9 contains the largest power of 3, and 10 contains the largest power of 5. Thus, our lcm is equal to

$$2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11$$

Since $n > 1$, our 2 smallest values of n are

$$n = 1 + (2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11)$$

and

$$n = 1 + 2(2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11)$$

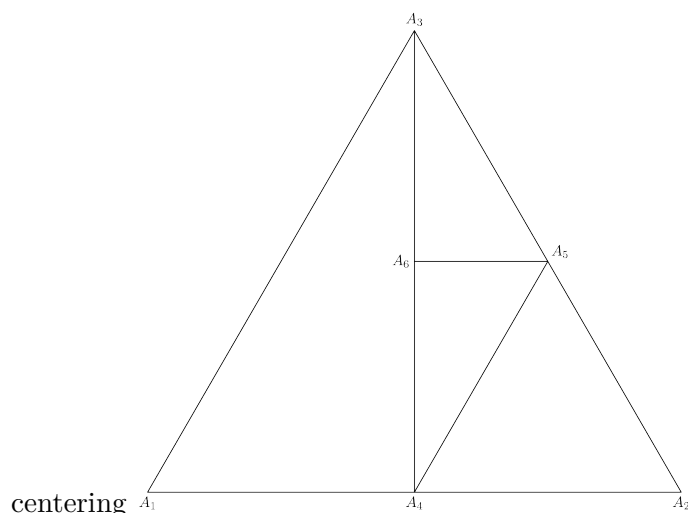
The difference between these values is simply the value of

$$(2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11) = 40 \cdot 99 \cdot 7$$

$$= 3960 \cdot 7$$

$$= 27720, \boxed{C}$$

28. (AHSME 1978)



If $\triangle A_1A_2A_3$ is equilateral and A_{n+3} is the midpoint of line segment A_nA_{n+1} for all positive integers n , then the measure of $\angle A_{44}A_{45}A_{43}$ equals

- (A) 30° (B) 45° (C) 60° (D) 90° (E) 120°

Answer: E

Solution (Repeating Cycles, Kinda)

Upon graphing more points according to the rule mentioned in the problem, you can visibly see the rule repeats every 8 points (i.e. A_9 is the same as A_1 , and A_{10} is the same as A_2 , etc.). Through very simple calculations, this means A_{44} becomes A_4 , A_{45} becomes A_5 , and A_{43} becomes A_3 , and you can clearly see on the diagram that this is a 120 degree angle, leading to $\boxed{E}$ as the final answer.

29. (AHSME 1978) Sides AB , BC , CD and DA , respectively, of convex quadrilateral $ABCD$ are extended past B , C , D and A to points B' , C' , D' and A' . Also, $AB = BB' = 6$, $BC = CC' = 7$, $CD = DD' = 8$ and $DA = AA' = 9$; and the area of $ABCD$ is 10. The area of $A'B'C'D'$ is

- (A) 20 (B) 40 (C) 45 (D) 50 (E) 60

Answer: D

Solution

Notice that the area of $\triangle DAB$ is the same as that of $\triangle A'AB$ (same base, same height). Thus, the area of $\triangle A'AB$ is twice that (same height, twice the base). Similarly, $[\triangle BB'C] = 2 \cdot [\triangle ABC]$, and so on.

Adding all of these, we see that the area the four triangles around $ABCD$ is twice $[\triangle DAB] + [\triangle ABC] + [\triangle BCD] + [\triangle CDA]$, which is itself twice the area of the quadrilateral $ABCD$. Finally, $[A'B'C'D'] = [ABCD] + 4 \cdot [ABCD] = 5 \cdot [ABCD] = \boxed{50}$.

Note: Anyone with a diagram would be of great help (still new to LaTeX).

30. (AHSME 1978) In a tennis tournament, n women and $2n$ men play, and each player plays exactly one match with every other player. If there are no ties and the ratio of the number of matches won by women to the number of matches won by men is $7/5$, then n equals
- (A) 2 (B) 4 (C) 6 (D) 7 (E) none of these

Answer: E

Solution 1

Since there are n women, the number of matches between only women is $\frac{n(n-1)}{2}$, and similarly, there are $(2n-1)n$ matches between only men. Since every woman plays every man exactly once, there are $2n \cdot n = 2n^2$ matches which are between a man and a woman. Call these $2n^2$ matches co-ed matches, and let w be the number of co-ed matches won by women.

Then it follows that

$$\frac{7}{5} = \frac{\frac{n(n-1)}{2} + w}{(2n-1)n + 2n^2 - w},$$

which can be simplified to

$$w = \frac{17}{8}n^2 - \frac{3}{8}n.$$

The number of matches won by women must be less than the total number of matches, so we obtain the inequality

$$\frac{17}{8}n^2 - \frac{3}{8}n \leq 2n^2.$$

Rearranging and factoring gives

$$0 \leq -n(n-3),$$

and the only integers which satisfy this inequality are $n = 0, 1, 2$, and 3 .

Clearly, there could not have been 0 people in the tournament, so $n \neq 0$. If $n = 1$, then there would have been only one woman and two men in the tournament, in which case the woman could not have won the majority of the matches.

We can now plug $n = 2$ back into the equation

$$\frac{7}{5} = \frac{\frac{n(n-1)}{2} + w}{(2n-1)n + 2n^2 - w},$$

and solving for w gives $w = \frac{31}{4}$. Since w must be an integer, n cannot be 2. It follows that $n = 3$, so the answer is (E). (When $n = 3$, solving gives $w = 18$.)

Solution 2

Since there are n women and $2n$ men, there are a total of $n + 2n = 3n$ players. Hence, the number of total matches must be $\binom{3n}{2} = \frac{3n(3n-1)}{2}$. We also know that the ratio of the number of matches won by women to the number of matches won

by men is $\frac{7}{5}$ and that there were no draws, so the total number of matches must be $(7 + 5)x = 12x$ for some value x . This gives

$$\frac{3n(3n - 1)}{2} = 12x$$

So

$$n(3n - 1) = 8x$$

It follows that $n = 8, 4, 3, 2, 1$ (various ways of splitting $8x$ into two factors). However, for solutions $n = 4, 2, 1$, x isn't an integer, therefore leaving solutions $n = 8$ and $n = 3$.

$n = 8$ is invalid. This can be shown as when $n = 8$,

$$\begin{aligned}\frac{24(23)}{2} &= 12x \\ x &= 23\end{aligned}$$

The ratio of matches won by gender is $\frac{7x}{5x} = \frac{161}{115}$, so the number of matches won by women must be at least 161. The maximum number of matches won by women is equivalent to the number of matches between men (generates 1 match won by men no matter the outcome) subtracted from the total number of matches, which is $\binom{24}{2} - \binom{16}{2} = 156$. As the maximum number of matches won is smaller than 161, the solution is extraneous/invalid.

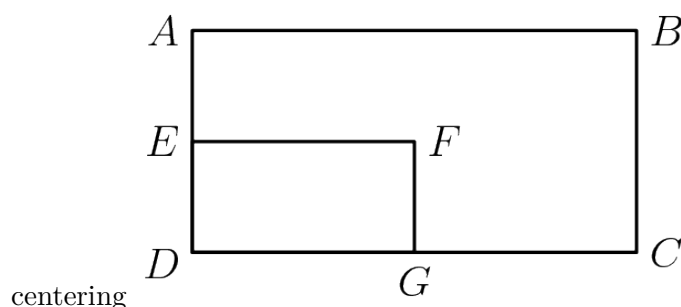
Hence, the answer must be $n = 3$, or (E) none of these

Solution 3 (answer choices)

Since $\frac{7}{12}$ of the total amount of games were won by women, the total amount of games has to be some multiple of 12. The total amount of games is given by $\frac{(3n-1)(3n)}{2}$, when there are $3n$ total people. Plugging in $n = 2$, we get $\frac{(5)(6)}{2} = 15$, not a multiple of 12. Likewise for $n = 4$, $\frac{(7)(8)}{2} = 28$, which doesn't work either. Next, $n = 6$ yields $\frac{(11)(12)}{2} = 66$. Lastly, after testing out $n = 7$ and obtaining $\frac{(13)(14)}{2} = 91$, we see that the answer must be E. martianrunner

§15 AHSME 1979

1. (AHSME 1979)



If rectangle ABCD has area 72 square meters and E and G are the midpoints of sides AD and CD, respectively, then the area of rectangle DEFG in square meters is

- (A) 8 (B) 9 (C) 12 (D) 18 (E) 24

Answer: D**Solution**

Solution by e_power_pi_times_i

Since the dimensions of $DEFG$ are half of the dimensions of $ABCD$, the area of $DEFG$ is $\frac{1}{2} \cdot \frac{1}{2}$ of $ABCD$, so the area of $ABCD$ is $\frac{1}{4} \cdot 72 = \boxed{\text{(D) } 18}$.

2. (AHSME 1979) For all non-zero real numbers x and y such that $x - y = xy$, $\frac{1}{x} - \frac{1}{y}$ equals

- (A)
- $\frac{1}{xy}$
- (B)
- $\frac{1}{x-y}$
- (C) 0 (D)
- -1
- (E)
- $y - x$

Answer: D**Solution 1**

Moving all variables to one side of the equation, we can use Simon's Favorite Factoring Trick to factor the equation into

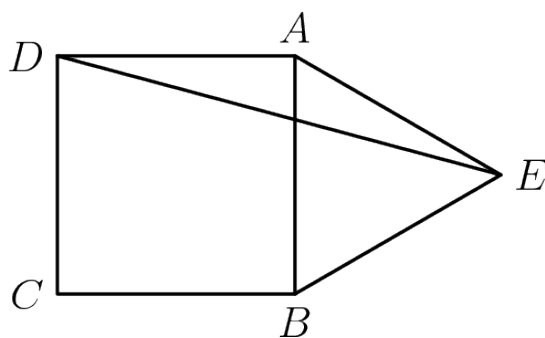
$$(x + 1)(y - 1) = -1$$

Plugging in -1 and 1 as the x and y sides respectively, we get $x = -2$ and $y = 2$. Plugging this in to $\frac{1}{x} - \frac{1}{y}$ gives us $\boxed{-1}$ as our final answer.

Solution 2

Notice that we can do $\frac{x-y}{xy} = \frac{xy}{xy}$. We are left with $\frac{1}{y} - \frac{1}{x} = 1$. Multiply by -1 to achieve $\frac{1}{x} - \frac{1}{y} = \boxed{-1}$.

3. (AHSME 1979)



centering

In the adjoining figure, $ABCD$ is a square, ABE is an equilateral triangle and point E is outside square $ABCD$. What is the measure of $\angle AED$ in degrees?

- (A) 10 (B) 12.5 (C) 15 (D) 20 (E) 25

Answer: C**Solution**

Solution by e_power_pi_times_i

Notice that $\angle DAE = 90^\circ + 60^\circ = 150^\circ$ and that $AD = AE$. Then triangle ADE is isosceles, so $\angle AED = \frac{180^\circ - 150^\circ}{2} = \boxed{\text{(C) } 15}$.

Solution 2 (Obnoxiously tedious)

WLOG, let the side length of the square and the equilateral triangle be 1. $\angle DAE = 90^\circ + 60^\circ = 150^\circ$. Apply the law of cosines then the law of sines, we find that $\angle AED = 15^\circ$. Select C.

Please don't try this solution. This is just obnoxiously tedious and impractical.

4. (AHSME 1979) For all real numbers x , $x[x\{x(2-x)-4\}+10]+1=$
 (A) $-x^4+2x^3+4x^2+10x+1$ (B) $-x^4-2x^3+4x^2+10x+1$
 (C) $-x^4-2x^3-4x^2+10x+1$ (D) $-x^4-2x^3-4x^2-10x+1$
 (E) $-x^4+2x^3-4x^2+10x+1$

Answer: E

5. (AHSME 1979) Find the sum of the digits of the largest even three digit number (in base ten representation) which is not changed when its units and hundreds digits are interchanged.
 (A) 22 (B) 23 (C) 24 (D) 25 (E) 26

Answer: D

Solution

Solution by e_power_pi_times_i

Since the number doesn't change when the units and hundreds digits are switched, the number must be of the form aba . We want to create the largest even 3-digit number, so $a = 8$ and $b = 9$. The sum of the digits is $8 + 9 + 8 =$ (D) 25.

6. (AHSME 1979) $\frac{3}{2} + \frac{5}{4} + \frac{9}{8} + \frac{17}{16} + \frac{33}{32} + \frac{65}{64} - 7 =$
 (A) $-\frac{1}{64}$ (B) $-\frac{1}{16}$ (C) 0 (D) $\frac{1}{16}$ (E) $\frac{1}{64}$

Answer: A

Solution

Solution by e_power_pi_times_i

Simplifying, we have $\frac{96}{64} + \frac{80}{64} + \frac{72}{64} + \frac{68}{64} + \frac{66}{64} + \frac{65}{64} - 7$, which is $\frac{96+80+72+68+66+65}{64} - 7 = \frac{447}{64} - 7 = 6\frac{63}{64} - 7 =$ (A) $-\frac{1}{64}$.

7. (AHSME 1979) The square of an integer is called a perfect square. If x is a perfect square, the next larger perfect square is
 (A) $x+1$ (B) x^2+1 (C) x^2+2x+1 (D) x^2+x (E) $x+2\sqrt{x}+1$

Answer: E

Solution

Solution by e_power_pi_times_i

Since x is a perfect square, denote k such that $k^2 = x$. Then the next perfect square is $(k+1)^2 = k^2 + 2k + 1$. Substituting back in the equation $k = \sqrt{x}$, the next square is (E) $x + 2\sqrt{x} + 1$.

8. (AHSME 1979) Find the area of the smallest region bounded by the graphs of $y = |x|$ and $x^2 + y^2 = 4$.
 (A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) π (D) $\frac{3\pi}{2}$ (E) 2π

Answer: C

Solution

Solution by e_power_pi_times_i

The graph of $x^2 + y^2 = 4$ is a circle with radius 2 centered at the origin. The graph of $y = |x|$ is the combined graphs of $y = x$ and $y = -x$ with a nonnegative y . Because the arguments of $y = x$ and $y = -x$ are 135° and 45° respectively, the angle between the graphs of $y = x$ and $y = -x$ is 90° . Thus, the smallest region bounded by the graphs is $\frac{1}{4} \cdot 2^2 \cdot \pi = \boxed{\text{(C)} \pi}$.

9. (AHSME 1979) The product of $\sqrt[3]{4}$ and $\sqrt[4]{8}$ equals

(A) $\sqrt[7]{12}$ (B) $2\sqrt[7]{12}$ (C) $\sqrt[7]{32}$ (D) $\sqrt[12]{32}$ (E) $2\sqrt[12]{32}$

Answer: E

Solution

Solution by e_power_pi_times_i

$\sqrt[3]{4}$ and $\sqrt[4]{8}$ can be expressed as $2^{\frac{2}{3}}$ and $2^{\frac{3}{4}}$, so their product is $2^{\frac{17}{12}} = \boxed{\text{(E)} 2\sqrt[12]{32}}$.

10. (AHSME 1979) If $P_1P_2P_3P_4P_5P_6$ is a regular hexagon whose apothem (distance from the center to midpoint of a side) is 2, and Q_i is the midpoint of side P_iP_{i+1} for $i = 1, 2, 3, 4$, then the area of quadrilateral $Q_1Q_2Q_3Q_4$ is

(A) 6 (B) $2\sqrt{6}$ (C) $\frac{8\sqrt{3}}{3}$ (D) $3\sqrt{3}$ (E) $4\sqrt{3}$

Answer: D

Solution

Solution by e_power_pi_times_i

Notice that quadrilateral $Q_1Q_2Q_3Q_4$ consists of 3 equilateral triangles with side length 2. Thus the area of the quadrilateral is $3 \cdot (\frac{2^2 \cdot \sqrt{3}}{4}) = 3 \cdot \sqrt{3} = \boxed{\text{(D)} 3\sqrt{3}}$.

11. (AHSME 1979) Find a positive integral solution to the equation $\frac{1+3+5+\dots+(2n-1)}{2+4+6+\dots+2n} = \frac{115}{116}$

(A) 110 (B) 115 (C) 116 (D) 231
(E) The equation has no positive integral solutions.

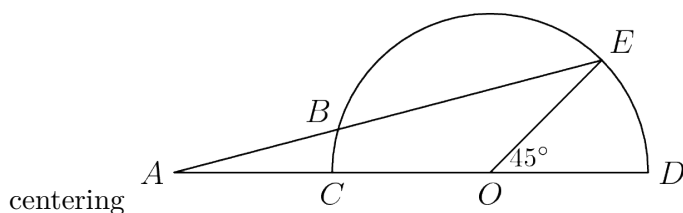
Answer: B

Solution

Solution by e_power_pi_times_i

Notice that the numerator and denominator are the sum of the first n odd and even numbers, respectively. Then the numerator is n^2 , and the denominator is $n(n+1)$. Then $\frac{n}{n+1} = \frac{115}{116}$, so $n = \boxed{\text{(B)} 115}$.

12. (AHSME 1979)



In the adjoining figure, CD is the diameter of a semi-circle with center O . Point A lies on the extension of DC past C ; point E lies on the semi-circle, and B is the point of intersection (distinct from E) of line segment AE with the semi-circle. If length AB equals length OD , and the measure of $\angle EOD$ is 45° , then the measure of $\angle BAO$ is

- (A) 10° (B) 15° (C) 20° (D) 25° (E) 30°

Answer: B

Solution

Solution by e_power_pi_times_i

Because $AB = OD$, triangles ABO and BOE are isosceles. Denote $\angle BAO = \angle AOB = \theta$. Then $\angle ABO = 180^\circ - 2\theta$, and $\angle EBO = \angle OEB = 2\theta$, so $\angle BOE = 180^\circ - 4\theta$. Notice that $\angle AOB + \angle BOE + 45^\circ = 180^\circ$. Therefore $\theta + 180 - 4\theta = 135^\circ$, and $\theta = \boxed{\text{(B)} 15^\circ}$.

Solution 2

Draw BO . Let $y = \angle BAO$. Since $AB = OD = BO$, triangle ABO is isosceles, so $\angle BOA = \angle BAO = y$. Angle $\angle EBO$ is exterior to triangle ABO , so $\angle EBO = \angle BAO + \angle BOA = y + y = 2y$.

Triangle BEO is isosceles, so $\angle BEO = \angle EBO = 2y$. Then $\angle EOD$ is external to triangle AEO , so $\angle EOD = \angle EAO + \angle AEO = y + 2y = 3y$. But $\angle EOD = 45^\circ$, so $\angle BAO = y = 45^\circ/3 = \boxed{15^\circ}$. That means the answer is **(B) 15°** .

13. (AHSME 1979) The inequality $y - x < \sqrt{x^2}$ is satisfied if and only if

- (A) $y < 0$ or $y < 2x$ (or both inequalities hold) (B) $y > 0$ or $y < 2x$ (or both inequalities hold)
 $2xy$ (D) $y < 0$ (E) $x > 0$ and $y < 2x$

Answer: A

Solution

Solution by e_power_pi_times_i

$\sqrt{x^2} = \pm x$, so the inequality is just $y - x < \pm x$. Therefore we get the two inequalities $y < 0$ and $y < 2x$. Checking the answer choices, we find that **(A) $y < 0$ or $y < 2x$ (or both inequalities hold)** is the answer.

14. (AHSME 1979) In a certain sequence of numbers, the first number is 1, and, for all $n \geq 2$, the product of the first n numbers in the sequence is n^2 . The sum of the third and the fifth numbers in the sequence is

- (A) $\frac{25}{9}$ (B) $\frac{31}{15}$ (C) $\frac{61}{16}$ (D) $\frac{576}{225}$ (E) 34

Answer: C

Solution

Solution by e_power_pi_times_i

Since the product of the first n numbers in the sequence is n^2 , the product of the first $n + 1$ numbers in the sequence is $(n + 1)^2$. Therefore the $n + 1$ th number in the sequence is $\frac{(n+1)^2}{n^2}$. Therefore the third and the fifth numbers are $\frac{9}{4}$ and $\frac{25}{16}$

respectively. The sum of those numbers is $\frac{36}{16} + \frac{25}{16} = \boxed{\text{(C)} \frac{61}{16}}$.

15. (AHSME 1979) Two identical jars are filled with alcohol solutions, the ratio of the volume of alcohol to the volume of water being $p : 1$ in one jar and $q : 1$ in the other jar. If the entire contents of the two jars are mixed together, the ratio of the volume of alcohol to the volume of water in the mixture is

(A) $\frac{p+q}{2}$ (B) $\frac{p^2+q^2}{p+q}$ (C) $\frac{2pq}{p+q}$ (D) $\frac{2(p^2+pq+q^2)}{3(p+q)}$ (E) $\frac{p+q+2pq}{p+q+2}$

Answer: E

Solution

Solution by e_power_pi_times_i

The amount of alcohol in the jars are $\frac{p}{p+1}$ and $\frac{q}{q+1}$, and the amount of water in the jars are $\frac{1}{p+1}$ and $\frac{1}{q+1}$. Then the total amount of alcohol is $\frac{p}{p+1} + \frac{q}{q+1} = \frac{p+q+2pq}{(p+1)(q+1)}$, and the total amount of water is $\frac{1}{p+1} + \frac{1}{q+1} = \frac{p+q+2}{(p+1)(q+1)}$. The ratio of the volume of alcohol to the volume of water in the mixture is $\frac{\frac{p+q+2pq}{(p+1)(q+1)}}{\frac{p+q+2}{(p+1)(q+1)}} = \boxed{\text{(E)} \frac{p+q+2pq}{p+q+2}}$.

16. (AHSME 1979) A circle with area A_1 is contained in the interior of a larger circle with area $A_1 + A_2$. If the radius of the larger circle is 3, and if $A_1, A_2, A_1 + A_2$ is an arithmetic progression, then the radius of the smaller circle is

(A) $\frac{\sqrt{3}}{2}$ (B) 1 (C) $\frac{2}{\sqrt{3}}$ (D) $\frac{3}{2}$ (E) $\sqrt{3}$

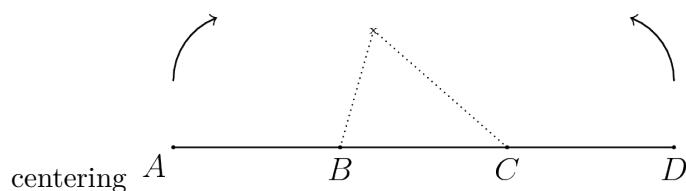
Answer: E

Solution

Solution by e_power_pi_times_i

The area of the larger circle is $A_1 + A_2 = 9\pi$. Then $A_1, 9\pi - A_1, 9\pi$ are in an arithmetic progression. Thus $9\pi - (9\pi - A_1) = 9\pi - A_1 - A_1$. This simplifies to $3A_1 = 9\pi$, or $A_1 = 3\pi$. The radius of the smaller circle is $\boxed{\text{(E)} \sqrt{3}}$.

17. (AHSME 1979)



Points A, B, C , and D are distinct and lie, in the given order, on a straight line. Line segments AB, AC , and AD have lengths x, y , and z , respectively. If line segments AB and CD may be rotated about points B and C , respectively, so that points A and D coincide, to form a triangle with positive area, then which of the following three inequalities must be satisfied?

I. $x < \frac{z}{2}$

II. $y < x + \frac{z}{2}$

III. $y < \frac{z}{2}$

- (A) I. only (B) II. only (C) I. and II. only (D) II. and III. only (E) I., II., and III.

Answer: C

Solution

Solution by e_power_pi_times_i

We know that this triangle has lengths of x , $y - x$, and $z - y$. Using the Triangle Inequality, we get 3 inequalities: $2y > z$, $z > 2x$, $2x + z > 0$. Therefore, we know that **I** is true and **III** is false. In **II**, we have to prove $2y < 2x + z$. We know that $2y > z$, so we have to prove $z < 2y < 2x + z$. $2x < z$, so we have to prove that $z < 2z$, which is true for all positive z . Therefore the answer is

(C) I. and II. only.

18. (AHSME 1979) To the nearest thousandth, $\log_{10} 2$ is .301 and $\log_{10} 3$ is .477. Which of the following is the best approximation of $\log_5 10$?

(A) $\frac{8}{7}$ (B) $\frac{9}{7}$ (C) $\frac{10}{7}$ (D) $\frac{11}{7}$ (E) $\frac{12}{7}$

Answer: C

Solution

Solution by e_power_pi_times_i

Notice that $\log_5 10 = \log_5 2 + 1 = \frac{1}{\log_2 5} + 1$. So we are trying to find $\log_2 5$. Denote $\log_{10} 2$ as x . Then $\frac{1}{x} = \log_2 10 = \log_2 5 + 1$. Therefore $\log_2 5 = \frac{1}{x} - 1$, and plugging this in gives $\log_5 10 = \frac{1}{\frac{1}{x}-1} + 1 = \frac{x}{1-x} + 1 = \frac{1}{1-x}$. Since x is around $\frac{3}{10}$, we substitute

and get **(C) $\frac{10}{7}$** .

19. (AHSME 1979) Find the sum of the squares of all real numbers satisfying the equation $x^{256} - 256^{32} = 0$.

(A) 8 (B) 128 (C) 512 (D) 65,536 (E) $2(256^{32})$

Answer: A

Solution

Solution by e_power_pi_times_i

Notice that the solutions to the equation $x^{256} - 1 = 0$ are the 256 roots of unity. Then the solutions to the equation $x^{256} - 256^{32} = 0$ are the 256 roots of unity dilated by $\sqrt[256]{256^{32}} = \sqrt[256]{2^{256}} = 2$. However, the only real solutions to the equation are the first root of unity and the root of unity opposite of it, as both are on the real axis in the complex plane. These two roots of unity are ± 1 , and dilating by 2 gives ± 2 . The sum of the squares is $(2)^2 + (-2)^2 = \mathbf{(A) 8}$

Solution 2

Notice that we can change the equation to be $x^{256} = 256^{32}$.

The RHS can be simplified to $x^{256} = 2^{256}$

Hence, the only real solutions are $x = -2, 2$

Hence, $(-2)^2 + (2)^2 = 8 = \mathbf{A}$.

20. (AHSME 1979) If $a = \frac{1}{2}$ and $(a+1)(b+1) = 2$ then the radian measure of $\arctan a + \arctan b$ equals

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{5}$ (E) $\frac{\pi}{6}$

Answer: C

Solution

Solution by e_power_pi_times_i

Since $a = \frac{1}{2}$, $b = \frac{1}{3}$. Now we evaluate $\arctan a$ and $\arctan b$. Denote x and θ such that $\arctan x = \theta$. Then $\tan(\arctan(x)) = \tan(\theta)$, and simplifying gives $x = \tan(\theta)$. So $a = \tan(\theta_a) = \frac{1}{2}$ and $b = \tan(\theta_b) = \frac{1}{3}$. The question asks for $\theta_a + \theta_b$, so we try to find $\tan(\theta_a + \theta_b)$ in terms of $\tan(\theta_a)$ and $\tan(\theta_b)$. Using the angle addition formula for $\tan(\alpha + \beta)$, we get that $\tan(\theta_a + \theta_b) = \frac{\tan(\theta_a) + \tan(\theta_b)}{1 - \tan(\theta_a)\tan(\theta_b)}$. Plugging $\tan(\theta_a) = \frac{1}{2}$ and $\tan(\theta_b) = \frac{1}{3}$ in, we have $\tan(\theta_a + \theta_b) = \frac{\frac{1}{2} + \frac{1}{3}}{1 - (\frac{1}{2})(\frac{1}{3})}$. Simplifying, $\tan(\theta_a + \theta_b) = 1$, so $\theta_a + \theta_b$ in radians is $\boxed{\text{(C)} \frac{\pi}{4}}$.

Solution 1.1 (Guiding through the thought process)

Thinking through the problem, we can see $\arctan a$ is the angle whose tangent is 0.5. $\arctan b$ is the angle whose tangent is $\frac{1}{3}$. Call the former angle θ_1 , the latter θ_2 . So we are trying to find $\theta_1 + \theta_2$. So given two tangent measures, it is natural for us to think about the sum of tangent measures (what else can we try? Remember: we are not allowed to use calculators). Plug in as above and continue on.

21. (AHSME 1979) The length of the hypotenuse of a right triangle is h , and the radius of the inscribed circle is r . The ratio of the area of the circle to the area of the triangle is

(A) $\frac{\pi r}{h+2r}$ (B) $\frac{\pi r}{h+r}$ (C) $\frac{\pi}{2h+r}$ (D) $\frac{\pi r^2}{r^2+h^2}$ (E) none of these

Answer: B

Solution

Solution by e_power_pi_times_i

Since $rs = A$, where r is the inradius, s is the semiperimeter, and A is the area, we have that the ratio of the area of the circle to the area of the triangle is $\frac{\pi r^2}{rs} = \frac{\pi r}{s}$. Now we try to express s as h and r . Denote the points where the incircle meets the triangle as X, Y, Z , where O is the incenter, and denote $AX = AY = z, BX = BZ = y, CY = CZ = x$. Since $XOZB$ is a square (tangents are perpendicular to radius), $r = BX = BZ = y$. The perimeter can be expressed as $2(x+y+z)$, so the semiperimeter is $x+y+z$. The hypotenuse is $AY + CY = z + x$.

Thus we have $s = x + y + z = (z + x) + y = h + r$. The answer is $\boxed{\text{(B)} \frac{\pi r}{h+r}}$.

22. (AHSME 1979) Find the number of pairs (m, n) of integers which satisfy the equation $m^3 + 6m^2 + 5m = 27n^3 + 9n^2 + 9n + 1$.

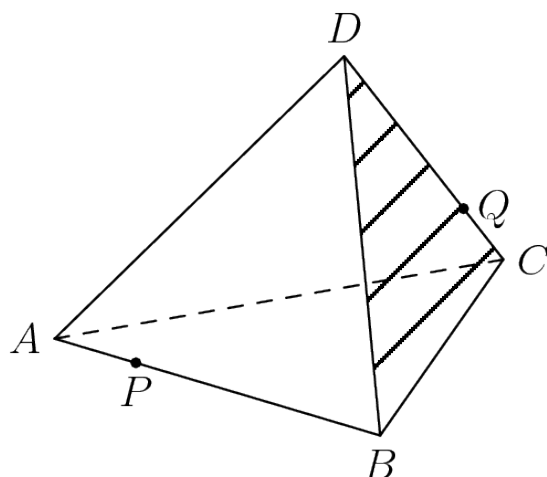
(A) 0 (B) 1 (C) 3 (D) 9 (E) ∞

Answer: A

Solution

The equation is equivalent to $m(m+1)(m+5) = 27n^3 + 27n^2 + 9n + 1$. Taking mod 3, we get $m(m+1)(m+2) = 1 \pmod{3}$. However, $m(m+1)(m+2)$ is always divisible by 3 for any integer m . Thus, the answer is $\boxed{\text{(A)} 0}$ Solution by mickyboy789

23. (AHSME 1979) The edges of a regular tetrahedron with vertices A, B, C , and D each have length one. Find the least possible distance between a pair of points P and Q , where P is on edge AB and Q is on edge CD .



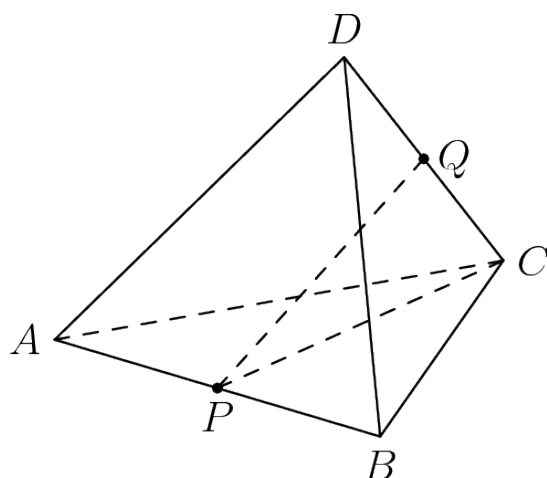
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- (A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) $\frac{\sqrt{2}}{2}$ (D) $\frac{\sqrt{3}}{2}$ (E) $\frac{\sqrt{3}}{3}$

Answer: C**Solution 1**

Note that the distance PQ will be minimized when P is the midpoint of AB and Q is the midpoint of CD .

To find this distance, consider triangle $\triangle PCQ$. Q is the midpoint of CD , so $CQ = \frac{1}{2}$. Additionally, since CP is the altitude of equilateral $\triangle ABC$, $CP = \frac{\sqrt{3}}{2}$.

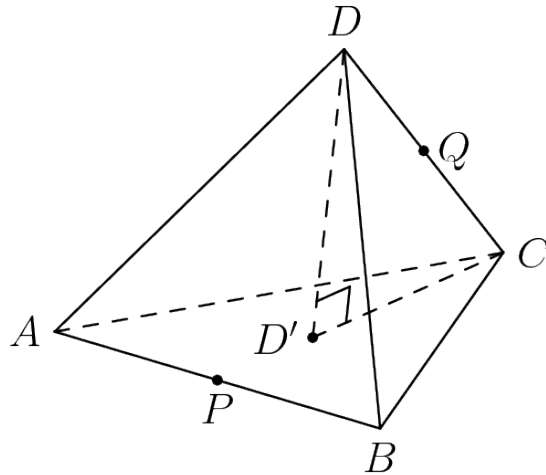


centering

Next, we need to find $\cos(\angle PCQ)$ in order to find PQ by the Law of Cosines. To do so, drop down D onto $\triangle ABC$ to get the point D' .

$\angle PCD$ is congruent to $\angle D'CD$, since P , D' , and C are collinear. Therefore, we can just find $\cos(\angle D'CD)$.

Note that $\triangle CD'D$ is a right triangle with $\angle CD'D$ as a right angle.



centering

As given by the problem, $CD = 1$.

Note that D' is the centroid of equilateral $\triangle ABC$. Additionally, since $\triangle ABC$ is equilateral, D' is also the orthocenter. Due to this, the distance from C to D' is $\frac{2}{3}$ of the altitude of $\triangle ABC$. Therefore, $CD' = \frac{\sqrt{3}}{3}$.

Since $\cos(\angle D'CD) = \cos(\angle PCQ) = \frac{CD'}{CD}$, $\cos(\angle PCQ) = \frac{\frac{\sqrt{3}}{3}}{1} = \frac{\sqrt{3}}{3}$

$$PQ^2 = CP^2 + CQ^2 - 2(CP)(CQ)\cos(\angle PCQ)$$

$$PQ^2 = \frac{3}{4} + \frac{1}{4} - 2\left(\frac{\sqrt{3}}{4}\right)\left(\frac{1}{2}\right)\left(\frac{\sqrt{3}}{3}\right)$$

Simplifying, $PQ^2 = \frac{1}{2}$. Therefore, $PQ = \frac{\sqrt{2}}{2} \Rightarrow \boxed{\text{C}}$

Solution by treetor10145

Solution 2 (less overkill)

Notice, like above said, that P is the midpoint of AB and Q is the midpoint of CD .

To find the length of PQ , first draw in lines CP and DP . Notice that DP is an altitude of $\triangle ADP$. We find that $\angle DAP = 60^\circ$ (since $\triangle ABD$ is equilateral), and $AD = \frac{1}{2}$. Use the properties of 30-60-90 triangles to get $DP = \frac{\sqrt{3}}{2}$. Since CP is an altitude of a congruent equilateral triangle, $CP = DP = \frac{\sqrt{3}}{2}$.

Notice that $\triangle CDP$ is isosceles with $CP = DP$. Also, since Q is the midpoint of base CD , we can conclude that PQ is an altitude. We can use Pythagorean theorem to get the following (taking into consideration $DQ = \frac{1}{2}$):

$$DQ^2 + PQ^2 = PD^2$$

$$\left(\frac{1}{2}\right)^2 + PQ^2 = \left(\frac{\sqrt{3}}{2}\right)^2$$

$$PQ^2 = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$$

$$PQ = \frac{\sqrt{2}}{2} \Rightarrow \boxed{\text{C}}$$

-WannabeCharmander

24. (AHSME 1979) Sides AB , BC , and CD of (simple*) quadrilateral $ABCD$ have lengths 4, 5, and 20, respectively. If vertex angles B and C are obtuse and $\sin C = -\cos B = \frac{3}{5}$, then side AD has length

- A polygon is called “simple” if it is not self intersecting.

(A) 24 (B) 24.5 (C) 24.6 (D) 24.8 (E) 25

Answer: E

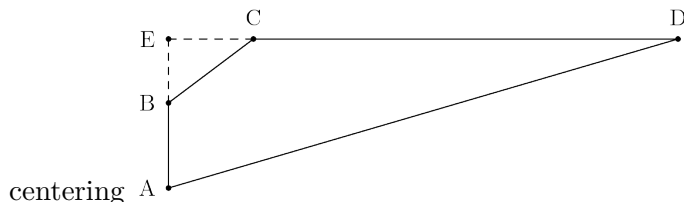
Solution

We know that $\sin(C) = -\cos(B) = \frac{3}{5}$. Since B and C are obtuse, we have $\sin(180 - C) = \cos(180 - B) = \frac{3}{5}$. It is known that $\sin(x) = \cos(90 - x)$, so $180 - C = 90 - (180 - B) = 180 - B$. We simplify this as follows:

$$-90 + C = 180 - B$$

$$B + C = 270^\circ$$

Since $B + C = 270^\circ$, we know that $A + D = 360 - (B + C) = 90^\circ$. Now extend AB and CD as follows:



Let AB and CD intersect at E . We know that $\angle AED = 90^\circ$ because $\angle E = 180 - (A + D) = 180 - 90 = 90^\circ$.

Since $\sin BCD = \frac{3}{5}$, we get $\sin ECB = \sin(180 - BCD) = \sin BCD = \frac{3}{5}$. Thus, $EB = 3$ and $EC = 4$ from simple sin application.

AD is the hypotenuse of right $\triangle AED$, with leg lengths $AB + BE = 7$ and $EC + CD = 24$. Thus, $AD = \boxed{\text{(E)}25}$

-WannabeCharmander

25. (AHSME 1979) If $q_1(x)$ and r_1 are the quotient and remainder, respectively, when the polynomial x^8 is divided by $x + \frac{1}{2}$, and if $q_2(x)$ and r_2 are the quotient and remainder, respectively, when $q_1(x)$ is divided by $x + \frac{1}{2}$, then find r_2 .

(A) $\frac{1}{256}$ (B) $-\frac{1}{16}$ (C) 1 (D) -16 (E) 256

Answer: B

Solution

Solution by e_power_pi_times_i

First, we divide x^8 by $x + \frac{1}{2}$ using synthetic division or some other method. The quotient is $x^7 - \frac{1}{2}x^6 + \frac{1}{4}x^5 - \frac{1}{8}x^4 + \frac{1}{16}x^3 - \frac{1}{32}x^2 + \frac{1}{64}x - \frac{1}{128}$, and the remainder is $\frac{1}{256}$. Then we plug the solution to $x + \frac{1}{2} = 0$ into the quotient to find the remainder. Notice that every term in the quotient, when $x = -\frac{1}{2}$, evaluates to $-\frac{1}{128}$. Thus

$$r_2 = -\frac{8}{128} = \boxed{\text{(B)} - \frac{1}{16}}.$$

Solution 2

Using the remainder theorem, we see that the remainder upon dividing x^8 by $x + \frac{1}{2}$ is equal to $\left(-\frac{1}{2}\right)^8 = \frac{1}{256}$. Thus, we have that

$$x^8 = \left(x + \frac{1}{2}\right) Q(x) + \frac{1}{256}.$$

Isolating $Q(x)$, we see

$$\frac{x^8 - \frac{1}{256}}{x + \frac{1}{2}} = Q(x).$$

Using the difference of squares factorization repeatedly, we get

$$\frac{\left(x - \frac{1}{2}\right)\left(x + \frac{1}{2}\right)\left(x^2 + \frac{1}{2^2}\right)\left(x^4 + \frac{1}{2^4}\right)}{x + \frac{1}{2}} = \left(x - \frac{1}{2}\right)\left(x^2 + \frac{1}{2^2}\right)\left(x^4 + \frac{1}{2^4}\right) = Q(x).$$

Finally, plugging in $x = -\frac{1}{2}$ again, the final answer is

$$\begin{aligned} & (-1)\left(\frac{1}{2}\right)\left(\frac{1}{8}\right) \\ &= \boxed{-\frac{1}{16}}. \end{aligned}$$

26. (AHSME 1979) The function f satisfies the functional equation $f(x) + f(y) = f(x + y) - xy - 1$ for every pair x, y of real numbers. If $f(1) = 1$, then the number of integers $n \neq 1$ for which $f(n) = n$ is

(A) 0 (B) 1 (C) 2 (D) 3 (E) ∞

Answer: B

Solution

We are given that $f(x) + f(y) = f(x + y) - xy - 1$ and $f(1) = 1$, so we can let $y = 1$. Thus we have:

$$f(x) + 1 = f(x + 1) - x - 1$$

Rearranging gives a recursive formula for f :

$$f(x + 1) = f(x) + x + 2$$

We notice that this is the recursive form for a quadratic, so $f(x)$ must be of the form $ax^2 + bx + c$. To solve for a, b , and c , we can first work backwards to solve for the values of $f(0)$ and $f(-1)$:

$$f(1) = f(0) + 0 + 2 \Rightarrow 1 = f(0) + 2 \Rightarrow f(0) = -1$$

$$f(0) = f(-1) - 1 + 2 \Rightarrow -1 = f(-1) + 1 \Rightarrow f(-1) = -2$$

Since $f(0) = -1$:

$$f(0) = a(0)^2 + b(0) + c = -1 \Rightarrow c = -1$$

Since $f(1) = 1$:

$$f(1) = a(1)^2 + b(1) + c = a + b - 1 = 1 \Rightarrow a + b = 2$$

Similarly, since $f(-1) = -2$:

$$f(-1) = a(-1)^2 + b(-1) + c = a - b - 1 = -2 \Rightarrow a - b = -1$$

Thus we have the system of equations:

$$a + b = 2$$

$$a - b = -1$$

Which can be solved to yield $a = \frac{1}{2}$, $b = \frac{3}{2}$. Therefore, $f(x) = \frac{1}{2}x^2 + \frac{3}{2}x - 1$.

Since we are searching for values for which $f(x) = x$, we have the equation $\frac{1}{2}x^2 + \frac{3}{2}x - 1 = x$. Subtracting x yields $\frac{1}{2}x^2 + \frac{1}{2}x - 1 = 0$, which we can simplify by dividing both sides by $\frac{1}{2}$: $x^2 + x - 2 = 0$. This factors into $(x + 2)(x - 1) = 0$, so therefore there are two solutions to $f(x) = x$: -2 and 1 . Since the problem asks only for solutions that do not equal 1 , the answer is (B) 1.

27. (AHSME 1979) An ordered pair (b, c) of integers, each of which has absolute value less than or equal to five, is chosen at random, with each such ordered pair having an equal likelihood of being chosen. What is the probability that the equation $x^2 + bx + c = 0$ will not have distinct positive real roots?

(A) $\frac{106}{121}$ (B) $\frac{108}{121}$ (C) $\frac{110}{121}$ (D) $\frac{112}{121}$ (E) none of these

Answer: E

Solution

E

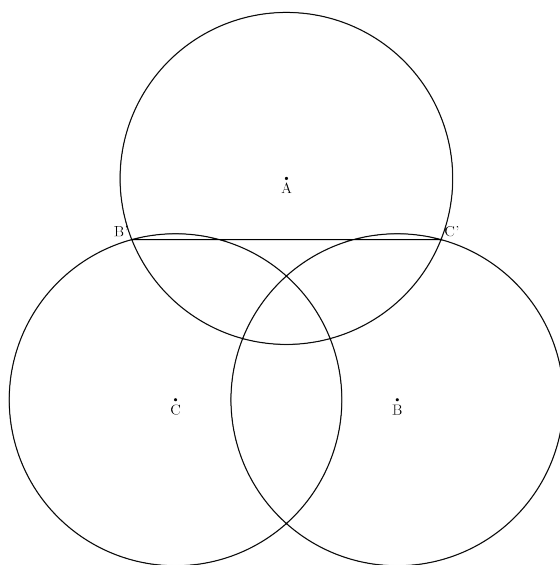
There are 10 cases where the roots are real, positive and distinct. The roots to the quadratic equation are $\frac{-b \pm \sqrt{b^2 - 4c}}{2}$.

In order for $-b - \sqrt{b^2 - 4c} > 0$ we certainly need $b < 0$. Additionally, for the roots to be both real and distinct we need $b^2 - 4c > 0$. And lastly, for both roots to be positive we need $-b > \sqrt{b^2 - 4c}$. Combining these inequalities we determine that $\frac{b^2}{4} > c > 0$ and $b < 0$.

This leaves just 10 cases $\{(-5, 1), (-5, 2), (-5, 3), (-5, 4), (-5, 5), (-4, 1), (-4, 2), (-4, 3), (-3, 1), (-3, 2)\}$ which can easily be checked by hand.

The answer is $\frac{111}{121}$

28. (AHSME 1979)



centering

Circles with centers A , B , and C each have radius r , where $1 < r < 2$. The distance between each pair of centers is 2. If B' is the point of intersection of circle A and circle C which is outside circle B , and if C' is the point of intersection of circle A and circle B which is outside circle C , then length $B'C'$ equals

- (A) $3r - 2$ (B) r^2 (C) $r + \sqrt{3(r-1)}$
 (D) $1 + \sqrt{3(r^2 - 1)}$ (E) none of these

Answer: D**Solution 1 (Coordinate Geometry)**

The circles can be described in the cartesian plane as being centered at $(-1, 0)$, $(1, 0)$ and $(0, \sqrt{3})$ with radius r by the equations

$$x^2 + (y - \sqrt{3})^2 = r^2$$

$$(x + 1)^2 + y^2 = r^2$$

$$(x - 1)^2 + y^2 = r^2.$$

Solving the first 2 equations gives $x = 1 - \sqrt{3} \cdot y$ which when substituted back in gives $y = \frac{\sqrt{3} \pm \sqrt{r^2 - 1}}{2}$.

The larger root $y = \frac{\sqrt{3} + \sqrt{r^2 - 1}}{2}$ is the point B' described in the question. This root corresponds to $x = -\frac{1 + \sqrt{3(r^2 - 1)}}{2}$.

By symmetry across the y -axis the length of the line segment $B'C'$ is $1 + \sqrt{3(r^2 - 1)}$ which is **D**.

Solution 2 (Synthetic)

Suppose $B'B$ and AC intersect at P . By the Pythagorean Theorem, $B'P = \sqrt{r^2 - 1}$ and by a 30-60-90 triangle, $PB = \sqrt{3}$. Using Ptolemy's Theorem on isosceles trapezoid $BCB'C'$, we get that

$$2(B'C') + r^2 = (\sqrt{3} + \sqrt{r^2 - 1})^2.$$

After a little algebra, we get that $B'C' = \boxed{1 + \sqrt{3(r^2 - 1)}}$ as desired. **Solasky (talk)**
 12:29, 27 May 2023 (EDT)

29. (AHSME 1979) For each positive number x , let $f(x) = \frac{(x+\frac{1}{x})^6 - (x^6 + \frac{1}{x^6}) - 2}{(x+\frac{1}{x})^3 + (x^3 + \frac{1}{x^3})}$. The minimum value of $f(x)$ is

(A) 1 (B) 2 (C) 3 (D) 4 (E) 6

Answer: E

Solution

Let $a = (x + \frac{1}{x})^3$ and $b = x^3 + \frac{1}{x^3}$. Then

$$\begin{aligned} f(x) &= \frac{(x + \frac{1}{x})^6 - (x^6 + \frac{1}{x^6}) - 2}{(x + \frac{1}{x})^3 + (x^3 + \frac{1}{x^3})} \\ &= \frac{(x + \frac{1}{x})^6 - (x^6 + 2 + \frac{1}{x^6})}{(x + \frac{1}{x})^3 + (x^3 + \frac{1}{x^3})} \\ &= \frac{(x + \frac{1}{x})^6 - (x^3 + \frac{1}{x^3})^2}{(x + \frac{1}{x})^3 + (x^3 + \frac{1}{x^3})} \\ &= \frac{a^2 - b^2}{a + b}. \end{aligned}$$

By difference of squares,

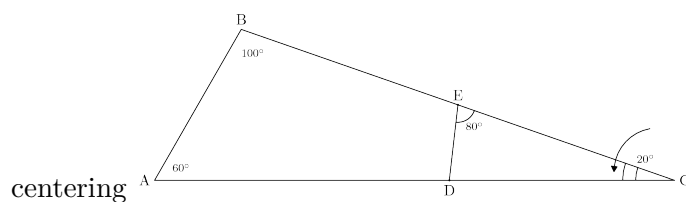
$$\begin{aligned} f(x) &= \frac{(a - b)(a + b)}{a + b} \\ &= a - b \\ &= \left(x + \frac{1}{x}\right)^3 - \left(x^3 + \frac{1}{x^3}\right) \\ &= \left(x^3 + 3x + \frac{3}{x} + \frac{1}{x^3}\right) - \left(x^3 + \frac{1}{x^3}\right) \\ &= 3x + \frac{3}{x} \\ &= 3\left(x + \frac{1}{x}\right). \end{aligned}$$

By the AM-GM inequality,

$$x + \frac{1}{x} \geq 2,$$

so $f(x) \geq 6$. Furthermore, when $x = 1$, $f(1) = 6$, so the minimum value of $f(x)$ is 6.

30. (AHSME 1979)



In $\triangle ABC$, E is the midpoint of side BC and D is on side AC . If the length of AC is 1 and $\angle BAC = 60^\circ$, $\angle ABC = 100^\circ$, $\angle ACB = 20^\circ$ and $\angle DEC = 80^\circ$, then the area of $\triangle ABC$ plus twice the area of $\triangle CDE$ equals

- (A) $\frac{1}{4} \cos 10^\circ$ (B) $\frac{\sqrt{3}}{8}$ (C) $\frac{1}{4} \cos 40^\circ$ (D) $\frac{1}{4} \cos 50^\circ$ (E) $\frac{1}{8}$

Answer: B

Solution

Let F be the point on the extension of side AB past B for which $AF = 1$. Since $AF = AC$ and $\angle FAC = 60^\circ$, $\triangle ACF$ is equilateral. Let G be the point on line segment BF for which $\angle BCG = 20^\circ$. Then $\triangle BCG$ is similar to $\triangle DCE$ and $BC = 2(EC)$. Also $\triangle FGC$ is congruent to $\triangle ABC$. Therefore, $[\triangle ACF] = ([\triangle ABC] + [\triangle GCF]) + [\triangle BCG]$. Plugging in the values that we know and then

dividing by 2 results in an answer of $(B) \frac{\sqrt{3}}{8}$.

This solution is from the solution manual but was typed here by alpha_2.

§16 AHSME 1980

1. (AHSME 1980) The largest whole number such that seven times the number is less than 100 is

- (A) 12 (B) 13 (C) 14 (D) 15 (E) 16

Answer: C

Solution

Let x be a whole number such that $7x < 100$. Dividing by 7 yields $x < \frac{100}{7} = 14\frac{2}{7}$, so $x = \boxed{(C) 14}$ is the maximum possible value.

2. (AHSME 1980) The degree of $(x^2 + 1)^4(x^3 + 1)^3$ as a polynomial in x is
- (A) 5 (B) 7 (C) 12 (D) 17 (E) 72

Answer: D

Solution 1

Expanding each factor yields $(x^8 + \dots)(x^9 + \dots)$. The degree of the first factor is 8, while the degree of the second factor is 9. Therefore, the degree of the polynomial is $8 + 9 = \boxed{(D) 17}$

Solution 2

Let $\deg P(x)$ be the degree of a polynomial in x . Recall that for any polynomials $P(x)$ and $Q(x)$, and nonnegative integer n , $\deg(P(x)^n) = n \deg P(x)$ and $\deg(P(x)Q(x)) = \deg P(x) + \deg Q(x)$.

So $\deg((x^2 + 1)^4(x^3 + 1)^3) = \deg((x^2 + 1)^4) + \deg((x^3 + 1)^3) = 4 \deg(x^2 + 1) + 3 \deg(x^3 + 1) = 4 \cdot 2 + 3 \cdot 3 = \boxed{(D) 17}$.

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3. (AHSME 1980) If the ratio of $2x - y$ to $x + y$ is $\frac{2}{3}$, what is the ratio of x to y ?
- (A) $\frac{1}{5}$ (B) $\frac{4}{5}$ (C) 1 (D) $\frac{6}{5}$ (E) $\frac{5}{4}$

Answer: E

Solution

$$\frac{2x-y}{x+y} = \frac{2}{3}$$

Cross multiplying yields

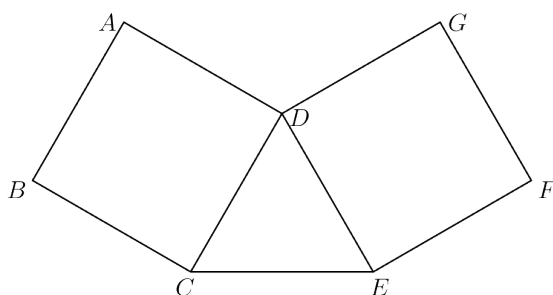
$$6x - 3y = 2x + 2y$$

$$4x = 5y$$

$$\frac{x}{y} = \boxed{\text{(E)} \frac{5}{4}}$$

4. (AHSME 1980) In the adjoining figure, CDE is an equilateral triangle and $ABCD$ and $DEFG$ are squares. The measure of $\angle GDA$ is

(A) 90° (B) 105° (C) 120° (D) 135° (E) 150°



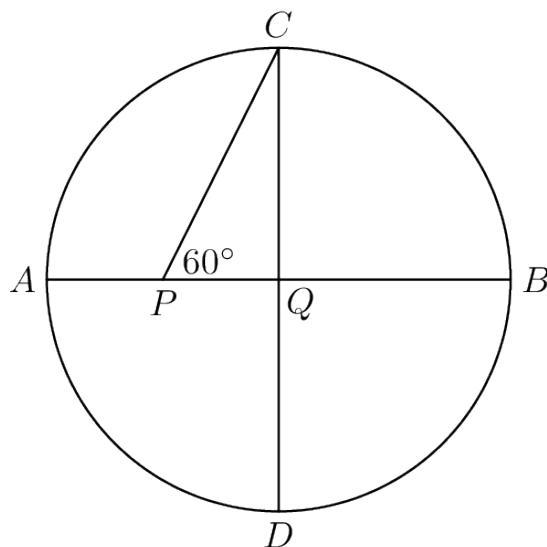
centering

Answer: C

Solution

Since $ABCD$ and $DEFG$ are squares, $\angle ADC = \angle GDE = 90^\circ$. Since CDE is equilateral, $\angle CDE = 60^\circ$. So $\angle GDA = 360^\circ - 90^\circ - 60^\circ - 90^\circ = \boxed{\text{(C)} 120^\circ}$.

5. (AHSME 1980) If AB and CD are perpendicular diameters of circle Q , and $\angle QPC = 60^\circ$, then the length of PQ divided by the length of AQ is



centering

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{\sqrt{3}}{3}$ (C) $\frac{\sqrt{2}}{2}$ (D) $\frac{1}{2}$ (E) $\frac{2}{3}$

Answer: B

Solution

Notice that $\angle PCQ = 30^\circ$, and therefore $\triangle PCQ$ is a $30^\circ - 60^\circ - 90^\circ$ right triangle.

Let $x = PQ$. Then $CQ = AQ = x\sqrt{3}$ and $\frac{PQ}{AQ} = \frac{x}{x\sqrt{3}} = \boxed{\text{(B)} \frac{\sqrt{3}}{3}}$.

6. (AHSME 1980) A positive number x satisfies the inequality $\sqrt{x} < 2x$ if and only if
 (A) $x > \frac{1}{4}$ (B) $x > 2$ (C) $x > 4$ (D) $x < \frac{1}{4}$ (E) $x < 4$

Answer: A

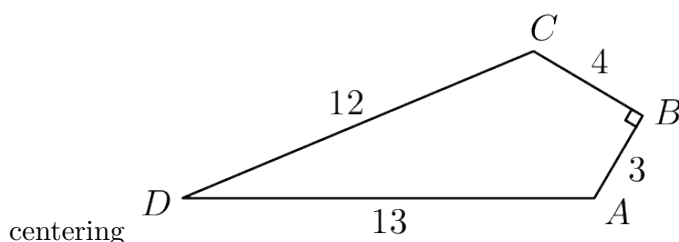
Solution 1

Since both sides of this inequality are positive, we may square both sides to get $x < 4x^2$. Since x is positive, we can divide both sides by x to get $1 < 4x$. Dividing both sides by 4 yields $\frac{1}{4} < x$, or $\boxed{\text{(A)} x > \frac{1}{4}}$.

Solution 2 (Graphing)

Graphing both $\sqrt{x}$ and $2x$ in Quadrant I of the coordinate plane, we see that the two curves intersect at $(\frac{1}{4}, \frac{1}{2})$, and $2x > \sqrt{x}$ whenever $\boxed{\text{(A)} x > \frac{1}{4}}$.

7. (AHSME 1980) Sides AB , BC , CD and DA of convex polygon $ABCD$ have lengths 3, 4, 12, and 13, respectively, and $\angle CBA$ is a right angle. The area of the quadrilateral is



- (A) 32 (B) 36 (C) 39 (D) 42 (E) 48

Answer: B

Solution

Draw $\overline{AC}$. By the Pythagorean Theorem, $AC = \sqrt{AB^2 + BC^2} = \sqrt{3^2 + 4^2} = 5$. Notice that $AD^2 = AC^2 + CD^2 = 169$, so by the converse of the Pythagorean Theorem, $\triangle ACD$ is right. Thus the area of $ABCD$ is $\frac{3 \cdot 4}{2} + \frac{5 \cdot 12}{2} = \boxed{\text{(B)} 36}$.

8. (AHSME 1980) How many pairs (a, b) of non-zero real numbers satisfy the equation

$$\frac{1}{a} + \frac{1}{b} = \frac{1}{a+b}$$

- (A) none (B) 1 (C) 2 (D) one pair for each $b \neq 0$ (E) two pairs for each $b \neq 0$

Answer: A

Solution

Adding the two fractions on the left side yields $\frac{a+b}{ab} = \frac{1}{a+b}$.

Cross-multiplying yields $a^2 + 2ab + b^2 = ab$.

Subtracting ab from both sides yields $a^2 + ab + b^2 = 0$.

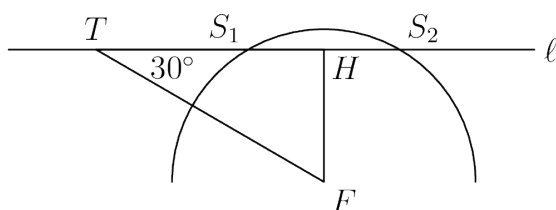
Suppose b is constant and a is variable. Then the discriminant of this quadratic equation is $b^2 - 4 \cdot 1 \cdot b^2 = -3b^2$, which is negative if $b \neq 0$. Therefore, for each $b \neq 0$, this equation has no real solutions, and the answer is **(A) none**.

9. (AHSME 1980) A man walks x miles due west, turns 150° to his left and walks 3 miles in the new direction. If he finishes at a point $\sqrt{3}$ from his starting point, then x is

(A) $\sqrt{3}$ (B) $2\sqrt{5}$ (C) $\frac{3}{2}$ (D) 3 (E) not uniquely determined

Answer: E

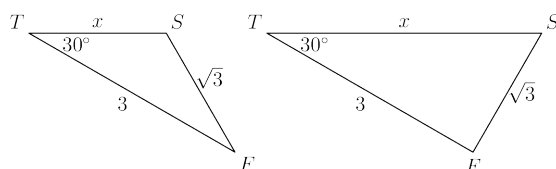
Solution 1



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Let ℓ be the line he was initially walking on, T be his turning point, F be his finishing point, and H be the foot of the perpendicular from F to ℓ . Since he turned 150° , $\angle HTF = 30^\circ$. Therefore, $HF = \frac{3}{2}$ miles. Since F is $\sqrt{3}$ miles from his starting point, his starting point lies on a circle of radius $\sqrt{3}$ miles centered at F . Since $\frac{3}{2} < \sqrt{3} < 3$, this circle must intersect ℓ once between T and H and once east of H . Let S_1 and S_2 be these two intersections, respectively. Both S_1 and S_2 are possible starting points, and clearly $TS_1 \neq TS_2$, so x is **(E) not uniquely determined**.

Solution 2



centering

Let S be his starting point, T be the point where he turns, and F be his finishing point. Since he turned 150° at T , $\angle STF = 30^\circ$. By the Law of Cosines, $FS^2 = FT^2 + ST^2 - 2 \cdot FT \cdot ST \cos \angle STF$. That is, $(\sqrt{3})^2 = 3^2 + x^2 - 2 \cdot 3 \cdot x \cdot \cos 30^\circ$. Combining all terms on one side yields $x^2 - 3x\sqrt{3} + 6 = 0$, which factors as $(x - \sqrt{3})(x - 2\sqrt{3}) = 0$. Therefore, $x = \sqrt{3}$ and $x = 2\sqrt{3}$ are both possible values of x , so x is **(E) not uniquely determined**.

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10. (AHSME 1980) The number of teeth in three meshed circular gears A , B , and C are x , y , and z , respectively. (The teeth on all gears are the same size and regularly spaced.) The angular speeds, in revolutions per minute, of A , B , and C are in the proportion

- (A) $x : y : z$ (B) $z : y : x$ (C) $y : z : x$ (D) $yz : xz : xy$ (E) $xz : yx : zy$

Answer: D

Solution

Since the gears are meshed, they must all rotate the same number of teeth per minute. Let k be this rate. Then gears A , B , and C must have angular speeds of $\frac{k}{x}$, $\frac{k}{y}$, and $\frac{k}{z}$, respectively. Thus the ratio of the three gears' angular speeds is $\frac{k}{x} : \frac{k}{y} : \frac{k}{z}$. Multiplying each term of this ratio by $\frac{xyz}{k}$ yields **(D)** $yz : xz : xy$.

11. (AHSME 1980) If the sum of the first 10 terms and the sum of the first 100 terms of a given arithmetic progression are 100 and 10, respectively, then the sum of first 110 terms is

- (A) 90 (B) -90 (C) 110 (D) -110 (E) -100

Answer: D

Solution 1

Let a and d be the first term and common difference of the sequence, respectively. The sum of the first 10 terms is $\frac{10}{2}(2a + 9d) = 100$. This equation can be simplified to $2a + 9d = 20$.

The sum of the first 100 terms is $\frac{100}{2}(2a + 99d) = 10$. This equation can be simplified to $2a + 99d = \frac{1}{5}$.

Solving the system of these two equations yields $(a, d) = \left(\frac{1099}{100}, -\frac{11}{50}\right)$. Therefore, the sum of the first 110 terms is $\frac{110}{2}(2a + 109d) = 55\left(2 \cdot \frac{1099}{100} + 109 \cdot -\frac{11}{50}\right) = 55 \cdot -2 = \boxed{\text{(D)} - 110}$.

Solution 2

Let $a_1, a_2, \dots, a_n$ be the terms of this arithmetic sequence, and let d be its common difference. Recall that in an arithmetic sequence with an even number of terms, the mean of the middle two terms equals the mean of all terms in the sequence.

The mean of the first 10 terms is $\frac{a_5 + a_6}{2} = \frac{100}{10} = 10$, while the mean of the first 100 terms is $\frac{a_{50} + a_{51}}{2} = \frac{10}{100} = \frac{1}{10}$.

Then $\frac{1}{10} = \frac{a_{50} + a_{51}}{2} = \frac{a_5 + 45d + a_6 + 45d}{2} = \frac{a_5 + a_6}{2} + 45d = 10 + 45d$, so $d = -\frac{11}{50}$. So the mean of the first 110 terms is $\frac{a_{55} + a_{56}}{2} = \frac{a_{50} + 5d + a_{51} + 5d}{2} = \frac{a_{50} + a_{51}}{2} + 5d = \frac{1}{10} + 5 \cdot -\frac{11}{50} = -1$. Therefore, the sum of the first 110 terms is $110 \cdot -1 = \boxed{\text{(D)} - 110}$.

-j314andrews

12. (AHSME 1980) The equations of L_1 and L_2 are $y = mx$ and $y = nx$, respectively. Suppose L_1 makes twice as large of an angle with the horizontal (measured counter-clockwise from the positive x-axis) as does L_2 , and that L_1 has 4 times the slope of L_2 . If L_1 is not horizontal, then mn is

- (A) $\frac{\sqrt{2}}{2}$ (B) $-\frac{\sqrt{2}}{2}$ (C) 2 (D) -2 (E) not uniquely determined

Answer: C

Solution 1 (Trigonometry)

Since L_1 has 4 times the slope of L_2 , $m = 4n$. Let $O = (0, 0)$, $A = (1, 0)$, $B = (1, n)$, $C = (1, m)$, and $\theta = \angle AOB = \angle BOC$. Then $m = \tan 2\theta$ and $n = \tan \theta$.

Since $m = 4n$, $\tan 2\theta = 4 \tan \theta$. By the tangent double-angle formula, $\frac{2 \tan \theta}{1 - \tan^2 \theta} = 4 \tan \theta$. Cross-multiplying and collecting terms on one side yields $4 \tan^3 \theta - 2 \tan \theta = 0$, which factors as $2 \tan \theta (2 \tan^2 \theta - 1) = 0$. Substituting $\tan \theta = n$ yields $2n(2n^2 - 1) = 0$.

Since line L_1 is not horizontal, $n \neq 0$. So $2n^2 - 1 = 0$, and thus $n^2 = \frac{1}{2}$. Therefore, $mn = 4n^2 = 4 \cdot \frac{1}{2} = \boxed{\text{(C) } 2}$.

-e_power_pi_times_i, edited by j314andrews

Solution 2 (Angle Bisector Theorem)

Since L_1 has 4 times the slope of L_2 , $m = 4n$. Let $O = (0, 0)$, $A = (1, 0)$, $B = (1, n)$, $C = (1, m) = (1, 4n)$. By the Angle Bisector Theorem, $\frac{OC}{OA} = \frac{BC}{AB}$. That is, $OC = \frac{3n}{n} = 3$.

By the Pythagorean Theorem, $AC = \sqrt{3^2 - 1^2} = 2\sqrt{2}$. So $m = 2\sqrt{2}$ and $n = \frac{2\sqrt{2}}{4} = \frac{\sqrt{2}}{2}$. Therefore, $mn = (2\sqrt{2}) \left(\frac{\sqrt{2}}{2}\right) = \boxed{\text{(C) } 2}$.

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13. (AHSME 1980) A bug (of negligible size) starts at the origin on the coordinate plane. First, it moves 1 unit right to $(1, 0)$. Then it makes a 90° counterclockwise and travels $\frac{1}{2}$ a unit to $\left(1, \frac{1}{2}\right)$. If it continues in this fashion, each time making a 90° degree turn counterclockwise and traveling half as far as in the previous move, to which of the following points will it come closest?

- (A) $\left(\frac{2}{3}, \frac{2}{3}\right)$ (B) $\left(\frac{4}{5}, \frac{2}{5}\right)$ (C) $\left(\frac{2}{3}, \frac{4}{5}\right)$ (D) $\left(\frac{2}{3}, \frac{1}{3}\right)$ (E) $\left(\frac{2}{5}, \frac{4}{5}\right)$

Answer: B

Solution 1

Each step the bug takes is half as long as its previous step. So each horizontal step is $\frac{1}{4}$ as long as and in the opposite direction of the previous horizontal step. Similarly, each vertical step is $\frac{1}{4}$ as long as and in the opposite direction the previous vertical step.

Therefore, the x -coordinate is the sum of an infinite geometric series with first term 1 and common ratio $-\frac{1}{4}$, so it must be $\frac{1}{1 - (-\frac{1}{4})} = \frac{4}{5}$. Similarly, the y -coordinate is the sum of an infinite geometric series with first term $\frac{1}{2}$ and common ratio $-\frac{1}{4}$, so it must be $\frac{\frac{1}{2}}{1 - (-\frac{1}{4})} = \frac{2}{5}$. Therefore, the bug's path approaches $\boxed{\text{(B) } \left(\frac{4}{5}, \frac{2}{5}\right)}$.

Solution 2 (Complex Plane)

Consider the bug's path as an infinite sequence of vectors in the complex plane. Then each step can be represented by a complex number, with the first step being 1 and the second step being $\frac{1}{2}i$.

Since each step is half as long as and rotated 90° counterclockwise from the previous step, the complex number representing a step must be $\frac{1}{2}i$ times the complex number representing the previous step.

Therefore, the bug's path can be represented by an infinite geometric series with first term 1 and common ratio $\frac{1}{2}i$. The sum of this series is $\frac{1}{1 - \frac{1}{2}i} = \frac{2}{2 - i} = \frac{2(2+i)}{(2-i)(2+i)} =$

$\frac{4}{5} + \frac{2}{5}i$, which corresponds to $\boxed{\text{(B) } \left(\frac{4}{5}, \frac{2}{5}\right)}$.

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14. (*AHSME 1980*) If the function f defined by

$$f(x) = \frac{cx}{2x+3}, \quad x \neq -\frac{3}{2},$$

satisfies $x = f(f(x))$ for all real numbers x except $-\frac{3}{2}$, then c is

- (A) -3 (B) $-\frac{3}{2}$ (C) $\frac{3}{2}$ (D) 3 (E) not uniquely determined

Answer: A

Solution 1

Since $f(x) = \frac{cx}{2x+3}$, $f(f(x)) = \frac{c^2x}{2cx+6x+9} = x$ for all $x \neq -\frac{3}{2}$.

Multiplying both sides by $\frac{2cx+6x+9}{x}$ yields $c^2 = 2cx + 6x + 9$. That is, $x(2c+6) + (3+c)(3-c) = 0$. Therefore, both $2c+6 = 0$ and $(3+c)(3-c) = 0$, so $c = \boxed{(\mathbf{A}) - 3}$.

Solution 2

Since $f(x) = \frac{cx}{2x+3}$, $f(f(x)) = \frac{c^2x}{2cx+6x+9} = x$. However, f is undefined at $x = -\frac{3}{2}$.

Substituting $x = -\frac{3}{2}$ into $\frac{c^2x}{2cx+6x+9} = x$ yields $\frac{c}{2} = -\frac{3}{2}$, so $c = \boxed{(\mathbf{A}) - 3}$.

Solution 3

Since $f(x) = \frac{cx}{2x+3} = \frac{\frac{1}{2}cx}{x+\frac{3}{2}} = \frac{\frac{1}{2}cx+\frac{3}{4}c-\frac{3}{4}c}{x+\frac{3}{2}} = \frac{\frac{1}{2}c(x+\frac{3}{2})-\frac{3}{4}c}{x+\frac{3}{2}} = \frac{1}{2}c - \frac{\frac{3}{4}c}{x+\frac{3}{2}}$, the graph of $f(x)$ must be a hyperbola with asymptotes $x = -\frac{3}{2}$ and $y = \frac{1}{2}c$.

Since $f(f(x)) = x$, $f(x) = f^{-1}(x)$ and the graph of $f(x)$ is symmetrical across the line $y = x$. The reflections of the asymptotes of $f(x)$ across $y = x$ are $y = -\frac{3}{2}$ and $x = \frac{1}{2}c$, so $-\frac{3}{2} = \frac{1}{2}c$ and thus $c = \boxed{(\mathbf{A}) - 3}$.

-j314andrews

Solution 4

We are given a function $f(x) = \frac{cx}{2x+3}$ for $x \neq -\frac{3}{2}$, and it satisfies the functional equation $x = f(f(x))$ for all real numbers $x \neq -\frac{3}{2}$. We are tasked with finding the value of c .

Step 1: Calculate $f(f(x))$

We begin by calculating $f(f(x))$, which is the composition of the function $f(x)$ with itself. To do this, we substitute $f(x) = \frac{cx}{2x+3}$ into itself:

$$f(f(x)) = f\left(\frac{cx}{2x+3}\right).$$

Substitute $\frac{cx}{2x+3}$ into the formula for f :

$$f\left(\frac{cx}{2x+3}\right) = \frac{c\left(\frac{cx}{2x+3}\right)}{2\left(\frac{cx}{2x+3}\right) + 3}.$$

Simplify the numerator:

$$\text{Numerator} = c \times \frac{cx}{2x+3} = \frac{c^2x}{2x+3}.$$

Now simplify the denominator:

$$\text{Denominator} = 2 \times \frac{cx}{2x+3} + 3 = \frac{2cx}{2x+3} + 3.$$

To combine the terms in the denominator, express 3 with a denominator of $2x+3$:

$$\frac{2cx}{2x+3} + 3 = \frac{2cx + 3(2x+3)}{2x+3} = \frac{2cx + 6x + 9}{2x+3} = \frac{(2c+6)x + 9}{2x+3}.$$

Thus, we have:

$$f(f(x)) = \frac{\frac{c^2x}{2x+3}}{\frac{(2c+6)x+9}{2x+3}} = \frac{c^2x}{(2c+6)x+9}.$$

Step 2: Set up the functional equation

We are given that $f(f(x)) = x$ for all $x \neq -\frac{3}{2}$. Therefore, we set the expression for $f(f(x))$ equal to x :

$$\frac{c^2x}{(2c+6)x+9} = x.$$

Step 3: Solve the equation

To eliminate the fraction, multiply both sides of the equation by $(2c+6)x+9$:

$$c^2x = x((2c+6)x+9).$$

Expand both sides:

$$c^2x = (2c+6)x^2 + 9x.$$

Now, move all terms to one side of the equation:

$$0 = (2c+6)x^2 + 9x - c^2x.$$

Factor out x :

$$0 = x((2c+6)x + 9 - c^2).$$

Since this equation must hold for all $x \neq 0$, the expression in parentheses must be equal to zero:

$$(2c+6)x + 9 - c^2 = 0.$$

This simplifies to:

$$(2c+6)x + (9 - c^2) = 0.$$

For this to hold for all $x \neq 0$, the coefficient of x must be zero, and the constant term must also be zero. Thus, we have the system of equations:

- $2c + 6 = 0$
- $9 - c^2 = 0$

Step 4: Solve for c

From $2c + 6 = 0$, we solve for c :

$$2c = -6 \quad \Rightarrow \quad c = -3.$$

Substitute $c = -3$ into $9 - c^2 = 0$:

$$9 - (-3)^2 = 9 - 9 = 0.$$

So, $c = -3$ satisfies both equations.

Final Answer:

The value of c is (A) -3.

15. (*AHSME 1980*) A store prices an item in dollars and cents so that when 4% sales tax is added, no rounding is necessary because the result is exactly n dollars where n is a positive integer. The smallest value of n is
 (A) 1 (B) 13 (C) 25 (D) 26 (E) 100

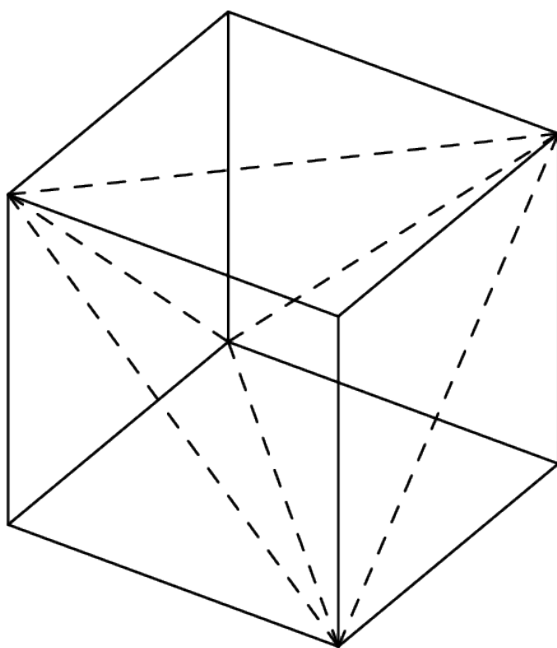
Answer: B

Solution

Let x be the price of the item in cents. The sales tax is $.04x = \frac{x}{25}$ cents, so $n = \frac{1}{100} \left(x + \frac{x}{25} \right) = \frac{26x}{2500} = \frac{13x}{1250}$.

Since x is positive integer, the smallest possible integer value of n occurs when $x = 1250$, and in this case n is (B) 13.

16. (*AHSME 1980*) Four of the eight vertices of a cube are the vertices of a regular tetrahedron. Find the ratio of the surface area of the cube to the surface area of the tetrahedron.



centering

- (A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) $\sqrt{\frac{3}{2}}$ (D) $\frac{2}{\sqrt{3}}$ (E) 2

Answer: B

Solution

Without loss of generality, suppose the cube has side length 1. By the Pythagorean Theorem, the side length of the tetrahedron is $\sqrt{2}$. Therefore, each face of the tetrahedron has area $\frac{(\sqrt{2})^2 \cdot \sqrt{3}}{4} = \frac{\sqrt{3}}{2}$ and its total surface area is $4 \cdot \frac{\sqrt{3}}{2} = 2\sqrt{3}$. The surface area of the cube is $6 \cdot 1^2 = 6$, so the ratio of the surface area of the cube to the surface area of the tetrahedron is $\frac{6}{2\sqrt{3}} = \boxed{\text{(B)} \sqrt{3}}$.

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17. (AHSME 1980) Given that $i^2 = -1$, for how many integers n is $(n+i)^4$ an integer?
 (A) none (B) 1 (C) 2 (D) 3 (E) 4

Answer: D

Solution 1

Expanding $(n+i)^4$ yields $n^4 + 4in^3 - 6n^2 - 4in + 1 = (n^4 - 6n^2 + 1) + (4n^3 - 4n)i$. This quantity is an integer if and only if $4n^3 - 4n = 4n(n+1)(n-1) = 0$, that is, if $n \in \{0, 1, -1\}$. Therefore, there are $\boxed{\text{(D)} 3}$ such values of n .

-aopsbandy, edited by j314andrews

Solution 2

For $(n+i)^4$ to be a real number, $n+i$ must be a scalar multiple of an eighth root of unity, that is, $n+i = r \left(\cos \frac{k\pi}{4} + i \sin \frac{k\pi}{4} \right)$ where $r \geq 0$ is a real number and k is an integer such that $0 \leq k \leq 7$. So $r \sin \frac{k\pi}{4} = 1$ and $\sin \frac{k\pi}{4}$ is positive. Therefore, $k \in \{1, 2, 3\}$.

If $k = 1$, $r = \frac{1}{\sin \frac{\pi}{4}} = \sqrt{2}$, $n = \sqrt{2} \cos \frac{\pi}{4} = 1$ and $(n+i)^4 = (1+i)^4 = -4$.

If $k = 2$, $r = \frac{1}{\sin \frac{\pi}{2}} = 1$, $n = \cos \frac{\pi}{2} = 0$ and $(n+i)^4 = i^4 = 1$.

If $k = 3$, $r = \frac{1}{\sin \frac{3\pi}{4}} = \sqrt{2}$, $n = \sqrt{2} \cos \frac{3\pi}{4} = -1$ and $(n+i)^4 = (-1+i)^4 = -4$.

In all these cases, $(n+i)^4$ is an integer, so $\{1, 0, -1\}$ are the $\boxed{\text{(D)} 3}$ possible values of n .

18. (AHSME 1980) If $b > 1$, $\sin x > 0$, $\cos x > 0$, and $\log_b \sin x = a$, then $\log_b \cos x$ equals
 (A) $2 \log_b(1 - b^{a/2})$ (B) $\sqrt{1 - a^2}$ (C) b^{a^2} (D) $\frac{1}{2} \log_b(1 - b^{2a})$ (E) none of these

Answer: D

Solution

Since $\log_b \sin x = a$, $b^a = \sin x$.

Let $c = \log_b \cos x$. Then $b^c = \cos x$.

Since $\sin^2 x + \cos^2 x = 1$,

$$(b^c)^2 + (b^a)^2 = 1$$

$$b^{2c} + b^{2a} = 1$$

$$b^{2c} = 1 - b^{2a}$$

$$\log_b(1 - b^{2a}) = 2c$$

$$c = \boxed{(\mathbf{D}) \frac{1}{2} \log_b(1 - b^{2a})}$$

-aopspandy

19. (*AHSME 1980*) Let C_1, C_2 and C_3 be three parallel chords of a circle on the same side of the center. The distance between C_1 and C_2 is the same as the distance between C_2 and C_3 . The lengths of the chords are 20, 16, and 8. The radius of the circle is

- (A) 12 (B) $4\sqrt{7}$ (C) $\frac{5\sqrt{65}}{3}$ (D) $\frac{5\sqrt{22}}{2}$ (E) not uniquely determined

Answer: D

Solution

Place this circle in the coordinate plane such that the center is at the origin all three chords are to the right of, and parallel to, the y -axis. The equation of this circle is $x^2 + y^2 = r^2$, where $r > 0$ is the radius of the circle.

Let A, B , and C be the upper ends of chords C_1, C_2 , and C_3 , respectively.

Since the x -axis is perpendicular to the three chords, it must bisect each chord. Let a be the x -coordinate A , and d be the distance between C_1 and C_2 , which is also the distance between C_2 and C_3 . Then A is at $(a, 10)$, B is at $(a + d, 8)$, and C is at $(a + 2d, 4)$.

Since A, B , and C all lie on this circle, the following equations must hold:

$$a^2 + 100 = r^2 \quad (\text{i})$$

$$(a + d)^2 + 64 = r^2 \quad (\text{ii})$$

$$(a + 2d)^2 + 16 = r^2 \quad (\text{iii})$$

Subtracting equation (i) from equation (ii) yields:

$$(a + d)^2 - a^2 - 36 = 0$$

$$2ad + d^2 = 36 \quad (\text{iv})$$

Subtracting equation (i) from equation (iii) yields:

$$(a + 2d)^2 - a^2 - 84 = 0$$

$$4ad + 4d^2 = 84$$

$$ad + d^2 = 21 \quad (\text{v})$$

Subtracting (v) from (iv) yields:

$$ad = 15 \quad (\text{vi})$$

Subtracting (vi) from (v) yields $d^2 - 6 = 0$, so $d = \sqrt{6}$.

Substituting $d = \sqrt{6}$ into (vi) yields $\sqrt{6}a = 15$, so $a = \frac{15}{\sqrt{6}} = \frac{5\sqrt{6}}{2}$.

Substituting $a = \frac{5\sqrt{6}}{2}$ yields $r^2 = \left(\frac{5\sqrt{6}}{2}\right)^2 + 100 = \frac{275}{2}$, so $r = \sqrt{\frac{275}{2}} = \boxed{(\mathbf{D}) \frac{5\sqrt{22}}{2}}$

20. (AHSME 1980) A box contains 2 pennies, 4 nickels, and 6 dimes. Six coins are drawn without replacement, with each coin having an equal probability of being chosen. What is the probability that the value of coins drawn is at least 50 cents?

(A) $\frac{37}{924}$ (B) $\frac{91}{924}$ (C) $\frac{127}{924}$ (D) $\frac{132}{924}$ (E) none of these

Answer: C

Solution 1

Each of the $\binom{12}{6} = 924$ possible combinations of 6 coins is equally likely to be selected.

If 6 dimes are selected, the total will be 60 cents. There is only $\binom{6}{6} = 1$ way to select 6 dimes.

If 5 dimes are selected, the total will be more than 50 cents. There are $\binom{6}{5} = 6$ ways to select the dimes, and $\binom{6}{1} = 6$ ways to select the other coin, for a total of $6 \cdot 6 = 36$ ways to select 5 dimes and 1 other coin.

If 4 dimes are selected, the total will be at least 50 cents if and only if the other 2 coins are both nickels. There are $\binom{6}{4} = 15$ ways to select 4 dimes and $\binom{4}{2} = 6$ ways to select 2 nickels, for a total of $15 \cdot 6 = 90$ ways to select 4 dimes and 2 nickels.

If 3 or fewer dimes are selected, the total will be less than 50 cents.

Therefore, there are $1 + 36 + 90 = 127$ ways to select 6 coins with a total of at least 50 cents, so the probability is $\boxed{\text{(C)} \frac{127}{924}}$.

-j314andrews

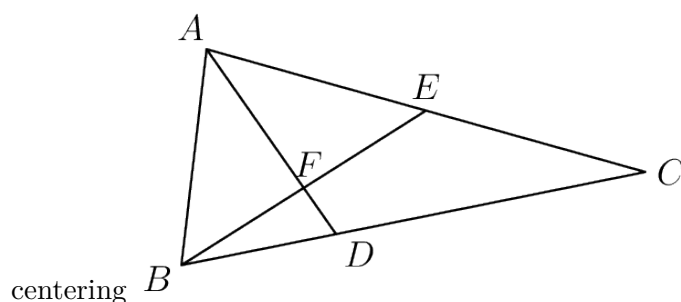
Solution 2

We want the number of Successful Outcomes over the number of Total Outcomes. We want to calculate the total outcomes first. Since we have 12 coins and we need to choose 6, we have $\binom{12}{6} = 924$ Total outcomes. For our successful outcomes, we can have (1)1 penny and 5 dimes, 2 nickels and 4 dimes, 1 nickel and 5 dimes, or 6 dimes.

For the case of 1 penny and 5 dimes, there are $\binom{6}{5}$ ways to choose the dimes and 2 ways to choose the pennies. That is $6 \cdot 2 = 12$ successful outcomes. For the case of 2 nickels and 4 dimes, we have $\binom{6}{4}$ ways to choose the dimes and $\binom{4}{2}$ ways to choose the nickels. We have $15 \cdot 6 = 90$ successful outcomes. For the case of 1 nickel and 5 dimes, we have $\binom{4}{1} \cdot \binom{6}{5} = 24$. Lastly, we have 6 dimes and 0 nickels, and 0 pennies,

so we only have one case. Therefore, we have $\frac{12 + 90 + 24 + 1}{924} = \frac{127}{924} = \boxed{C}$

21. (AHSME 1980)

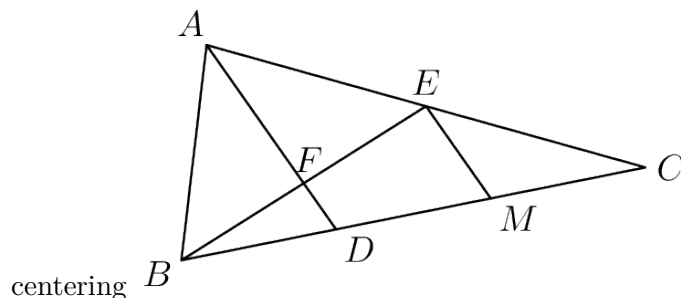


In triangle ABC , $\angle CBA = 72^\circ$, E is the midpoint of side AC , and D is a point on side BC such that $2BD = DC$; AD and BE intersect at F . The ratio of the area of $\triangle BDF$ to the area of quadrilateral $FDCE$ is

- (A) $\frac{1}{5}$ (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{2}{5}$ (E) none of these

Answer: A

Solution



Let M be the midpoint of $\overline{DC}$. Then $\triangle ECM \sim \triangle ACD$ and $\overline{EM} \parallel \overline{AD}$. Since $\overline{EM} \parallel \overline{FD}$, it follows that $\triangle BFD \sim \triangle BEM$.

Let a be the area of $\triangle BFD$. Since the sides of $\triangle BEM$ are twice as long as the corresponding sides of $\triangle BFD$, the area of $\triangle BEM$ must be $2^2 = 4$ times the area of $\triangle BFD$, that is, $4a$.

Since the height of $\triangle BEC$ is the same as the height of $\triangle BEM$ and the base of $\triangle BEC$ is $\frac{3}{2}$ times the base of $\triangle BEM$, the area of $\triangle BEC$ is $\frac{3}{2}$ times the area of $\triangle BEM$, or $\frac{3}{2} \cdot 4a = 6a$.

Thus the area of quadrilateral $FDCE$ is $6a - a = 5a$, so the ratio of the area of $\triangle BFD$ to the area of quadrilateral $FDCE$ is $\frac{a}{5a} = \boxed{\text{(A)} \frac{1}{5}}$.

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22. (AHSME 1980) For each real number x , let $f(x)$ be the minimum of the numbers $4x + 1$, $x + 2$, and $-2x + 4$. Then the maximum value of $f(x)$ is
- (A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{5}{2}$ (E) $\frac{8}{3}$

Answer: E

Solution

Let $a(x) = 4x + 1$, $b(x) = x + 2$, and $c(x) = -2x + 4$. Then $f(x) = \min(a(x), b(x), c(x))$.

$a(x) - b(x) = 3x - 1$, so $a(x) > b(x)$ when $x > \frac{1}{3}$, $a(x) = b(x)$ when $x = \frac{1}{3}$, and $a(x) < b(x)$ when $x < \frac{1}{3}$.

$a(x) - c(x) = 6x - 3$, so $a(x) > c(x)$ when $x > \frac{1}{2}$, $a(x) = c(x)$ when $x = \frac{1}{2}$, and $a(x) < c(x)$ when $x < \frac{1}{2}$.

$b(x) - c(x) = 3x - 2$, so $b(x) > c(x)$ when $x > \frac{2}{3}$, $b(x) = c(x)$ when $x = \frac{2}{3}$, and $b(x) < c(x)$ when $x < \frac{2}{3}$.

If $x \leq \frac{1}{3}$, $a(x) \leq b(x) < c(x)$, so $f(x) = a(x) = 4x + 1$.

If $\frac{1}{3} \leq x \leq \frac{1}{2}$, $b(x) \leq a(x) \leq c(x)$, so $f(x) = b(x) = x + 2$.

If $\frac{1}{2} \leq x \leq \frac{2}{3}$, $b(x) \leq c(x) \leq a(x)$, so $f(x) = b(x) = x + 2$.

If $x \geq \frac{2}{3}$, $c(x) \leq b(x) < a(x)$, so $f(x) = c(x) = -2x + 4$.

So $f(x)$ is increasing while $x < \frac{2}{3}$ and decreasing while $x > \frac{2}{3}$, and therefore, $f(x)$ is maximized at $x = \frac{2}{3}$, with a value of $f\left(\frac{2}{3}\right) = \frac{2}{3} + 2 = \boxed{\text{(E)} \frac{8}{3}}$

23. (AHSME 1980) Line segments drawn from the vertex opposite the hypotenuse of a right triangle to the points trisecting the hypotenuse have lengths $\sin x$ and $\cos x$, where x is a real number such that $0 < x < \frac{\pi}{2}$. The length of the hypotenuse is
- (A) $\frac{4}{3}$ (B) $\frac{3}{2}$ (C) $\frac{3\sqrt{5}}{5}$ (D) $\frac{2\sqrt{5}}{3}$ (E) not uniquely determined

Answer: C

Solution 1

Let A , B , and C be the vertices of the right triangle, where C is the right angle. Let a , b , and c be the lengths of the sides opposite A , B , and C , respectively. Place $\triangle ABC$ in the coordinate plane such that C is at the origin, B lies on the positive x -axis, and A lies on the positive y -axis. Then A has coordinates $(0, b)$ and B has coordinates $(a, 0)$.

Let M and N be the trisection points of $\overline{AB}$, with M closer to A and N closer to B . Then M has coordinates $\left(\frac{1}{3}a, \frac{2}{3}b\right)$ and N has coordinates $\left(\frac{2}{3}a, \frac{1}{3}b\right)$.

Let $m = CM$ and $n = CN$. Then $m^2 = \frac{1}{9}a^2 + \frac{4}{9}b^2$ and $n^2 = \frac{4}{9}a^2 + \frac{1}{9}b^2$. Adding these two equations yields $m^2 + n^2 = \frac{5}{9}(a^2 + b^2) = \frac{5}{9}c^2$.

Either $m = \sin x$ and $n = \cos x$, or $m = \cos x$ and $n = \sin x$. In both cases, $m^2 + n^2 = \sin^2 x + \cos^2 x = 1$. Therefore, $\frac{5}{9}c^2 = 1$ and $c = \sqrt{\frac{9}{5}} = \boxed{\text{(C)} \frac{3\sqrt{5}}{5}}$.

-j314andrews

Solution 2 (Stewart's Theorem)

Let A , B , and C be the vertices of the right triangle, where C is the right angle. Let a , b , and c be the lengths of the sides opposite A , B , and C , respectively. Let M and N be the trisection points of $\overline{AB}$. Let $m = CM$ and $n = CN$.

Using Stewart's Theorem on $\triangle ABC$ with segment CM yields:

$$m^2 \cdot c + \frac{2}{3}c \cdot \frac{1}{3}c \cdot c = a^2 \cdot \frac{1}{3}c + b^2 \cdot \frac{2}{3}c$$

Dividing both sides by c and isolating m^2 yields:

$$m^2 = \frac{1}{3}a^2 + \frac{2}{3}b^2 - \frac{2}{9}c^2 \quad (\text{i})$$

Using Stewart's Theorem on $\triangle ABC$ with segment CN yields:

$$n^2 \cdot c + \frac{2}{3}c \cdot \frac{1}{3}c \cdot c = a^2 \cdot \frac{2}{3}c + b^2 \cdot \frac{1}{3}c$$

Dividing both sides by c and isolating n^2 yields:

$$n^2 = \frac{2}{3}a^2 + \frac{1}{3}b^2 - \frac{2}{9}c^2 \quad (\text{ii})$$

Adding equations (i) and (ii) yields:

$$m^2 + n^2 = a^2 + b^2 - \frac{4}{9}c^2$$

Either $m = \sin x$ and $n = \cos x$, or $m = \cos x$ and $n = \sin x$. In both cases, $m^2 + n^2 = \sin^2 x + \cos^2 x = 1$. Also, by the Pythagorean theorem, $a^2 + b^2 = c^2$.

Therefore, $1 = c^2 - \frac{4}{9}c^2 = \frac{5}{9}c^2$, so $c^2 = \frac{9}{5}$ and $c = \sqrt{\frac{9}{5}} = \boxed{\text{(C)} \frac{3\sqrt{5}}{5}}$.

24. (AHSME 1980) For some real number r , the polynomial $8x^3 - 4x^2 - 42x + 45$ is divisible by $(x - r)^2$. Which of the following numbers is closest to r ?

(A) 1.22 (B) 1.32 (C) 1.42 (D) 1.52 (E) 1.62

Answer: D

Solution 1

Since $(x - r)^2$ is a factor of $8x^3 - 4x^2 - 42x + 45$, r is a double root. Let s be the third root of $8x^3 - 4x^2 - 42x + 45$.

By Vieta's formulas, $2r + s = \frac{1}{2}$, so $s = \frac{1}{2} - 2r$.

Also, by Vieta's formulas, $r^2 + 2rs = -\frac{21}{4}$. Substituting $s = \frac{1}{2} - 2r$ yields $r^2 + 2r\left(\frac{1}{2} - 2r\right) = -3r^2 + r = -\frac{21}{4}$. So $12r^2 - 4r - 21 = 0$, that is, $(6r + 7)(2r - 3) = 0$.

Therefore, (r, s) is either $\left(\frac{3}{2}, -\frac{5}{2}\right)$ or $\left(-\frac{7}{6}, \frac{17}{6}\right)$.

Finally, by Vieta's formulas, $r^2s = -\frac{45}{8}$. Though $(r, s) = \left(-\frac{7}{6}, \frac{17}{6}\right)$ does not satisfy this equation, $(r, s) = \left(\frac{3}{2}, -\frac{5}{2}\right)$ does.

Therefore, $r = \frac{3}{2} = 1.5$, so $\boxed{\text{(D)} 1.52}$ is closest to r .

-e_pi_times_i, edited by j314andrews

Solution 2

By polynomial long division, $\frac{8x^3 - 4x^2 - 42x + 45}{x^2 - 2rx + r^2} = 8x + (16r - 4) + \frac{(24r^2 - 8r - 42)x + (-16r^3 + 4r^2 + 45)}{x^2 - 2rx + r^2}$.

So $(24r^2 - 8r - 42)x + (-16r^3 + 4r^2 + 45) = 0$, that is, r is a root of both $24r^2 - 8r - 42$ and $-16r^3 + 4r^2 + 45$.

Since $24r^2 - 8r - 42 = 2(6r + 7)(2r - 3)$, either $r = -\frac{7}{6}$ or $r = \frac{3}{2}$. By the Rational Root Theorem, $-\frac{7}{6}$ is not a root of $-16r^3 + 4r^2 + 45$. However, $-16\left(\frac{3}{2}\right)^3 + 4\left(\frac{3}{2}\right)^2 + 45 = -16 \cdot \frac{27}{8} + 4 \cdot \frac{9}{4} + 45 = -54 + 9 + 45 = 0$, so $\frac{3}{2}$ is a root of $-16r^3 + 4r^2 + 45$.

Therefore, $r = \frac{3}{2} = 1.5$, so $\boxed{\text{(D)} 1.52}$ is closest to r .

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Solution 3

Let $y = 2x$. Then $x = \frac{y}{2}$, so $8x^3 - 4x^2 - 42x + 45 = y^3 - y^2 - 21y + 45$.

By the Rational Root Theorem, any rational root of $y^3 - y^2 - 21y + 45$ is a factor of 45. Of the factors of 45, 3 and 5 are roots, and $y^3 - y^2 - 21y + 45 = (y - 3)^2(y - 5)$.

So $8x^3 - 4x^2 - 42x + 45 = (2x - 3)^2(2x - 5) = 8\left(x - \frac{3}{2}\right)^2\left(x - \frac{5}{2}\right)$, and $r = \frac{3}{2} = 1.5$.

So $\boxed{\text{(D)} 1.52}$ is closest to r .

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25. (AHSME 1980) In the non-decreasing sequence of odd integers $\{a_1, a_2, a_3, \dots\} = \{1, 3, 3, 3, 5, 5, 5, 5, 5, \dots\}$ each odd positive integer k appears k times. It is a fact that there are integers b, c , and d such that for all positive integers n , $a_n = b\lfloor\sqrt{n+c}\rfloor + d$, where $\lfloor x \rfloor$ denotes the largest integer not exceeding x . The sum $b + c + d$ equals
- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Answer: C

Solution 1

Define the sequence $\{x_n\} = \lfloor\sqrt{n}\rfloor$ for all non-negative integers n . Then for any non-negative integer m , $x_n = m$ if and only if $m \leq \sqrt{n} < m+1$, that is, $m^2 \leq n < (m+1)^2$. So each non-negative integer m appears $(m+1)^2 - m^2 = 2m+1$ times. So $\{x_0, x_1, x_2, x_3, \dots\} = \{0, 1, 1, 1, 2, 2, 2, 2, \dots\}$.

In the sequence a_n , each positive odd integer $2m+1$ appears $2m+1$ times. Notice that $\{2x_0+1, 2x_1+1, 2x_2+1, 2x_3+1, \dots\} = \{1, 3, 3, 3, 5, 5, 5, 5, \dots\} = \{a_1, a_2, a_3, a_4, \dots\}$.

Therefore, $a_n = 2x_{n-1} + 1 = 2\lfloor\sqrt{n-1}\rfloor + 1$, so $b = 2$, $c = -1$, and $d = 1$, and $b + c + d = \boxed{\text{(C) } 2}$.

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Solution 2

Because the set consists of odd numbers, and since $\lfloor\sqrt{n+c}\rfloor$ is an integer and can be odd or even, $b = 2$ and $|a| = 1$. However, given that $\lfloor\sqrt{n+c}\rfloor$ can be 0, $a = 1$. Then, $a_1 = 1 = 2\lfloor\sqrt{1+c}\rfloor + 1$, and $\lfloor\sqrt{1+c}\rfloor = 0$, and $c = -1$ because c is an integer. $b + c + d = 2 + (-1) + 1 = \boxed{\text{(C) } 2}$

-e_power_pi_times_i

26. (AHSME 1980) Four balls of radius 1 are mutually tangent, three resting on the floor and the fourth resting on the others. A tetrahedron, each of whose edges have length s , is circumscribed around the balls. Then s equals
- (A) $4\sqrt{2}$ (B) $4\sqrt{3}$ (C) $2\sqrt{6}$ (D) $1 + 2\sqrt{6}$ (E) $2 + 2\sqrt{6}$

Answer: E

Solution

Let A, B, C , and D be the centers of the four spheres of radius 1. These points must form a regular tetrahedron of side length 2. Let O be the center of $\triangle ABC$. Then $OA = \frac{2\sqrt{3}}{3}$. By the Pythagorean Theorem, the height DO of tetrahedron $ABCD$ must be $\sqrt{2^2 - \left(\frac{2\sqrt{3}}{3}\right)^2} = \frac{2\sqrt{6}}{3}$. Let I be the center of tetrahedron $ABCD$. Then $IABC, IABD, IACD$, and $IBCD$ are all congruent tetrahedra, each with $\frac{1}{4}$ of the volume of $ABCD$. Since tetrahedra $IABC$ and $ABCD$ share base ABC , the height IO of tetrahedron $IABC$, and therefore the inradius of tetrahedron $ABCD$, must be $\frac{1}{4} \cdot \frac{2\sqrt{6}}{3} = \frac{\sqrt{6}}{6}$.

Let $EFGH$ be the tetrahedron circumscribed around the spheres, with AE, BF, CG , and DH intersecting at I . Planes ABC and EFG must be parallel and a distance 1 from each other. So the distance from I to plane EFG , and therefore the inradius of tetrahedron $EFGH$, must be $1 + \frac{\sqrt{6}}{6}$. Therefore, the ratio of the inradius of tetrahedron $EFGH$ to the inradius of tetrahedron $ABCD$ is $\frac{1+\sqrt{6}/6}{\sqrt{6}/6} = 1 + \sqrt{6}$.

The ratio of the side length of tetrahedron $EFGH$ to the side length of tetrahedron $ABCD$ must also be $1 + \sqrt{6}$, so $s = 2(1 + \sqrt{6}) = 2 + 2\sqrt{6}$. (E)

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27. (AHSME 1980) The sum $\sqrt[3]{5 + 2\sqrt{13}} + \sqrt[3]{5 - 2\sqrt{13}}$ equals

(A) $\frac{3}{2}$ (B) $\frac{\sqrt[3]{65}}{4}$ (C) $\frac{1 + \sqrt[6]{13}}{2}$ (D) $\sqrt[3]{2}$ (E) none of these

Answer: E

Solution

Let $y = \sqrt[3]{5 + 2\sqrt{13}}$, $z = \sqrt[3]{5 - 2\sqrt{13}}$, and $x = \sqrt[3]{5 + 2\sqrt{13}} + \sqrt[3]{5 - 2\sqrt{13}} = y + z$. Then $x^3 = (y + z)^3 = y^3 + 3y^2z + 3yz^2 + z^3 = y^3 + z^3 + 3yz(y + z) = y^3 + z^3 + 3yzx$.

Note that $y^3 + z^3 = 5 + 2\sqrt{13} + 5 - 2\sqrt{13} = 10$ and $yz = \sqrt[3]{(5 + 2\sqrt{13})(5 - 2\sqrt{13})} = \sqrt[3]{5^2 - (2\sqrt{13})^2} = \sqrt[3]{-27} = -3$.

So $x^3 = 10 + 3 \cdot -3 \cdot x = 10 - 9x$, that is $x^3 + 9x - 10 = 0$. This factors to $(x - 1)(x^2 + x + 10) = 0$, which has $x = 1$ as its only real solution.

Therefore, $\sqrt[3]{5 + 2\sqrt{13}} + \sqrt[3]{5 - 2\sqrt{13}} = 1$ and the answer is (E) none of these.

28. (AHSME 1980) The polynomial $x^{2n} + 1 + (x + 1)^{2n}$ is not divisible by $x^2 + x + 1$ if n equals

(A) 17 (B) 20 (C) 21 (D) 64 (E) 65

Answer: C

Solution 1

The roots of $x^2 + x + 1$ are $-\frac{1}{2} + \frac{\sqrt{3}}{2}i = \cos \frac{2}{3}\pi + i \sin \frac{2}{3}\pi$ and $-\frac{1}{2} - \frac{\sqrt{3}}{2}i = \cos \frac{4}{3}\pi + i \sin \frac{4}{3}\pi$, which are the primitive third roots of unity.

Let $w = -\frac{1}{2} + \frac{\sqrt{3}}{2}i = \cos \frac{2}{3}\pi + i \sin \frac{2}{3}\pi$. Note that $-\frac{1}{2} - \frac{\sqrt{3}}{2}i = \cos \frac{4}{3}\pi + i \sin \frac{4}{3}\pi = w^2 = \bar{w}$. So by the conjugate root theorem, $x^{2n} + 1 + (1 + x)^{2n}$ is divisible by $x^2 + x + 1$ if and only if w is a root of $x^{2n} + 1 + (1 + x)^{2n}$.

Also, $w^{2n} + 1 + (w + 1)^{2n} = w^{2n} + 1 + (-w^2)^{2n} = w^{2n} + 1 + w^{4n} = w^{2n} + 1 + w^n = w^{2n} + w^n + 1$.

If $n \equiv 0 \pmod{3}$, $w^{2n} + w^n + 1 \equiv 1 + 1 + 1 = 3$.

If $n \equiv 1 \pmod{3}$, $w^{2n} + w^n + 1 \equiv w^2 + w + 1 = 0$.

If $n \equiv 2 \pmod{3}$, $w^{2n} + w^n + 1 \equiv w + w^2 + 1 = 0$.

Therefore, $x^{2n} + 1 + (1 + x)^{2n}$ is not divisible by $x^2 + x + 1$ if and only if n is divisible by 3. Of the answer choices, only (C) 21 is divisible by 3.

Solution 2

Let $h(x) = x^2 + x + 1$.

Then $(x + 1)^{2n} = (x^2 + 2x + 1)^n = (x + h(x))^n = \sum_{k=0}^n \binom{n}{k} x^k (h(x))^{n-k} = \sum_{k=0}^{n-1} \binom{n}{k} x^k (h(x))^{n-k} + x^n = h(x) \left(\sum_{k=0}^{n-1} \binom{n}{k} x^k (h(x))^{n-k-1} \right) + x^n$.

Let $g(x) = \sum_{k=0}^{n-1} \binom{n}{k} x^k (h(x))^{n-k-1}$. Then $x^{2n} + 1 + (x + 1)^{2n} = x^{2n} + 1 + g(x)h(x) + x^n$, so $x^{2n} + 1 + (x + 1)^{2n}$ is divisible by $x^2 + x + 1$ if and only if $x^{2n} + x^n + 1$ is divisible by $x^2 + x + 1$.

Let k be any non-negative integer, and let $r \in \{0, 1, 2\}$. Then $x^{3k+r} - x^r = x^r(x^{3k} - 1) = x^r(x^3 - 1) \left(\sum_{j=0}^{k-1} x^{3j} \right) = x^r(x-1)(x^2+x+1) \left(\sum_{j=0}^{k-1} x^{3j} \right)$.

Therefore, $x^{3k+r} - x^r$ is divisible by $x^2 + x + 1$.

If n is a nonnegative integer, then n can be written as $3k + r$, where nonnegative integer k and $r \in \{0, 1, 2\}$ are the quotient and remainder when n is divided by 3.

If $r = 2$, then $x^{2n} + x^n + 1 = x^{6k+4} + x^{3k+2} + 1 = x^{3(2k+1)+1} + x^{3k+2} + 1 = (x^{3(2k+1)+1} - x) + (x^{3k+2} - x^2) + x^2 + x + 1$. Since $x^{3(2k+1)+1} - x$ and $(x^{3k+2} - x^2)$ are divisible by $x^2 + x + 1$, $x^{2n} + x^n + 1$ is divisible by $x^2 + x + 1$.

If $r = 1$, then $x^{2n} + x^n + 1 = x^{6k+2} + x^{3k+1} + 1 = x^{3(2k)+2} + x^{3k+1} + 1 = (x^{3(2k)+2} - x^2) + (x^{3k+1} - x) + x^2 + x + 1$. Since $x^{3(2k)+2} - x^2$ and $(x^{3k+1} - x)$ are divisible by $x^2 + x + 1$, $x^{2n} + x^n + 1$ is divisible by $x^2 + x + 1$.

If $r = 0$, then $x^{2n} + x^n + 1 = x^{6k} + x^{3k} + 1 = (x^{6k} - 1) + (x^{3k} - 1) + 3$. Since $x^{6k} - 1$ and $x^{3k} - 1$ are divisible by $x^2 + x + 1$, $x^{2n} + x^n + 1$ is not divisible by $x^2 + x + 1$.

So $x^{2n} + 1 + (1+x)^{2n}$ is not divisible by $x^2 + x + 1$ if and only if $r = 0$, that is, n is divisible by 3. The only answer choice that is divisible by 3 is (C) 21.

-Wei, edited by j314andrews

Solution 3

We start by noting that

$$x + 1 \equiv -x^2 \pmod{x^2 + x + 1}.$$

Let $n = 3k + r$, where $r \in \{0, 1, 2\}$.

Thus we have

$$x^{4n} + x^{2n} + 1 \equiv x^{4r} + x^{2r} + 1 \pmod{x^3 - 1}.$$

When $r = 0$,

$$x^{4n} + x^{2n} + 1 \equiv 3 \pmod{x^3 - 1}.$$

When $r = 1$,

$$x^{4n} + x^{2n} + 1 \equiv x^2 + x + 1 \pmod{x^3 - 1},$$

which will be divisible by $x^2 + x + 1$.

When $r = 2$,

$$x^{4n} + x^{2n} + 1 \equiv x^2 + x + 1 \pmod{x^3 - 1},$$

which will also be divisible by $x^2 + x + 1$.

Thus $r \neq 0$, so n cannot be divisible by 3, and the answer is (C).

29. (AHSME 1980) How many ordered triples (x, y, z) of integers satisfy the system of equations below?

$$\begin{aligned} x^2 - 3xy + 2y^2 - z^2 &= 31 \\ -x^2 + 6yz + 2z^2 &= 44 \\ x^2 + xy + 8z^2 &= 100 \end{aligned}$$

- (A) 0 (B) 1 (C) 2 (D) a finite number greater than 2 (E) infinitely many

Answer: A

Solution 1 (Sum of Squares)

We are considering the sum of three equations:

$$x^2 - 2xy + 2y^2 + 6yz + 9z^2 = (x - y)^2 + (y + 3z)^2 = 175.$$

Since x, y, z are integers, this implies that 175 can be written as the sum of two squares, i.e.,

$$175 = a^2 + b^2,$$

where a and b are integers.

However, $175 \equiv 3 \pmod{4}$, but a^2 and b^2 must each be congruent to either 0 or 1 (mod 4). So this equation, and therefore this system, has 0 integer solutions. (A)

Wwei.yu (talk) 22:09, 28 March 2020 (EDT)

Solution 2 (Eliminate Cross Terms)

First, to eliminate the xy term, multiply both sides of the third equation by 3 to get $3x^2 + 3xy + 24z^2 = 300$, and then add it to the first equation to get $4x^2 + 2yz + 23z^2 = 331$.

Next, to eliminate the yz term, multiply both sides of the new equation by -3 to get $-12x^2 - 6yz - 69z^2 = -993$, and then add it to the second equation to get $-13x^2 - 67z^2 = -949$.

This equation can be rearranged to get $67z^2 = 949 - 13x^2 = 13(73 - x^2)$. Since 67 is not divisible by 13, z must be divisible by 13 and there exists an integer n such that $z = 13n$. Substituting $z = 13n$ and dividing both sides of this equation by 13 yields $871n^2 = 73 - x^2$. So $871n^2 \leq 73$, which is only true if $n = 0$. But $n = 0$ would require $x^2 = 73$, which is impossible since 73 is not a perfect square. Therefore, this equation (and also this system) has 0 integer solutions. (A)

-j314andrews, based on solution by Farenhajt

Solution 3 (Modular Arithmetic)

Reducing the second equation mod 2 yields $x^2 \equiv 0 \pmod{2}$. Therefore, $x \equiv 0 \pmod{2}$ and x is even.

Reducing the first equation mod 2 and substituting yields $z^2 \equiv 1 \pmod{2}$. Therefore, $z \equiv 1 \pmod{2}$ and z is odd.

Since x is even, $x^2 \equiv 0 \pmod{4}$, and since z is odd, $z^2 \equiv 1 \pmod{4}$. Also $6z \equiv 2z \equiv 2 \pmod{4}$. Reducing the second equation mod 4 and substituting yields $2y + 2 \equiv 0 \pmod{4}$, so y is odd.

Reducing the third equation mod 4 and substituting yields $xy \equiv 0 \pmod{4}$. Since y is odd, $x \equiv 0 \pmod{4}$.

Since y is odd, $y^2 \equiv 1 \pmod{4}$. Reducing the first equation mod 4 and substituting yields $1 \equiv 3 \pmod{4}$, which is impossible.

Therefore, this system has 0 integer solutions. (A)

-j314andrews

30. (*AHSME 1980*) A six digit number (base 10) is squarish if it satisfies the following conditions:

- (i) none of its digits are zero;
- (ii) it is a perfect square; and
- (iii) the first of two digits, the middle two digits and the last two digits of the number are all perfect squares when considered as two digit numbers.

How many squarish numbers are there?

- (A) 0 (B) 2 (C) 3 (D) 8 (E) 9

Answer: B

Solution

If N is a squarish number, then $N = 10000a^2 + 100b^2 + c^2$, where a , b , and c are all integers between 4 and 9 inclusive. The only way that N could be a perfect square would be if $N = (100a + c)^2$. Expansion yields $N = 10000a^2 + 200ac + c^2$, so $2ac = b^2$, and therefore b must be even.

If $b = 4$, then $ac = 8$, which is impossible.

If $b = 6$, then $ac = 18$, which is impossible.

If $b = 8$, then $ac = 32$, which is possible if (a, c) is either $(8, 4)$ or $(4, 8)$. This yields squarish numbers $646416 = 804^2$ and $164664 = 408^2$.

Therefore, there are 2 squarish numbers. (B)

Wwei.yu (talk)Wei

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